1 Einstein Equivalence Principle: from Gravity to Geometry

1.1 Motivation: The Einstein Equivalence Principle

The highly successful Newtonian theory of gravity can be succinctly summarised by two sets of differential equations, one describing the dynamics (motion) of particles in a given gravitational field (described by a potential ϕ), and the other describing the dynamics of the gravitational field itself, namely how ϕ is to be determined from a given mass configuration. The former takes the standard Newtonian form

$$m\ddot{\vec{x}} = \vec{F}_g = -m\vec{\nabla}\phi \tag{1.1}$$

(but we will come back in some detail below to the question if/why the same mass parameter m appears on both sides of this equation, so as to incorporate the observation, going back to Galileo, that "all bodies fall at the same rate in a a gravitational field"). The latter is the Poisson equation

$$\Delta\phi = 4\pi G_N \mu = (4\pi G_N/c^2)\rho \ , \tag{1.2}$$

with G_N denoting, here and throughout, Newton's constant, i.e. the gravitational coupling constant, and where μ is the mass density, and $\rho = \mu c^2$ the associated rest mass energy density - I will set $c = 1$ in the following and use ρ.

Let us start with the field equation. It is immediately evident that this cannot be the final story. Not only is this equation not Lorentz invariant. Because of the absence of time-derivatives in (1.2), it actually describes an "action at a distance" and an instantaneous propagation of the gravitational field to every point in space (if you wiggle your mass distribution here now, this will immediately effect the gravitational potential arbitrarily far away). This is something that Einstein had just successfully exorcised from other aspects of physics, and clearly Newtonian gravity had to be revised as well.

It is then also immediately clear that what would have to replace Newton's theory is something rather more complicated. The reason for this is that, according to Special Relativity, mass is just another form of energy. Then, since gravity couples to masses, in a relativistically invariant theory gravity will also have to couple to energy. In particular, therefore, gravity would have to couple to gravitational energy, i.e. to itself. As a consequence, the new gravitational field equations will, unlike Newton's, have to be non-linear: the field of the sum of two masses cannot equal the sum of the gravitational fields of the two masses because it should also take into account the gravitational energy of the two-body system.

Now, having realised that Newton's theory cannot be the final word on the issue, how does one go about finding a better theory?

I will first very briefly discuss (and then dismiss) what at first sight may appear to be the most natural and naive approach to formulating a relativistic theory of gravity,

namely the simple replacement of Newton's field equation (1.2) by its relativistically covariant version

$$\Delta\phi = 4\pi G_N \rho \quad \longrightarrow \quad \Box\phi = 4\pi G_N \rho \ , \tag{1.3}$$

where $\Box$ is the Lorentz invariant d'Alembert or wave operator. While this looks promising, something can't be quite right about this equation. We already know (from Special Relativity) that ρ is not a scalar but rather the 00-component of a tensor, the energy-momentum tensor T_{ab}, so if actually ρ appears on the right-hand side, ϕ cannot be a scalar, while if ϕ is a scalar something needs to be done to fix the right-hand side.

Turning first to the latter possibility, one option that suggests itself is to replace ρ by the trace $T = T^a_{\ a}$ of the energy-momentum tensor. This is by definition / construction a scalar, and it will agree with ρ in the non-relativistic limit (where rest mass dominates over other contributions). With the sign conventions that we will use, one has $T_{00} = \rho \Rightarrow T = -\rho + \ldots$, and thus a first attempt at fixing the above equation might look like

$$\Box\phi = -4\pi G_N T \ . \tag{1.4}$$

This is certainly an attractive equation, but it definitely has the drawback that it is too linear. Recall from the discussion above that the universality of gravity (coupling to all forms of matter) and the equivalence of mass and energy lead to the conclusion that gravity should couple to gravitational energy, invariably predicting non-linear (self-interacting) equations for the gravitational field. However, the left hand side could be such that it only reduces to $\Box$ or Δ of the Newtonian potential in the Newtonian limit of weak time-independent fields. Thus a second attempt at fixing the above equation might look like

$$\Box\Phi(\phi) = -4\pi G_N T \ , \tag{1.5}$$

where $\Phi(\phi) \approx \phi$ for weak fields.

Such scalar relativistic theories of gravity (or rather some minor variants thereof) were proposed and studied among others by Abraham, Mie, and Nordstrøm. As it stands, the field equation appears to be perfectly consistent (and it may be interesting to discuss if/how the Einstein equivalence principle, which will put us on our route towards metrics and space-time curvature is realised in such a theory).

However, regardless of this, this theory is incorrect simply because it is ruled out experimentally. The easiest way to see this (with hindsight) is to note that the energy-momentum tensor of Maxwell theory (7.47) is traceless (7.121), and thus the above equation would predict no coupling of gravity to the electro-magnetic field, in particular to light. Hence in such a theory there would be no deflection of light by the sun etc.[1]

[1] For more on the history and properties of scalar theories of gravity see the review by D. Giulini, *What is (not) wrong with scalar gravity?*, `arXiv:gr-qc/0611100`.

The other possibility to render (1.3) consistent is the, a priori perhaps much less compelling, option to think of ϕ and $\Delta\phi$ or $\Box\phi$ not as scalars but as (00)-components of some tensor, in which case one could try to salvage (1.3) by promoting it to a tensorial equation

$$\{\text{Some tensor generalising } \Delta\phi\}_{ab} \sim 4\pi G_N T_{ab} \ . \tag{1.6}$$

This appears to require not just one potential, but actually 10 of them,

$$\phi \to \phi_{ab} = \phi_{ba} \tag{1.7}$$

and this seems to be a rather crazy thing to do at this stage (in particular, without any insight into the nature of these potentials). However, this is indeed the form of the field equations for gravity (the Einstein equations) we will ultimately be led to (see section 19.4), but Einstein arrived at this in a completely different, and much more insightful, way.

Let us now, very briefly and in a streamlined way, try to retrace (one aspect of) Einstein's thoughts, namely on the relation between inertial and gravitational mass, which, as we will see, will lead us rather quickly to the geometric picture of gravity sketched in the Introduction.

To that end we return to the Newtonian equation of motion (1.1). Recall that in this Newtonian theory, there are two *a priori* completely independent concepts of mass:

- inertial mass m_i (or acceleration mass), which accounts for the resistance of a body or particle against acceleration and appears universally on the left-hand-side of the Newtonian equation of motion

  $$m_i \vec{a} = \vec{F} \tag{1.8}$$

  in conjunction with the acceleration $\vec{a}$;

- gravitational mass m_g which is the mass the gravitational field couples to, i.e. it is the gravitational charge of a particle,

  $$\vec{F}_g = -m_g \vec{\nabla}\phi \ . \tag{1.9}$$

Now it is an important empirical fact that the inertial mass of a body is equal to its gravitational mass. This realisation, at least with this clarity, is usually attributed to Newton, although it goes back to experiments and observations by Galileo usually paraphrased as "all bodies fall at the same rate in a gravitational field". (It is not true, though, that Galileo dropped objects from the leaning tower of Pisa to test this - he used an inclined plane, a water clock and a pendulum).

These experiments were later on improved, in various forms, by Huygens, Newton, Bessel and others and reached unprecedented accuracy with the work of Baron von

Eötvös (starting in 1889), who was able to show that inertial and gravitational mass of different materials (like wood and platinum) agree to one part in 10^9. In the 1950/60's, this was still further improved by R. Dicke to something like one part in 10^{11}. More recently, rumours of a 'fifth force', based on a reanalysis of Eötvös' data (but buried in the meantime) motivated experiments with even higher accuracy and no difference between m_i and m_g was found.

Newton's theory would in principle be perfectly consistent with $m_i \neq m_g$, just as the formally analogous equation for an electrically charged particle with charge q_e in an electrostatic field $\vec{E} = -\vec{\nabla}\phi$,

$$m_i \ddot{\vec{x}} = -q_e \vec{\nabla}\phi \quad , \tag{1.10}$$

is perfectly acceptable for any ratio q_e/m_i, and Einstein was very impressed with the observed equality of m_i and m_g. This should, he reasoned, not be a mere coincidence but is probably trying to tell us something rather deep about the nature of gravity.

To see what this could be, let us recall that there is a very common class of "forces" for which the equality between the inertial mass and the coupling constant is actually built in and automatic. These are the "pseudo-forces" or "fictitious forces" $\vec{P}$ (like centrifugal forces) which arise when one transforms the Newtonian equations of motion via a non-linear coordinate transformation to accelerated (or other non-Cartesian) coordinates,

$$x^i \to z^m = z^m(x^i) \tag{1.11}$$

(like spherical coordinates). These "forces" arise from the non-trivial transformation behaviour of the acceleration $\ddot{\vec{x}}$ under such non-linear coordinate transformations, and are therefore inevitably and automatically proportional to m_i,

$$m_i \ddot{\vec{x}} = \vec{F} \quad \Rightarrow \quad m_i \ddot{\vec{z}} = \vec{F} + \vec{P} \quad \text{with} \quad \vec{P} \sim m_i \quad . \tag{1.12}$$

To be explicit, note that if we perform such a time-independent coordinate transformation, the velocity and acceleration of a particle with trajectory $x^i(t)$ transform as

$$\begin{aligned} \dot{z}^m &= \frac{\partial z^m}{\partial x^i} \dot{x}^i \\ \ddot{z}^m &= \frac{\partial z^m}{\partial x^i} \ddot{x}^i + \frac{\partial^2 z^m}{\partial x^i \partial x^j} \dot{x}^i \dot{x}^j \end{aligned} \tag{1.13}$$

(cf. section 1.5 for an analogous calculation in relativistic mechanics in Minkowski space). The first line expresses the fact that velocities transform linearly (with the Jacobi matrix) under arbitrary coordinate transformations (and are thus the protyotype of what we will call vectors or tensors later on). In the coordinates z^m, the equations of motion thus take the form

$$m_i \ddot{z}^m = \frac{\partial z^m}{\partial x^i} F^i + P^m \equiv F^m + P^m \quad , \tag{1.14}$$

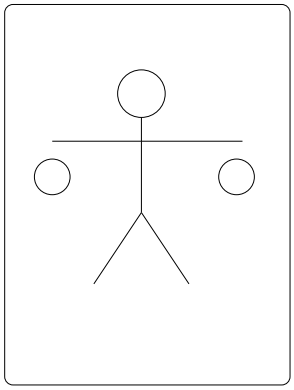


Figure 1: Experimenter and his two stones freely floating somewhere in outer space, i.e. in the absence of forces.

where it is the second term in the second line of (1.13) that gives rise to the (centrifugal etc.) pseudo-force

$$P^m = m_i \frac{\partial^2 z^m}{\partial x^i \partial x^j} \dot{x}^i \dot{x}^j \quad , \tag{1.15}$$

manifestly proportional to m_i. If one considers 3-dimensional coordinate transformations that depend explicitly on time, $z^m = z^m(x^i, t)$ (such as a transformation to a rotating reference system), then one finds additional contributions to the pseudo-force, like Coriolis forces etc. Conversely, any such pseudo-forces can be eliminated by transforming the equations of motion to a suitable (inertial) coordinate system or reference system.

With his unequalled talent for discovering profound truths in such simple observations, he concluded (calling this "der glücklichste Gedanke meines Lebens" (the happiest thought of my life)) that the equality of inertial and gravitational mass suggests a close relation between inertia and gravity itself, suggests, in fact, that locally effects of gravity and acceleration (or non-linear transformations of the reference system) are indistinguishable,

gravitational mass = inertial mass <u>because</u> (locally) GRAVITY = ACCELERATION

He substantiated this with some classical thought experiments, *Gedankenexperimente*, as he called them, which in the meantime have morphed into and have come to be known as the *elevator thought experiments*, which we will now discuss.

1. Consider somebody in a small sealed box (elevator) somewhere in outer space. In the absence of any forces, this person will float. Likewise, two stones he has just dropped (see Figure 1) will float with him.

2. Now assume (Figure 2) that somebody on the outside suddenly pulls the box up with a constant acceleration. Then of course, our friend will be pressed to the

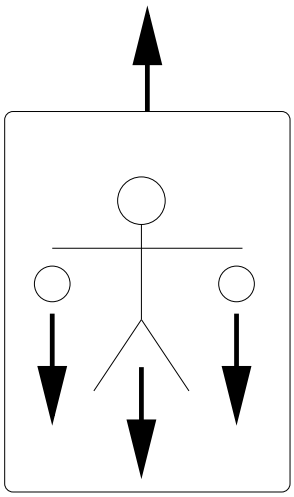


Figure 2: Constant acceleration upwards mimics the effect of a gravitational field: experimenter and stones drop to the bottom of the box.

bottom of the elevator with a constant force and he will also see his stones drop to the floor.

3. Now consider (Figure 3) this same box brought into a constant gravitational field. Then again, the experimenter will be pressed to the bottom of the elevator with a constant force and will see the stones drop to the floor. There is no experiment inside the elevator that permits him to decide if this is actually due to a gravitational field or due to the fact that somebody is pulling the elevator upwards.

   Thus our first lesson is that, indeed, locally the effects of acceleration and gravity are indistinguishable.

4. Now consider somebody cutting the cable of the elevator (Figure 4). Then the elevator will fall freely downwards but, as in Figure 1, our experimenter and his stones will float as in the absence of gravity.

   Thus lesson number two is that, locally the effect of gravity can be eliminated by going to a freely falling reference frame (or coordinate system). This should not come as a surprise, since this is of course built into the Newtonian theory: free fall in a constant gravitational field (in the vertial z-direction, say) is described by the equation

$$\ddot{z} = -g \ \ (+ \ \text{other forces}) \ . \tag{1.16}$$

   In the accelerated (freely falling) coordinate system

$$Z(z,t) = z + gt^2/2 \tag{1.17}$$

   the same physics is described by the equation

$$\ddot{Z} = 0 \ \ (+ \ \text{other forces}) \ , \tag{1.18}$$

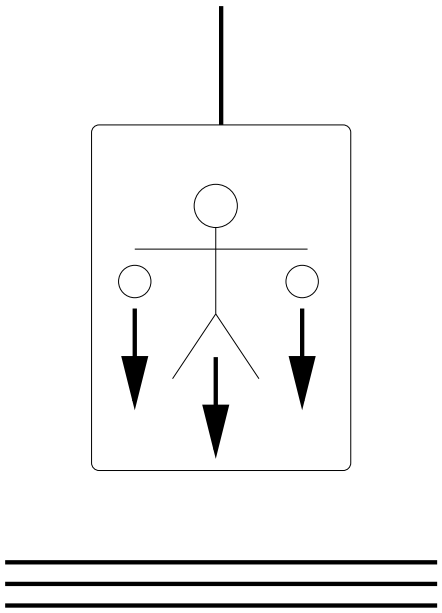


Figure 3: Effect of a constant gravitational field: indistinguishable for our experimenter from that of a constant acceleration in Figure 2.

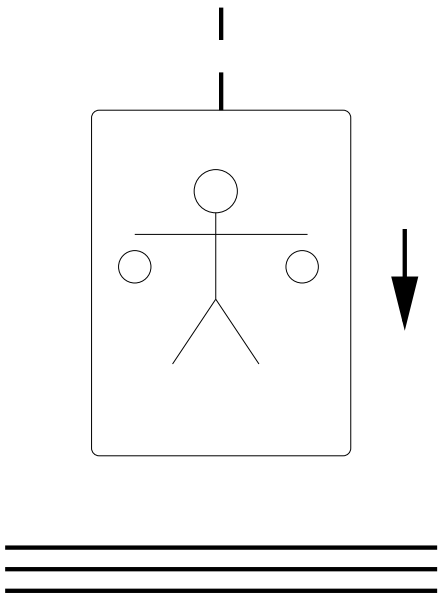


Figure 4: Free fall in a gravitational field has the same effect as no gravitational field (Figure 1): experimenter and stones float.

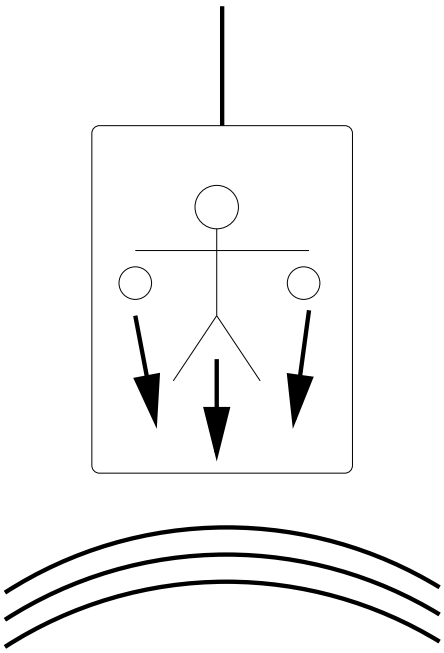


Figure 5: Experimenter and his stones in a non-uniform gravitational field: the stones will approach each other slightly as they fall to the bottom of the elevator.

and the effect of gravity has been eliminated completely by going to the freely falling reference system described by Z. The crucial point here is that in such a reference frame not only our observer will float freely, but because of the equality of inertial and gravitational mass he will also observe all other objects obeying the usual laws of motion in the absence of gravity.

5. In the above discussion, I have put the emphasis on constant accelerations and on 'locally'. To see the significance of this, consider our experimenter with his elevator in the gravitational field of the earth (Figure 5). This gravitational field is not constant but spherically symmetric, pointing towards the center of the earth. Therefore the stones will slightly approach each other as they fall towards the bottom of the elevator, in the direction of the center of the gravitational field.

6. Thus, if somebody cuts the cable now and the elevator is again in free fall (Figure 6), our experimenter will float again, so will the stones, but our experimenter will also notice that the stones move closer together for some reason. He will have to conclude that there is some force responsible for this.

   This is lesson number three: in a non-uniform gravitational field the effects of gravity cannot be eliminated by going to a freely falling coordinate system. This is only possible locally, on such scales on which the gravitational field is essentially constant.

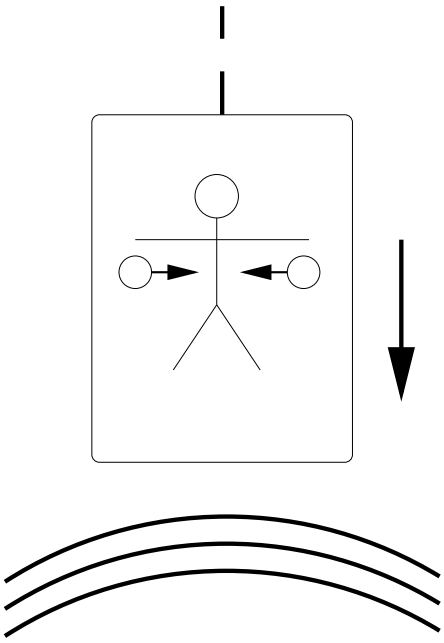


Figure 6: Experimentator and stones freely falling in a non-uniform gravitational field. The experimenter floats, so do the stones, but they move closer together, indicating the presence of some external force.

Einstein formalised the outcome of these thought experiments in what is now known as the *Einstein Equivalence Principle* which roughly states that physics in a freely falling frame in a gravitational field is the same as physics in an inertial frame in Minkowski space in the absence of gravitation. Two formulations are

> At every space-time point in an arbitrary gravitational field it is possible to choose a *locally inertial* (or *freely falling*) *coordinate system* such that, within a sufficiently small region of this point, the laws of nature take the same form as in unaccelerated Cartesian coordinate systems in the absence of gravitation.[2]

and

> Experiments in a sufficiently small freely falling laboratory, over a sufficiently short time, give results that are indistinguishable from those of the same experiments in an inertial frame in empty space.[3]

There are different versions of this principle depending on what precisely one means by 'the laws of nature'. If one just means the laws of Newtonian (or relativistic) mechanics,

[2] S. Weinberg, *Gravitation and Cosmology.*
[3] J. Hartle, *Gravity. An Introduction to Einstein's General Relativity.*

then this principle essentially reduces to the statement that inertial and gravitational mass are equal. Usually, however, this statement is taken to imply also Maxwell's theory, quantum mechanics etc.[4] What it pragmatically asserts in one of its stronger forms is that

> [...] there is no experiment that can distinguish a uniform acceleration from a uniform gravitational field. (J. Hartle, loc. cit.)

The power of the above principle, which we will regard as a heuristic guideline, rather than trying to (prematurely) give it a mathematically precise formulation, lies in the fact that we can combine it with our understanding of physics in accelerated reference systems to gain insight into the physics in a gravitational field. Two immediate consequences of this (which cannot be derived on the basis of Newtonian physics or Special Relativity alone) are

- light is deflected by a gravitational field just like material objects;
- clocks run slower in a gravitational field than in the absence of gravity.

To see the inevitability of the first assertion, imagine a lightray entering the rocket / elevator in Figure 1 horizontally through a window on the left hand side and exiting again at the same height through a window on the right. Now imagine, as in Figure 2, accelerating the elevator upwards. Then clearly the lightray that enters on the left will exit at a lower point of the elevator on the right because the elevator is accelerating upwards. By the equivalence principle one should observe exactly the same thing in a constant gravitational field (Figure 3). It follows that in a gravitational field the lightray is bent downwards, i.e. it experiences a downward acceleration with the (locally constant) gravitational acceleration g.

To understand the second assertion, one can e.g. simply appeal to the so-called "twin-paradox" of Special Relativity: the accelerated twin is younger than his unaccelerated inertial sibling. Hence accelerated clocks run slower than inertial clocks. Hence, by the equivalence principle, clocks in a gravitational field run slower than clocks in the absence of gravity.

Alternatively, one can imagine two observers at the top and bottom of the elevator, having identical clocks and sending light signals to each other at regular intervals as determined by their clocks. Once the elevator accelerates upwards, the observer at the bottom will receive the signals at a higher rate than he emits them (because he is accelerating towards the signals he receives), and he will interpret this as his clock

[4] For a discussion of different formulations of the equivalence principle and the logical relations among them, see E. di Casola, S. Liberati, S. Sonego, *Nonequivalence of equivalence principles*, `arXiv:1310.7426 [gr-qc]`.

running more slowly than that of the observer at the top. By the equivalence principle, the same conclusion now applies to two observers at different heights in a gravitational field. This can also be interpreted in terms of a gravitational redshift or blueshift (photons losing or gaining energy by climbing or falling in a gravitational field), and we will return to a more quantitative discussion of this effect in section 3.5.

1.2 Lorentz-Covariant Formulation of Special Relativity (Review)

What the equivalence principle tells us is that we can expect to learn something about the effects of gravitation by transforming the laws of nature (equations of motion) from an inertial Cartesian coordinate system to other (accelerated, curvilinear) coordinates. As a first step, we will, in section 1.3 below, discuss the above example of an observer undergoing constant acceleration in the context of special relativity.

As a preparation for this, and the remainder of the course, this section will provide a lightning review of the Lorentz-covariant formulation of special relativity, mainly to set the notation and conventions that will be used throughout, and only to the extent that it will be used in the following.

1. Minkowski space(-time)

   (a) The arena of special relativity is Minkowski space-time [henceforth *Minkowski space* for short, the union of space and time is implied by uttering the word "Minkowski"]. It is the space of events, labelled by 3 Cartesian spatial coordinates x^k and a time-coordinate t or, more usefully, by the coordinates

   $$(\xi^a) = (\xi^0 = ct, \xi^k = x^k) \ , \tag{1.19}$$

   where c is the speed of light. Typically in these notes ξ^a will indicate such a (locally) inertial coordinate system, whereas generic coordinates will be called x^μ etc. We will almost always work in units in which $c = 1$.

   (b) Minkowski space is equipped with a prescription for measuring distances, encoded in a line-element which, in these coordinates, takes the form

   $$ds^2 = -(d\xi^0)^2 + \sum_k (d\xi^k)^2 \ . \tag{1.20}$$

   (c) This can be written as

   $$ds^2 = -(d\xi^0)^2 + \sum_k (d\xi^k)^2 \equiv \eta_{ab}\, d\xi^a d\xi^b \ . \tag{1.21}$$

   with metric $(\eta_{ab}) = \mathrm{diag}(-1, +1, +1, +1)$ or, more explicitly,

   $$\eta_{ab} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix} \tag{1.22}$$

(thus we are using the "mostly plus" convention).

2. Lorentz Transformations

   (a) Lorentz transformations are by definition those linear transformations

$$\xi^a \mapsto \bar{\xi}^a = L^a_{\ b}\xi^b \tag{1.23}$$

   that leave the Minkowski line-element invariant,

$$d\bar{s}^2 \equiv \eta_{ab}d\bar{\xi}^a d\bar{\xi}^b = \eta_{ab}d\xi^a d\xi^b = ds^2 \quad \Leftrightarrow \quad \eta_{ab}L^a_{\ c}L^b_{\ d} = \eta_{cd} \ . \tag{1.24}$$

   In matrix notation this can also be written as

$$\bar{\xi} = L\xi \quad , \quad L^t\eta L = \eta \tag{1.25}$$

   where L^t is the transpose of L. This is the defining condition for Lorentz transformations, and the Lorentz signature analogue of the condition $R^t\mathbb{1}R = \mathbb{1}$ for an orthogonal transformation (rotation or reflection) in Euclidean space, with metric $\eta_{ab} \to \mathbb{1}_{ik} = \delta_{ik}$.
   Alternative notation:

$$\bar{\xi}^{\bar{a}} = L^{\bar{a}}_{\ b}\xi^b \quad \text{or} \quad \xi^{\bar{a}} = L^{\bar{a}}_{\ b}\xi^b \tag{1.26}$$

   Strictly speaking $\bar{\xi}^a$ and $\xi^{\bar{a}}$ may be considered to refer to two different quantities, to the coordinates of the new point $\bar{\xi}$ in the old coordinate system versus the coordinates of the old point ξ in the new coordinate system. However, for many elementary purposes this difference between what is known as the active (moving points) versus the passive (relabelling points) point of view is not crucial, and one should not be hung-up on notation: coordinates (and in particular indices referring to them) are fundamentally just bookkeeping devices so use whatever is convenient for current bookkeeping or other purposes.

   (b) Infinitesimal Lorentz rotations, i.e. Lorentz transformations with L of the form $L = \mathbb{1} + \omega$, ω infinitesimal, are characterised by

$$(\mathbb{1} + \omega)^t\eta(\mathbb{1} + \omega) = \eta \quad \Rightarrow \quad (\eta\omega) + (\eta\omega)^t = 0 \ . \tag{1.27}$$

   Thus the matrix $\eta\omega$ is anti-symmetric. In components, an infinitesimal Lorentz transformation therefore has the form

$$\delta\xi^a = \omega^a_{\ b}\xi^b \quad \text{with} \quad \omega_{ab} \equiv \eta_{ac}\omega^c_{\ b} = -\omega_{ba} \ . \tag{1.28}$$

   (c) Poincaré transformations are those affine transformations that leave the Minkowski line-element invariant. They are composed of Lorentz transformations and arbitrary constant translations and thus have the form

$$\xi^a \mapsto \bar{\xi}^a = L^a_{\ b}\xi^b + \zeta^a \ , \tag{1.29}$$

infinitesimally

$$\delta\xi^a = \omega^a_{\ b}\xi^b + \epsilon^a \quad . \tag{1.30}$$

Any two inertial systems in the sense of the equivalence principle of special relativity are related by a Poincaré transformation.

3. Distance & Causal Structure

   (a) The Minkowski metric defines the Lorentz (and Poincaré) invariant distance

   $$(\Delta\xi)^2 = \eta_{ab}(\xi^a_P - \xi^a_Q)(\xi^b_P - \xi^b_Q) \tag{1.31}$$

   betwen two events P and Q with coordinates ξ^a_P and ξ^a_Q respectively.

   (b) Depending on the sign of $(\Delta\xi)^2$, the two events P, Q are called, spacelike, lightlike (null) or timelike separated,

   $$(\Delta\xi)^2 = \begin{cases} > 0 & \text{spacelike} \quad \text{separated} \\ = 0 & \text{lightlike} \quad \text{separated} \\ < 0 & \text{timelike} \quad \text{separated} \end{cases} \tag{1.32}$$

   (c) The set of events that are lightlike separated from P define the lightcone at P. It consists of two components (joined at P), the future and the past lightcone, distinguished by the sign of $\xi^0_Q - \xi^0_P$ (positive for Q on the future lightcone, $\xi^0_Q > \xi^0_P$, negative for Q on the past lightcone).

4. Curves and Tangent Vectors

   (a) A parametrised curve is given by a map $\lambda \mapsto \xi^a(\lambda)$. The tangent vector to the curve at the point $\xi(\lambda_0)$ has components

   $$\xi'^a(\lambda_0) = \frac{d}{d\lambda}\xi^a(\lambda)|_{\lambda=\lambda_0} \quad . \tag{1.33}$$

   It is called spacelike, lightlike (null) or timelike, depending on the sign of $\eta_{ab}\xi'^a\xi'^b$,

   $$\eta_{ab}\xi'^a\xi'^b \quad \begin{cases} > 0 & \text{spacelike} \\ = 0 & \text{lightlike} \\ < 0 & \text{timelike} \end{cases} \tag{1.34}$$

   This sign (and hence this classification) depends only on the image of the curve, not its parametrisation.

   (b) A curve whose tangent vector is everywhere timelike is called a timelike curve (and likewise for lightlike and spacelike curves). A curve whose tangent vector is everywhere timelike or null (i.e. non-spacelike) is called a causal curve. Worldlines of massive particles are timelike curves, those of massless particles (light) are null curves.

(c) A natural Lorentz-invariant parametrisation of timelike curves is provided by the Lorentz-invariant proper time τ along the curves,

$$\xi^a = \xi^a(\tau) \ , \tag{1.35}$$

with

$$\begin{aligned} & c d\tau = \sqrt{-ds^2} = \sqrt{-\eta_{ab} d\xi^a d\xi^b} = \sqrt{-\eta_{ab} \xi'^a \xi'^b} d\lambda \\ \Rightarrow \quad & \eta_{ab} \frac{d\xi^a(\tau)}{d\tau} \frac{d\xi^b(\tau)}{d\tau} = -c^2 \ . \end{aligned} \tag{1.36}$$

Likewise spacelike curves are naturally parametrised by proper distance ds. The derivative with respect to proper time will be denoted by an overdot,

$$\dot{\xi}^a(\tau) = \frac{d}{d\tau} \xi^a(\tau) \ . \tag{1.37}$$

Because τ is Lorentz-invariant, $\bar{\tau} = \tau$, tangent vectors $\dot{\xi}^a$ of τ-parametrised curves transform linearly under Lorentz transformations,

$$\dot{\bar{\xi}}^a(\tau) = \frac{d}{d\tau} \bar{\xi}^a(\tau) = \frac{\partial \bar{\xi}^a}{\partial \xi^b} \frac{d}{d\tau} \xi^b(\tau) = L^a_{\ b} \dot{\xi}^b(\tau) \ . \tag{1.38}$$

These are the prototypes of what are called Lorentz vectors or, more generally, Lorentz tensors.

5. Lorentz Vectors

(a) Lorentz vectors (or 4-vectors) are objects with components v^a which transform under Lorentz transformations with the matrix $L^a_{\ b}$ (to be thought of as the Jacobian of the transformation relating $\bar{\xi}^a$ and ξ^a),

$$\bar{v}^a = L^a_{\ b} v^b \ . \tag{1.39}$$

(b) η_{ab} can be regarded as defining an indefinite scalar product on the space of Lorentz vectors, and the Minkowski norm $\eta_{ab} v^a v^b$ and the Minkowski scalar product $\eta_{ab} v^a w^b$ of Lorentz vectors are invariant under Lorentz transformations,

$$\eta_{ab} \bar{v}^a \bar{v}^b = \eta_{ab} v^a v^b \quad , \quad \eta_{ab} \bar{v}^a \bar{w}^b = \eta_{ab} v^a w^b \ . \tag{1.40}$$

A vector is called, spacelike, lightlike (null) or timelike depending on the sign of its Minkowski norm.

(c) One can identify Minkowski space with its "tangent space", i.e. with the vector space $\mathbb{V} = \mathbb{R}^{1,3}$ of 4-vectors equipped with the quadratic form or scalar product η_{ab} with signature (1,3).

6. Other Lorentz Tensors

(a) Lorentz scalars are objects that are invariant under Lorentz transformations. Examples are scalar products and norms of Lorentz vectors.

(b) Lorentz covectors are objects u_a that transform under Lorentz transformations with the dual (or contragredient = transpose inverse) representation

$$\Lambda = (L^t)^{-1} = \eta L \eta^{-1} \ , \tag{1.41}$$

i.e.

$$\bar{u}_a = \Lambda_a{}^b u_b \quad , \quad \Lambda_a{}^b = \eta_{ac} L^c{}_d \eta^{db} \ , \tag{1.42}$$

where η^{ab} denotes the components of the inverse metric η^{-1}. In terms of Λ, the condition that L is a Lorentz transformation (i.e. preserves η_{ab}) can evidently be written as

$$L^t \eta L = \eta \quad \Leftrightarrow \quad \Lambda \eta \Lambda^t = \eta \quad \Leftrightarrow \quad \Lambda_a^c \Lambda_b{}^d \eta_{cd} = \eta_{ab} \ . \tag{1.43}$$

Covectors can be regarded as elements of the dual $\mathbb{V}^*$ of the space $\mathbb{V}$ of 4-vectors, with u_a defining the Lorentz-invariant linear mapping

$$u : \ v \in \mathbb{V} \mapsto u(v) = u_a v^a \in \mathbb{R} \ . \tag{1.44}$$

Examples are $u_a = \eta_{ab} v^b \equiv v_a$ with v^a a Lorentz vector, the scalar product η_{ab} defining the isomorphism $\mathbb{V}^* \cong \mathbb{V}$.

(c) Lorentz (p,q)-tensors are objects that transform under Lorentz transformations like a product of p vectors and q covectors,

$$T^{a_1 \ldots a_p}_{c_1 \ldots c_q} \to \bar{T}^{a_1 \ldots a_p}_{c_1 \ldots c_q} = L^{a_1}_{\ b_1} \ldots L^{a_p}_{\ b_p} \Lambda_{c_1}^{\ d_1} \ldots \Lambda_{c_q}^{\ d_q} \ T^{b_1 \ldots b_p}_{d_1 \ldots d_q} \ . \tag{1.45}$$

In particular, direct products of vectors and covectors like $V^a W^b U_c$ are tensors. A special case is η_{ab}, which is a Lorentz-invariant $(0,2)$-tensor by definition,

$$\bar{\eta}_{ab} = \Lambda_a^c \Lambda_b{}^d \eta_{cd} = \eta_{ab} \ . \tag{1.46}$$

Linear combinations of (p,q)-tensors are again (p,q)-tensors. Arbitrary products and contractions of Lorentz tensors are again Lorentz tensors (and the tensor type can be read off from the number and position of the "free" indices).

7. Tensor Fields

(a) Lorentz tensor fields are assignments of Lorentz tensors to each point of Minkowski space,

$$T : \xi \mapsto T^{a_1 \ldots a_p}_{c_1 \ldots c_q}(\xi) \ . \tag{1.47}$$

(b) Given a vector field $V^a(\xi)$, $\eta_{ab}V^a(\xi)V^b(\xi)$ is an example of a scalar field, and given a scalar field $f(\xi)$, its partial derivatives give a covector field

$$U_a(\xi) = \partial_{\xi^a} f(\xi) \equiv \partial_a f(\xi) \tag{1.48}$$

(providing the justification for abbreviating $\partial_{\xi^a} = \partial_a$). More generally, the partial derivatives of the components of a (p,q)-tensor,

$$T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \quad \to \quad \partial_a T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \tag{1.49}$$

are the components of a $(p, q+1)$-tensor, and the wave operator

$$\Box = \eta^{ab}\partial_a\partial_b \tag{1.50}$$

is a Lorentz-invariant differential operator mapping (p,q) tensor fields to (p,q) tensor fields.

(c) Tensorial equations of the form

$$T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) = 0 \tag{1.51}$$

are Lorentz invariant in the sense that they are satisfied in one inertial system iff they are satisfied in all inertial systems. (Here and throughout these notes "iff" is the usual mathematicians' shorthand for "if and only if".)

8. Worldlines of Massive Particles

(a) In the covariant formulation, the timelike worldline of a massive particle is parametrised by proper time, $\xi^a = \xi^a(\tau)$. The velocity (tangent) vector

$$u^a \equiv \dot{\xi}^a(\tau) \tag{1.52}$$

is a Lorentz vector, normalised as

$$u^a u_a \equiv \eta_{ab}u^a u^b = -c^2 \ . \tag{1.53}$$

(b) The Lorentz-covariant acceleration is the 4-vector

$$a^c = \frac{d}{d\tau}u^c = \frac{d^2}{d\tau^2}\xi^c \ , \tag{1.54}$$

and the equation of motion of a massive free particle is

$$a^c = \frac{d^2}{d\tau^2}\xi^c(\tau) = 0 \ . \tag{1.55}$$

We will study this equation further (in any arbitrary coordinate system) in section 1.5. For observers with non-zero acceleration it follows from (1.53) by differentiation that a^c is orthogonal to u^b,

$$a^c u_c \equiv \eta_{cb}a^c u^b = 0 \ , \tag{1.56}$$

and therefore spacelike,

$$\eta_{cb}a^c a^b \equiv \mathsf{a}^2 > 0 \ . \tag{1.57}$$

Observers with constant acceleration will be the subject of section 1.3.

(c) The action for a free massive particle with worldline $\xi^a(\tau)$ is essentially the total proper time along the path,

$$S[\xi] = -mc^2 \int d\tau = -mc \int \sqrt{-\eta_{ab} d\xi^a d\xi^b} \ , \tag{1.58}$$

worldlines of free massive particles extremising (maximising) the proper time. In terms of an arbitrary parametrisation $\xi^a = \xi^a(\lambda)$ of the path, this action can be written as

$$S[\xi] = \int d\lambda \, L_\lambda \quad , \quad L_\lambda = -mc \left(-\eta_{ab} \frac{d\xi^a}{d\lambda} \frac{d\xi^b}{d\lambda} \right)^{1/2} \ . \tag{1.59}$$

A special choice is $\lambda = t$, for which

$$L_t = -mc^2 \sqrt{1 - \vec{v}^2/c^2} \qquad \vec{v} = d\vec{\xi}/dt = d\vec{x}/dt \ . \tag{1.60}$$

9. Energy-Momentum 4-Vector

(a) The covariant momenta p_a are defined by

$$p_a = \frac{\partial L_\lambda}{\partial (d\xi^a/d\lambda)} = m \eta_{ab} u^b \quad \Rightarrow \quad p^a = m u^a = m(d\xi^a/d\tau) \tag{1.61}$$

(independently of the choice of λ).

(b) Its components are

$$p^0 = E/c \quad , \quad p^k = p^{(c)k} \tag{1.62}$$

where $p^{(c)k}$ are the canonical momenta associated to the Lagrangian L_t,

$$p_k^{(c)} = \frac{\partial L_t}{\partial v^k} = m\gamma(v) v_k \ , \tag{1.63}$$

with $\gamma(v) = (1 - \vec{v}^2/c^2)^{-1/2}$, and $E = H$ is the corresponding energy or Hamiltonian

$$H = p_k^{(c)} v^k - L_t = m\gamma(v) c^2 \ . \tag{1.64}$$

(c) The p^a are the components of a Lorentz vector, the energy-momentum 4-vector. It satisfies the mass-shell relation

$$\eta_{ab} p^a p^b = -m^2 c^2 \quad \Leftrightarrow \quad E^2 = m^2 c^4 + \vec{p}^2 c^2 \ . \tag{1.65}$$

1.3 Accelerated Observers in Minkowski Space and the Rindler Metric

We return to the issue discussed in the context of the Einstein equivalence principle in section 1.1, namely physics as experienced by an observer undergoing constant acceleration (as a precursor to studying this observer in a genuine gravitational field), now specifically within the framework of special relativity.

Specialising (1.56) to an observer accelerating in the ξ^1-direction (so that in the momentary restframe of this observer one has $u^a = (1,0,0,0), a^a = (0,\mathsf{a},0,0)$), we will say that the observer undergoes constant acceleration if a is time-independent. To determine the worldline of such an observer, we note that the general solution to (1.53) with $u^2 = u^3 = 0$,

$$\eta_{ab}u^a u^b = -(u^0)^2 + (u^1)^2 = -1 \quad , \tag{1.66}$$

is

$$u^0 = \cosh F(\tau) \quad , \quad u^1 = \sinh F(\tau) \tag{1.67}$$

for some function $F(\tau)$. Thus the acceleration is

$$a^a = \dot{F}(\tau)(\sinh F(\tau), \cosh F(\tau), 0, 0) \quad , \tag{1.68}$$

with norm

$$\eta_{ab}a^a a^b = \mathsf{a}^2 = \dot{F}^2 \quad , \tag{1.69}$$

and an observer with constant acceleration is characterised by $F(\tau) = \mathsf{a}\tau$,

$$u^a(\tau) = (\cosh \mathsf{a}\tau, \sinh \mathsf{a}\tau, 0, 0) \quad . \tag{1.70}$$

This can now be integrated, and in particular

$$\xi^a(\tau) = (\mathsf{a}^{-1}\sinh \mathsf{a}\tau, \mathsf{a}^{-1}\cosh \mathsf{a}\tau, 0, 0) \tag{1.71}$$

is the worldline of an observer with constant acceleration a and initial condition $\xi^a(\tau = 0) = (0, \mathsf{a}^{-1}, 0, 0)$. The worldlines of this observer is the hyperbola

$$\eta_{ab}\xi^a\xi^b = -(\xi^0)^2 + (\xi^1)^2 = \mathsf{a}^{-2} \tag{1.72}$$

in the quadrant $\xi^1 > |\xi^0|$ of Minkowski space-time.

We can now ask the question what the Minkowski metric or line-element looks like in the restframe of such an observer. Note that one cannot expect this to be again the constant Minkowski metric η_{ab}: the transformation to an accelerated reference system, while certainly allowed in special relativity, is not a Lorentz transformation, while η_{ab} is, by definition, invariant under Lorentz-transformations.

We are thus looking for coordinates that are adapted to these accelerated observers in the same way that the inertial coordinates are adapted to static observers (ξ^0 is proper time, and the spatial components ξ^i remain constant). In other words, we seek a coordinate transformation $(\xi^0, \xi^1) \to (\eta, \rho)$ such that the worldlines of these accelerated observers are characterised by $\rho =$ constant (this is what we mean by restframe, the observer stays at a fixed value of ρ) and ideally such that then η is proportional to the proper time of the observer.

Comparison with (1.71) suggests the coordinate transformation

$$\xi^0(\eta, \rho) = \rho \sinh \eta \qquad \xi^1(\eta, \rho) = \rho \cosh \eta \quad . \tag{1.73}$$

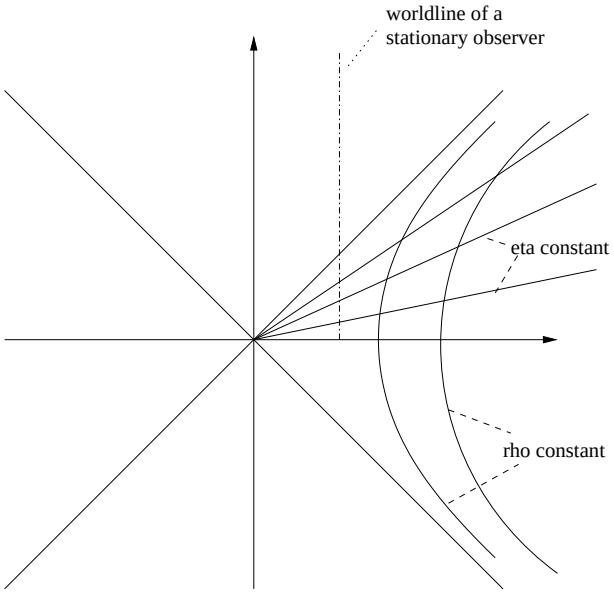


Figure 7: Rindler metric: Rindler coordinates (η, ρ) cover the first quadrant $\xi^1 > |\xi^0|$. Indicated are lines of constant ρ (hyperbolas, worldlines of constantly accelerating observers) and lines of constant η (straight lines through the origin). The quadrant is bounded by the lightlike lines $\xi^0 = \pm\xi^1 \Leftrightarrow \eta = \pm\infty$. An inertial observer reaches and crosses the line $\eta = \infty$ in finite proper time $\tau = \xi^0$.

It is now easy to see that in terms of these new coordinates the 2-dimensional Minkowski metric $ds^2 = -(d\xi^0)^2 + (d\xi^1)^2$ (we are now suppressing, here and in the remainder of this subsection, the transverse spectator dimensions 2 and 3) takes the form

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad . \tag{1.74}$$

This is the so-called *Rindler metric*.

Let us try to gain a better understanding of these Rindler coordinates (illustrated in Figure 7 - see also Figure 25 in section 28.4 for a *Penrose Diagram* illustration).

- The Rindler coordinates ρ and η are obvisouly in some sense hyperbolic (Lorentzian) analogues of polar coordinates ($x = r\cos\phi, y = r\sin\phi, ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2$). In particular, since

  $$(\xi^1)^2 - (\xi^0)^2 = \rho^2 \qquad , \qquad \frac{\xi^0}{\xi^1} = \tanh\eta \quad , \tag{1.75}$$

  by construction the lines of constant ρ, $\rho = \rho_0$, are hyperbolas, $(\xi^1)^2 - (\xi^0)^2 = \rho_0^2$, while the lines of constant $\eta = \eta_0$ are straight lines through the origin, $\xi^0 = (\tanh\eta_0)\xi^1$.

- The null lines $\xi^0 = \pm\xi^1$ correspond to $\eta = \pm\infty$. Thus the Rindler coordinates cover the first quadrant $\xi^1 > |\xi^0|$ of Minkowski space and can be used as coordinates there.

- The metric in these new coordinates is time-independent, where time means η, and time-independent means that the coefficients of the metric or line-element in (1.74) do not depend on η. This is due to the fact that the generator ∂_η of η-"time evolution" is actually the generator of a Lorentz boost in the (ξ^0, ξ^1)-plane in Minkowski space,

$$\partial_\eta = (\partial_\eta \xi^0)\partial_{\xi^0} + (\partial_\eta \xi^1)\partial_{\xi^1} = \xi^1 \partial_{\xi^0} + \xi^0 \partial_{\xi^1} \quad . \tag{1.76}$$

  Since a Lorentz boost leaves the Minkowski metric invariant, the latter has to be invariant under translations in η, i.e. it has to be η-independent, as is indeed the case.

- Along the worldline of an observer with constant ρ one has $d\tau = \rho_0 d\eta$, so that his proper time parametrised path is

$$\xi^0(\tau) = \rho_0 \sinh \tau/\rho_0 \qquad \xi^1(\tau) = \rho_0 \cosh \tau/\rho_0 \quad , \tag{1.77}$$

  and his 4-velocity is given by

$$u^0 = \frac{d}{d\tau}\xi^0(\tau) = \cosh \tau/\rho_0 \qquad u^1 = \frac{d}{d\tau}\xi^1(\tau) = \sinh \tau/\rho_0 \quad . \tag{1.78}$$

  These satisfy $-(u^0)^2+(u^1)^2 = -1$ (as they should), and comparison with (1.70,1.71) shows that the observer's (constant) acceleration is $\mathsf{a} = 1/\rho_0$.

Even though (1.74) is just the metric of Minkowski space-time, written in accelerated coordinates, this metric exhibits a number of interesting features that are prototypical of more general metrics that one encounters in general relativity:

1. First of all, we notice that the coefficients of the line element (metric) in (1.74) are no longer constant (space-time independent). Since in the case of constant acceleration we are just describing a "fake" gravitational field, this dependence on the coordinates is such that it can be completely and globally eliminated by passing to appropriate new coordinates (namely inertial Minkowski coordinates). Since, by the equivalence principle, locally an observer cannot distinguish between a fake and a "true" gravitational field, this now suggests that a "true" gravitational field can be described in terms of a space-time coordinate dependent line-element

$$ds^2 = g_{\alpha\beta}(x) dx^\alpha dx^\beta \quad , \tag{1.79}$$

   where the coordinate dependence on the x^α is now such that it cannot be eliminated globally by a suitable choice of coordinates.

2. We observe that (1.74) appears to be ill-defined at $\rho = 0$. However, in this case we already know that this is a mere *coordinate singularity* at $\rho = 0$ (akin to the

coordinate singularity at the origin of standard polar coordinates in the Cartesian plane). More generally, whenever a metric written in some coordinate system appears to exhibit some singular behaviour, one needs to investigate whether this is just a coordinate singularity or a true singularity of the gravitational field itself.

3. The above coordinates do not just fail at $\rho = 0$, they actually fail to cover large parts of Minkowski space. Thus the next lesson is that, given a metric in some coordinate system, one has to investigate if the space-time described in this way needs to be extended beyond the range of the original coordinates. One way to analyse this question (which we will make extensive use of in sections 26 and 27 when trying to understand and come to terms with black holes) is to study lightrays or the worldlines of freely falling (inertial) observers.

   In the present case, an example of an inertial observer is a static observer in Minkowski space, i.e. an observer at a fixed value of ξ^1, say, with $\xi^0 = \tau$ his proper time. In Rindler coordinates this is described by the condition that $\xi^1 = \rho \cosh \eta$ is a constant, so this is most certainly not a straight line in an (η, ρ)-diagram.

   Such an observer will of course "discover" that $\eta = +\infty$ is not the end of the world (indeed, he crosses this line at finite proper time $\tau = \xi^1$) and that Minkowski space continues (at the very least) into the quadrant $\xi^0 > |\xi^1|$ (see Figure 7 for an illustration of this).

4. Related to this is the behaviour of lightcones when expressed in terms of the coordinates (η, ρ) or when drawn in the (η, ρ)-plane (do this!). These lightcones satisfy $ds^2 = 0$, i.e.

$$\rho^2 d\eta^2 = d\rho^2 \quad \Rightarrow \quad d\eta = \pm \rho^{-1} d\rho \ . \tag{1.80}$$

   describing outgoing (ρ grows with η) respectively ingoing (ρ decreases with increasing η) lightrays. These lightcones have the familiar Minkowskian shape at $\rho = 1$, but the lightcones open up for $\rho > 1$ and become more and more narrow for $\rho \to 0$, once again exactly as we will find for the Schwarzschild black hole metric (see Figure 16 in section 26).

5. It follows from (1.76) that the Minkowski norm of ∂_η is

$$|\partial_\eta|^2 = (\xi^0)^2 - (\xi^1)^2 \ . \tag{1.81}$$

   Thus this generator of Rindler time-translations really is timelike in the region $\xi^1 > |\xi^0|$ covered by the Rindler coordinates, but it actually becomes lightlike on the lightlike boundary $\xi^1 = |\xi^0|$ of that region. As we will discuss in section 27.10, such a *Killing horizon* also happens to be one of the characteristic properties of a black hole.

6. Finally we note that there is a large region of Minkowski space that is "invisible" to the constantly accelerated observers. While a static observer will eventually receive information from any event anywhere in space-time (his past lightcone will eventually cover all of Minkowski space ...), the past lightcone of one of the Rindler accelerated observers (whose worldlines asymptote to the lightcone direction $\xi^0 = \xi^1$) will asymptotically only cover one half of Minkowski space, namely the region $\xi^0 < \xi^1$. Thus any event above the line $\xi^0 = \xi^1$ will forever be invisible to this class of observers. Such an observer-dependent horizon has some similarities with the *event horizon* characterising a black hole (see section 27.5 for a first encounter with such an object, and section 32 for a detailed discussion).

For more on Rindler coordinates, see sections 3.4 and 7.8.

1.4 General Coordinate Transformations in Minkowski Space I: Metric

In order to move away from constant accelerations (as models of observers in constant gravitational fields only), we now consider the effect of arbitrary (general) coordinate transformations on the laws of special relativity and the geometry of Minkowski space. This may look like a somewhat exaggerated move at this point (should we perhaps not just look at coordinate transformations to coordinates that somehow correspond to adapted coordinates for some arbitrary accelerated observer?), but

- it is actually easier to just do this than to understand what is meant precisely by this parenthetical remark and how to implement it;
- there are many useful things that one can learn from doing this;
- and we will see later (when discussing the relation between the *Einstein Equivalence Principle* and the *Principle of General Covariance* in section 4.1), that the relation between the description of physics in an arbitrary gravitational field and the behaviour of this description under arbitrary coordinate transformations is much closer and more far-reaching than we perhaps have the right to expect at the moment.

Let us see what things look like when written in some other (non-inertial, accelerating) coordinate system. It is extremely useful for bookkeeping purposes and for avoiding algebraic errors to use different kinds of indices for different coordinate systems. Thus we will usually call the new coordinates $x^\mu(\xi^b)$ or $x^\alpha(\xi^a)$, and not, say, $x^a(\xi^b)$ (although there would be nothing wrong with that).

We start with the definition of proper time, as described in inertial coordinates by the Minkowski line element,

$$d\tau^2 = -\eta_{ab} d\xi^a d\xi^b \tag{1.82}$$

First of all, this proper time should not depend on which coordinates we use to describe the motion of the particle, but only on the world line of the particle itself. After all, the particle could not care less what coordinates we experimenters or observers use to describe the particle's proper time.

[By the way: this is the best way to resolve the so-called "twin-paradox", which should really be referred to (and presented) as the "twin-non-paradox" or simply the "twin-fact" - everything else is deliberate and unhelpful obfuscation! It does not matter which reference system you use - the accelerating twin in the rocket will always be younger than her brother when they meet again.]

Thus all we need to know is how the same proper time τ is expressed in terms of the new coordinates, which simply follows from

$$d\tau^2 = -\eta_{ab} d\xi^a d\xi^b = -\eta_{ab} \frac{\partial \xi^a}{\partial x^\mu} \frac{\partial \xi^b}{\partial x^\nu} dx^\mu dx^\nu \quad . \tag{1.83}$$

Here

$$J^a_\mu(x) = \frac{\partial \xi^a}{\partial x^\mu} \tag{1.84}$$

is the Jacobi matrix associated to the coordinate transformation $\xi^a = \xi^a(x^\mu)$, and we will make the assumption that (locally) this matrix is non-degenerate, thus has an inverse $J^\mu_a(x)$ or $J^\mu_a(\xi)$ which is the Jacobi matrix associated to the inverse coordinate transformation $x^\mu = x^\mu(\xi^a)$,

$$J^a_\mu J^\mu_b = \delta^a_b \qquad J^\mu_a J^a_\nu = \delta^\mu_\nu \quad . \tag{1.85}$$

We see that in the new coordinates, proper time and distance are no longer measured by the Minkowski metric in its standard form (the constant matrix η_{ab}), but by

$$d\tau^2 = -\eta_{\mu\nu}(x) dx^\mu dx^\nu \quad , \tag{1.86}$$

where the *metric tensor* (or *metric* for short) $\eta_{\mu\nu}(x)$, the Minkowski metric in the coordinates x^μ, is

$$\eta_{\mu\nu}(x) = \eta_{ab} \frac{\partial \xi^a}{\partial x^\mu} \frac{\partial \xi^b}{\partial x^\nu} \quad . \tag{1.87}$$

Remarks:

1. The fact that the Minkowski metric written in the coordinates x^μ in general depends on x should not come as a surprise - after all, this also happens when one writes the Euclidean metric in spherical coordinates, $ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2$ etc.

2. Even though the components of the metric are not those of the Minkowski metric in inertial coordinates, this metric or line element (with its associated presciption for time- and space-measurements) still describes exactly the same Minkowskian

geometry as the standard Minkowski metric η_{ab}: just as passing from Cartesian to spherical coordinates in $\mathbb{R}^3$ does not change the Euclidean geometry of the space, using some other than inertial coordinates does not change the Minkowskian geometry of Minkowski space.

3. It is easy to check, using (1.85), that the inverse metric, which we will denote by $\eta^{\mu\nu}$,

$$\eta^{\mu\nu}(x)\eta_{\nu\lambda}(x) = \delta^{\mu}_{\lambda} \ , \tag{1.88}$$

is given by

$$\eta^{\mu\nu}(x) = \eta^{ab}\frac{\partial x^{\mu}}{\partial \xi^{a}}\frac{\partial x^{\nu}}{\partial \xi^{b}} \ . \tag{1.89}$$

In terms of Jacobi matrices we have

$$\eta_{\mu\nu} = J^{a}_{\mu}J^{b}_{\nu}\eta_{ab} \quad , \quad \eta^{\mu\nu} = J^{\mu}_{a}J^{\nu}_{b}\eta^{ab} \ . \tag{1.90}$$

The rationale for using a different positioning of the indices for the metric and its inverse is that, as is apparent form the above equations, they transform in a *different* way under coordinate transformations: one with the Jacobi matrices, the other with their inverses. Thus geometrically they are different kinds of objects, and (as in the tensor algebra for Lorentz tensors of special relativity) the positioning of the indices is used to indicate and keep track of this fact in a simple and intuitive manner.

We will have much more to say about the metric below and, indeed, throughout this course.

1.5 General Coordinate Transformations in Minkowski Space II: Free Particle

We now turn to the equation of motion of a free particle, given in inertial coordinates by

$$\frac{d^2}{d\tau^2}\xi^{a}(\tau) = 0 \ . \tag{1.91}$$

The usual rules for a change of variables give

$$\frac{d}{d\tau}\xi^{a} = \frac{\partial \xi^{a}}{\partial x^{\mu}}\frac{dx^{\mu}}{d\tau} \ , \tag{1.92}$$

where $J^{a}_{\mu} = \frac{\partial \xi^{a}}{\partial x^{\mu}}$ is the invertible Jacobi matrix. This shows that, as usual, velocities transform in a particularly simple (linear, vectorial) way under arbitrary coordinate transformations, namely just with the Jacobi matrix,

$$\dot{\xi}^{a} = J^{a}_{\mu}\dot{x}^{\mu} \ . \tag{1.93}$$

Differentiating once more, one finds

$$\begin{aligned}\frac{d^2}{d\tau^2}\xi^a &= \frac{\partial\xi^a}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2\xi^a}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} \\ &= \frac{\partial\xi^a}{\partial x^\mu}\frac{d^2x^\mu}{d\tau^2} + \delta^a_{\ b}\frac{\partial^2\xi^b}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} \\ &= \frac{\partial\xi^a}{\partial x^\mu}\left[\frac{d^2x^\mu}{d\tau^2} + \frac{\partial x^\mu}{\partial\xi^b}\frac{\partial^2\xi^b}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \ . \end{aligned} \tag{1.94}$$

Thus, since the matrix appearing outside the square bracket is invertible, in terms of the coordinates x^μ the equation of motion, or the equation for a straight (and, in the case at hand, timelike) line in Minkowski space, becomes

$$\frac{d^2x^\mu}{d\tau^2} + \frac{\partial x^\mu}{\partial\xi^a}\frac{\partial^2\xi^a}{\partial x^\nu\partial x^\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ . \tag{1.95}$$

We will write this as

$$\frac{d^2x^\mu}{d\tau^2} + \gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ , \tag{1.96}$$

or just

$$\ddot{x}^\mu + \gamma^\mu_{\ \nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{1.97}$$

where

$$\gamma^\mu_{\ \nu\lambda} = \frac{\partial x^\mu}{\partial\xi^a}\frac{\partial^2\xi^a}{\partial x^\nu\partial x^\lambda} \ . \tag{1.98}$$

REMARKS:

1. Of course the statement (1.93) regarding the linear transformation rule of velocities under coordinate transformations remains true for transformations between arbitary coordinate systems in Minkowski space, $\{x^\mu\}$ and $\{y^\alpha\}$, say, i.e. one has

$$\dot{y}^\alpha = \frac{\partial y^\alpha}{\partial x^\mu}\dot{x}^\mu \equiv J^\alpha_\mu\dot{x}^\mu \ . \tag{1.99}$$

In general this simply follows from the chain rule. It can also be deduced (in a somewhat unnecessarily long way included here only for later reference purposes) from what we have already done, namely by simply repeating the calculation leading to (1.93), but now for the coordinates y^α,

$$\dot{\xi}^a = J^a_\alpha\dot{y}^\alpha \ . \tag{1.100}$$

This now implies

$$\dot{y}^\alpha = J^\alpha_a\dot{\xi}^a = J^\alpha_a J^a_\mu\dot{x}^\mu = J^\alpha_\mu\dot{x}^\mu \tag{1.101}$$

where

$$J^\alpha_\mu = \frac{\partial y^\alpha}{\partial\xi^a}\frac{\partial\xi^a}{\partial x^\mu} = \frac{\partial y^\alpha}{\partial x^\mu} \tag{1.102}$$

and in the last step the chain rule (or multiplicativity of the Jacobi matrix under consecutive coordinate transformations) was used.

2. While (1.97) looks a bit complicated and unattractive, it is simply the general variant of a calculation that you have probably done numerous times before in various specific contexts. In particular, the second term in this equation, is just the general expression for a pseudo-force or fictitious gravitational force (like a centrifugal force or the Coriolis force) that arises whenever one describes inertial motion in non-inertial coordinates.

3. More compactly, this pseudo-force term can be written as

$$\gamma^{\mu}_{\nu\lambda} = J^{\mu}_{a}\partial_{\nu}J^{a}_{\lambda} = J^{\mu}_{a}\partial_{\lambda}J^{a}_{\nu} \equiv J^{\mu}_{a}J^{a}_{\nu\lambda} \ . \tag{1.103}$$

It is absent precisely for linear coordinate transformations $\xi^{a}(x^{\mu}) = M^{a}_{\mu}x^{\mu}$,

$$\gamma^{\mu}_{\nu\lambda} = 0 \quad (\forall\, ^{\mu}, _{\nu}, _{\lambda}) \quad \Leftrightarrow \quad \xi^{a} = M^{a}_{\mu}x^{\mu} \tag{1.104}$$

for some constant matrix M^{a}_{μ}. In particular, this means that the equation of motion for a free particle is invariant under Lorentz transformations, as it should be.

4. By the same reasoning, the quantity $\gamma^{\mu}_{\nu\lambda}$ is independent of the choice of reference inertial coordinate system ξ^{a}. I.e. if $\zeta^{a} = L^{a}_{b}\xi^{b}$ for some Lorentz (or more general linear) transformation matrix L^{a}_{b}, then

$$\frac{\partial x^{\mu}}{\partial \xi^{a}}\frac{\partial^{2}\xi^{a}}{\partial x^{\nu}\partial x^{\lambda}} = \frac{\partial x^{\mu}}{\partial \zeta^{a}}\frac{\partial^{2}\zeta^{a}}{\partial x^{\nu}\partial x^{\lambda}} \tag{1.105}$$

5. In the same way that the equation of motion for a free particle in inertial coordinates follows from the extremisation of the proper time (written in inertial coordinates),

$$\delta\int d\tau = \delta\int\sqrt{-\eta_{ab}d\xi^{a}d\xi^{b}} = 0 \quad \Rightarrow \quad \ddot{\xi}^{a} = 0 \ , \tag{1.106}$$

the equation of motion for a free particle in noninertial coordinates follows from the extremisation of the proper time (written in these noninertial coordinates),

$$\delta\int d\tau = \delta\int\sqrt{-\eta_{\mu\nu}dx^{\mu}dx^{\nu}} = 0 \quad \Rightarrow \quad \ddot{x}^{\mu} + \gamma^{\mu}_{\nu\lambda}\dot{x}^{\nu}\dot{x}^{\lambda} = 0 \ . \tag{1.107}$$

It is a straightforward exercise to establish this, and simply reflects the well-known covariance of the Euler-Lagrange equations under coordinate transformations. The proof will not be given here also because we will straightaway establish a more general statement (for a space equipped with an arbitrary metric) in section 2.3 below.

6. It is immediately apparent from this action principle, that the (pseudo-)force terms in the equation of motion will arise from the fact that in general the components of the metric $\eta_{\mu\nu}(x)$ are not constant in these cooordinates. In particular, therefore, it must be possible to write the pseudo-force terms $\gamma^{\mu}_{\nu\lambda}$ in terms of the first partial derivatives of the metric $\eta_{\mu\nu}$. We will come back to this in section 1.6 below.

Even though the resulting equations look a bit uninviting at the moment, that is just what you get when you do write things in arbitrary coordinates. Moreover, there are at least two very useful things that we can extract or anticipate from this, namely

1. candidates for the appropriate generalisation of the Newtonian gravitational potential
2. the prototypical *general covariance* of physical equations

in *any* theory of gravity satisfying the Einstein equivalence principle. Let us now discuss these features in turn (relegating some uninspiring calculational details to the end of this subsection):

1. the Metric as a Candidate for the Gravitational Potential

   Recall that in non-inertial coordinates the metric takes the form

   $$\eta_{\mu\nu} = J^a_\mu J^b_\nu \eta_{ab} \tag{1.108}$$

   and the equation of motion of a free particle is (1.97) with the pseudo-force term (1.103)

   $$\gamma^\mu_{\ \nu\lambda} = J^\mu_a \partial_\nu J^a_\lambda \ . \tag{1.109}$$

   It turns out that this term can be expressed in terms of the partial derivatives of the metric (as anticipated from the action principle above). Indeed, given any symmetric and invertible objectg $g_{\mu\nu}(x)$ (a "metric"), with inverse $g^{\mu\nu}$, define

   $$\begin{aligned} \Gamma^\mu_{\ \nu\lambda} &= g^{\mu\rho}\Gamma_{\rho\nu\lambda} \\ \Gamma_{\rho\nu\lambda} &= \tfrac{1}{2}(g_{\rho\nu,\lambda} + g_{\rho\lambda,\nu} - g_{\nu\lambda,\rho}) \end{aligned} \tag{1.110}$$

   (this definition is such that it remains valid for an arbitary metric). Then one finds

   $$g_{\mu\nu} = \eta_{\mu\nu} \equiv J^a_\mu J^b_\nu \eta_{ab} \quad \Rightarrow \quad \Gamma^\mu_{\ \nu\lambda} = \gamma^\mu_{\ \nu\lambda} \ . \tag{1.111}$$

   It is an elementary but nevertheless useful exercise to check this (see below - but do try this yourself as well).

   This shows that the components $\eta_{\mu\nu}$ of the metric appear to play the role of "potentials" for the gravitational pseudo-force. In particular, since in principle all components of the metric can contribute to $\gamma_{\rho\nu\lambda}$, we learn the interesting fact that in order to achieve this a single scalar potential is completely insufficient (and one could have discovered the possibility or necessity of a multitude of potentials simply by the study of pseudo-forces in non-gravitational Newtonian mechanics).

It is enormously pleasing to note that the "number" of potentials that we seem to have discovered, namely 10 (for the symmetric (4×4)-matrix $g_{\mu\nu}(x)$), agrees with the number of potentials anticipated in our discussion of section 1.1 when we contemplated a tensorial generalisations (1.6) of the Poisson equation (with source T_{ab}).

If the metric indeed plays the role of the gravitational potential, as suggested by these considerations, then it will play the role of the fundamental dynamical variable of gravity. Since the metric encodes what one usually refers to as the geometry of a space(-time), as we will discuss in much more detail below, namely the information required to determine distances, areas, volumes etc., this means that we are being led to the conclusion that any theory of gravity based on the equivalence principle is a theory of dynamical geometry. Wow ...

2. the General Covariance of the Equation of Motion

   The equation of motion (1.97) has one other fundamental redeeming and attractive feature which will also make it the prototype of the kind of equations that we will be looking for in general. This feature is its *covariance* under general coordinate transformations, i.e. its *general covariance*, which means that the equation takes the same form in any coordinate system. Equivalently, but somewhat more to the point, as discussed in more detail in section 4.1 below, this is the statement that the equation is satisified in one coordinate system if and only if it is satisfied in all coordinate systems.

   This covariance is in some sense tautologically true since the coordinate system $\{x^\mu\}$ that we have chosen is indeed arbitrary. However, it is instructive to see how this comes about by explicitly transforming (1.97) from one coordinate system x^μ to another, say y^α.

   If one does this (cf. below for a proof), one finds that the equations of motion (1.97) in the coordinates x^μ and y^α are related by

$$\frac{d^2y^\alpha}{d\tau^2} + \gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \tag{1.112}$$

   Thus the geodesic equation transforms in the simplest possible non-trivial way under coordinate transformations $x \to y$, namely with the Jacobi matrix

$$J^\alpha_\mu = \frac{\partial y^\alpha}{\partial x^\mu} \ . \tag{1.113}$$

   We will see later that this transformation behaviour characterises/defines tensors, in this particular case a vector (or contravariant tensor of rank 1).

   In particular, since this matrix is assumed to be invertible, we reach the conclusion that the left hand side of (1.112) is zero if and only if the term in square brackets

on the right hand side is zero,

$$\frac{d^2y^\alpha}{d\tau^2} + \gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = 0 \quad \Leftrightarrow \quad \frac{d^2x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \tag{1.114}$$

This is what is meant by the statement that the equation takes the same form in any coordinate system, and is therefore satisfied in one coordinate system if and only if it is satisfied in all coordinate systems. We see that in this case this is achieved by having the equation transform in a particularly simple way under coordinate transformations, namely as a tensor. We will discuss this in more generality in section 4.1.

---

1. Proof of (1.111):

   - From
   $$\eta_{\mu\nu} = \eta_{ab}J^a_\mu J^b_\nu \tag{1.115}$$
   one deduces
   $$\eta_{\mu\nu,\lambda} = \eta_{ab}(J^a_{\mu\lambda}J^b_\nu + J^a_\mu J^b_{\nu\lambda}) \tag{1.116}$$
   where
   $$J^a_{\mu\lambda} = \partial_\lambda J^a_\mu = \frac{\partial^2\xi^a}{\partial x^\mu \partial x^\lambda} = J^a_{\lambda\mu} \ . \tag{1.117}$$
   - Therefore, form the definition (1.110) of the Γ-symbols, one has (for $g_{\mu\nu} = \eta_{\mu\nu}$)
   $$\begin{aligned}\Gamma_{\mu\nu\lambda} &= \tfrac{1}{2}(\eta_{\mu\nu,\lambda} + \eta_{\mu\lambda,\nu} - \eta_{\nu\lambda,\mu}) \\ &= \tfrac{1}{2}\eta_{ab}(J^a_{\mu\lambda}J^b_\nu + J^a_\mu J^b_{\nu\lambda} + J^a_{\mu\nu}J^b_\lambda + J^a_\mu J^b_{\lambda\nu} - J^a_{\nu\mu}J^b_\lambda - J^a_\nu J^b_{\lambda\mu}) \\ &= \eta_{ab}J^a_\mu J^b_{\nu\lambda} \ ,\end{aligned} \tag{1.118}$$
   where the cancellations in passing to the last line arise from the symmetries $\eta_{ab} = \eta_{ba}$, $J^b_{\lambda\mu} = J^b_{\mu\lambda}$ etc.
   - Thus, finally (and writing out everything in detail for once),
   $$\begin{aligned}\Gamma^\mu_{\nu\lambda} &= \eta^{\mu\rho}\Gamma_{\rho\nu\lambda} = \eta^{cd}J^\mu_c J^\rho_d \eta_{ab}J^a_\rho J^b_{\nu\lambda} = \eta^{cd}J^\mu_c \delta^a_d \eta_{ab}J^b_{\nu\lambda} \\ &= \eta^{ca}J^\mu_c \eta_{ab}J^b_{\nu\lambda} = \delta^c_b J^\mu_c J^b_{\nu\lambda} = J^\mu_b J^b_{\nu\lambda} \ ,\end{aligned} \tag{1.119}$$
   as was to be shown.

2. Proof of (1.112):

   - We proceed as in the proof of (1.99). Thus consider transforming the free particle equation of motion in inertial coordinates (1.55) not to the coordinate systsem x^μ, as we did before, but to another coordinate system $\{y^\alpha\}$.

Following the same steps as above, one arrives at the y-version of (1.94), namely

$$\frac{d^2}{d\tau^2}\xi^a = \frac{\partial \xi^a}{\partial y^\alpha}\left[\frac{d^2 y^\alpha}{d\tau^2} + \frac{\partial y^\alpha}{\partial \xi^b}\frac{\partial^2 \xi^b}{\partial y^\beta \partial y^\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau}\right] \quad . \tag{1.120}$$

- Equating this result to (1.94) and using the chain rule for partial derivatives

$$\frac{\partial y^\alpha}{\partial x^\mu} = \frac{\partial y^\alpha}{\partial \xi^a}\frac{\partial \xi^a}{\partial x^\mu} \quad , \tag{1.121}$$

one finds

$$\frac{d^2 y^\alpha}{d\tau^2} + \gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2 x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \tag{1.122}$$

as claimed.

2 Metrics, Geometry and Geodesics

Above we saw that the motion of free particles in Minkowski space in curvilinear coordinates is described in terms of a modified metric, $g_{\mu\nu}$, and a force term $\gamma^{\mu}_{\nu\lambda}$ representing the "pseudo-force" on the particle. Thus the Einstein Equivalence Principle suggests that an appropriate description of true gravitational fields is in terms of a metric tensor $g_{\mu\nu}(x)$ (and its associated Γ-symbols) which can only locally be related to the Minkowski metric via a suitable coordinate transformation (to locally inertial coordinates). We adopt this as our working hypothesis.

2.1 Metrics and Geometry I: Definition and Examples

Thus our starting point will now be a space-time equipped with some metric $g_{\mu\nu}(x)$, which (by analogy with the Euclidean and Minkowski metrics) we will assume to be symmetric and non-degenerate, i.e.

$$g_{\mu\nu}(x) = g_{\nu\mu}(x) \qquad \mathsf{det}(g_{\mu\nu}(x)) \neq 0 \ . \tag{2.1}$$

The metric encodes the information how to measure (spatial and temporal) distances, as well as areas, volumes etc., via the associated line element

$$g_{\mu\nu}(x) \quad \Rightarrow \quad ds^2 = g_{\mu\nu}(x) dx^{\mu} dx^{\nu} \ . \tag{2.2}$$

As an example, the most general 2-dimensional line element (on a space with local coordinates $(x^i) = (x^1, x^2)$) has the form

$$ds^2 = g_{ij} dx^i dx^j = g_{11}(dx^1)^2 + (g_{22})(dx^2)^2 + 2g_{12} dx^1 dx^2 \tag{2.3}$$

(which is non-degenerate if $g_{11}g_{22} - (g_{12})^2 \neq 0$).

A metric determines a geometry (in the literal sense of a prescription for measuring distances etc.), but different metrics may well determine the same geometry, namely those metrics which are just related by coordinate transformations. In particular, distances should not depend on which coordinate system is used. Hence, changing coordinates from the $\{x^{\mu}\}$ to new coordinates $\{y^{\alpha}(x^{\mu})\}$ and demanding that

$$g_{\mu\nu}(x) dx^{\mu} dx^{\nu} = g_{\alpha\beta}(y) dy^{\alpha} dy^{\beta} \ , \tag{2.4}$$

one finds that under a coordinate transformation a metric transforms as

$$g_{\alpha\beta}(y) = g_{\mu\nu}(x) \frac{\partial x^{\mu}}{\partial y^{\alpha}} \frac{\partial x^{\nu}}{\partial y^{\beta}} \equiv J^{\mu}_{\alpha} J^{\nu}_{\beta} \ g_{\mu\nu} \ . \tag{2.5}$$

Objects which transform in such a nice and simple way under coordinate transformations are known as *tensors* - the metric is an example of what is known as (and we will get to

know as) a covariant symmetric rank two tensor. We will study tensors in much more detail and generality later, starting in section 4.

REMARKS:

1. Here I have denoted the components of the metric in the new coordinates y^α simply by $g_{\alpha\beta}$. Occasionally it is more convenient to use a more elaborate notation, such as
$$x^\mu \to x'^\alpha = y^\alpha \quad \Rightarrow \quad g_{\mu\nu} \to g'_{\alpha\beta} = J^\mu_\alpha J^\nu_\beta g_{\mu\nu} \ , \tag{2.6}$$
which allows one to distinguish notationally specific components of the metric in 2 different coordinate systems, such as g'_{11} (the (11)-component of the metric in the y-coordinates) from g_{11} (the (11)-component of the metric in the x-coordinates). As mentioned before, indices and other decorations are primarily bookkeeping devices; therefore I will usually not be overly-pedantic about these things in the following and will use whatever notation is more convenient in the case at hand.

2. As a consequence of the non-degeneracy condition, pointwise $g_{\mu\nu}(x)$ possesses an inverse, whose components we will denote by $g^{\mu\nu}(x)$, i.e.
$$g^{\mu\nu}(x)g_{\nu\lambda}(x) = \delta^\mu_\lambda \quad , \quad g_{\mu\nu}(x)g^{\nu\lambda}(x) = \delta^\lambda_\mu \ . \tag{2.7}$$
Clearly, the inverse metric then transforms inversely, i.e. with the inverse Jacobi matrices J^α_μ, and this is now nicely compatible with the convention to denote the inverse metric by upper indices,
$$g^{\alpha\beta} = J^\alpha_\mu J^\beta_\nu g^{\mu\nu} \ . \tag{2.8}$$
This is also the rationale for writing the inverse metric with "upper" indices: the positioning of indices is used to indicate how an object transforms under coordinate transformations (and we will formalise this in the discussion of section 4 on tensor algebra).

3. A space-time equipped with a metric tensor $g_{\mu\nu}(x)$ is called a metric space-time or (pseudo-)Riemannian space-time. Here "Riemannian" usually refers to a space equipped with a positive-definite metric (all eigenvalues positive), while pseudo-Riemannian (or Lorentzian) refers to a space-time with a metric with one negative and 3 (or 27, or whatever) positive eigenvalues.

4. One point to note about the tensorial transformation behaviour is that pointwise it is a similarity transformation in the sense of linear algebra, in matrix notation
$$g \mapsto J^t g J \ . \tag{2.9}$$
In particular, therefore, if in one coordinate system the space-time metric tensor has one negative and three positive eigenvalues (as in a locally inertial coordinate

system), then the same will be true in any other coordinate system (even though the eigenvalues themselves will in general be different) - this statement should be familiar from linear algebra (e.g. as Sylvester's law, but it also goes under various other names).

Here are some examples of Riemannian metrics that you may already be familiar with.

EXAMPLES:

1. The Euclidean metrics or line-elements on $\mathbb{R}^2$ or $\mathbb{R}^3$, but written in polar or spherical coordinates,

$$\begin{aligned} ds^2(\mathbb{R}^2) &= dx^2 + dy^2 = dr^2 + r^2 d\phi^2 \\ ds^2(\mathbb{R}^3) &= dx^2 + dy^2 + dz^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \end{aligned} \tag{2.10}$$

   E.g. for the latter case one has

$$(x, y, z) = (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta) \ , \tag{2.11}$$

   and plugging this into the Euclidean line-element $dx^2 + dy^2 + dz^2$, one finds the above result.

   Denoting the Cartesian coordinates by x^α and the spherical coordinates by y^α, with $(y^1 = r, y^2 = \theta, y^3 = \phi)$, the non-vanishing components of the metric in the two coordinate systems are thus (using the prime notation (2.6))

$$g_{11} = g_{22} = g_{33} = 1 \quad , \quad g'_{11} = 1 \ , \ g'_{22} = r^2 \ , \ g'_{33} = r^2\sin^2\theta \ . \tag{2.12}$$

   Alternatively, it is often more informative (and very common) to use the coordinates themselves, rather than indices, as the labels of the components of the metric tensor. In this case one can dispense with the prime notation and simply write the components of the metric in spherical coordinates as

$$g_{rr} = 1 \ , \ g_{\theta\theta} = r^2 \ , \ g_{\phi\phi} = r^2\sin^2\theta \ . \tag{2.13}$$

2. Restricting the first example above to constant radius $r = R$, this gives us the line-element on the circle S^1_R of radius R,

$$ds^2(S^1_R) = R^2 d\phi^2 \ . \tag{2.14}$$

   Restricting the second to the 2-sphere S^2_R of radius R,

$$x^2 + y^2 + z^2 = r^2 = R^2 \quad \text{or} \quad r = R \ , \tag{2.15}$$

   one finds the line-element

$$ds^2(S^2_R) = R^2(d\theta^2 + \sin^2\theta d\phi^2) \equiv R^2 d\Omega^2 \ . \tag{2.16}$$

Here

$$d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2 \tag{2.17}$$

is usually called the solid angle, and we can now interpret it as the line element on the unit 2-sphere. We will use the notation / abbrevation $d\Omega^2$ for this line element throughout the notes.

This example provides a nice illustration of the fact that by drawing the coordinate grid / infinitesimal parallelograms determined by the metric tensor, one can get a feeling for the geometry and can in particular convince oneself that in general a metric space or space-time need not or cannot be flat, i.e. is not the flat Euclidean space of Euclidean geometry.

Indeed, the coordinate grid of the metric $d\theta^2 + \sin^2\theta d\phi^2$ cannot be drawn in flat space because the infinitesimal parallelograms described by ds^2 degenerate to triangles not just at $\theta = 0$ (as would also be the case for the flat metric $ds^2 = dr^2 + r^2 d\phi^2$ in polar coordinates at $r = 0$), but also at $\theta = \pi$. This coordinate grid can, on the other hand, of course be drawn on the 2-sphere.

3. This line-element on the unit 2-sphere generalises to the line-element on a unit 3-sphere,

$$ds^2(S^3) = d\psi^2 + \sin^2\psi(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{2.18}$$

This can be obtained by simply generalising the construction of spherical coordinates from $\mathbb{R}^3$ to $\mathbb{R}^4$, and (if required) this can be continued iteratively to yet higher-dimensional spheres.

Alternatively, by thinking of the 3-sphere as the locus

$$x^2 + y^2 + z^2 + w^2 = 1 \tag{2.19}$$

in $\mathbb{R}^4$, and "solving" this equation by first setting

$$x^2 + y^2 = \sin^2\alpha \quad , \quad z^2 + w^2 = \cos^2\alpha \ , \tag{2.20}$$

and then refining this to

$$x = \sin\alpha\cos\beta \quad , \quad y = \sin\alpha\sin\beta \quad , \quad z = \cos\alpha\cos\gamma \quad , \quad w = \cos\alpha\sin\gamma \ , \tag{2.21}$$

one finds that the standard Euclidean line-element

$$ds^2 = dx^2 + dy^2 + dz^2 + dw^2 \tag{2.22}$$

induces the line-element

$$ds^2(S^3) = d\alpha^2 + \sin^2\alpha d\beta^2 + \cos^2\alpha d\gamma^2 \tag{2.23}$$

on the sphere. This is the same metric on S^3 as above (2.18), namely the one induced from the Euclidean metric on $\mathbb{R}^4$, but written in different coordinates.

In particular, both are invariant under 4-dimensional rotations, i.e. under $SO(4)$-transformations.

However, we can obtain genuinely different metrics on the 3-sphere e.g. by starting with different metrics on $\mathbb{R}^4$. One of the simplest possibilities is to replace (2.22) by

$$d\tilde{s}^2 = a^2(dx^2 + dy^2) + b^2(dz^2 + dw^2) \ , \tag{2.24}$$

with a, b real non-zero parameters. Then the induced metric on the 3-sphere $x^2 + y^2 + z^2 + w^2 = 1$ is

$$d\tilde{s}^2(S^3) = (a^2\cos^2\alpha + b^2\sin^2\alpha)d\alpha^2 + a^2\sin^2\alpha d\beta^2 + b^2\cos^2\alpha d\gamma^2 \ . \tag{2.25}$$

For $a^2 \neq b^2$, this metric is not invariant under full 4-dimensional rotations, but only under rotations in the (x, y) and (z, w) planes, i.e. under $SO(2) \times SO(2)$ transformations. Thus this equips the 3-sphere with a genuinely different geometry (and is an example of what is sometimes referred to as a "squashed 3-sphere geometry").

4. If instead of the unit 2-sphere one considers the "unit" hyperboloid H^2, defined by

$$x^2 + y^2 + z^2 = +1 \quad \longrightarrow \quad x^2 + y^2 - z^2 = -1 \ , \tag{2.26}$$

then this is naturally thought of as being embedded not in $\mathbb{R}^3$ but in $\mathbb{R}^{1,2}$, i.e. into the 3-dimensional vector space with line-element

$$ds^2 = dx^2 + dy^2 - dz^2 \ . \tag{2.27}$$

The hyperbolic analogues (r, σ, ϕ) of the spherical coordinates, defined by

$$(x, y, z) = (r\sinh\sigma\cos\phi, r\sinh\sigma\sin\phi, r\cosh\sigma) \ , \tag{2.28}$$

are naturally adapted to this situation, because

$$x^2 + y^2 - z^2 = -r^2 \tag{2.29}$$

so that the unit hyperboloid is evidently just the surface $r = 1$. In these coordinates, the metric (2.27) takes the form

$$ds^2 = -dr^2 + r^2(d\sigma^2 + \sinh^2\sigma\, d\phi^2) \ , \tag{2.30}$$

and therefore the induced metric on the unit hyperboloid $r = 1$ is

$$ds^2(H^2) = d\sigma^2 + \sinh^2\sigma\, d\phi^2 \ . \tag{2.31}$$

2.2 Metrics and Geometry II: Lorentzian (Pseudo-Riemannian) Metrics

We now turn to Lorentzian (pseudo-Riemannian) metrics and geometries. These will of course occupy and accompany us throughout these notes, so this section is meant to just provide a first brief encounter with these objects.

For a metric with Lorentzian signature, and with coordinates $x^\alpha = (x^0 = t, x^k)$, say, the metric has components $g_{00}, g_{0k} = g_{k0}$ and $g_{ik} = g_{ki}$, and the corresponding line element has the form

$$ds^2 = g_{00}dt^2 + 2g_{0k}dt\,dx^k + g_{ik}dx^i dx^k \tag{2.32}$$

Without any further conditions on the coefficients (except those ensuring non-degeneracy), this could a priori be a metric of any signature, and the signature of the metric may not always be readily apparent even when the coefficients of the metric are given explicitly.

Before looking at this in somewhat more detail, here are some simple examples, where the Lorentzian signature of the metrics is reasonably manifest:

Examples:

1. Of course any of the Riemannian metrics of the previous section can be promoted to space-time metrics by simply adding a $(-dt^2)$ (i.e. by taking the direct product with the time-axis). Thus the Minkowski metric in spatial spherical coordinates has the form
$$ds^2(\mathbb{R}^{1,3}) = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{2.33}$$

2. A generalisation of this is provided by the so-called *ultrastatic* metrics, i.e. metrics that are just a product of the standard metric $-dt^2$ along the time-direction and a spatial metric $\tilde{g}_{ij}(x)$
$$ds^2 = -dt^2 + \tilde{g}_{ij}(x)dx^i dx^j \tag{2.34}$$
(i.e. the components depend only on the spatial coordinates x^i, not on t).

3. Somewhat more generally, the spatial components of the metric can depend non-trivially on time. For example, a space-time metric describing a spatially spherical universe with a time-dependent radius (expansion of the universe!) might be described by the line element
$$ds^2 = -dt^2 + a(t)^2\left(d\psi^2 + \sin^2\psi(d\theta^2 + \sin^2\theta d\phi^2)\right) \ , \tag{2.35}$$
and more generally one can consider the corresponding generalisation of (2.34), namely metrics of the form
$$ds^2 = -dt^2 + a(t)^2\tilde{g}_{ij}(x)dx^i dx^j \ . \tag{2.36}$$
This describes a space-time with spatial metric $\tilde{g}_{ij}(x)dx^i dx^j$ and a time-dependent overall scale factor $a(t)$; in particular, such a space-time metric can describe an

expanding universe in cosmology. We will discuss such metrics in detail later on in the context of cosmology, sections 33-38.

4. The (time-time)-component of the metric can of course in general depend non-trivially on the spatial coordinates. We already encountered this in the example of the Rindler metric (1.74), which has the form

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad . \tag{2.37}$$

5. A particularly prominent example is the *Schwarzschild Metric*

$$ds^2 = -(1-2m/r)dt^2 + (1-2m/r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad . \tag{2.38}$$

   It is of fundamental importance for General Relativity, and perhaps the most important exact solution of the Einstein field equations for the gravitational field, as it describes the gravitational field outside a spherical star (as well as black holes, as it turns out ...). We will discuss this metric in great detail in sections 24-27.

The characteristic feature of metrics with Lorentzian signature is of course the presence of timelike and null (lightlike) directions, and thus in a pseudo-Riemannian space-time one has the same distinction between spacelike, timelike and lightlike separations as in Minkowski space(-time). Infinitesimal

- spacelike distances correspond to $ds^2 > 0$,
- timelike distances to $d\tau^2 = -ds^2 > 0$,
- and null or lightlike distances to $ds^2 = d\tau^2 = 0$.

Likewise, a vector $V^\mu(x)$ at a point x is called

- spacelike if $g_{\mu\nu}(x)V^\mu(x)V^\nu(x) > 0$,
- timelike if $g_{\mu\nu}(x)V^\mu(x)V^\nu(x) < 0$,
- and null or lightlike if $g_{\mu\nu}(x)V^\mu(x)V^\nu(x) = 0$,

and a curve $x^\mu(\lambda)$ is called spacelike if its tangent vector is everywhere spacelike etc.

Using the definition of a vector in general relativity (to be introduced in section 4), namely an object that transforms in the obvious way, with the Jacobi matrix, under coordinate transformations, one sees that $g_{\mu\nu}(x)V^\mu(x)V^\nu(x)$ is a scalar, i.e. invariant under coordinate transformations, and hence the statement that a vector is, say, spacelike is a coordinate-independent statement, as it should be.

REMARKS:

1. When the metric (2.32) is (time-space) block-diagonal, i.e. when the mixed components $g_{0k} = 0$ (as in all of the above examples), then the timelike and spacelike directions are easy to distinguish by inspection. Typically then the "spatial" metric g_{ik} is positive definite, and thus necessarily $g_{00} < 0$.

2. When some of the g_{0k} are non-zero, on the other hand, one has a more intricate mixing of time- and space-directions. For example, consider the simple 2-dimensional metric

$$ds^2 = dx^2 + dy^2 + 2a\, dx\, dy \tag{2.39}$$

for some real parameter a. For any a, the coordinate lines of x and y are spacelike curves. However, this does not imply all by itself that the metric is Euclidean (just like the sum of two spacelike vectors in Minkowski space is not necessarily spacelike: it may be spacelike, timelike or null: e.g. if $v = (0,1)$, choose $w = (0,1), (1/2,-1), (1,-2)$ respectively).

Indeed, by calculating the determinant of the metric,

$$\det \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix} = 1 - a^2 \ , \tag{2.40}$$

one sees that

- when $a^2 < 1$, the metric has Euclidean signature
  (it is actually related to the standard 2-dimensional Euclidean metric by a linear transformation)
- when $a^2 = 1$, the metric is degenerate
  (in this case, the line element can be written as $ds^2 = d(x \pm y)^2$, which is a 1-dimensional metric for the single coordinate $x \pm y$)
- when $a^2 > 1$, the metric has Lorentzian signature
  (it is actually related to the standard 2-dimensional Minkowski metric by a linear transformation)

3. This mixing of time- and space-directions for a metric which is not block-diagonal can also be seen from the components of the inverse metric. Indeed, from (2.7), one finds

$$g_{0\nu} g^{\nu k} = g_{00} g^{0k} + g_{0i} g^{ik} = \delta_0^k = 0 \ , \tag{2.41}$$

and thus (for $g_{00} \neq 0$)

$$g^{0k} = -\frac{1}{g_{00}} g_{0i} g^{ik} \ . \tag{2.42}$$

Likewise from (2.7) one deduces

$$g_{i\nu} g^{\nu k} = g_{i0} g^{0k} + g_{ij} g^{jk} = \delta_i^k \ . \tag{2.43}$$

In particular, this shows that in general (i.e. unless the off-diagonal components g_{0k} are all zero), the spatial components g^{ik} of the inverse metric are not the inverse of the spatial components g_{ij} of the metric. Rather, using (2.42) one has

$$\left(g_{ij} - \frac{1}{g_{00}} g_{i0} g_{j0}\right) g^{jk} = \delta_i^k \quad . \tag{2.44}$$

Outlook:

This ends our first brief encounter with metrics and geometries. At this point the question naturally arises how one can tell whether a given (perhaps complicated looking) metric is just the "flat" (Euclidean or Minkowski) metric written in other coordinates or whether it describes a genuinely new geometry.

We will see later that there is an object, the *Riemann curvature tensor*, constructed from the metric and its 1st and 2nd derivatives, which has the property that all of its components vanish if and only if the metric is a coordinate transform of the flat space Minkowski metric. Thus, given a metric, by calculating its curvature tensor one can decide if the metric is just the flat metric in disguise or not. The curvature tensor will be introduced in section 8, and the above statement will be established in section 11.2.

2.3 Geodesic Equation from the Extremisation of Proper Time

We have seen that the equation for a straight line in Minkowski space, written in arbitrary coordinates, is

$$\frac{d^2 x^\mu}{d\tau^2} + \gamma^\mu_{\nu\lambda} \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} = 0 \quad , \tag{2.45}$$

where the pseudo-force term $\gamma^\mu_{\nu\lambda}$ is given by (1.98). We have also seen in (1.111) (provided you checked this) that $\gamma^\mu_{\nu\lambda}$ can be expressed in terms of the metric (1.87) as

$$\gamma^\mu_{\nu\lambda} = \tfrac{1}{2} \eta^{\mu\rho} (\eta_{\rho\nu,\lambda} + \eta_{\rho\lambda,\nu} - \eta_{\nu\lambda,\rho}) \quad . \tag{2.46}$$

This gravitational force term is fictitious since it can globally be transformed away by going to the global inertial coordinates ξ^a. The equivalence principle suggests, however, that in general the equation for the worldline of a massive particle, i.e. a path that extremises proper time, in a true gravitational field described by a non-trivial metric $g_{\mu\nu}(x)$ (not related to the Minkowski metric by a coordinate transformation) is also of the above form.

We will now confirm this by deriving the equations for a timelike path that extremises proper time from a variational principle. These paths will be referred to as (timelike) *geodesics*. We will briefly return below to the (delicate) issue to which extent these can be regarded as world lines of actual massive particles.

Recall first of all from special relativity that the Lorentz-covariant description of the dynamics of a massive particle is based on describing the timelike worldline of the particle in the parametric form

$$\xi^a = \xi^a(\tau) \tag{2.47}$$

where τ is the proper time along the worldline,

$$d\tau^2 = -\eta_{ab} d\xi^a d\xi^b \ . \tag{2.48}$$

In particular, the 4-velocity

$$u^a = \frac{d\xi^a(\tau)}{d\tau} \tag{2.49}$$

is normalised as

$$\eta_{ab} u^a u^b = -1 \ . \tag{2.50}$$

The Lorentz-invariant action for a free massive particle with mass m is

$$S_0 = -m \int d\tau \ . \tag{2.51}$$

We can adopt the same set-up and action in the present setting. Thus we parametrise the worldlines by

$$x^\mu = x^\mu(\tau) \ , \tag{2.52}$$

with τ the proper time

$$d\tau^2 = -g_{\mu\nu}(x) dx^\mu dx^\nu \ , \tag{2.53}$$

invariant under general coordinate transformations (provided that one transforms the metric appropriately). The corresponding 4-velocity

$$u^\mu = \frac{dx^\mu}{d\tau} \tag{2.54}$$

is again normalised as

$$g_{\mu\nu} u^\mu u^\nu = -1 \ , \tag{2.55}$$

and we are led to consider the coordinate-invariant Lagrangian

$$S_0[x] = -m \int d\tau = -m \int \sqrt{-g_{\mu\nu}(x) dx^\mu dx^\nu} \ . \tag{2.56}$$

Of course m drops out of the variational equations (as it should by the equivalence principle) and we will therefore ignore m in the following.

In order to perform the variation, it is useful to introduce an arbitrary auxiliary parameter λ in the initial stages of the calculation via

$$d\tau = (-g_{\mu\nu} \tfrac{dx^\mu}{d\lambda} \tfrac{dx^\nu}{d\lambda})^{1/2} d\lambda \ , \tag{2.57}$$

and to write

$$\int d\tau = \int (d\tau/d\lambda) d\lambda = \int (-g_{\mu\nu} \tfrac{dx^\mu}{d\lambda} \tfrac{dx^\nu}{d\lambda})^{1/2} d\lambda \ . \tag{2.58}$$

We are varying the paths

$$x^\mu(\tau) \to x^\mu(\tau) + \delta x^\mu(\tau) \tag{2.59}$$

keeping the end-points fixed, and will denote the τ-derivatives by $\dot{x}^\mu(\tau)$. Under this variation, the metric $g_{\mu\nu}(x)$ varies as

$$\delta g_{\mu\nu} = g_{\mu\nu,\lambda}\, \delta x^\lambda \quad . \tag{2.60}$$

By the standard variational procedure one then finds, first of all,

$$\delta \int d\tau = \tfrac{1}{2} \int (-g_{\mu\nu} \tfrac{dx^\mu}{d\lambda} \tfrac{dx^\nu}{d\lambda})^{-1/2} d\lambda \left[-\delta g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - 2 g_{\mu\nu} \frac{d\delta x^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right] \quad . \tag{2.61}$$

Already at this stage we can revert from λ to τ, and the expression simplifies to

$$\delta \int d\tau = \tfrac{1}{2} \int d\tau \left[-(\delta g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu - 2 g_{\mu\nu} \frac{d\delta x^\mu}{d\tau} \dot{x}^\nu \right] \quad . \tag{2.62}$$

Integration by parts of the 2nd term (in order to eliminate the derivative of the variation) and use of (2.60) then leads to

$$\begin{aligned} \delta \int d\tau &= \frac{1}{2} \int d\tau \left[-g_{\mu\nu,\lambda}\, \dot{x}^\mu \dot{x}^\nu \delta x^\lambda + 2 g_{\mu\nu} \ddot{x}^\nu \delta x^\mu + 2 g_{\mu\nu,\lambda}\, \dot{x}^\lambda \dot{x}^\nu \delta x^\mu \right] \\ &= \int d\tau \left[g_{\mu\nu} \ddot{x}^\nu + \tfrac{1}{2} (g_{\mu\nu,\lambda} + g_{\mu\lambda,\nu} - g_{\nu\lambda,\mu}) \dot{x}^\nu \dot{x}^\lambda \right] \delta x^\mu \end{aligned} \tag{2.63}$$

after a suitable relabelling of the indices.

If we now adopt the definition (2.46) for an arbitrary metric,

$$\Gamma^\mu_{\ \nu\lambda} = g^{\mu\rho} \Gamma_{\rho\nu\lambda} = \tfrac{1}{2} g^{\mu\rho} (g_{\rho\nu,\lambda} + g_{\rho\lambda,\nu} - g_{\nu\lambda,\rho}) \quad , \tag{2.64}$$

we can write the result as

$$\delta \int d\tau = \int d\tau g_{\mu\nu} (\ddot{x}^\nu + \Gamma^\nu_{\ \rho\lambda} \dot{x}^\rho \dot{x}^\lambda) \delta x^\mu \quad . \tag{2.65}$$

Thus we see that indeed the equations for a timelike geodesic in an arbitrary gravitational field are

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda} \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} = 0 \quad . \tag{2.66}$$

REMARKS:

1. Given a metric $g_{\mu\nu}$, the quantities $\Gamma_{\rho\nu\lambda}$ and $\Gamma^\mu_{\ \nu\lambda}$ defined in (2.64) are known as the *Christoffel Symbols* of the 1st kind and 2nd kind.

2. The Christoffel symbols (2.64) play the role of the gravitational force term, and thus in this sense the components of the metric play the role of the gravitational potential. These Christoffel symbols play an important role not just in the geodesic equation but, as we will see later on, more generally in the definition of a covariant derivative operator and the construction of the curvature tensor, and thus ultimately also in the generally covariant description of the dynamics of the gravitational field itself.

3. Two elementary important properties of the Christoffel symbols are that they are symmetric in the second and third indices,

$$\Gamma_{\mu\nu\lambda} = \Gamma_{\mu\lambda\nu} \quad , \quad \Gamma^{\mu}_{\ \nu\lambda} = \Gamma^{\mu}_{\ \lambda\nu} \tag{2.67}$$

(this follows simply from the definition), and that symmetrising $\Gamma_{\mu\nu\lambda}$ over the first pair of indices one finds

$$\Gamma_{\mu\nu\lambda} + \Gamma_{\nu\mu\lambda} = g_{\mu\nu,\lambda} \tag{2.68}$$

(and this follows from noting that 4 of the 6 partial derivative terms of the metric cancel in this linear combination while 2 add up)

4. One can also consider spacelike paths that extremise (minimise) proper distance, by using the action

$$S_0 \sim \int ds \tag{2.69}$$

where

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu \tag{2.70}$$

is the proper distance (or arc-length in the traditional terminology of the differnetial geometry of curves).

One should also consider massless particles, whose worldlines will be null (or lightlike) paths. However, in that case one can evidently not use proper time or proper distance, since these are by definition zero along a null path, $ds^2 = 0$. We will come back to this special case, and a unified description of the massive and massless case, below (section 2.5). In all cases, we will refer to the resulting paths as *geodesics*. If required, we add the qualifier "timelike", "spacelike" or "null", and this is meaningful and unambiguous since, as we will see below, a geodesic that is initially timelike will always remain timelike etc.

5. By definition, massive test particles are those particles that satisfy the above geodesic equation, i.e. that follow timelike geodesics in space-time. However, it needs to be borne in mind that this notion of a test particle is a fiction, in particular as it neglects the backreaction, i.e. the change in the background gravitational field due to the mass of the particle. Moreover, real particles either have a finite extent (in which case this finite size should play a role in their equations of motion) or are considered to be point-like. However, the notion of a point-like particle is extremely dangerous and delicate in general relativity: as we will see later, if a given total mass is concentrated in a sufficiently small region of space-time (and "point-like" certainly qualifies as "sufficiently small"), then one will end up with a black hole rather than with the description of a particle. The correct description of point particles in general relativity is a complicated issue and an active area of research.[5]

[5] See e.g. E. Poisson, A. Pound, I. Vega, *The Motion of Point Particles in Curved Spacetime*,

2.4 Geodesic Equation and Coordinate Transformations

As we have seen, in a coordinate system x^μ, with metric $g_{\mu\nu}$ and Christoffel symbols $\Gamma^\mu_{\ \nu\lambda}$ the geodesic equation takes the form (2.66)

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ . \tag{2.71}$$

Since the coordinate system x^μ was arbitrary, evidently this equation is true for an arbitrary coordinate system. Nevertheless, when one changes from coordinates x^μ to y^α, the components of the metric and the Christoffel symbols will change (the latter in a somewhat complicated way). It is therefore of interest to find out explicitly how it comes about that the validity of (2.71) is independent of the coordinate system. It turns out that the geodesic equations in the two coordinate systems are simply related by the Jacobi matrix,

$$\frac{d^2y^\alpha}{d\tau^2} + \Gamma^\alpha_{\ \beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \ . \tag{2.72}$$

This generalises the result (1.112) that we had obtained before regarding coordinate transformations in Minkowski space, and the same remarks about covariance and tensors etc. apply. In particular, the invertibility of the Jacobi matrix allows us to conclude that

$$\frac{d^2y^\alpha}{d\tau^2} + \Gamma^\alpha_{\ \beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = 0 \quad \Leftrightarrow \quad \frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau} = 0 \ , \tag{2.73}$$

as anticipated. I will now give 3 arguments to establish (2.72).

1. General Argument at the Level of the Variation of the Action

   That the geodesic equation transforms in this simple way (namely as a vector) should not come as a surprise. We obtained this equation as a variational equation. The Lagrangian itself is a scalar (invariant under coordinate transformations),

   $$d\tau = \sqrt{-g_{\alpha\beta}dy^\alpha dy^\beta} = \sqrt{-g_{\mu\nu}dx^\mu dx^\nu} \ . \tag{2.74}$$

   Moreover, the variations of y^α and x^μ are related by

   $$\delta y^\alpha = \frac{\partial y^\alpha}{\partial x^\mu}\delta x^\mu = J^\alpha_\mu \delta x^\mu \ . \tag{2.75}$$

   Putting these pieces together, one finds the desired result (2.72).

2. General Argument at the Level of the Euler-Lagrange Equations

   A general feature of the Lagrange formalism and the Euler-Lagrange equations is their covariance under what are known as "point-transformations" in classical

`arXiv:1102.0529 [gr-qc]` for a detailed discussion and many references (but you will need to acquire a solid understanding of tensor analysis first).

mechanics, arbitrary (possibly even time-dependent) coordinate transformations of the configuration space variables. In the case at hand, this reduces to the statement that the Euler-Lagrange equations with respect to x^μ and y^α respectively are related by the Jacobi matrix,

$$\frac{d}{d\tau}\frac{\partial L}{\partial x'^\mu} - \frac{\partial L}{\partial x^\mu} = J^\alpha_\mu \left[\frac{d}{d\tau}\frac{\partial L}{\partial y'^\alpha} - \frac{\partial L}{\partial y^\alpha}\right] \tag{2.76}$$

where $x'^\mu = dx^\mu/d\lambda$ etc. This also implies (2.72).

3. Specific Argument at the Level of the Geodesic Equation

   Knowing how the metric transforms under coordinate transformations,

$$g_{\alpha\beta} = J^\mu_\alpha J^\nu_\beta g_{\mu\nu} \ , \tag{2.77}$$

   we can now also determine how the Christoffel symbols (2.64) and the geodesic equation transform. A straightforward but not particularly inspiring calculation (which you should nevertheless do) shows that under $x^\mu \to y^\alpha$ the Christoffel symbols are related by

$$\Gamma^\alpha_{\beta\gamma} = \Gamma^\mu_{\nu\lambda}\frac{\partial y^\alpha}{\partial x^\mu}\frac{\partial x^\nu}{\partial y^\beta}\frac{\partial x^\lambda}{\partial y^\gamma} + \frac{\partial y^\alpha}{\partial x^\mu}\frac{\partial^2 x^\mu}{\partial y^\beta \partial y^\gamma} \ , \tag{2.78}$$

   or

$$\Gamma^\alpha_{\beta\gamma} = J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \Gamma^\mu_{\nu\lambda} + J^\alpha_\mu \partial_\beta J^\mu_\gamma \ . \tag{2.79}$$

   Thus, $\Gamma^\mu_{\nu\lambda}$ transforms inhomogenously under coordinate transformations. If only the first term on the right hand side were present, then $\Gamma^\mu_{\nu\lambda}$ would be a tensor. However, the second term is there precisely to compensate for the fact that $\ddot{x}^\mu$ is also not a tensor - the combined geodesic equation turns out to transform in a nice way under coordinate transformations.

   Namely, after another not terribly inspiring calculation (which you should nevertheless also do at least once in your life) , one finds precisely (2.72),

$$\frac{d^2 y^\alpha}{d\tau^2} + \Gamma^\alpha_{\beta\gamma}\frac{dy^\beta}{d\tau}\frac{dy^\gamma}{d\tau} = \frac{\partial y^\alpha}{\partial x^\mu}\left[\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}\right] \ . \tag{2.80}$$

   An explicit proof of (2.79) and (2.80) is given at the end of this subsection. A more general result along these lines will be established in section 5.1 below, when we introduce the covariant derivative of a vector field.

REMARKS:

1. Argument 3 in particular highlights the importance and significance of the Christoffel symbols: they are not a nuisance but a blessing, because they allow one to modify the definition of the acceleration $\ddot{x}^\mu = (d/d\tau)\dot{x}^\mu$ in such a way that it

behaves nicely under coordinate transformations, namely just like the velocity $\dot{x}^\mu$ itself. We will see later on, in section 5, that the Christoffel symbols allow us to accomplish this much more generally. Namely, with their aid we can define a (generally covariant) notion of a derivative that generalises the ordinary (or partial) derivative and maps tensors to tensors.

2. General covariance, i.e. invariance of the equations of motion under general coordinate transformations, as exhibited e.g. by the geodesic equation, is of course a desirable feature regardless of whether or not one is attempting to describe gravity. After all, the particle could not care less which coordinates we use to describe its motion, and therefore we should also formulate the equations of motion for a particle in a way that does not single out some preferred coordinate system or class of coordinate systems. This is precisely what is achieved by general covariance.

   However, here general covariance seems to have arisen somewhat coincidentally and spontaneously, and the relation between general covariance and gravity, or general covariance and the equivalence principle, may still appear to be somewhat mysterious at this point. The precise relation between the two concepts will be explained in section 4.1.

3. There is of course a very good physical reason for why the force term in the geodesic equation (quadratic in the 4-velocities) is not tensorial. This simply reflects the equivalence principle that locally, at a point (or in a sufficiently small neighbourhood of a point) you can eliminate the gravitational force by going to a freely falling (inertial) coordinate system. This would not be possible if the gravitational force term in the equation of motion for a particle were tensorial.

---

1. Proof of (2.79)

   For partial derivatives one has the chain rule $\partial_\gamma = J^\lambda_\gamma \partial_\lambda$ ("∂_λ is a covector"). Therefore for the partial derivatives of the metric one has

$$g_{\alpha\beta,\gamma} = (J^\mu_\alpha J^\nu_\beta g_{\mu\nu})_{,\gamma} = g_{\mu\nu,\lambda} J^\mu_\alpha J^\nu_\beta J^\lambda_\gamma + (J^\mu_{\alpha\gamma} J^\nu_\beta + J^\mu_\alpha J^\nu_{\beta\gamma}) g_{\mu\nu} \quad . \tag{2.81}$$

   Adding up the 3 terms comprising the Christoffel symbol $\Gamma_{\alpha\beta\gamma}$, one obtains

$$\begin{aligned} 2\Gamma_{\alpha\beta\gamma} &= g_{\alpha\beta,\gamma} + g_{\alpha\gamma,\beta} - g_{\beta\gamma,\alpha} \\ &= 2 J^\mu_\alpha J^\nu_\beta J^\lambda_\gamma \Gamma_{\mu\nu\lambda} \\ &\quad + (J^\mu_{\alpha\gamma} J^\nu_\beta + J^\mu_\alpha J^\nu_{\beta\gamma} + J^\mu_{\alpha\beta} J^\nu_\gamma + J^\mu_\alpha J^\nu_{\gamma\beta} - J^\mu_{\beta\alpha} J^\nu_\gamma - J^\mu_\beta J^\nu_{\gamma\alpha}) g_{\mu\nu} \quad . \end{aligned} \tag{2.82}$$

   In the last line, the 3rd term cancels against the 5th (because $J^\mu_{\alpha\beta}$ is symmetric), the 1st term cancels against the 6th (because $J^\mu_{\alpha\gamma}$ and $g_{\mu\nu}$ are symmetric), while

the 2nd and 4th term add up, so that one finds

$$\Gamma_{\alpha\beta\gamma} = J^\mu_\alpha J^\nu_\beta J^\lambda_\gamma \Gamma_{\mu\nu\lambda} + J^\mu_\alpha J^\nu_{\beta\gamma} g_{\mu\nu} \quad . \tag{2.83}$$

Now the hard work has been done. Raising the 1st index of the Christoffel symbol, using the inverse metric

$$g^{\alpha\delta} = g^{\sigma\rho} J^\alpha_\sigma J^\delta_\rho \quad , \tag{2.84}$$

it is now simple to see that one obtains the claimed result (4),

$$\Gamma^\alpha_{\beta\gamma} = g^{\alpha\delta}\Gamma_{\delta\beta\gamma} = J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \Gamma^\mu_{\ \nu\lambda} + J^\alpha_\mu J^\mu_{\beta\gamma} \quad . \tag{2.85}$$

For example, for the 2nd term one has (just using properties of inverse Jacobi matrices and metrics)

$$\begin{aligned} g^{\alpha\delta} J^\mu_\delta J^\nu_{\beta\gamma} g_{\mu\nu} &= g^{\sigma\rho} J^\alpha_\sigma J^\delta_\rho J^\mu_\delta J^\nu_{\beta\gamma} g_{\mu\nu} = g^{\sigma\rho} J^\alpha_\sigma \delta^\mu_\rho J^\nu_{\beta\gamma} g_{\mu\nu} \\ &= g^{\sigma\mu} J^\alpha_\sigma J^\nu_{\beta\gamma} g_{\mu\nu} = \delta^\sigma_\nu J^\alpha_\sigma J^\nu_{\beta\gamma} = J^\alpha_\nu J^\nu_{\beta\gamma} \end{aligned} \tag{2.86}$$

2. Proof of (2.72)

   The 4-velocities transform as vectors (the chain rule again), $\dot{y}^\alpha = J^\alpha_\mu \dot{x}^\mu$. Therefore for the acceleration one has

$$\ddot{y}^\alpha = J^\alpha_\mu \ddot{x}^\mu + J^\alpha_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \quad . \tag{2.87}$$

   Therefore

$$\begin{aligned} \ddot{y}^\alpha + \Gamma^\alpha_{\beta\gamma} \dot{y}^\beta \dot{y}^\gamma &= J^\alpha_\mu (\ddot{x}^\mu + \Gamma^\mu_{\ \nu\lambda} J^\nu_\beta J^\lambda_\gamma \dot{y}^\beta \dot{y}^\gamma) + J^\alpha_\mu J^\mu_{\beta\gamma} \dot{y}^\beta \dot{y}^\gamma + J^\alpha_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \\ &= J^\alpha_\mu (\ddot{x}^\mu + \Gamma^\mu_{\ \nu\lambda} \dot{x}^\nu \dot{x}^\lambda) + (J^\alpha_\mu J^\mu_{\beta\gamma} + J^\alpha_{\mu\nu} J^\mu_\beta J^\nu_\gamma) \dot{y}^\beta \dot{y}^\gamma \end{aligned} \tag{2.88}$$

   The 1st term will give us the desired result, and cooperatively the 2nd term is identically zero because (use $\partial_\gamma = J^\nu_\gamma \partial_\nu$ again)

$$0 = (\delta^\alpha_\beta)_{,\gamma} = (J^\alpha_\mu J^\mu_\beta)_{,\gamma} = J^\alpha_{\mu\nu} J^\nu_\gamma J^\mu_\beta + J^\alpha_\mu J^\mu_{\beta\gamma} \quad . \tag{2.89}$$

Apology and Outlook:

You may feel that, after a promising start, some of the things that we have done subsequently (in particular in this subsection) look terribly messy. I agree, indeed they are! However, I can assure you that this is by far the messiest part of the entire lecture notes and that things will improve dramatically rather quickly.

Indeed, the main purpose and benefit of developing tensor calculus in the next couple of sections is to develop a formalism in the framework of which (among other things)

- one can avoid having to deal explicitly with objects that transform in complicated ways under coordinate transformations
- the transformation behaviour of any object is manifest (and does not have to be checked)
- it is straightforward to write down equations that are *generally covariant*, i.e. independent of the coordinate system in the sense that they are satisfied in all coordinate systems if and only if they are satisfied in one.

This tensor calculus formalism is simple, elegant and efficient and will then allow us to make rapid progress towards describing the dynamics in a (and subsequently of the) gravitational field in a way compatible with the Einstein equivalence principle.

2.5 Alternative Action Principles for Geodesics

As we have already noted in section 2.3, there is a problem with the action $S \sim m\int d\tau$ (2.56) for massless particles (null geodesics). For this reason and many other practical purposes (the square root in the action is awkward) it is much more convenient to use, instead of the action

$$S_0[x] = -m\int d\tau = -m\int d\lambda\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \equiv \int d\lambda\ \mathcal{L}_0^\lambda \tag{2.90}$$

the simpler Lagrangian

$$\mathcal{L} = \tfrac{1}{2}g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \tag{2.91}$$

and action

$$S_1[x] = \int d\lambda\ \mathcal{L}\ . \tag{2.92}$$

Let us first verify that S_1 really leads to the same equations of motion as S_0. Either by direct variation of the action, or by using the Euler-Lagrange equations

$$\frac{d}{d\lambda}\frac{\partial\mathcal{L}}{\partial(dx^\gamma/d\lambda)} - \frac{\partial\mathcal{L}}{\partial x^\gamma} = 0\ , \tag{2.93}$$

one finds that the action is extremised by solutions to the equation

$$\frac{d}{d\lambda}\left(g_{\gamma\beta}\frac{dx^\beta}{d\lambda}\right) = \tfrac{1}{2}g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\ . \tag{2.94}$$

The terms involving first derivatives of the metric cooperatively combine into the Christoffel symbols,

$$\left(\frac{d}{d\lambda}g_{\gamma\beta}\right)\frac{dx^\beta}{d\lambda} - \tfrac{1}{2}g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = g_{\gamma\beta,\alpha}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} - \tfrac{1}{2}g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = \Gamma_{\gamma\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\ . \tag{2.95}$$

Here we have used the fact that we can write

$$g_{\gamma\beta,\alpha}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = \tfrac{1}{2}\left(g_{\gamma\beta,\alpha} + g_{\gamma\alpha,\beta}\right)\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \tag{2.96}$$

because

$$\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = \frac{dx^\beta}{d\lambda}\frac{dx^\alpha}{d\lambda} \tag{2.97}$$

is symmetric. Therefore one has

$$g_{\gamma\beta}\frac{d^2x^\beta}{d\lambda^2} + \Gamma_{\gamma\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \ . \tag{2.98}$$

By raising the index (or multiplying with the inverse metric) one can write this as

$$\frac{d^2x^\gamma}{d\lambda^2} + \Gamma^\gamma_{\ \alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \ . \tag{2.99}$$

This is identical to the geodesic equation derived from S_0 (with $\lambda \to \tau$, the proper time).

We will make extensive use of this simpler Lagrangian for geodesics throughout these notes. In particular, in practice the version (2.94) of the geodesic action is much more efficient and user-friendly than the standard form, because everything is expressed directly in terms of the metric and its first derivatives (neither does one need the inverse metric, nor does one have to assemble the derivatives of the metric into Christoffel symbols first).

Moreover, as will be explained in section 3.1 below, (2.94) actually also provides one with a fairly efficient method to determine (essentially read off) the Christoffel symbols, simply by comparing (2.94) with (2.98) or (2.99).

One important consequence of (2.99) is that the quantity $\mathcal{L}$ is a constant of motion, i.e. constant along the geodesic,

$$\frac{d^2x^\gamma}{d\lambda^2} + \Gamma^\gamma_{\ \alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \quad \Rightarrow \quad \frac{d}{d\lambda}\left(g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\right) = 0 \ . \tag{2.100}$$

This useful result can be understood and derived in a variety of ways:

- The least insightful way is just direct calculation. Nevertheless, this is straightforward and a good exercise in Γ-ology (and as such is left as an exercise).
- Alternatively, noting that $\mathcal{L}$ does not depend explicitly on λ, this result can be derived (as the corresponding conserved "energy") from Noether's theorem (cf. section 2.6 below for this argument).
- Yet another derivation will be given in section 5.8, using the concept of "parallel transport".

One obvious consequence of (2.100) is that, if one imposes the initial condition

$$g_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}\bigg|_{\lambda=0} = \epsilon \ , \tag{2.101}$$

then this condition will be satisfied for all λ. This is as it should be. After all, something that starts off as a massless particle will remain a massless particle etc. In particular, therefore, even though with this choice of Lagrangian λ is a priori unrelated to proper time, say, this shows that one can choose $\epsilon = \mp 1$ for timelike (spacelike) geodesics, and λ can then be identified with proper time (proper distance), while the choice $\epsilon = 0$ sets the initial conditions appropriate to massless particles (for which λ is then not related to proper time or proper distance).

Moreover, the constancy of $\mathcal{L}$ for solutions of the Euler-Lagrange equations is the reason (or one explanation for) why $\mathcal{L}$ and $\sqrt{\pm\mathcal{L}}$ (more generally any monotonic function $f(\mathcal{L})$ of $\mathcal{L}$) give rise to equivalent equations of motion.

Indeed, this can be seen by simply comparing the Euler-Lagrange equations for $f(\mathcal{L})$ with those for $\mathcal{L}$. Denoting the Euler-Lagrange equations for a Lagrangian $\mathcal{L}$ by

$$\mathcal{E}^{\mathcal{L}}_\gamma = \frac{d}{d\lambda}\frac{\partial\mathcal{L}}{\partial(dx^\gamma/d\lambda)} - \frac{\partial\mathcal{L}}{\partial x^\gamma} \ , \tag{2.102}$$

for any $\mathcal{L}$ and any f one has

$$\mathcal{E}^{f(\mathcal{L})}_\gamma = f'(\mathcal{L})\mathcal{E}^{\mathcal{L}}_\gamma + f''(\mathcal{L})\left(\frac{d}{d\lambda}\mathcal{L}\right)\frac{\partial\mathcal{L}}{\partial(dx^\gamma/d\lambda)} \ . \tag{2.103}$$

Thus, if

$$\mathcal{E}^{\mathcal{L}}_\gamma = 0 \quad \Rightarrow \quad \frac{d}{d\lambda}\mathcal{L} = 0 \ , \tag{2.104}$$

then one has

$$\mathcal{E}^{\mathcal{L}}_\gamma = 0 \quad \Rightarrow \quad \mathcal{E}^{f(\mathcal{L})}_\gamma = 0 \ , \tag{2.105}$$

and if f is monotonic ($f' \neq 0$ everywhere), then one also has the converse,

$$f' \neq 0 \quad \Rightarrow \quad \left(\mathcal{E}^{\mathcal{L}}_\gamma = 0 \quad \Leftrightarrow \quad \mathcal{E}^{f(\mathcal{L})}_\gamma = 0\right) \ . \tag{2.106}$$

2.6 On the Relation between the two Action Principles

Even though not strictly required in the following, it is nevertheless quite instructive in its own right to try to understand and establish the precise relation between the two actions S_0 and S_1, and this is the subject of this subsection.

The first thing to notice is that S_0 is manifestly parametrisation-invariant, i.e. independent of how one parametrises the path. The reason for this is that

$$d\tau = (d\tau/d\lambda)d\lambda \tag{2.107}$$

is evidently independent of λ. This is not the case for S_1, which changes under parametrisations or, put more positively, singles out a preferred parametrisation (more precisely, as we will see below, a special class of parametrisations).

Thus, what is the relation (if any) between the two actions? In order to explain this, it will be useful to introduce an additional field $e(\lambda)$ (i.e. in addition to the $x^\alpha(\lambda)$), and a "master action" (or parent action) S which we can relate to both S_0 and S_1. Consider the action

$$S[x,e] = \tfrac{1}{2}\int d\lambda \left(e(\lambda)^{-1} g_{\alpha\beta} \frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} - m^2 e(\lambda)\right) = \int d\lambda \left(e(\lambda)^{-1}\mathcal{L} - \tfrac{1}{2}m^2 e(\lambda)\right) \tag{2.108}$$

The crucial property of this action is that it is parametrisation invariant provided that one declares $e(\lambda)$ to transform appropriately. It is easy to see that under a transformation $\lambda \to \bar\lambda = f(\lambda)$, with

$$\bar x^\alpha(\bar\lambda) = x^\alpha(\lambda) \qquad d\bar\lambda = f'(\lambda) d\lambda \tag{2.109}$$

the action $S[x,e]$ is invariant provided that $e(\lambda)$ transforms such that $e(\lambda)d\lambda$ is invariant, i.e.

$$\bar e(\bar\lambda) d\bar\lambda \stackrel{!}{=} e(\lambda) d\lambda \quad \Rightarrow \quad \bar e(\bar\lambda) = e(\lambda)/f'(\lambda) \ . \tag{2.110}$$

Indeed, this is evident when one writes the action (2.108) in the form

$$S[x,e] = \tfrac{1}{2}\int e(\lambda) d\lambda \left(e(\lambda)^{-2} g_{\alpha\beta} \frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} - m^2\right) \tag{2.111}$$

and notes that $d\lambda$ and $e(\lambda)$ only appear in the combinations $e(\lambda)d\lambda$ and $e(\lambda)^{-1}(d/d\lambda)$.

Now what is the relation between the action $S[x,e]$ and the two "standard" actions $S_0[x]$ and $S_1[x]$?

- Courtesy of this parametrisation invariance, we can always choose a "gauge" in which $e(\lambda) = 1$. With this choice, the action $S[x,e]$ manifestly reduces to the action $S_1[x]$ modulo an irrelevant field-independent constant,

  $$S[x, e=1] = \int d\lambda\ \mathcal{L} - \tfrac{1}{2}m^2 \int d\lambda = S_1[x] + \text{const.} \ . \tag{2.112}$$

  Thus we can regard S_1 as a gauge-fixed version of S (no wonder it is not parametrisation invariant ...). We will come back to the small residual gauge invariance (reparametrisations that preserve the gauge condition $e(\lambda) = 1$) below.

- Alternatively, instead of fixing the gauge, we can try to eliminate $e(\lambda)$ (which appears purely algebraically, i.e. without derivatives, in the action) by its equation of motion. Varying $S[x,e]$ with respect to $e(\lambda)$, one finds the constraint

  $$g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} + m^2 e(\lambda)^2 = 0 \ . \tag{2.113}$$

This is just the usual mass-shell condition in disguise. It suggests that a better gauge fixing than $e(\lambda) = 1$ would have been $e(\lambda) = m^{-1}$. However, the sole effect of this would have been to replace $\mathcal{L}$ in (2.112) by $m\mathcal{L}$,

$$e(\lambda) = 1 \to e(\lambda) = m^{-1} \quad \Rightarrow \quad \mathcal{L} \to m\mathcal{L} \ . \tag{2.114}$$

In any case, for a massive particle, $m^2 \neq 0$, one can alternatively solve (2.113) for $e(\lambda)$,

$$e(\lambda) = m^{-1}\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \ . \tag{2.115}$$

Using this to eliminate $e(\lambda)$ from the action, one finds

$$S[x, e = m^{-1}\sqrt{\dots}] = -m\int d\lambda\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} = -m\int d\tau = S_0[x] \ . \tag{2.116}$$

Thus for $m^2 \neq 0$ we find exactly the original action (integral of the proper time) $S_0[x]$ (and since we have not touched or fixed the parametrisation invariance, no wonder that S_0 is parametrisation invariant).

Thus we have elucidated the common origin of the actions S_0 and S_1 for a massive particle.

The perspective provided by the parent action $S[x, e]$ also gives some further insights. For example, an added benefit of the parent action $S[x, e]$ is that it also makes perfect sense for a massless particle. For $m^2 = 0$, the mass shell condition

$$g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \tag{2.117}$$

says that these particles move along null lines, and the action reduces to

$$S[x, e] = \tfrac{1}{2}\int d\lambda\ e(\lambda)^{-1} g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \tag{2.118}$$

which is parametrisation invariant but can (as in the massive case) be fixed to $e(\lambda) = 1$, upon which the action reduces to $S_1[x]$. Thus we see that $S_1[x]$ indeed provides a simple and unified action for both massive and massless particles, and in both cases the resulting equation of motion is the (affinely parametrised) geodesic equation (2.99),

$$\frac{d^2x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\beta\gamma}\frac{dx^\beta}{d\lambda}\frac{dx^\gamma}{d\lambda} = 0 \ . \tag{2.119}$$

REMARKS:

1. The infinitesimal form of the invariance of the action $S[x, e]$ under (2.109) and (2.110) is obtained by considering the infinitesimal transformation of $x^\alpha(\lambda)$ and

$e(\lambda)$ induced by an infinitesimal transformation $\bar{\lambda} = \lambda + \epsilon(\lambda)$,

$$\begin{aligned} \delta\lambda = \epsilon(\lambda) \quad \Rightarrow \quad \delta x^\alpha(\lambda) &= \epsilon(\lambda)\frac{dx^\alpha(\lambda)}{d\lambda} \\ \delta e(\lambda) &= \frac{d\epsilon(\lambda)}{d\lambda}e(\lambda) + \epsilon(\lambda)\frac{de(\lambda)}{d\lambda} \quad . \end{aligned} \tag{2.120}$$

Here the (at first perhaps somewhat peculiar looking) transformation behaviour of the auxiliary field $e(\lambda)$ arises from the transformation behaviour (2.110) by setting

$$\delta e(\lambda) = e(\lambda) - \bar{e}(\lambda) \tag{2.121}$$

and calculating (keeping at most linear terms in $\epsilon(\lambda)$)

$$\begin{aligned} e(\lambda) &= f'(\lambda)\bar{e}(\bar{\lambda}) = (1 + \epsilon'(\lambda))\ \bar{e}(\lambda + \epsilon(\lambda)) \\ &= (1 + \epsilon'(\lambda))\ (\bar{e}(\lambda) + \epsilon(\lambda)\bar{e}'(\lambda)) \\ &= \bar{e}(\lambda) + \epsilon'(\lambda)\bar{e}(\lambda) + \epsilon(\lambda)\bar{e}'(\lambda) \quad . \end{aligned} \tag{2.122}$$

Under this infinitesimal transformation, the Lagrangian

$$\mathcal{L}_e = e(\lambda)^{-1}\mathcal{L} - \tfrac{1}{2}m^2 e(\lambda) \tag{2.123}$$

of the action $S[x, e]$ (2.108) transforms as

$$\delta\mathcal{L}_e = \frac{d}{d\lambda}(\epsilon(\lambda)\mathcal{L}_e) \quad , \tag{2.124}$$

implying the invariance of the action.

2. We saw in (2.100), that the Lagrangian $\mathcal{L}$ itself is a constant of motion,

$$\frac{d}{d\lambda}\left(g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}\right) = 0 \tag{2.125}$$

for a solution to the geodesic equation. From the present (action-based) perspective it is most useful to think of this as the conserved quantity associated (via Noether's theorem) to the invariance of the action $S_1[x]$ under translations in λ. Note that evidently $S_1[x]$ has this invariance (as there is no explicit dependence on λ) and that this invariance is precisely the residual parametrisation invariance $f(\lambda) = \lambda + a$, $f'(\lambda) = 1$, that leaves invariant the "gauge" condition $e(\lambda) = 1$. For an infinitesimal constant λ-translation one has $\delta x^\alpha(\lambda) = dx^\alpha/d\lambda$ etc., so that

$$\frac{\partial}{\partial\lambda}\mathcal{L} = 0 \quad \Rightarrow \quad \delta\mathcal{L} = \frac{d}{d\lambda}\mathcal{L} = \frac{d}{d\lambda}\left(\frac{\partial\mathcal{L}}{\partial(dx^\alpha/d\lambda)}\frac{dx^\alpha}{d\lambda}\right) + \quad \text{Euler} - \text{Lagrange} \quad . \tag{2.126}$$

Thus via Noether's theorem the associated conserved charge for a solution to the Euler-Lagrange equations is the Legendre transform

$$\mathcal{H} = \left(\frac{\partial\mathcal{L}}{\partial(dx^\alpha/d\lambda)}\frac{dx^\alpha}{d\lambda}\right) - \mathcal{L} \tag{2.127}$$

of the Lagrangian (also known as the Hamiltonian, once expressed in terms of the momenta). In the case at hand, with the Lagrangian $\mathcal{L}$ consisting of a purely quadratic term in the velocities (the $dx^\alpha/d\lambda$), the Hamiltonian is equal to the Lagrangian, and hence the Lagrangian $\mathcal{L}$ itself is conserved,

$$\mathcal{H} = \mathcal{L} \quad , \quad \frac{d}{d\lambda}\mathcal{L}|_{\text{solution}} = 0 \ . \tag{2.128}$$

3. The above non-trivial Hamiltonian associated to the Lagrangian $\mathcal{L}$ should be contrasted with the Hamiltonian associated to the action S_0 with Lagrangian (2.90)

$$\mathcal{L}_0^\lambda = -m\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \ , \tag{2.129}$$

which turns out to be zero identically. Indeed, the canonical momenta are

$$p_\alpha = mg_{\alpha\beta}(dx^\beta/d\lambda)/\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} = mg_{\alpha\beta}(dx^\beta/d\tau) \ , \tag{2.130}$$

which evidently satisfy the mass shell condition

$$p_\alpha p^\alpha + m^2 = 0 \tag{2.131}$$

and lead to the Hamiltonian

$$\mathcal{H}_0 = p_\alpha(dx^\alpha/d\lambda) - \mathcal{L}_0^\lambda = 0 \ . \tag{2.132}$$

This vanishing of the Hamiltonian is strictly related to the reparametrisation invariance of the action S_0.

2.7 Affine and Non-affine Parametrisations

To understand the significance of how one parametrises the geodesic, observe that the geodesic equation itself,

$$\ddot{x}^\mu + \Gamma^\mu_{\ \nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{2.133}$$

is not reparametrisation invariant. Indeed, consider a change of parametrisation $\tau \to \sigma = f(\tau)$. Then

$$\frac{dx^\mu}{d\tau} = \frac{df}{d\tau}\frac{dx^\mu}{d\sigma} \ , \tag{2.134}$$

and therefore the geodesic equation written in terms of σ reads

$$\frac{d^2x^\mu}{d\sigma^2} + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\sigma}\frac{dx^\lambda}{d\sigma} = -\frac{\ddot{f}}{\dot{f}^2}\frac{dx^\mu}{d\sigma} \ . \tag{2.135}$$

Thus the geodesic equation retains its form only under affine changes of the proper time parameter τ, $f(\tau) = a\tau + b$, and parameters $\sigma = f(\tau)$ related to τ by such an affine transformation are known as affine parameters.

From the first variational principle, based on S_0, the term on the right hand side arises in the calculation of (2.63) from the integration by parts if one does not switch back from λ to the affine parameter τ. The second variational principle, based on S_1 and the Lagrangian $\mathcal{L}$, on the other hand, always and automatically yields the geodesic equation in affine form.

Conversely, if we find a curve that satisfies

$$\frac{d^2x^\mu}{d\sigma^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\sigma}\frac{dx^\lambda}{d\sigma} = \kappa(\sigma)\frac{dx^\mu}{d\sigma} \quad , \tag{2.136}$$

for some function $\kappa(\sigma)$ (the *inaffinity*), we can deduce that this curve is the trajectory of a geodesic, but that it is simply not parametrised by an affine parameter (like proper time in the case of a timelike curve). Comparison of (2.135) and (2.136) shows that, given $\kappa(\sigma)$, an affine parameter τ is determined by

$$\kappa(f(\tau)) = -\frac{\ddot{f}}{\dot{f}^2} \quad \Leftrightarrow \quad \kappa(\sigma) = \frac{d}{d\sigma}\ln\frac{d\tau}{d\sigma} \tag{2.137}$$

or

$$\frac{d\tau}{d\sigma} = \mathrm{e}^{\int^\sigma ds\, \kappa(s)} \quad . \tag{2.138}$$

The two integration constants, the first hidden in the lower limit of integration in the exponent and the second in the additive constant arising from integrating $d\tau/d\sigma$, are precisely the two constants a, b that parametrise the freedom in the choice of affine parameter, $\tau \to a\tau + b$.

In the following, whenever we talk about geodesics we will practically always have in mind the variational principle based on S_1 leading to the geodesic equation (2.119) in affinely parametrised form.

However, it should be kept in mind that sometimes non-affine parameters appear naturally. For instance, it is occasionally convenient to parametrise timelike geodesics in a geometry with coordinates $x^\alpha = (x^0 = t, x^k)$ not by $x^\alpha = x^\alpha(\tau)$, where τ is the proper time along the geodesic, but rather as $x^k = x^k(t)$. This is the same curve, but described with respect to coordinate time (which could for instance agree with the proper time of some other, perhaps static, observer). The curve $t \to (t, x^k(t))$ will not be an affinely parametrised curve unless t itself satisfies the geodesic equation

$$\ddot{t} = 0 \quad \Leftrightarrow \quad t = a\tau + b \ . \tag{2.139}$$

One occasion where this will play a role (and from where I have borrowed the symbol κ for the "inaffinity") is in our discussion, much later, of the horizon of a black hole, where the lack of a certain coordinate to be an affine parameter is directly related to the physical properties of black holes (see section 27.10). In this context κ is known as the surface gravity of a black hole.

2.8 Example: Geodesics in $\mathbb{R}^2$ in Polar Coordinates

It is high time to consider an example. We will consider the simplest non-trivial metric, namely the standard Euclidean metric on $\mathbb{R}^2$ in polar coordinates. Thus the line element is

$$ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2 \tag{2.140}$$

and the non-zero components of the metric are

$$g_{xx} = g_{yy} = 1 \tag{2.141}$$

and

$$g_{rr} = 1 \quad , \quad g_{\phi\phi} = r^2 \ . \tag{2.142}$$

respectively. Since this metric is diagonal, the non-zero components of the inverse metric $g^{\mu\nu}$ are

$$g^{xx} = g^{yy} = 1 \tag{2.143}$$

and

$$g^{rr} = 1 \quad , \quad g^{\phi\phi} = r^{-2} \tag{2.144}$$

respectively.

A reminder on notation (cf. the discussion leading to (2.13)): since μ, ν in $g_{\mu\nu}$ are coordinate indices, we should really have called $x^1 = r$, $x^2 = \phi$, say, and written $g_{11} = 1, g_{22} = r^2$, etc. However, writing g_{rr} etc. is more informative and useful since one then knows that this is the (rr)-component of the metric without having to remember if one called $r = x^1$ or $r = x^2$. In the following we will frequently use this kind of notation when dealing with a specific coordinate system, while we retain the index notation $g_{\mu\nu}$ etc. for general purposes.

Let us now look at the geodesic equations for this metric, first in the Cartesian coordinates (x, y) and then in the polar coordinates (r, ϕ).

1. Cartesian coordinates

   Since the metric in Cartesian coordinates is the constant Euclidean metric $g_{\mu\nu} = \delta_{\mu\nu}$, all the partial derivatives of the metric are zero, and therefore also all the Christoffel symbols are zero. The geodesic equations thus take the form

   $$\ddot{x} = \ddot{y} = 0 \ . \tag{2.145}$$

   These equations could also have been obtained as the Euler-Lagrange equations of the Lagrangian

   $$\mathcal{L} = \tfrac{1}{2}(\dot{x}^2 + \dot{y}^2) \ . \tag{2.146}$$

   The general solution is

   $$x(s) = as + b \quad , \quad y(s) = cs + d \ . \tag{2.147}$$

Combining these two, one finds the standard representation

$$y = kx + e \tag{2.148}$$

for a straight line in $\mathbb{R}^2$, with slope k and intercept e.

2. Polar coordinates

Now let us consider the same problem in polar coordinates. The crucial point here is that in these coordinates the geodesic equations will not simply be $\ddot{r} = \ddot{\phi} = 0$, but that there are additional terms arising

- either from the non-linear coordinate transformation between Cartesian and polar coordinates
- or equivalently from the fact that the coefficients of the metric are not constant in polar coordinates.

Taking the latter point of view, the Christoffel symbols of this metric are to be calculated from

$$\Gamma_{\mu\nu\lambda} = \tfrac{1}{2}(g_{\mu\nu,\lambda} + g_{\mu\lambda,\nu} - g_{\nu\lambda,\mu}) \ . \tag{2.149}$$

Since the only non-trivial derivative of the metric is $g_{\phi\phi,r} = 2r$, only Christoffel symbols with exactly two ϕ's and one r are non-zero,

$$\begin{aligned} \Gamma_{r\phi\phi} &= \tfrac{1}{2}(g_{r\phi,\phi} + g_{r\phi,\phi} - g_{\phi\phi,r}) = -r \\ \Gamma_{\phi\phi r} &= \Gamma_{\phi r\phi} = r \ . \end{aligned} \tag{2.150}$$

Thus, since the metric is diagonal, the non-zero $\Gamma^{\mu}_{\ \nu\lambda}$ are

$$\begin{aligned} \Gamma^{r}_{\ \phi\phi} &= g^{r\mu}\Gamma_{\mu\phi\phi} = g^{rr}\Gamma_{r\phi\phi} = -r \\ \Gamma^{\phi}_{\ r\phi} &= \Gamma^{\phi}_{\ \phi r} = g^{\phi\mu}\Gamma_{\mu r\phi} = g^{\phi\phi}\Gamma_{\phi r\phi} = \frac{1}{r} \ . \end{aligned} \tag{2.151}$$

Note that here it was even convenient to use a hybrid notation, as in $g^{r\mu}$, where r is a coordinate and μ is a coordinate index. Once again, it is very convenient to permit oneself to use such a mixed notation.

In any case, having assembled all the Christoffel symbols, we can now write down the geodesic equations (once again in the convenient hybrid notation). For r one has

$$\ddot{r} + \Gamma^{r}_{\ \mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} = 0 \ , \tag{2.152}$$

which, since the only non-zero $\Gamma^{r}_{\ \mu\nu}$ is $\Gamma^{r}_{\ \phi\phi}$, reduces to

$$\ddot{r} - r\dot{\phi}^2 = 0 \ . \tag{2.153}$$

Likewise for ϕ one finds

$$\ddot{\phi} + \frac{2}{r}\dot{\phi}\dot{r} = 0 \ . \tag{2.154}$$

Here the factor of 2 arises because both $\Gamma^{\phi}_{r\phi}$ and $\Gamma^{\phi}_{\phi r} = \Gamma^{\phi}_{r\phi}$ contribute.

REMARKS:

(a) This equation is supposed to describe geodesics in $\mathbb{R}^2$, i.e. straight lines. This can be verified in general (but, in general, polar coordinates are of course not particularly well suited to describe straight lines). However, it is easy to find a special class of solutions to the above equations, namely curves with $\dot{\phi} = \ddot{r} = 0$. These correspond to paths of the form

$$(r(s), \phi(s)) = (s, \phi_0) \ , \tag{2.155}$$

which are a special case of straight lines, namely straight lines through the origin.

(b) The geodesic equations can of course also be derived as the Euler-Lagrange equations of the Lagrangian

$$\mathcal{L} = \tfrac{1}{2}(\dot{r}^2 + r^2\dot{\phi}^2) \ . \tag{2.156}$$

Indeed, one has

$$\begin{aligned} \frac{d}{ds}\frac{\partial \mathcal{L}}{\partial \dot{r}} - \frac{\partial \mathcal{L}}{\partial r} &= \ddot{r} - r\dot{\phi}^2 = 0 \\ \frac{d}{ds}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{\partial \mathcal{L}}{\partial \phi} &= r^2\ddot{\phi} + 2r\dot{r}\dot{\phi} = 0 \ , \end{aligned} \tag{2.157}$$

which are obviously identical to the equations derived above.

(c) You may have the impression that getting the geodesic equation in this way, rather than via calculation of the Christoffel symbols first, is much simpler. I agree wholeheartedly. Not only is the Lagrangian approach the method of choice to determine the geodesic equations. It is also frequently the most efficient method to determine the Christoffel symbols. This will be described in section 3.1.

(d) Another advantage of the Lagrangian formulation is, as in classical mechanics, that it makes it much easier to detect and exploit symmetries. Indeed, you may have already noticed that the above second-order equation for ϕ is overkill. Since the Lagrangian does not depend on ϕ (i.e. it is invariant under rotations), one has

$$\frac{d}{ds}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 0 \ , \tag{2.158}$$

which means that $\partial\mathcal{L}/\partial\dot{\phi}$ is a constant of motion, the angular momentum L,

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = r^2\dot{\phi} = L \ . \tag{2.159}$$

This equation is a first integral of the second-order equation for ϕ. We will come back to this in somewhat more generality below.

The next simplest example to discuss would be the two-sphere with its standard metric $d\theta^2 + \sin^2\theta d\phi^2$. It will appear, in bits and pieces, in section 3.1 to illustrate the general remarks.

2.9 Example: Geodesics for Ultrastatic and Direct Product Metrics

As another example, let us consider the ultrastatic metrics introduced in (2.34) with coordinates $x^\mu = (t, x^k)$ and line-element

$$ds^2 = -dt^2 + \tilde{g}_{ik}(x)dx^i dx^k \ . \tag{2.160}$$

Because $g_{00} = -1$, $g_{0k} = 0$, and the $g_{ik} = \tilde{g}_{ik}$ are time-independent, all Christoffel symbols with at least one x^0- or t-index are zero,

$$\Gamma_{0\mu\nu} = \Gamma_{\mu 0\nu} = \Gamma_{\mu\nu 0} = 0 \ , \tag{2.161}$$

and the purely spatial components of the Christoffel symbols agree with those of the spatial metric,

$$\Gamma^i_{\ jk} = \tilde{\Gamma}^i_{\ jk} \ . \tag{2.162}$$

Therefore the geodesic equations read, for the t-component,

$$\ddot{t} = 0 \ , \tag{2.163}$$

and for the spatial components

$$\ddot{x}^i + \tilde{\Gamma}^i_{\ jk}\dot{x}^j\dot{x}^k = 0 \tag{2.164}$$

where the dot denotes a derivative with respect to the affine parameter τ. The first equation tells us that

$$\ddot{t} = 0 \quad \Leftrightarrow \quad t(\tau) = a\tau + t_0 \ . \tag{2.165}$$

Thus provided that $a \neq 0$ we can use t instead of τ to parametrise the paths (and in the present case t is then also an affine parameter, cf. the discussion in section 2.7 in connection with (2.139)), and then one can rewrite the spatial equations as equations for $x^i = x^i(t)$,

$$\frac{d^2x^i}{dt^2} + \tilde{\Gamma}^i_{\ jk}\frac{dx^j}{dt}\frac{dx^k}{dt} = 0 \ . \tag{2.166}$$

Therefore the solutions to the space-time geodesic equations have the form

$$x^\alpha(t) = (t, x^i(t)) \tag{2.167}$$

where $x^i(t)$ is an affinely parametrised geodesic for the metric $\tilde{g}_{ij}$. When $a = 0$, one cannot change variables from τ to t because $t = t_0$ is fixed. One is then necessarily dealing with spacelike geodesics in space-time and the solutions have the form

$$x^\alpha(\tau) = (t_0, x^i(\tau)) \tag{2.168}$$

where $x^i(\tau)$ is again an affinely parametrised geodesic for the metric $\tilde{g}_{ij}$.

These sorts of considerations evidently generalise to more general metrics of this direct product form,

$$ds^2 = g_{ab}(y)dy^a dy^b + g_{ik}(x)dx^i dx^k \quad , \tag{2.169}$$

with the conclusion that geodesics in such space-times have the form $(y^a(\tau), x^i(\tau))$ with $y^a(\tau)$ and $x^i(\tau)$ individually solutions of the geodesic equations for the metric $g_{ab}(y)$ respectively $g_{ik}(x)$.

3 Geodesics and Motion in a Gravitational Field

3.1 Consequences and Uses of the Euler-Lagrange Equations

Recall from above that the geodesic equation for a metric $g_{\mu\nu}$ can be derived from the Lagrangian $\mathcal{L} = (1/2)g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$

$$\frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial\dot{x}^\mu} - \frac{\partial\mathcal{L}}{\partial x^\mu} = 0 \ . \tag{3.1}$$

This has several immediate consequences which are useful for the determination of Christoffel symbols and geodesics in practice.

1. Conserved charges / first integrals of the geodesic equation

   Just as in classical mechanics, a coordinate the Lagrangian does not depend on explicitly (a cyclic coordinate) leads to a conserved quantity, associated with the translation invariance of the system in that direction. In the present context this means that if, say, $\partial\mathcal{L}/\partial x^1 = 0$ (this means that the coeffcients of the metric do not depend on x^1), then the corresponding momentum

   $$p_1 = \partial\mathcal{L}/\partial\dot{x}^1 = g_{1\nu}\dot{x}^\nu \tag{3.2}$$

   is conserved along the geodesic.

   Remarks:

   (a) One might perhaps have wanted to argue that the definition (and interpretation) of conserved momenta should be based on the physical Lagrangian (2.90)

   $$\mathcal{L}_0^\lambda = -m\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \tag{3.3}$$

   with action $S = -m\int d\tau$, but this makes no difference since the two momenta are essentially equal: one has

   $$\frac{\partial\mathcal{L}_0^\lambda}{\partial(dx^1/d\lambda)} = mp_1 \tag{3.4}$$

   with p_1 as defined in (3.2), so that this just supplies us with the additional information that the momenta obtained from the Lagrangian $\mathcal{L}$ should (for a massive particle) be interpreted as momenta per unit mass. This discrepancy could have been avoided by working with the Lagrangian $m\mathcal{L}$ (alternatively: fixing the gauge $e(\lambda) = m^{-1}$ in section 2.5, see (2.114)), but unless or until one starts coupling the particle to fields other than the gravitational field it is unnecessary (and a nuisance) to carry m around all the time.

(b) For example, on the two-sphere the Lagrangian reads

$$\mathcal{L} = \tfrac{1}{2}(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2) \ . \tag{3.5}$$

The angle ϕ is a cyclic variable and the angular momentum (actually angular momentum per unit mass for a massive particle)

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \sin^2\theta\dot{\phi} \tag{3.6}$$

is a conserved quantity. This generalises to conservation of angular momentum for a particle moving in an arbitrary spherically symmetric gravitational field.

(c) Likewise, if the metric is independent of the time coordinate $x^0 = t$, the corresponding conserved quantity

$$p_0 = g_{0\nu}\dot{x}^\nu \equiv -E \tag{3.7}$$

has the interpretation as minus the energy (per unit mass) of the particle, "minus" because, with our sign conventions, $p_0 = -E$ in special relativity. We will discuss the relation between this notion of energy and the notion of energy familiar from special relativity (this requires an asymptotically Minkowski-like metric) in more detail in section 25.3.

(d) We will discuss in more detail in section 3.2 (and then again in sections 9 and 10) how to detect and describe symmetries and conserved charges in coordinate systems in which the symmetries are not as manifest (via cyclic variables) as above.

2. Reading off (some) geodesics directly from the metric

Another immediate consequence is the following: consider a space or space-time with coordinates $\{y, x^\mu\}$ and a metric of the form

$$ds^2 = \pm dy^2 + g_{\mu\nu}(x, y)dx^\mu dx^\nu \ . \tag{3.8}$$

Then the coordinate lines of y are geodesics.

The quickest way to see this is (as usual) from the Lagrangian point of view. Indeed, since the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}(\pm\dot{y}^2 + g_{\mu\nu}(y, x)\dot{x}^\mu\dot{x}^\nu) \ , \tag{3.9}$$

the Euler-Lagrange equations are of the form

$$\begin{aligned} &\pm\ddot{y} - \tfrac{1}{2}g_{\mu\nu,y}\,\dot{x}^\mu\dot{x}^\nu = 0 \\ &\ddot{x}^\mu + \text{terms proportional to } \ \dot{x} = 0 \ . \end{aligned} \tag{3.10}$$

Therefore $\dot{x}^\mu = 0$, $\ddot{y} = 0$ is a solution of the geodesic equation, and it describes motion along the coordinate lines of y.

Alternatively, this special form of the metric implies that any Christoffel symbol with at least two y-indices is zero, and the conclusion then follows in the same way as above.

REMARKS:

(a) In the case of the two-sphere, with its metric $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$, this translates into the familiar statement that the great circles, the coordinate lines of $y = \theta$, are geodesics.

(b) The result is also valid when y is a timelike coordinate. For example, consider a space-time with coordinates (t, x^i) and metric (2.36)

$$ds^2 = -dt^2 + a(t)^2 \tilde{g}_{ij}(x) dx^i dx^j \quad . \tag{3.11}$$

In such a cosmological space-time, there is, according to the above result, a privileged class of freely falling (i.e. geodesic) observers, namely those that stay at fixed values of the spatial coordinates x^i. For such *comoving observers*, the coordinate-time t coincides with their proper time τ.

(c) In general, these preferred geodesics are orthogonal to the hypersurfaces of constant y, and coordinates in which the metric (locally) takes such a form in the neighbourhood of some timelike or spacelike hypersurface are occasionally called *Gaussian normal coordinates*.

3. Using the Euler-Lagrange equations to determine the Christoffel symbols

Finally, as mentioned and observed above, the Euler-Lagrange form of the geodesic equations frequently provides the most direct way of calculating Christoffel symbols - by comparing the Euler-Lagrange equations with the expected form of the geodesic equation in terms of Christoffel symbols. More precisely, by rewriting the Euler-Lagrange equations (2.94)

$$\frac{d}{d\lambda}\left(g_{\gamma\beta}\frac{dx^\beta}{d\lambda}\right) = \tfrac{1}{2} g_{\alpha\beta,\gamma}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} \quad . \tag{3.12}$$

in the form

$$\ddot{x}^\gamma + \text{terms proportional to } \dot{x}\dot{x} = 0 \ , \tag{3.13}$$

and comparing with the geodesic equation

$$\frac{d^2x^\gamma}{d\lambda^2} + \Gamma^\gamma_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \ , \tag{3.14}$$

one can read off the $\Gamma^\gamma_{\alpha\beta}$.

REMARKS:

(a) Careful - in this and similar calculations beware of factors of 2:

$$\Gamma^{\mu}_{\ \nu\lambda}\dot{x}^{\nu}\dot{x}^{\lambda} = \Gamma^{\mu}_{\ 11}(\dot{x}^1)^2 + 2\Gamma^{\mu}_{\ 12}\dot{x}^1\dot{x}^2 + \ldots \tag{3.15}$$

(b) For example, once again in the case of the two-sphere, for the θ-equation one has

$$\frac{d}{d\tau}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \ddot{\theta} \quad , \quad \frac{\partial \mathcal{L}}{\partial \theta} = \sin\theta\cos\theta\dot{\phi}^2 \ . \tag{3.16}$$

Comparing the resulting Euler-Lagrange equation

$$\ddot{\theta} - \sin\theta\cos\theta\dot{\phi}^2 = 0 \tag{3.17}$$

with the geodesic equation

$$\ddot{\theta} + \Gamma^{\theta}_{\ \theta\theta}\dot{\theta}^2 + 2\Gamma^{\theta}_{\ \theta\phi}\dot{\theta}\dot{\phi} + \Gamma^{\theta}_{\ \phi\phi}\dot{\phi}^2 = 0 \ , \tag{3.18}$$

one can immediately read off that

$$\Gamma^{\theta}_{\ \phi\phi} = -\sin\theta\cos\theta \quad , \quad \Gamma^{\theta}_{\ \theta\theta} = \Gamma^{\theta}_{\ \theta\phi} = 0 \ . \tag{3.19}$$

Likewise, from

$$\frac{d}{d\tau}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \quad \Leftrightarrow \quad \sin^2\theta(\ddot{\phi} + 2\cot\theta\dot{\theta}\dot{\phi}) = 0 \tag{3.20}$$

one deduces that

$$\Gamma^{\phi}_{\ \theta\phi} = \Gamma^{\phi}_{\ \phi\theta} = \cot\theta \quad , \quad \Gamma^{\phi}_{\ \theta\theta} = \Gamma^{\phi}_{\ \phi\phi} = 0 \ . \tag{3.21}$$

(c) As another example, which will turn out to be of considerable importance later on, consider a space-time metric of the form

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega^2 \ . \tag{3.22}$$

As will be discussed in section 24.2, this is the general form of a static spherically symmetric metric, and as such will provide us with the starting point for describing the gravitational field of a star. The corresponding Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\left(-A(r)\dot{t}^2 + B(r)\dot{r}^2 + r^2(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2)\right) \ , \tag{3.23}$$

and therefore the Euler-Lagrange equations for t (a cyclic variable) are

$$0 = \frac{d}{d\tau}(-A(r)\dot{t}) = -A(r)\left(\ddot{t} + \frac{A'(r)}{A(r)}\dot{r}\dot{t}\right) \tag{3.24}$$

(a prime denoting an r-derivative), from which one can immediately read off

$$\Gamma^{t}_{\ rt} = \Gamma^{t}_{\ tr} = \frac{A'}{2A} \quad , \quad \Gamma^{t}_{\ \mu\nu} = 0 \quad \text{otherwise} \ . \tag{3.25}$$

Likewise, the equation for r takes the form

$$\ddot{r} + \frac{B'}{2B}\dot{r}^2 + \frac{A'}{2B}\dot{t}^2 + \cdots = 0 \ , \tag{3.26}$$

and from this one can read off that

$$\Gamma^r{}_{rr} = \frac{B'}{2B} \quad , \quad \Gamma^r{}_{tt} = \frac{A'}{2B} \quad , \quad \ldots \tag{3.27}$$

As we will need them anyway in section 24.3, it is a good exercise to determine all the Christoffel symbols in this way.

3.2 Conserved Charges and (a first encounter with) Killing Vectors

In the previous section we have seen that cyclic coordinates, i.e. coordinates the metric does not depend on, lead to conserved charges, as in (3.2). As nice and useful as this may be (and it is nice and useful), it is obvioulsy somewhat unsatisfactory because it is an explicitly coordinate-dependent statement: the metric may well be independent of one coordinate in some coordinate system, but if one now performs a coordinate transformation which depends on that coordinate, then in the new coordinate system the metric will typically depend on all the new coordinates. Nevertheless,

- the statement that a metric has a certain symmetry (a translational symmetry in the first coordinate system) should be coordinate-independent, and
- thus there should be a corresponding first integral of the geodesic equation in any coordinate system.

To see how this works, let us reconsider the situation discussed in the previous section, namely a metric which in some coordinate system, we will now call it $\{y^\mu\}$, has components $g_{\mu\nu}$ which are independent of y^1, say. Translation invariance of the geodesic Lagrangian is the statement that the Lagrangian is invariant under the infinitesimal variation $\delta y^1 = \epsilon$, $\delta y^\mu = 0$ otherwise, and via Noether's theorem this leads to a conserved charge $g_{1\mu}\dot{y}^\mu$, as in (3.2).

Now we ask ourselves what this statement corresponds to in another coordinate system. Note that in the y-coordinates, invariance is the statement that the metric is invariant under the (infinitesimal) coordinate transformation $y^1 \to y^1 + \epsilon$ or $\delta y^1 = \epsilon$, $\delta y^\mu = 0$ otherwise,

$$\delta g_{\mu\nu} \equiv \epsilon \partial_{y^1} g_{\mu\nu} = 0 \ . \tag{3.28}$$

It is then clear that in another coordinate system, infinitesimal y^1-translations must also correspond to some infinitesimal coordinate transformation (but not necessarily just a translation),

$$\delta x^\alpha = \epsilon V^\alpha(x) \ . \tag{3.29}$$

In particular, if (as in the above example) in y-coordinates V^μ has the components $V^1 = 1$, $V^\mu = 0$ otherwise, then in any other coordinate system one has

$$\delta x^\alpha = (\partial x^\alpha / \partial y^\mu)\delta y^\mu = \epsilon(\partial x^\alpha / \partial y^1) \tag{3.30}$$

so that

$$V^\alpha = J_1^\alpha \tag{3.31}$$

is just the corresponding column of the Jacobi matrix.

In order to determine how to characterise the translational symmetry (3.28) of the metric in an arbitrary coordinate system, we will now proceed in two (as it turns out ultimately equivalent) ways.

1. We can investigate directly, under which conditions on the V^α the transformation (3.29) leads to an invariance of the Lagrangian (2.91). Using

   $$\delta\dot{x}^\alpha = \epsilon\dot{x}^\gamma\partial_\gamma V^\alpha \tag{3.32}$$

   one straightforwardly finds

   $$\delta\left(g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta\right) = \epsilon\left(\delta_V g_{\alpha\beta}\right)\dot{x}^\alpha\dot{x}^\beta \tag{3.33}$$

   where

   $$\delta_V g_{\alpha\beta} = V^\gamma\partial_\gamma g_{\alpha\beta} + (\partial_\alpha V^\gamma)g_{\gamma\beta} + (\partial_\beta V^\gamma)g_{\alpha\gamma} \tag{3.34}$$

   Thus the condition for the infinitesimal transformation (3.29) to leave the Lagrangian invariant is

   $$\delta_V g_{\alpha\beta} = 0 \ . \tag{3.35}$$

   Noether's theorem then leads to the corresponding conserved charge

   $$Q_V = p_\alpha V^\alpha = g_{\alpha\beta}V^\alpha\dot{x}^\beta \ . \tag{3.36}$$

   Note that for constant components V^α, (3.35) is simply the statement that the metric is constant in the direction V, $V^\gamma\partial_\gamma g_{\alpha\beta} = 0$.

2. Alternatively, we can determine the variation $\delta_V g_{\alpha\beta}$ of the components $g_{\alpha\beta}$ of the metric in x-coordinates from the variation (3.28) of the components $g_{\mu\nu}$ of the metric in y-coordinates by demanding that under a coordinate transformation the variation (3.28) of the metric transforms like the metric. Since we know how the metric transforms (2.5), and we also know how ∂_{y^1} transforms,

   $$g_{\mu\nu} = J_\mu^\alpha J_\nu^\beta g_{\alpha\beta} \quad , \quad \partial_{y^1} = (\partial_{y^1}x^\alpha)\partial_\alpha \equiv J_1^\alpha\partial_\alpha \equiv V^\alpha\partial_\alpha \tag{3.37}$$

   we find the condition

   $$\begin{aligned} &\partial_{y^1}g_{\mu\nu} = J_1^\gamma\partial_\gamma(J_\mu^\alpha J_\nu^\beta g_{\alpha\beta}) \stackrel{!}{=} J_\mu^\alpha J_\nu^\beta\delta_V g_{\alpha\beta} \\ \Leftrightarrow\quad &\delta_V g_{\alpha\beta} = J_\alpha^\mu J_\beta^\nu J_1^\gamma\partial_\gamma(J_\mu^\delta J_\nu^\epsilon g_{\delta\epsilon}) \ . \end{aligned} \tag{3.38}$$

In order to disentangle this, one can make use of identities such as

$$J_1^\gamma \partial_\gamma J_\mu^\delta = \partial_1 J_\mu^\delta = \partial_\mu J_1^\delta = \partial_\mu V^\delta = J_\mu^\alpha \partial_\alpha V^\delta \tag{3.39}$$

to show that this expression for $\delta_V g_{\alpha\beta}$ is identical to that given in (3.34).

All of this may seem a bit ham-handed at this point, and indeed it is. However, we will see later how these results can be written and understood in a much more pleasing and covariant way. In particular, we will see in section 5.5 how to write (3.34) in a way that makes it completely manifest that it transforms like the metric under coordinate transformations. Moreover, we will discover in section 9 that (3.34) is a special case of the *Lie derivative* of a tensor field along a vector field V, denoted by L_V. Continuous symmetries of a metric correspond to vector fields along which the Lie derivative of the metric vanishes. Such vectors are known as *Killing vectors*, and are thus vectors V^α satisfying the *Killing equation* (3.35),

$$L_V g_{\alpha\beta} \equiv \delta_V g_{\alpha\beta} = 0 \ . \tag{3.40}$$

3.3 Newtonian Limit of the Geodesic Equation

We saw that the 10 components of the metric $g_{\mu\nu}$ appear to play the role of potentials for the gravitational force. In order to substantiate this, and to show that in an appropriate limit this setting is able to reproduce the Newtonian results, we now want to find the relation of these potentials to the Newtonian potential, and the relation between the geodesic equation and the Newtonian equation of motion for a particle moving in a gravitational field.

First let us determine the conditions under which we might expect the general relativistic equation of motion (namely the non-linear coupled set of partial differential geodesic equations) to reduce to the linear equation of motion

$$\frac{d^2}{dt^2}\vec{x} = -\vec{\nabla}\phi \tag{3.41}$$

of Newtonian mechanics, with ϕ the gravitational potential, e.g.

$$\phi = -\frac{G_N M}{r} \ . \tag{3.42}$$

Thus we are trying to characterise the circumstances in which we know and can trust the validity of Newton's equations, such as those provided e.g. by the gravitational field of the earth or the sun, the gravitational fields in which Newton's laws were discovered and tested. Two of these are fairly obvious:

1. Weak Fields: our first plausible assumption is that the gravitational field is in a suitable sense sufficiently weak. We will need to make more precise by what we mean by this, and we will come back to this below.

2. Slow Motion: our second, equally reasonable and plausible, assumption is that the test particle moves at speeds at which we can neglect special relativistic effects, so "slow" should be taken to mean that its velocity is small compared to the velocity of light.

Interestingly, it turns out that one more condition is required. Note that the gravitational fields we have access to are not only quite weak but also only very slowly varying in time, and we will add this condition,

3. Stationary Fields: we will assume that the gravitational field does not vary significantly in time (over the time scale probed by our test particle).

The very fact that we have to add this condition in order to find Newton's equations (as will be borne out by the calculations below) is interesting in its own right, because it also shows that general relativity predicts phenomena deviating from the Newtonian picture even for weak fields, provided that they vary sufficiently rapidly (e.g. quickly oscillating fields), and one such phenomenon is that of gravitational waves (see section 23).

Now, having formulated in words the conditions that we wish to impose, we need to translate these conditions into equations that we can then use in conjunction with the geodesic equation.

1. In order to define a notion of weak fields, we need to keep in mind that this is not a coordinate-independent statement since we can simulate arbitrarily strong gravitational fields even in Minkowski space by going to suitably accelerated coordinates, and therefore a "weak field" condition will be a condition not only on the metric but also on the choice of coordinates. Thus we assume that we can choose coordinates $\{x^\mu\} = \{t, x^i\}$ in such a way that *in these coordinates* the metric differs from the standard constant Minkowski metric $\eta_{\mu\nu}$ only by a small amount,
$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \tag{3.43}$$
where we will implement "by a small amount" in the calculations below by dropping all terms that are at least quadratic in $h_{\mu\nu}$ (and/or its derivatives).

2. The second condition is obviously (with the coordinates chosen above) $dx^i/dt \ll 1$ or, expressed in terms of proper time,
$$\frac{dx^i}{d\tau} \ll \frac{dt}{d\tau} \ . \tag{3.44}$$

3. The third condition of stationarity we implement simply by considering time-independent fields,
$$g_{\alpha\beta,0} = 0 \quad \Rightarrow \quad h_{\alpha\beta,0} = 0 \tag{3.45}$$

(for a discussion and explanation of the difference betwen the term "stationary" used here and the term "static" used e.g. to describe the metric (3.22), see section 16.4 - it is not crucial here).

Before embarking on the calculation, we note that for the inverse $g^{\alpha\beta}$ of the metric $g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$, with $h_{\alpha\beta}$ "small", one evidently has

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \quad \Rightarrow \quad g^{\alpha\beta} = \eta^{\alpha\beta} + \mathcal{O}(h) \ , \tag{3.46}$$

where $\eta^{\alpha\beta}$ is just the inverse Minkowski metric. The explicit expression of the order h term (which we will not need) is given in (3.65) below.

Now we look at the geodesic equation

$$\ddot{x}^\mu + \Gamma^\mu_{\nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ . \tag{3.47}$$

From the decomposition $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ we see that $\Gamma^\mu_{\nu\lambda}$ is at least linear in $h_{\mu\nu}$, and by the weak field condition (condition 1) we will only retain the terms linear in $h_{\mu\nu}$. Then the condition of slow motion (condition 2) implies that among the quadratic terms $\dot{x}^\nu\dot{x}^\lambda$ we need to only retain the leading term, namely $\dot{t}\dot{t}$. Thus the geodesic equation can be approximated by

$$\ddot{x}^\mu + \Gamma^\mu_{00}\dot{t}^2 = 0 \ . \tag{3.48}$$

Thus we need to determine

$$\Gamma^\mu_{00} = \tfrac{1}{2}g^{\mu\nu}(g_{\nu 0,0} + g_{\nu 0,0} - g_{00,\nu}) \ . \tag{3.49}$$

Stationarity (condition 3) tells us that the first two terms are zero, and

$$\Gamma^\mu_{00} = -\tfrac{1}{2}g^{\mu\nu}\partial_\nu g_{00} = -\tfrac{1}{2}g^{\mu i}\partial_i g_{00} \ . \tag{3.50}$$

Now $\partial_i g_{00} = \partial_i h_{00}$ is already of order h. Therefore, by the weak field condition, working to linear order in h we can can replace the inverse metric $g^{\mu i}$ by the inverse Minkowski metric $\eta^{\mu i}$, so that in this approximation

$$\Gamma^\mu_{00} = -\tfrac{1}{2}\eta^{\mu i}\partial_i h_{00} \ . \tag{3.51}$$

Thus the relevant Christoffel symbols are

$$\Gamma^0_{00} = 0 \ , \qquad \Gamma^i_{00} = -\tfrac{1}{2}\partial^i h_{00} \ , \tag{3.52}$$

and the geodesic equation splits into

$$\begin{aligned} \ddot{t} &= 0 \\ \ddot{x}^i &= \tfrac{1}{2}\partial_i h_{00}\dot{t}^2 \ . \end{aligned} \tag{3.53}$$

The first of these just says that $\dot{t}$ is constant, or that t is also an affine parameter,

$$t(\tau) = a\tau + b \ . \tag{3.54}$$

In other words, in the Newtonian limit there is essentially (up to a choice of scale/units) no difference between coordinate time and proper time. We can use this in the second equation to convert the τ-derivatives into derivatives with respect to the coordinate time t,

$$\ddot{t} = 0 \quad \Rightarrow \quad \frac{1}{\dot{t}^2}\frac{d^2}{d\tau^2} = \frac{1}{\dot{t}}\frac{d}{d\tau}\frac{1}{\dot{t}}\frac{d}{d\tau} = \frac{d^2}{dt^2} \ . \tag{3.55}$$

Hence we obtain

$$\frac{d^2x^i}{dt^2} = \tfrac{1}{2}h_{00,i} \tag{3.56}$$

(the spatial index i in this expression is raised or lowered with the Kronecker symbol, $\eta^{ik} = \delta^{ik}$). Comparing this with the Newtonian equation (3.41),

$$\frac{d^2x^i}{dt^2} = -\phi_{,i} \tag{3.57}$$

leads us (with the constant of integration absorbed into an arbitrary constant term in the gravitational potential) to the key identification

$$h_{00} = -2\phi \tag{3.58}$$

between the Newtonian gravitational potential and the (00)-component of the deviation of the space-time metric from the Minkowski metric. By relating this back to $g_{\alpha\beta}$,

$$g_{00} = -(1 + 2\phi) \ . \tag{3.59}$$

we find the sought-for relation between the Newtonian potential and the space-time metric. Thus Newtonian gravity can be captured or described by a space-time metric of the form

$$ds^2 = -(1 + 2\phi(\vec{x}))dt^2 + d\vec{x}^2 \ . \tag{3.60}$$

For a radial gravitational field, with $\phi = \phi(r)$, it is also natural to write this in terms of spatial spherical coordinates as

$$ds^2 = -(1 + 2\phi(r))dt^2 + dr^2 + r^2 d\Omega_2^2 \ . \tag{3.61}$$

Remarks:

1. With the speed of light not set equal to $c = 1$, the dimensionally correct form of this identification is (recall that kinetic and potential energy have the same dimension so that the dimension of ϕ, the energy per unit mass, is that of a velocity-squared; thus ϕ/c^2 is dimensionless)

$$g_{00} = -(1 + 2\phi/c^2) \ . \tag{3.62}$$

2. For the gravitational field of isolated systems, it makes sense to choose the integration constant in such a way that the potential goes to zero at infinity, and this choice also ensures that the metric approaches the flat Minkowski metric at infinity.

3. Restoring the appropriate units, in particular the above factor of c^2, one finds that the dimensionless factor $\phi/c^2 \sim 10^{-9}$ on the surface of the earth, 10^{-6} on the surface of the sun (see section 24.4 for some more details), so that the distortion in the space-time geometry produced by gravitation is in general quite small (justifying our approximations).

4. Just for the record, here is the explicit expression for the inverse $g^{\alpha\beta}$ of a metric of the form $g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$, with $h_{\alpha\beta}$ "small". In analogy with the series expansion

$$(1+x)^{-1} = 1 - x + x^2 \mp \dots \tag{3.63}$$

for a real number x with $|x| < 1$, the exact result for $g^{\alpha\beta}$ can be written as an infinite power series in $h_{\alpha\beta}$. We will not need the exact result here, but only the result to linear order in $h_{\alpha\beta}$.

In linear algebra notation, if I is an invertible matrix and A is sufficiently small so that $I + A$ is still invertible, one has (as a matrix generalisation of the power series (3.63) for $(1+x)^{-1}$)

$$(I+A)^{-1} = I^{-1} - I^{-1}AI^{-1} + \mathcal{O}(A^2) \ . \tag{3.64}$$

In the case at hand (with $I \to \eta, A \to h$), this is

$$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \quad \Rightarrow \quad g^{\alpha\beta} = \eta^{\alpha\beta} - h^{\alpha\beta} + \mathcal{O}(h^2) \ , \tag{3.65}$$

where

$$h^{\alpha\beta} = \eta^{\alpha\gamma}\eta^{\beta\delta}h_{\gamma\delta} \ . \tag{3.66}$$

Indeed it is now easily verified that this satisfies

$$g^{\alpha\beta}g_{\beta\gamma} = \delta^\alpha_\gamma + \mathcal{O}(h^2) \ . \tag{3.67}$$

5. Within this Newtonian approximation, we cannot distinguish the above result $g_{00} = -(1+2\phi)$ from $g_{00} = -(1+\phi)^2$, say (or a host of other possibilities).

6. Likewise, in this approximation it does not make sense to inquire about the other subleading components of the metric. As we have seen, a slowly moving particle in a weak static gravitational field is not sensitive to them, and hence can also not be used to probe or determine these components.

7. In this approximation, the modification of the space-time geometry can equivalently be described as, or attributed to, a space-time dependent speed of light in Minkowski space, along the lines of

$$ds^2 = -c(x)^2dt^2 + d\vec{x}^2 \ , \tag{3.68}$$

with

$$c^2(x) = (1 + 2\phi(x)/c^2)c^2 \ . \tag{3.69}$$

Einstein realised fairly early on (1911) in his search for a relativistic theory of gravity that this would have to be part of the story. However, this interpretation is neither useful nor tenable when considering gravitational fields beyond the static Newtonian approximation (which requires one to go beyond a theory with a single scalar potential).

8. Later on, we will determine the exact solution of the Einstein equations (the field equations for the gravitational field, i.e. for the metric) for the gravitational field outside a spherically symmetric mass distribution with mass M (the Schwarzschild metric). The metric turns out to have the simple form (24.31)

$$ds^2 = -\left(1 - \frac{2G_N M}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2G_N M}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad . \tag{3.70}$$

From this expression one can read off that the leading correction to the flat metric indeed arises from the 00-component of the metric,

$$\begin{aligned} ds^2 &\approx -c^2 dt^2 + dr^2 + r^2 d\Omega^2 + \frac{2G_N M}{r} dt^2 + \dots \\ &= \eta_{\alpha\beta} dx^\alpha dx^\beta + \frac{2G_N M}{r c^2} (dx^0)^2 + \dots \quad . \end{aligned} \tag{3.71}$$

This is indeed precisely of the above Newtonian form, with the standard Newtonian potential

$$\phi(r) = -\frac{G_N M}{r} \quad . \tag{3.72}$$

One can then also determine the subleading (known as "post-Newtonian") corrections to the general relativistic gravitational field, which are evidently suppressed by additional inverse powers of c^2.

9. The key relation (3.58) can also be obtained at the level of the action. Starting with the action S_0 (the integral of the proper time), and using the time-coordinate t as the parameter, using the same approximations as above one finds that the action can be written as (keeping c explicit for a change, so that $x^0 = ct$)

$$\begin{aligned} S_0[x] &= -mc \int dt \sqrt{-g_{\mu\nu}(dx^\mu/dt)(dx^\nu/dt)} \\ &= -mc \int dt \sqrt{-\eta_{\mu\nu}(dx^\mu/dt)(dx^\nu/dt) - h_{\mu\nu}(dx^\mu/dt)(dx^\nu/dt)} \\ &= -mc \int dt \sqrt{c^2 - \eta_{ik}(dx^i/dt)(dx^k/dt) - h_{00} c^2} \\ &= -mc^2 \int dt \sqrt{1 - \vec{v}^2/c^2 - h_{00}} \quad . \end{aligned} \tag{3.73}$$

Expanding the square root and dropping the first (irrelevant) term, one finds that in this limit the action reduces to

$$S_0[x] \quad \to \quad \int dt \, \left(\frac{m}{2}\vec{v}^2 + \frac{mc^2}{2} h_{00}\right) \tag{3.74}$$

which is precisely the Newtonian action for a particle in a gravitational potential ϕ,

$$S_N[x] = \int dt \ \left(\frac{m}{2}\vec{v}^2 - m\phi\right) \tag{3.75}$$

provided that one makes the identification (3.58),

$$h_{00} = -2\phi/c^2 \ . \tag{3.76}$$

In this compact (but slightly dubious) derivation of this relation, the significance of the stationarity condition is not manifest: it enters through the condition of the equivalence of the 4-dimensional and 3-dimensional variational principles (with respect to the fields $x^\mu(\tau)$ and $x^k(t)$ respectively), guaranteed by the affine relation between t and τ implied by requiring in addition stationarity.

3.4 Rindler Coordinates Revisited

In section 1.3 we had discussed the Minkowski metric in Rindler coordinates, i.e. in coordinates adapted to a constantly accelerating observer. For an observer accelerating in the x^1-direction, the metric took the form (1.74),

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 + d\vec{y}^2 \ , \tag{3.77}$$

with $\vec{y} = (x^2, x^3)$ denoting the transverse spectator coordinates (which will again be suppressed in the following).

What is the relation, if any, between this metric and the metric describing a weak gravitational field, as derived above (after all, small accelerations should mimic weak gravitational fields)? At first sight, the only thing they appear to have in common is that the departure from what would be the Minkowski-metric in these coordinates is encoded in the time-time component of the metric, ρ^2 in one case, $(1 + 2\phi)$ in the other, but apart from that ρ^2 and $(1 + 2\phi)$ look quite different. This difference is, however, again a coordinate artefact and the Rindler metric can be made to look like the weak-field metric with the help of a suitable further redefinition of the coordinates.

For starters, it will be convenient, for this purpose and for a generalisation which we will discuss below, to introduce the acceleration a explicitly into the coordinates by redefining the coordinate transformation (1.73) to (I will now also call the Minkowski coordinates $\xi^0 = t$ and $\xi^1 = x$)

$$t(\eta, \rho) = (\rho/a)\sinh a\eta \qquad x(\eta, \rho) = (\rho/a)\cosh a\eta \tag{3.78}$$

(so this differs by $\rho \to \rho/a, \eta \to a\eta$ from the transformation given in (1.73)). Thus, now it is the observer at $\rho = 1$ who has acceleration a and whose proper time is $\tau = \eta$. The Rindler metric now has the form

$$ds^2 = -dt^2 + dx^2 = -\rho^2 d\eta^2 + a^{-2} d\rho^2 \ . \tag{3.79}$$

Now the transformation $\rho = 1 + a\hat{x}$ (reminding us that we are talking about acceleration in the $x = x^1$ direction), leads to

$$ds^2 = -(1 + a\hat{x})^2 d\eta^2 + d\hat{x}^2 \tag{3.80}$$

and it is the observer at $\hat{x} = 0$ who has proper time $\tau = \eta$ and constant acceleration a. If one now assumes that the acceleration a is sufficiently small, one can approximate

$$(1 + a\hat{x})^2 \approx 1 + 2a\hat{x} \equiv 1 + 2\phi(\hat{x}) \ , \tag{3.81}$$

and in $\phi = a\hat{x}$ we recognise precisely the Newtonian potential for a constant force a in the $\hat{x}$-direction. Thus the Rindler and weak field form of the metric agree in this case.

Remarks:

1. Remarkably, this same form of the metric remains valid for an arbitrary time-dependent acceleration $a = a(\tau)$, and thus is capable of reproducing the weak field form of the metric for general potentials. To see this, consider the worldline $(t(\tau), x(\tau))$ with general 4-velocity (actually 2-velocity in this case)

$$u^0 = \dot{t}(\tau) = \cosh v(\tau) \quad , \quad u^1 = \dot{x}(\tau) = \sinh v(\tau) \ , \tag{3.82}$$

which satisfies

$$(u^0)^2 - (u^1)^2 = 1 \ , \tag{3.83}$$

as it should, and has the time-dependent acceleration $a(\tau) = \dot{v}(\tau)$,

$$(\dot{u}^1)^2 - (\dot{u}^0)^2 = \ddot{x}^2 - \ddot{t}^2 = \dot{v}(\tau)^2 \equiv a(\tau)^2 \ . \tag{3.84}$$

We can pass to adapted coordinates $(\eta, \hat{x})$, as above, by setting

$$(t(\eta, \hat{x}), x(\eta, \hat{x})) = (t(\eta) + \hat{x} \sinh v(\eta), x(\eta) + \hat{x} \cosh v(\eta)) \ , \tag{3.85}$$

leading to the metric

$$ds^2 = -dt^2 + dx^2 = -(1 + a(\eta)\hat{x})^2 d\eta^2 + d\hat{x}^2 \ . \tag{3.86}$$

This is indeed the direct generalisation of (3.80) to arbitrary accelerations, the original worldline manifestly corresponding to the observer at fixed $\hat{x} = 0$, with $\eta = \tau$, and the same remarks regarding the weak field limit / small accelerations apply.

2. Another useful alternative coordinate transformation for constant a is $\rho = \exp a\xi$, leading to

$$ds^2 = \mathrm{e}^{2a\xi}(-d\eta^2 + d\xi^2) \ . \tag{3.87}$$

Note that in these coordinates the metric is *conformally flat*, i.e. it differs from the flat Minkowski metric $-d\eta^2 + d\xi^2$ in these coordinates only by an overall factor.

Moreover, for small values of a the time-component of the metric again reduces to something like the Newtonian limit expression (3.81), namely

$$\mathrm{e}^{2a\xi} \approx 1 + 2a\xi \quad . \tag{3.88}$$

For the record, and for later use, we note that the complete coordinate transformation between the Minkowski coordinates (t, x) and the conformally flat Rindler coordinates (η, ξ) is

$$(t, x) = (a^{-1}\mathrm{e}^{a\xi} \sinh a\eta, a^{-1}\mathrm{e}^{a\xi} \cosh a\eta) \quad . \tag{3.89}$$

The generator of Rindler time evolution in these coordinates is

$$\partial_\eta = a(t\partial_x + x\partial_t) \quad . \tag{3.90}$$

This is a boost in the (t, x)-plane, but the limit $a \to 0$ appears to be singular.

3. A simple and useful way to rectify this is to introduce a further constant shift of x, $x \to x - 1/a$, into the 1-parameter family (3.89) of coordinate transformations,

$$(t, x) = (a^{-1}\mathrm{e}^{a\xi} \sinh a\eta, a^{-1}\mathrm{e}^{a\xi} \cosh a\eta - a^{-1}) \quad . \tag{3.91}$$

This transformation now has the desirable property that as $a \to 0$ it continuously connects the Rindler and Minkowski coordinates,

$$a \to 0 \quad \Rightarrow \quad t \to \eta \quad , \quad x \to \xi \quad . \tag{3.92}$$

As a consequence, also the Rindler time evolution generator

$$\partial_\eta = \partial_t + a(t\partial_x + x\partial_t) \tag{3.93}$$

now has a non-singular limit as $a \to 0$, namely the Minkowski time generator ∂_t.

4. In terms of the Minkowski null (or advanced and retarded time) coordinates

$$u_M = t - x \quad , \quad v_M = t + x \quad \Rightarrow \quad ds^2 = -du_M\, dv_M \quad , \tag{3.94}$$

and their Rindler counterparts

$$u_R = \eta - \xi \quad , \quad v_R = \eta + \xi \quad \Rightarrow \quad ds^2 = -\mathrm{e}^{a(v_R - u_R)} du_R\, dv_R \quad , \tag{3.95}$$

the transformation (3.89) takes the form

$$t \mp x = a^{-1}\mathrm{e}^{a\xi} \left(\sinh a\eta \mp \cosh a\eta\right) = \mp a^{-1}\mathrm{e}^{a(\xi \mp \eta)} \tag{3.96}$$

or, compactly,

$$u_M = -a^{-1}\mathrm{e}^{-au_R} \quad , \quad v_M = +a^{-1}\mathrm{e}^{+av_R} \quad . \tag{3.97}$$

Note that the range of the coordinates is $-\infty < \eta, \xi < +\infty$ or $-\infty < u_R, v_R < +\infty$, and that the coordinates (η, ξ) or (u_R, v_R) cover (and can be used in) the right-hand quadrant $x > |t|$ of Minkowski space-time, corresponding to $-\infty < u_M = t - x < 0$ and $0 < v_M = t + x < +\infty$, the so called (right) *Rindler wedge*.

As we will see in section 7.8, these null Rindler coordinates are particularly useful for studying the solutions of the scalar wave equation in the Rindler wedge.

Let me close this section with some comments on other versions of $(3+1)$-dimensional Rindler space. First of all, instead of looking at acceleration in the x^1-direction, say, one can consider radial accelerations. To that end one first writes the metric in spatial spherical coordinates,

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega_2^2 \ , \tag{3.98}$$

and introduces Rindler coordinates (ρ, τ), say, via

$$(t, r) = (\rho \sinh \tau, \rho \cosh \tau) \ , \tag{3.99}$$

leading to the $(3+1)$-dimensional spherical Rindler metric

$$ds^2 = -\rho^2 d\tau^2 + d\rho^2 + \rho^2 \cosh^2 \tau \, d\Omega_2^2 \ . \tag{3.100}$$

This form of the metric is adapted to the hyperboloids

$$r^2 - t^2 = \rho^2 \tag{3.101}$$

i.e. to a family of constantly radially accelerating observers whose worldlines asymptotically approach the lightcone through the origin $(t = 0, r = 0)$. These coordinates cover precisely the region of Minkowski space outside this lightcone, i.e. the region of events at spacelike distance from the origin while the region that is not covered is the past and future of the origin.

The "complementary" metric (adapted to the hyperboloids with $r^2 - t^2 < 0$ and covering precisely the interior of the lightcone) is the so-called *Milne metric* to be discussed in section 37.1.

A non-trivial variant of this metric[6] is obtained by shifting $r \to r - r_0$ in the above coordinate transformation,

$$(t, r) = (\rho \sinh \tau, r_0 + \rho \cosh \tau) \ , \tag{3.102}$$

leading to the metric

$$ds^2 = -\rho^2 d\tau^2 + d\rho^2 + (r_0 + \rho \cosh \tau)^2 d\Omega_2^2 \tag{3.103}$$

[6] V. Balasubramanian, B. Czech, B. Chowdhury, J. de Boer, *The entropy of a hole in spacetime*, `arXiv:1305.0856 [hep-th]`, V. Balasubramanian, B. Chowdhury, B. Czech, J. de Boer, M. Heller, *A hole-ographic spacetime*, `arXiv:1310.4204 [hep-th]`.

(an analogous shift $x \to x - x_0$ for acceleration in the x-direction would have had no effect on the metric since such a translation is a symmetry of the Minkowski metric, whereas a translation in the radial direction is not). This form of the metric is adapted to the hyperboloids

$$(r - r_0)^2 - t^2 = \rho^2 \tag{3.104}$$

and now describes radially accelerating observers, each one asymptotically approaching the radial lightray emanating from a distance r_0 from the origin (and correspondingly the region of space-time covered by these coordinates is the complement of the past and future of the 2-sphere of radius r_0 at the origin, a "hole" in space-time).

3.5 Gravitational Redshift

Following Einstein, the gravitational redshift (i.e. the fact that photons appear to lose or gain energy when rising or falling in a gravitational field) is usually presented as a direct consequence of the Einstein Equivalence Principle (and is therefore also said to provide an experimental test of the Einstein Equivalence Principle itself). It (or, better, its Newtonian weak field limit) can indeed be derived in this way (see Remark 1 at the end of this section for one such argument, albeit not the original one). However, here we will derive this effect without any approximation within the framework that we have already adopted, inspired by the equivalence principle, namely in terms of the description of the gravitational field by a metric.

This has several advantages. It allows us to further familiarise ourselves with the formalism and to illustrate how to extract physical effects from our description of lightrays as null geodesics (much as we employed timelike geodesics above to study the Newtonian limit). Moreover, it allows us to derive formulae for this effect in quite some generality and I will actually give 3 different derivations in increasing order of generality. In conjunction with the Newtonian approximation to the gravitational field these then reduce to the result in the form in which it is usually presented, e.g. as in (3.130) or (3.131) (and as then rederived on the basis of the equivalence principle in (3.137)).

To set the stage, note that it is manifest from the expression

$$d\tau^2 = -g_{\alpha\beta}(x)dx^\alpha dx^\beta \tag{3.105}$$

for the proper time that e.g. the rate of clocks is affected by where one is in a gravitational field. However, as by the universality of gravity everything is (and in particular all ideal clocks are) affected in the same way by gravity, it is impossible to measure this effect locally, at a fixed point in a gravitational field. In order to find an observable effect, one needs to compare data from two different points in a gravitational potential.

The situation we could consider is that of two observers A and B moving on worldlines (paths) γ_A and γ_B, A sending light signals to B. In general the frequency, measured

in the observers rest-frame at A (or in a locally inertial coordinate system there) will differ from the frequency measured by B upon receiving the signal.

In order to separate out Doppler-like effects due to relative velocities, we consider two observers A and B at rest radially to each other, at radii r_A and r_B, in a static spherically symmetric gravitational field. This means that the metric depends only on a radial coordinate r and we can choose it to be of the form

$$ds^2 = g_{00}(r)dt^2 + g_{rr}(r)dr^2 + r^2 d\Omega^2 \quad , \tag{3.106}$$

where $d\Omega^2$ is the standard volume element on the two-sphere (see section 24 for a more detailed justification of this ansatz for the metric).

Observer A sends out light of a given frequency ν_A, say n pulses per proper time unit $\Delta\tau_A$. Observer B receives these n pulses in his proper time $\Delta\tau_B$ and interprets this as a frequency ν_B. Thus the relation between the frequency ν_A emitted at A and the frequency ν_B observed at B is

$$\frac{\nu_A}{\nu_B} = \frac{\Delta\tau_B}{\Delta\tau_A} \quad . \tag{3.107}$$

I will now give two arguments to show that this ratio depends on the metric (i.e. the gravitational field) at r_A and r_B through

$$\frac{\nu_A}{\nu_B} = \frac{(-g_{00}(r_B))^{1/2}}{(-g_{00}(r_A))^{1/2}} \quad , \tag{3.108}$$

and then a 3rd argument establishing a slightly more general result.

1. The first argument is essentially one based on geometric optics (and is best accompanied by drawing a (1+1)-dimensional space-time diagram of the lightrays and worldlines of the observers).

   The geometry of the situation dictates that the coordinate time intervals recorded at A and B are equal, $\Delta t_A = \Delta t_B$ as nothing in the metric actually depends on t. In equations, this can be seen as follows. First of all, the equation for a radial lightray is

   $$-g_{00}(r)dt^2 = g_{rr}(r)dr^2 \quad , \tag{3.109}$$

   or

   $$\frac{dt}{dr} = \pm\left(\frac{g_{rr}(r)}{-g_{00}(r)}\right)^{1/2} \quad . \tag{3.110}$$

   From this we can calculate the coordinate time for the lightray to go from A to B. Say that the first light pulse is emitted at point A at time $t(A)_1$ and received at B at coordinate time $t(B)_1$. Then

   $$t(B)_1 - t(A)_1 = \int_{r_A}^{r_B} dr(-g_{rr}(r)/g_{00}(r))^{1/2} \tag{3.111}$$

The right hand side obviously does not depend on t, so we also have

$$t(B)_2 - t(A)_2 = \int_{r_A}^{r_B} dr(-g_{rr}(r)/g_{00}(r))^{1/2} \tag{3.112}$$

where t_2 denotes the coordinate time for the arrival of the n-th pulse. Therefore,

$$t(B)_1 - t(A)_1 = t(B)_2 - t(A)_2 \ , \tag{3.113}$$

or

$$t(A)_2 - t(A)_1 = t(B)_2 - t(B)_1 \ , \tag{3.114}$$

as claimed. Thus the coordinate time intervals recorded at A and B between the first and last pulse are equal. However, to convert this to proper time, we have to multiply the coordinate time intervals by an r-dependent function,

$$\Delta\tau_{A,B} = (-g_{\alpha\beta}(r_{A,B})\frac{dx^\alpha}{dt}\frac{dx^\beta}{dt})^{1/2}\Delta t_{A,B} \ , \tag{3.115}$$

and therefore the proper time intervals will not be equal. For observers at rest, $dx^i/dt = 0$, one has

$$\Delta\tau_{A,B} = (-g_{00}(r_{A,B}))^{1/2}\Delta t_{A,B} \ . \tag{3.116}$$

Since $\Delta t_A = \Delta t_B$, (3.108) now follows from (3.107).

2. The second argument uses the null geodesic equation, in particular the conserved quantity associated to time-translations (recall that we have assumed that the metric (3.106) is time-independent), as well as a somewhat more covariant looking, but equivalent, notion of frequency.

   First of all, let the lightray be described by the wave vector k^α. In special relativity, we would parametrise this as $k^\alpha = (\omega, \vec{k})$ with $\omega = 2\pi\nu$ the frequency. This is the frequency observed by an inertial observer at rest, with 4-velocity $u^\alpha = (1, 0, 0, 0)$. A Lorentz-invariant, and in our context now coordinate-independent, notion of the frequency as measured by an observer with velocity u^α is thus

   $$\omega = -u^\alpha k_\alpha \ . \tag{3.117}$$

   This includes as special cases

   - the standard (special) relativistic Doppler effect (where one compares ω with $\bar{\omega} = -\bar{u}^\alpha k_\alpha$, $\bar{u}^\alpha$ the 4-velocity of a boosted observer),
   - and the gravitational redshift between static observers we want to discuss here,

   but more generally also the redshift for observers with arbitrary 4-velocity u^α. And indeed we will employ this method in section 26.5 to look at

- the redshift between a static and a freely falling observer in the Schwarzschild geometry.

Returning to the case at hand, a static observer in the spherically-symmetric and static gravitational field (3.106) is described by the 4-velocity

$$u^\alpha = (u^0, 0, 0, 0) \qquad g_{\alpha\beta} u^\alpha u^\beta = g_{00} (u^0)^2 = -1 \quad . \tag{3.118}$$

Thus for the static observer at $r = r_A$, say, one has

$$u^0_A = (-g_{00}(r_A))^{-1/2} \tag{3.119}$$

(and likewise for the observer at $r = r_B$). The wave vector k^μ is a null tangent vector, $k^\mu k_\mu = 0$, to a null geodesic corresponding to the Lagrangian

$$\mathcal{L} = \tfrac{1}{2} g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = \tfrac{1}{2} g_{00}(r) \dot{t}^2 + \dots \tag{3.120}$$

Since the metric is time-independent, there is (cf. the discussion in section 3.1) the corresponding conserved quantity

$$E = -\frac{\partial \mathcal{L}}{\partial \dot{t}} = -g_{00}(r) \dot{t} \tag{3.121}$$

(the minus sign serving only to make this quantity positive for $\dot{t} > 0$). Then one finds that the frequency measured by the static observer at $r = r_A$ is

$$\begin{aligned} \omega_A = -u^\alpha_A k_\alpha &= -\left(-g_{00}(r_A)\right)^{-1/2} k_0 = -\left(-g_{00}(r_A)\right)^{-1/2} g_{0\alpha}(r_A) \dot{x}^\alpha \\ &= -\left(-g_{00}(r_A)\right)^{-1/2} g_{00}(r_A) \dot{t} = E \left(-g_{00}(r_A)\right)^{-1/2} \end{aligned} \tag{3.122}$$

Since E is a conserved quantity, i.e. the same for the lightray at $r = r_A$ or $r = r_B$, one sees that $\omega_A / \omega_B = \nu_A / \nu_B$ is given by (3.108), as claimed.

Note that this derivation shows that the relation between ω and E is exactly like the relation (3.116) between $(\Delta\tau)^{-1}$ and $(\Delta t)^{-1}$, which provides us with an interpretation of the conserved quantity E for a massless particle / photon: it is the frequency measured with respect to coordinate time (as the momentum conjugate to the time-coordinate t this should not be too surprising).

3. Even with the restriction to static observers in static gravitational fields, the above derivation is not completely general, and still not completely covariant, because we used the explicit form of the metric (which is the general form of a metric with a time-translation invariance in spherical symmetry, but not in general). We can improve this somewhat by using the more general characterisation of time-translation invariance in terms of Killing vectors (section 3.2) and the associated conserved charge (3.36).

   Thus assume that we have a timelike Killing vector V^α. Then by definition a static observer is one whose 4-velocity u^α is proportional to V^α,

$$u^\alpha \sim V^\alpha \quad . \tag{3.123}$$

For $V = \partial_t$ this evidently reduces to the statement that only t changes along the worldline, i.e. that the observer remains at fixed values of the spatial coordinates, and this is the sense in which we have informally used the term "static observer" so far. Denoting the norm of V by

$$\mathbb{V} = (-V^\alpha V_\alpha)^{1/2} \ , \tag{3.124}$$

the normalisation condition $u^\alpha u_\alpha = -1$ fixes the proportionality factor between u^α and V^α to be

$$u^\alpha = V^\alpha / \mathbb{V} \ . \tag{3.125}$$

Given the null wave vector k^α, we have the conserved energy (3.36),

$$E = -k_\alpha V^\alpha \ . \tag{3.126}$$

Therefore, adopting the definition (3.117), the frequency observed by a static observer is

$$\omega = -u^\alpha k_\alpha = -k_\alpha V^\alpha / \mathbb{V} = E/\mathbb{V} \ . \tag{3.127}$$

Since E is constant along the lightray, frequencies observed by two different static observers are related by

$$\frac{\omega_A}{\omega_B} = \frac{\mathbb{V}_B}{\mathbb{V}_A} \ . \tag{3.128}$$

For this reason, the norm $\mathbb{V}$ is also known as the *redshift factor* associated with a timelike Killing vector.

Note that this result reduces to (3.108) if the metric has the form (3.106) and $V = \partial_t$ since then

$$V = \partial_t \quad \Rightarrow \quad \mathbb{V}(r) = (-g_{00}(r))^{1/2} \ . \tag{3.129}$$

Having derived (3.108) in 3 different ways, let us now look at what the result tells us in specific situations of interest. Since on earth and in the solar system we only have access to gravitational fields that are to a reasonably high degree of precision well described by Newtonian gravity, we can use the Newtonian approximation (3.59). Then (3.108) becomes

$$g_{00} = -(1 + 2\phi) \quad \Rightarrow \quad \frac{\nu_A}{\nu_B} \sim 1 + \phi(r_B) - \phi(r_A) \ , \tag{3.130}$$

or, with $\phi(r) = -G_N M/r$,

$$\frac{\nu_A - \nu_B}{\nu_B} = \frac{G_N M (r_B - r_A)}{r_A r_B} \tag{3.131}$$

Thus for $r_B > r_A$ one has

$$r_B > r_A \quad \Rightarrow \quad \nu_B < \nu_A \ , \tag{3.132}$$

so that the frequency of a lightray appears to be redshifted when rising (moving upwards) in a gravitational field, and one has a gravitational blueshift effect

$$r_B < r_A \quad \Rightarrow \quad \nu_B > \nu_A \ . \tag{3.133}$$

for photons falling (moving downwards) in a gravitational field.

So far these are the objective facts, just the results of the equation (3.108) that we have derived. Let me now turn to the more subtle question of how to interpret this result, or what words to use to best describe this result.

It is tempting (and unfortunately quite common) to interpret these results as saying that "as expected, a photon loses energy when rising in (and against the pull of) a gravitational field". This kind of non-relativistic Newtonian reasoning, wildly extrapolated to include the energy of lightrays, can to a certain extent be justified by appealing to the Einstein Equivalence Principle (see Remark 1 below), but I believe that this is misleading and provides the wrong intuition for the physics of the situation, and I will subsequently provide a different explanation that (to me) appears to be much more to the point (Remark 4).

REMARKS:

1. The Newtonian limit (3.130) of the exact result (3.108) can to some extent also be deduced from energy conservation applied to Newtonian gravity.[7] By the Einstein Equivalence Principle a local observer at the emitter A will see a change in the internal mass of the emitter $\Delta m_A = -h\nu_A$ when a photon of frequency of ν_A is emitted. Likewise, the absorber at point B will experience an increase in inertial mass by $\Delta m_B = h\nu_B$, but the total internal plus gravitational potential energy

$$mc^2 + m\phi \equiv m + m\phi = m(1+\phi) \tag{3.134}$$

must be conserved, i.e.

$$m_A(1+\phi(r_A))+m_B(1+\phi(r_B)) = (m_A+\Delta m_A)(1+\phi(r_A))+(m_B+\Delta m_B)(1+\phi(r_B)) \ . \tag{3.135}$$

Thus

$$0 = \Delta m_A(1+\phi(r_A)) + \Delta m_B(1+\phi(r_B)) \ , \tag{3.136}$$

leading to (3.130),

$$\frac{\nu_A}{\nu_B} = \frac{1+\phi(r_B)}{1+\phi(r_A)} \sim 1 + \phi(r_B) - \phi(r_A) \ . \tag{3.137}$$

2. This "derivation" (in quotes, because we are wildly mixing Newtonian gravity, special relativity and quantum mechanics - do take this "derivation" with an

[7] See S. Weinberg, *Gravitation and Cosmology* section 3.5 for a careful discussion.

appropriately sized grain of salt, please) shows that gravitational redshift experiments test the Einstein Equivalence Principle in its strong form, in which the term 'laws of nature' is not restricted to mechanics (inertial = gravitational mass), but also includes quantum mechanics in the sense that it tests if in an inertial frame the relation between photon energy and frequency is unaffected by the presence of a gravitational field.

3. One significant shortcoming of the above "explanation" is that it relies on the Newtonian concept of a gravitational potential energy, and that it therefore applies at best to the Newtonian limit (3.130) of the exact result (3.108). It also seems to suggest that something is happening to the photon / lightray as it travels from A to B (namely that it "loses energy"), and it is in this sense that the Newtonian intuition is misleading.

   Indeed, note that the simple general result (3.108) depends only on the value of the gravitational field at the points r_A and r_B, not on the gravitational field inbetween. This is at it should be. After all, this gravitational time delay between clocks at r_A and r_B is all that we have used is the derivation (and not some new law of physics that causes photons to lose energy in some unexplained way). This strongly suggest the interpretation, that the gravitational redshift is only due to the different rate of clocks / proper time at the positions r_A and r_B, and not due to the fact that "something happens to the lightray as it travels through a gravitational field" (which should lead to a cumulative effect depending also on the intermediate gravitational field).

4. Indeed, let us reanalyse the gravitational redshift (without any approximation) from this point of view. We note that the observer A at r_A uses his (or her) proper time τ_A to define a notion of frequency ν_A, and hence a notion of energy $E_A = h\nu_A$. Likewise the observer B defines an energy $E_B = h\nu_B$ based on his (or her) proper time τ_B. The fact that $E_B \neq E_A$ has nothing to do with the photon or lightray having lost or gained energy, it is simply due to the fact that the observers A and B define energy in a different way!

   Moreover, there is a rigorous and well-defined sense in which the energy of a photon moving in a time-independent gravitational field is conserved. Indeed, recall that in such a case there is a conserved quantity $p_0 = -E$ (3.7). In the case at hand it takes the form (3.121),

$$E = -g_{00}(r)\dot{t} \tag{3.138}$$

   and (as the momentum conjugate to the time coordinate t) has the interpretation as (Planck's constant times) the frequency ν_t of the lightray with respect to this (observer-independent) coordinate time t,

$$E = E_t = h\nu_t \quad . \tag{3.139}$$

Putting all these facts together, one sees that the gravitational redshift can and should be attributed entirely to the gravitational time-delay between observers A and B, and thus to the different conventions for measuring frequencies and energies, and not to some kind of "loss of energy of a photon in a gravitational field".

Let me close this section with a brief comment on the observational tests of the redshift prediction (3.131) (but you should consult other sources for a more detailed discussion of experimental tests of General Relativity). While difficult to observe directly (by looking at light from the sun), the prediction (3.131) has been verified in the laboratory, first by Pound and Rebka (1960), and subsequently, with one percent accuracy, by Pound and Snider in 1964 (using the Mössbauer effect).

Here is a rough estimate of the expected effect. We first consider light reaching us (B) from the sun (A). In this case, we have $r_B \gg r_A$, where r_A is the radius of the sun, and (also inserting a so far suppressed factor of c^2) we obtain

$$\frac{\nu_A - \nu_B}{\nu_B} = \frac{G_N M(r_B - r_A)}{c^2 r_A r_B} \simeq \frac{G_N M}{c^2 r_A} \quad . \tag{3.140}$$

Using the approximate values

$$\begin{aligned} r_A &\simeq 0.7 \times 10^6 \ \mathsf{km} \\ M_{\rm sun} &\simeq 2 \times 10^{33} \ \mathsf{g} \\ G_N &\simeq 7 \times 10^{-8} \ \mathsf{g}^{-1}\mathsf{cm}^3\mathsf{s}^{-2} \\ G_N c^{-2} &\simeq 7 \times 10^{-29} \ \mathsf{g}^{-1}\mathsf{cm} = 7 \times 10^{-34} \ \mathsf{g}^{-1}\mathsf{km} \ , \end{aligned} \tag{3.141}$$

one finds

$$\frac{\Delta\nu}{\nu} \simeq 2 \times 10^{-6} \quad . \tag{3.142}$$

In principle, such a frequency shift should be observable. In practice, however, the spectral lines of light emitted by the sun are strongly effected e.g. by convection in the atmosphere of the sun (Doppler effect), and this makes it difficult to measure this effect with the required precision.

In the Pound-Snider experiment, the actual value of $\Delta\nu/\nu$ is much smaller. In the original set-up one has $r_B - r_A \simeq 20\mathsf{m}$ (the distance from floor to ceiling of the laboratory), and $r_A = r_{\rm earth} \simeq 6.4 \times 10^6 \mathsf{m}$, leading to

$$\frac{\Delta\nu}{\nu} \simeq 2.5 \times 10^{-15} \quad . \tag{3.143}$$

However, here the experiment is much better controlled, and the gravitational redshift was verified with 1% accuracy.

3.6 Equivalence Principle Revisited: Existence of Locally Inertial Coordinates

Central to our initial discussion of gravity was the Einstein Equivalence Principle which postulates the existence of locally inertial (or freely falling) coordinate systems in which locally at (or around) a point the effects of gravity are absent. Now that we have decided that the arena of gravity is a general metric space-time, we should establish that such coordinate systems indeed exist. Looking at the geodesic equation, it is clear that at least in this context "absence of gravitational effects" is tantamount to the existence of a coordinate system $\{\xi^a\}$ in which at a given point p the metric is the Minkowski metric, $g_{ab}(p) = \eta_{ab}$ and the Christoffel symbols are zero, $\Gamma^a_{\ bc}(p) = 0$,

$$g_{ab}(p) = \eta_{ab} \quad , \quad \Gamma^a_{\ bc}(p) = 0 \ . \tag{3.144}$$

Owing to the identity

$$g_{\mu\nu\,,\lambda} = \Gamma_{\mu\nu\lambda} + \Gamma_{\nu\mu\lambda} \ , \tag{3.145}$$

the latter condition is equivalent to $g_{ab\,,\,c}(p) = 0$. Below (after the Remarks) I will sketch three arguments establishing the existence of such coordinate systems, each one having its own virtues and providing its own insights into the issue.

Remarks:

1. Actually it is physically plausible (and fortuitously moreover true) that one can always find coordinates which embody the equivalence principle in the stronger sense that the metric is the flat metric η_{ab} and the Christoffel symbols are zero not just at a point but along the entire worldline of an inertial (freely falling) observer, i.e. along a geodesic γ,

$$g_{ab}|_\gamma = \eta_{ab} \quad , \quad \Gamma^a_{\ bc}|_\gamma = 0 \ . \tag{3.146}$$

   Such coordinates, based on a geodesic rather than on a point, are known as Fermi normal coordinates. The construction is similar to that of Riemann normal coordinates (based at a point) to be discussed below.[8]

2. In this mathematically idealised realisation of the equivalence principle, nothing is said about the metric and the Christoffel symbols in a neighbourhood of that point (or of the geodesic), and nothing is said about the 2nd and higher derivatives of the metric at that point.

[8] Most discussions of Fermi coordinates in the literature follow the presentation given in F. Menasse, C. Misner, *Fermi normal coordinates and some basic concepts in differential geometry*, J. Math. Phys. 4 (1963) 735-745; for a geometrically transparent treatment see also section 1.11 of E. Poisson, *A Relativist's Toolkit*; Fermi coordinates for *null* geodesics are constructed in M. Blau, D. Frank, S. Weiss, *Fermi Coordinates and Penrose Limits*, `arXiv:hep-th/0603109`.

3. In particular, thinking of the 1st derivatives of the metric as encoding the gravitational force, the 2nd derivatives of the metric must then correspond to gravitational *tidal forces*. (We will see this in more detail in section 8.4, where these tidal forces are related to components of the Riemann curvature tensor, a tensor that involves up to 2nd derivatives of the metric.) Such tidal forces are objective, i.e. physically real, as they lead to stresses in (or deformations of) extended bodies. One can and should therefore not expect to be able to eliminate such tidal forces by a suitable choice of reference system.

4. Therefore, physically what the equivalence principle says (or should say) is that in a gravitational field there is locally a reference system in which the effects of gravity are absent, provided that you choose the spacetime region to be sufficently small so that you can neglect the effect of gravitational tidal forces.

Here is a sketch of 3 arguments establishing the existence of locally inertial coordinate systems:

1. Direct Construction

   We know that given a coordinate system $\{\xi^a\}$ that is inertial at a point p, the metric and Christoffel symbols at p in a new coordinate system $\{x^\mu\}$ are determined by (1.87,1.98). Conversely, we will now see that knowledge of the metric and Christoffel symbols at a point p is sufficient to construct a locally inertial coordinate system at p.

   We will construct this coordinate system $\xi^a = \xi^a(x)$ locally around the point p (with coordinates x_0^α, say, in the original coordinate system) by a Taylor series expansion,

   $$\xi^a(x) = d^a + (x - x_0)^\alpha e^a_\alpha + \tfrac{1}{2}(x - x_0)^\beta (x - x_0)^\gamma f^a_{\beta\gamma} + \dots \quad . \tag{3.147}$$

   Here

   $$d^a = \xi^a(x_0) \equiv \xi^a_0 \tag{3.148}$$

   are the (arbitrary) coordinate values of the point p in the new coordinates ξ^a,

   $$e^a_\alpha = \frac{\partial \xi^a}{\partial x^\alpha}(x_0) \tag{3.149}$$

   is the Jacobi matrix of the coordinate transformation at $x = x_0$, and

   $$f^a_{\beta\gamma} = \frac{\partial^2 \xi^a}{\partial x^\beta \partial x^\gamma}(x_0) \tag{3.150}$$

   is its 1st derivative at x_0.

   Form the tensorial transformation behaviour of the metric we know that

   $$g_{\alpha\beta}(x_0) = g_{ab}(\xi_0) e^a_\alpha e^b_\beta \quad . \tag{3.151}$$

Requiring that $g_{ab}(\xi_0) = \eta_{ab}$ leads to the condition

$$g_{ab}(\xi_0) = \eta_{ab} \quad \Rightarrow \quad g_{\alpha\beta}(x_0) = \eta_{ab} e^a_\alpha e^b_\beta \ . \tag{3.152}$$

This shows that the e^a_α, thought of as a matrix, are an invertible (4×4)-matrix in $\mathsf{GL}(4, \mathbb{R})$. Denoting its inverse by e^α_a, with

$$e^\alpha_a e^a_\beta = \delta^\alpha_\beta \quad , \quad e^\alpha_a e^b_\alpha = \delta^a_b \ , \tag{3.153}$$

we see that the inverse matrix diagonalises (and scales) the metric at the point p in such a way that

$$g_{\alpha\beta}(x_0) e^\alpha_a e^\beta_b = \eta_{ab} \ . \tag{3.154}$$

Since $g_{\alpha\beta}(x_0)$ is a symmetric non-degenerate matrix, such matrices always exist (and are unique up to similarity transformations that leave η_{ab} invariant, i.e. up to Lorentz transformations). The notation e^a_α and e^α_a reflects the fact that these matrices are the components of an orthonormal vierbein (or vielbein) at the point p, which are traditionally denoted this way (cf. the discussion in section 4.8 below).

Taking stock, we see that the condition $g_{ab}(p) = \eta_{ab}$ determines the coordinate system to 1st order in a Taylor series expansion, up to translations (the choice of d^a) and Lorentz transformations, i.e. up to Poincaré transformation.

We now turn to the 2nd condition characterising a locally inertial coordinate system, namely $\Gamma^a_{\ bc}(p) = 0$. We can write the inhomogeneous transformation behaviour of the Christoffel symbols as

$$\frac{\partial \xi^a}{\partial x^\alpha} \Gamma^\alpha_{\ \beta\gamma} = \Gamma^a_{\ bc} \frac{\partial \xi^b}{\partial x^\beta} \frac{\partial \xi^c}{\partial x^\gamma} + \frac{\partial^2 \xi^a}{\partial x^\beta \partial x^\gamma} \ . \tag{3.155}$$

Thus at the point p we have

$$e^a_\alpha \Gamma^\alpha_{\ \beta\gamma}(x_0) = \Gamma^a_{\ bc}(\xi_0) e^b_\beta e^c_\gamma + f^a_{\beta\gamma} \ . \tag{3.156}$$

Requiring $\Gamma^a_{\ bc}(p) = 0$ now uniquely determines the 2nd order Taylor coefficients,

$$\Gamma^a_{\ bc}(\xi_0) = 0 \quad \Rightarrow \quad f^a_{\beta\gamma} = e^a_\alpha \Gamma^\alpha_{\ \beta\gamma}(x_0) \ . \tag{3.157}$$

Thus to 2nd order in a Taylor series expansion, the transformation from arbitrary coordinates x^α to inertial coordinates ξ^a at the point p is given by

$$\xi^a(x) = \xi^a_0 + (x - x_0)^\alpha e^a_\alpha + \tfrac{1}{2}(x - x_0)^\beta (x - x_0)^\gamma e^a_\alpha \Gamma^\alpha_{\ \beta\gamma}(x_0) + \ldots \ . \tag{3.158}$$

To this order we can also write the inverse coordinate transformation as

$$x^\alpha(\xi) = x^\alpha_0 + (\xi - \xi_0)^a e^\alpha_a - \tfrac{1}{2}(\xi - \xi_0)^b (\xi - \xi_0)^c \Gamma^\alpha_{\ \beta\gamma}(x_0) e^\beta_b e^\gamma_c + \ldots \ . \tag{3.159}$$

We have therefore established that for an arbitrary point p in an arbitrary gravitational field one can always introduce local coordinates which are inertial at that

point, and that up to 2nd order in a Taylor series expansion such a coordinate system is unique up to Poincaré transformations.

Since this leaves the infinite number of higher-order terms of the Taylor expansion undetermined, this shows that inertial coordinate systems are highly non-unique, and raises the following questions:

- Can one continue in this vein and choose the (so far undetermined) higher-order terms in the Taylor expansion such that also e.g. the 2nd derivatives of the metric at p are equal to zero,

$$\exists? \quad \xi^a(x): \quad g_{ab,\,cd}(p) = 0 \ ? \tag{3.160}$$

  The answer to this is a resounding "no", as the 3rd (numerological) argument below will show. In fact, as we will see (and study in detail) later on, the 2nd derivatives of the metric contain important coordinate-invariant information about the curvature of the metric.

- Are there nevertheless preferred inertial coordinate systems, i.e. preferred choices for the higher-order terms in the Taylor expansion? The answer to this is "yes". One such preferred and geometrically natural class of inertial coordinate systems are e.g. Riemann normal coordinates, based on geodesics at the point p, and briefly discussed below.

2. Geodesic (or Riemann Normal) Coordinates

   A slightly more insightful way of constructing a locally inertial coordinate system, rather than by directly solving the relevant differential equation, makes use of geodesics at p. Recall that in Minkowski space the metric takes the simplest possible form in coordinates whose coordinate lines are (orthogonal) geodesics. One might thus suspect that in a general metric space-time the metric will also (locally) look particularly simple when expressed in terms of such *geodesic coordinates*.

   Roughly speaking (I will give a more detailed argument below), since locally around p we can solve the geodesic equation with four linearly independent initial conditions, we can assume the existence of a coordinate system $\{\xi^a\}$ in which the coordinate lines are geodesics $\xi^a(\tau) = \xi^a\tau$. This means that in these coordinates geodesics satisfy $\ddot{\xi}^a = 0$. Comparing with the full geodesic equation in these coordinates, one sees that this implies that

$$\Gamma^a_{\ bc}\dot{\xi}^b\dot{\xi}^c = 0 \ . \tag{3.161}$$

   As at p the $\dot{\xi}^a$ were chosen to be linearly independent, this implies $\Gamma^a_{\ bc}(p) = 0$, as desired. It is easy to see that the coordinates ξ^a can also be chosen in such a way that $g_{ab}(p) = \eta_{ab}$ (by choosing the four directions at p to be orthonormal unit vectors).

Before turning to the more detailed construction, let us look at an example. Consider the standard metric $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$ on the two-sphere. Any point is as good as any other point, and one can construct an inertial coordinate system at the north pole $\theta = 0$ in terms of geodesics shot off from the north pole into the $\phi = 0$ (ξ^1) and $\phi = \pi/2$ (ξ^2) directions. The affine parameter along a great circle (geodesic) connecting the north pole to a point (θ, ϕ) is θ, and thus θ is also the geodesic distance, and the coordinates of the point (θ, ϕ) are $(\xi^1 = \theta\cos\phi, \xi^2 = \theta\sin\phi)$. In particular, the north pole is the origin $\xi^1 = \xi^2 = 0$. Note that one could have guessed these coordinates from the fact that near $\theta = 0$ the metric is $d\theta^2 + \theta^2 d\phi^2$, which is the Euclidean metric in polar coordinates $(\theta\cos\phi, \theta\sin\phi)$.

Calculating the metric in these new components, using

$$(\xi^1 = \theta\cos\phi, \xi^2 = \theta\sin\phi) \quad \Rightarrow \quad ((\xi^1)^2 + (\xi^2)^2 = \theta^2, \xi^2/\xi^1 = \tan\phi) \tag{3.162}$$

and thus

$$d\theta = \frac{\xi^1 d\xi^1 + \xi^2 d\xi^2}{\sqrt{(\xi^1)^2 + (\xi^2)^2}} \quad , \quad d\phi = \frac{\xi^1 d\xi^2 - \xi^2 d\xi^1}{(\xi^1)^2 + (\xi^2)^2} \ , \tag{3.163}$$

one finds

$$d\theta^2 + \sin^2\theta d\phi^2 = (d\xi^1)^2 + (d\xi^2)^2 + \mathcal{O}(\xi^2 d\xi^2) \ , \tag{3.164}$$

i.e.

$$g_{ab}(\xi) = \delta_{ab} + \mathcal{O}(\xi^2) \ . \tag{3.165}$$

Therefore

$$g_{ab}(\xi = 0) = \delta_{ab} \quad , \quad g_{ab,c}(\xi = 0) = 0 \ , \tag{3.166}$$

as required.

We now (re)turn to the general construction of such coordinates, starting with the geodesic equation

$$\ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma = 0 \ . \tag{3.167}$$

We consider geodesics passing through (or emanating from) the point p with coordinates x_0^α at $\tau = 0$, and with initial 4-velocity u_0^α,

$$x^\alpha(\tau = 0) = x_0^\alpha \quad , \quad \dot{x}^\alpha(\tau = 0) = u_0^\alpha \ . \tag{3.168}$$

It then follows that the 2nd derivative at $\tau = 0$ is given by

$$\ddot{x}^\alpha(\tau = 0) = -\Gamma^\alpha_{\beta\gamma}(x_0)u_0^\beta u_0^\gamma \ . \tag{3.169}$$

Hence in a Taylor expansion around $\tau = 0$ we can write the solution to the geodesic equation as

$$x^\alpha(\tau) = x_0^\alpha + \tau u_0^\alpha - \tfrac{1}{2}\tau^2\Gamma^\alpha_{\beta\gamma}(x_0)u_0^\beta u_0^\gamma + \dots \ . \tag{3.170}$$

We can expand the (arbitrary) initial 4-velocity u_0^α in terms of 4 linearly independent (and orthonormal, say) vectors at p as

$$u_0^\alpha = \lambda^a e_a^\alpha \quad , \quad g_{\alpha\beta}(x_0) e_a^\alpha e_b^\beta = \eta_{ab} \ . \tag{3.171}$$

We can then think of the Taylor expansion (3.170) as defining a coordinate transformation

$$x^\alpha(\xi) = x_0^\alpha + (\xi - \xi_0)^a e_a^\alpha - \tfrac{1}{2}(\xi - \xi_0)^b (\xi - \xi_0)^c \Gamma^\alpha_{\ \beta\gamma}(x_0) e_b^\beta e_c^\gamma + \dots \quad , \tag{3.172}$$

which has the following properties:

(a) First of all, for

$$\xi^a(\tau) = \xi_0^a + \tau \lambda^a \tag{3.173}$$

this reduces to the Taylor-expanded solution (3.170) of the geodesic equation with $u_0^\alpha = \lambda^a e_a^\alpha$. Thus in this coordinate system in particular the 4 coordinate lines

$$\xi^a_{(b)}(\tau) = \xi_0^a + \tau \delta_b^a \quad , \quad b = 0, 1, 2, 3 \ , \tag{3.174}$$

are (affinely parametrised) geodesics, as desired.

(b) Moreover, up to quadratic order in the Taylor expansion (3.172) is identical to the coordinate transformation (3.159). In particular, this establishes that the geodesic coordinates constructed here are a special class of inertial coordinates, with

$$g_{ab}(\xi_0) = \eta_{ab} \quad , \quad \Gamma^a_{\ bc}(\xi_0) = 0 \ . \tag{3.175}$$

(c) From the present point of view, the 2nd condition arises from the fact (mentioned above) that in these coordinates the geodesic equation for the above geodesics reduces to

$$\ddot{\xi}^a + \Gamma^a_{\ bc} \dot{\xi}^b \dot{\xi}^c = 0 \quad \Rightarrow \quad \Gamma^a_{\ bc}(\xi_0^a + \tau\lambda^a)\lambda^b\lambda^c = 0 \ . \tag{3.176}$$

At ξ_0, i.e. for $\tau = 0$, the Christoffel symbols are independent of the λ^a, and therefore

$$\Gamma^a_{\ bc}(\xi_0)\lambda^b\lambda^c = 0 \quad \forall \lambda^a \quad \Rightarrow \Gamma^a_{\ bc}(\xi_0) = 0 \ , \tag{3.177}$$

as claimed.

(d) In contrast to the previous construction leading to (3.159), here the higher-order terms in the Taylor expansion of the coordinate transformation are now determined by the higher-order terms in the Taylor expansion of the solution (3.170) of the geodesic equation. These higher-order terms will depend on 2nd and higher derivatives of the metric $g_{\alpha\beta}(x)$ at x_0, and these in turn will

also determine the quadratic and higher terms of the Taylor expansion of the metric in these coordinates,

$$\begin{aligned} g_{ab}(\xi) &= g_{ab}(\xi_0) + (\xi-\xi_0)^c g_{ab,\,c}(\xi_0) + \tfrac{1}{2}(\xi-\xi_0)^c(\xi-\xi_0)^d g_{ab,\,cd}(\xi_0) + \dots \\ &= \eta_{ab} + \tfrac{1}{2}(\xi-\xi_0)^c(\xi-\xi_0)^d g_{ab,\,cd}(\xi_0) + \dots \quad . \end{aligned} \tag{3.178}$$

We will determine the quadratic term in this expansion (expressed in terms of the Riemann curvature tensor) in section 8.9.

3. A Numerological Argument

This is my favourite argument because it requires no calculations and at the same time provides additional insight into the nature of curved space-times.

Assuming that the local existence of solutions to differential equations is guaranteed by some mathematical theorems, it is frequently sufficient to check that one has enough degrees of freedom to satisfy the desired initial conditions (one may also need to check integrability conditions). Here we are looking at something even more elementary, namely the functional freedom contained in the coordinate transformations to impose certain conditions at one point. In the present context, this argument is useful because it also reveals some information about the 'true' curvature hidden in the second derivatives of the metric. It works as follows:

(a) Zero'th Derivatives:

Consider a Taylor expansion of the metric around p in the sought-for new coordinates. Then the metric at p will transform with the matrix $(\partial x^\mu/\partial\xi^a)(p)$. This matrix has $(4\times 4) = 16$ independent components, precisely enough to impose the 10 conditions $g_{ab}(p) = \eta_{ab}$ up to Lorentz transformations.

(b) First Derivatives:

The derivative of the metric at p, $g_{ab,c}\,(p)$, will appear in conjunction with the second derivative $\partial^2 x^\mu/\partial\xi^a\partial\xi^b$. The $4\times(4\times 5)/2 = 40$ coefficients are precisely sufficient to impose the 40 conditions $g_{ab,c}\,(p) = 0$.

(c) Second Derivatives:

Now let us look at the second derivatives of the metric. $g_{ab,cd}$ has $(10\times 10) = 100$ independent components, while the third derivative of $x^\mu(\xi)$ at p, $\partial^3 x^\mu/\partial\xi^a\partial\xi^b\partial\xi^c$ has $4\times(4\times 5\times 6)/(2\times 3) = 80$ components. Thus 20 linear combinations of the second derivatives of the metric at p cannot in general be set to zero by a coordinate transformation. Thus these encode the information about the real curvature at p. This agrees nicely with the fact that the Riemann curvature tensor we will construct later turns out to have precisely 20 independent components.

Repeating this argument in space-time dimension $D = d + 1$, one finds that the number of 2nd derivatives of the metric modulo coordinate transformations is

$$\left(\frac{D(D+1)}{2}\right)^2 - D\frac{D(D+1)(D+2)}{6} = \frac{1}{12}D^2(D^2-1) \ . \tag{3.179}$$

Again this turns out to agree with the number of independent components (8.27) of the curvature tensor in D dimensions.

Note:

At this point in the course I find it useful to develop in parallel (and suggest to read in parallel)

- the more formal material on tensor analysis in sections 4, 5, 6, 7, 8 and 11, say (and then moving on to the Einstein equations themselves)
- and a detailed discussion of the basic properties of the Schwarzschild metric (sections 25 - 27),

since much of the latter (in particular geodesics, solar system tests of general relativity, even the issues that arise in connection with the Schwarzschild radius) can be understood just on the basis of what has been done so far (if, for the time being, one accepts on faith that the Schwarzschild metric is the unique spherically symmetric vacuum solution of the Einstein field equations).

Not only is this an interesting and physically relevant application of the machinery developed so far, it also provides an appropriate balance between physics and formalism in the lectures. More advanced material in the intervening sections can then be covered and dealt with if and when needed or desired (or, ideally, both).

4 Tensor Algebra

4.1 Principle of General Covariance

The Einstein Equivalence Principle tells us that the laws of nature (including the effects of gravity) should be such that in an inertial frame they reduce to the laws of Special Relativity.

A prime example of this is the geodesic equation

$$\ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma = 0 \ , \tag{4.1}$$

describing the motion of a particle in a gravitational field. Using the Einstein Equivalence Principle, and the *General Covariance* of the geodesic equation established in section 1.6, one can argue as follows to establish that this is the "correct" equation at an aribitrary point p in an arbitrary gravitational field described by a metric $g_{\alpha\beta}(x)$:

- Choose an inertial coordinate system z^a centered at p, i.e. with $g_{ab}(p) = \eta_{ab}$ and $\Gamma^a_{bc}(p) = 0$. Then one has

$$\left(\ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma\right)(p) = \left(\frac{\partial x^\alpha}{\partial z^a}\right)\left(\ddot{z}^a + \Gamma^a_{bc}\dot{z}^b\dot{z}^c\right)(p) = \left(\frac{\partial x^\alpha}{\partial z^a}\right)\ddot{z}^a(p) \ . \tag{4.2}$$

- By the Equivalence Principle, $\ddot{z}^a = 0$ is the correct equation for a particle in a freely falling reference system (such as at the origin p of an inertial coordinate system). Therefore, by the above identity, at the point p the geodesic equation is also satisfied in arbitrary coordinates x^α.

- Both the point p and the metric $g_{\alpha\beta}(x)$ are arbitrary, and therefore the geodesic equation holds at an arbitrary point of an arbitrary gravitational field.

This argument generalises and shows that the Einstein Equivalence Principle provides us with a general link between equations describing physics in an arbitrary gravitational field and general covariance. In other words, general covariance provides us with a concrete way of implementing the Einstein Equivalence Principle. One classical (but not entirely satisfactory - see the discussion below) textbook formulation of this *Principle of General Covariance* is[9]

Principle of General Covariance

> By virtue of the Einstein Equivalence Principle, a physical equation holds in an arbitrary gravitational field if

[9] S. Weinberg, *Gravitation and Cosmology* section 4.1

1. the equation holds in the absence of gravity, i.e. when $g_{\mu\nu} = \eta_{\mu\nu}$, $\Gamma^{\mu}_{\nu\lambda} = 0$, and
2. the equation is *generally covariant*, i.e. preserves its form under a general coordinate transformation.

We will turn momentarily to a proof of (a slightly modified version of) this statement. First, however, I would like to add a *caveat* to the 1st and a clarification to the 2nd condition (starting with the latter):

ad 2: We first need to clarify (and then reformulate slightly) the 2nd condition, as the statement "preserves its form under a general coordinate transformation" is neither completely unambiguous (without further explanation or definitions) nor totally to the point.

Concretely, the 2nd condition means the following: assume that you have some physical equation that in some coordinate system takes the form $T = 0$, where $T = 0$ could be some multi-component (thus T is adorned with various indices) differential equation. Now perform a coordinate transformation $x \to y(x)$, and assume that the new object T' has the form

$$T' = (\ldots)T + \text{junk} \tag{4.3}$$

where the term in brackets is some invertible matrix or operator. Then clearly the presence of the junk-terms means that the equation $T' = 0$ is not equivalent to the equation $T = 0$. An example of an object that transform in this way is, as we have seen, the Christoffel symbols. On the other hand, if these junk terms are absent, so that we have

$$T' = (\ldots)T \tag{4.4}$$

then one might like to say that the equation has preserved its form under a general coordinate transformation.

As a consequence of this, one also has $T' = 0$ if and only if $T = 0$, i.e. the equation is satisfied in one coordinate system if and only if it is satisfied in any other (or all) coordinate systems. An example of this is the geodesic equation which, as we have seen, transform precisely in such a way, with the term $(\ldots)$ in brackets being the Jacobi matrix.

However, if this is what one desires (and it is), then one may as well say this directly. Thus, to be more concrete, we can replace the 2nd condition above by

2' the equation is *generally covariant*, i.e. it is satisfied in one coordinate system iff it is satisfied in all coordinate systems.

ad 1: The argument below will invoke general covariance in order to be able to look at a given equation at the origin of an inertial (freely falling) coordinate system. As we have seen in various ways in section 3.6, at that point p, the effects of gravity are absent to the extent that the metric at that point is the Minkowski metric, $g_{ab}(p) = \eta_{ab}$, and that the derivatives of the metric (or Christoffel symbols) at that point are zero, $g_{ab,c}(p) = 0$. However, as also noted there, higher derivatives of the metric can in general not also be chosen to be zero at that point (and indeed the second derivatives of the metric at that point will turn out to contain coordinate independent information about the curvature of the space-time).

In that sense, looking at an equation at the origin of an inertial coordinate system is not strictly identical to looking at that same equation in the absence of gravity (i.e. in Minkowski space), where all these higher derivatives of the metric are also zero (in inertial coordinates). For the purposes of the argument below we will ignore this difference, and thus allow for the possibility that the Einstein Equivalence Principle holds only to first order in derivatives of the metric. Indeed, as discussed in more detail in section 8.4, and as could be anticipated from our identification of the metric with the potential of the gravitational field, second derivatives of the metric encode tidal gravitational forces (which one cannot expect to be able to eliminate by passing to a freely falling reference system), so this is a plausible relaxation of some stricter interpretation of the Einstein Equivalence Principle. We will look at the implications of this in the remarks below.

With these remarks in mind, let us now establish the above statement, namely that the Einstein equivalence principle implies that an equation that satisfies the conditions 1 and 2 (or 2') is valid in an arbitrary gravitational field:

- consider some equation that satisfies these conditions, and assume that we are in an arbitrary gravitational field;
- condition 2' implies that this equation is true (or satisfied) in all coordinate systems if it is satisfied just in one coordinate system;
- now we know that we can always (locally) construct a freely falling coordinate system in which the effects of gravity are absent;
- the Einstein Equivalence Principle now posits that in such a reference system the physics is that of Minkowski space-time;
- condition 1 means that the equation is true (satisfied) there;
- thus it is valid in all coordinate systems;
- since we started off by considering an arbitrary gravitational field, it follows that the equation is now valid in an arbitrary gravitational field, as claimed in the Principle of General Covariance.

Remarks:

1. Note that general covariance alone is an empty statement since any equation (whether correct or not) can be made generally covariant simply by writing it in an arbitrary coordinate system (cf. also the discussion in section 6.4). It develops its power only when used in conjunction with the Einstein Equivalence Principle as a statement about physics in a gravitational field, namely that by virtue of its general covariance an equation will be true in a gravitational field if it is true in the absence of gravitation.

2. As alluded to above, the principle of general covariance does not fix the equations uniquely because there are generally covariant objects that one can construct e.g. from the (second) derivatives of the metric (via the Riemann curvature tensor to be introduced in section 8) that can therefore be added to an equation and which vanish for Minkowski space, i.e. in the absence of gravitation.

3. In section 6.1 we will introduce a recipe / algorithm, the *principle of minimal coupling*, that allows us to produce generally covariant equations from those of special relativity. However, as we will discuss in section 8.10, also this description is ambiguous. The upshot is that there is no unique way of implementing the principle of general covariance, but this was probably too much to hope for anyway.

4.2 Tensors and Tensor Fields

In order to construct generally covariant equations, we need objects that transform in a simple way under coordinate transformations

$$x^\alpha \to y^\mu(x) = \bar{x}^\mu(x) \tag{4.5}$$

(it will be convenient to be able to switch between denoting the new coordinates by either y^μ or $\bar{x}^\mu$, depending on what is easier to read or write in a given situation). The prime examples of such objects are *tensors*, which transform multi-linearly with the Jacobi matrices

$$J^\mu_\alpha = \frac{\partial y^\mu}{\partial x^\alpha} \quad , \quad J^\alpha_\mu = \frac{\partial x^\alpha}{\partial y^\mu} \quad , \tag{4.6}$$

satisfying

$$J^\alpha_\mu J^\mu_\beta = \delta^\alpha_\beta \quad , \quad J^\mu_\alpha J^\alpha_\nu = \delta^\mu_\nu \quad . \tag{4.7}$$

If you are already familiar with Lorentz tensors from special relativity (as briefly recalled in section 1.2, these are objects which transform in a particularly simple multi-linear way under Lorentz transformations), then hardly anything in this or the subsequent section 4.3 should be new or unexpected (but interesting new features will arise in particular when we move on from tensor algebra to tensor analysis in section 5).

1. Scalars

   The simplest example of a tensor is a function (or scalar) f which under a coordinate transformation $x^\alpha \to \bar{x}^\mu(x)$ simply transforms as

   $$\bar{f}(\bar{x}(x)) = f(x) \ . \tag{4.8}$$

   One frequently suppresses the argument, and thus writes simply $\bar{f} = f$, expressing the fact that, up to the obvious and familiar change of argument, functions are invariant under coordinate transformations.

   REMARKS:

   (a) Just in case this transformation behaviour is not as familiar as it should be, consider the transformation from Cartesian (x^1, x^2) to polar $(y^1 = r, y^2 = \phi)$ coordinates, and the function $f(x^1, x^2) = (x^1)^2 + (x^2)^2$. In terms of polar coordinates, this function can be written as $\bar{f}(r, \phi) = r^2$. While f and $\bar{f}$ are evidently different functions of 2 variables, they are equal up to the change of coordinates, and this is how functions or scalars transform in general.

   (b) Note that the coordinates themselves are functions (that assign coordinates to a point p). Thus the x-coordinates can be regarded as 4 functions $f^{(\alpha)}$ defined in the coordinate system x by $f^{(\alpha)}(x) = x^\alpha$. Likewise, the new coordinates y^μ can be regarded as being defined by 4 in general genuinely *different* functions of x, say $g^{(\mu)}(x) = y^\mu(x)$.

   Under changes of coordinates, these functions transform as they should, i.e. one has $\bar{f}^{(\alpha)}(y) = f^{(\alpha)}(x) = x^\alpha$, and likewise for the functions $g^{(\mu)}(x)$. In the example above, the former would simply be the Cartesian coordinates expressed in terms of polar coordinates, $\bar{f}^{(1)}(r, \phi) = r\cos\phi = x^1 = f^{(1)}(x^1, x^2)$. On the other hand, $g^{(1)}(r, \phi) = r$, which is quite distinct from $\bar{f}^{(1)}(r, \phi) = r\cos\phi$. In that sense, allowing oneself to also denote the new coordinates y^μ by $\bar{x}^\mu$ may initially be a source of confusion (but is frequently so useful that we will continue to do this anyway ...).

   For a slightly more sophisticated perspective on these things, which disentangles functions from their representatives in terms of local coordinates, see the discussion of manifolds and local charts in section 5.11.

2. Vectors

   The next simplest case are objects that transform under coordinate transformations with the Jacobi matrix J^μ_α. Such objects are called vectors and have components $V^\alpha(x)$ transforming as

   $$\bar{V}^\mu(y(x)) = J^\mu_\alpha(x) V^\alpha(x) \ . \tag{4.9}$$

These are the prototypes of tensors and indeed all the other tensorial objects that we will encounter in the following are rather straightforward generalisations of such vectors.

REMARKS:

(a) One comment on notation: since already the index μ on $\bar{V}^\mu$ indicates that these are the components of a vector with resepct to the coordinates $y^\mu = \bar{x}^\mu$, it is not strictly speaking necessary to indicate this as well by putting a bar on the vector. Therefore one could (and commonly just does) write simply

$$V^\mu(y) \equiv \bar{V}^\mu(y) \quad . \tag{4.10}$$

(b) One comment on terminology: it is sometimes useful to distinguish *vectors* from *vector fields* and, likewise, *tensors* from *tensor fields*. A vector is then just a vector $V^\alpha(x)$ at some point x of space-time whereas a vector field is something that assigns a vector to each point of space-time,

$$\text{vector field}: \quad x \mapsto V^\alpha(x) \tag{4.11}$$

and likewise for scalars and scalar fields, and more general tensors and tensor fields.

(c) An intermediate situation is that of a vector or tensor defined neither just at a single point nor in the entire space-time but for example along some curve (representing e.g. the worldline of a particle). A prime example is the tangent vector $\dot{x}^\alpha$ to a curve, which indeed transforms as a vector,

$$\dot{x}^\alpha \to \dot{y}^\mu = \frac{\partial y^\mu}{\partial x^\alpha}\dot{x}^\alpha = J^\mu_a \dot{x}^\alpha \quad . \tag{4.12}$$

And while the naive acceleration $\ddot{x}^\alpha$ is *not* a vector, as we have seen on several occasions before, we can define a vectorial acceleration by

$$a^\alpha = \ddot{x}^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma \quad . \tag{4.13}$$

We will rederive this expression for a covariant acceleration from a different perspective in section 5.7.

(d) One way of thinking about vector fields is as tangent vector fields to families of curves on a space or space-time which arise as the solutions to the differential equation

$$\frac{d}{d\sigma}x^\alpha(\sigma) = V^\alpha(x(\sigma)) \tag{4.14}$$

(and we take local existence and uniqueness of these solutions under suitable regularity and differentiability conditions for granted). These curves $x^\alpha(s)$ are the *integral curves* (or *orbits*) of the vector field V^α, and by by construction they are characterised by the fact that at any point x the tangent

vector to the curve passing through that point is the vector $V^\alpha(x)$ at that point. Thus vector fields also generate a *flow* on the space(-time), namely the motion of points along these integral curves, $x^\alpha(\sigma) \mapsto x^\alpha(\sigma+s)$ for $s \in \mathbb{R}$.

(e) An extremely useful related way of thinking about vectors (vector fields) is as first order differential operators, via the correspondence

$$V^\alpha \Leftrightarrow V := V^\alpha \partial_\alpha \ . \tag{4.15}$$

This operator gives the directional derivative along the flow of the vector field V, as in

$$\frac{d}{d\sigma} f(x(\sigma)) = V^\alpha \partial_\alpha f(x(\sigma)) \tag{4.16}$$

(by the chain rule). One of the advantages of this point of view is that the object V is completely invariant under coordinate transformations as the *components* V^α of V transform inversely to the *basis vectors* ∂_α. For more on this see sections 4.6 and 4.8 on the coordinate-independent interpretation of tensors below.

3. Covectors

A covector (field) is an object $U_\alpha(x)$ which under a coordinate transformation transforms inversely to a vector, i.e. as

$$\bar{U}_\mu(y(x)) \equiv U_\mu(y(x)) = J^\alpha_\mu(y(x)) U_\alpha(x) \ . \tag{4.17}$$

A familiar example of a covector is the derivative $U_\alpha = \partial_\alpha f$ of a function (scalar) which of course transforms as

$$\partial_\mu \bar{f}(y(x)) = J^\alpha_\mu \partial_\alpha f(x) \ , \tag{4.18}$$

because that is how partial derivatives transform.

REMARKS:

(a) As in the case of covectors of special relativity (1.44), one should think of covectors pointwise as elements of the dual vector space $\mathbb{V}^*$ to the space of vectors $\mathbb{V}$, i.e. as linear functionals on the space of vectors, given by

$$U(x): \quad V^\alpha(x) \ \mapsto \ U_\alpha(x) V^\alpha(x) \ \in \ \mathbb{R} \ . \tag{4.19}$$

The transformation properties of U_α and V^α guarantee that the result is a scalar (function) under coordinate transformations,

$$U_\alpha V^\alpha = \bar{U}_\mu \bar{V}^\mu \ . \tag{4.20}$$

(b) Just as it was useful to think of a vector field $V^\alpha(x)$ in a more coordinate independent way as the components of the coordinate-independent object $V = V^\alpha \partial_\alpha$ with respect to the basis ∂_α, one can think of covector fields U_α as the components of an object

$$U = U_\alpha dx^\alpha \tag{4.21}$$

(which is invariant under coordinate transformations) with respect to the basis dx^α. The prime example is again the *differential*

$$df = \partial_\alpha f(x) dx^\alpha = \partial_\mu \bar{f}(y) dy^\mu \tag{4.22}$$

of a scalar.

(c) Combining the two points of view in the remarks above, one can thus think of df as the linear functional on vector fields that assigns to a vector field V the scalar which is the derivative of f along V,

$$\begin{aligned} df: &\quad \text{vector fields} \longrightarrow \text{scalar fields} \\ &\quad df(V) = V^\alpha \partial_\alpha f \equiv Vf \ . \end{aligned} \tag{4.23}$$

4. Covariant 2-Tensors

Clearly, given the above objects, we can construct more general objects which transform in a nice way under coordinate transformations by taking products of them. Tensors in general are objects which transform like (but need not be equal to) products of vectors and covectors.

In particular, a covariant 2-tensor, or (0,2)-tensor, is an object $A_{\alpha\beta}$ that transforms under coordinate transformations like the product of two covectors, i.e.

$$\bar{A}_{\mu\nu}(y(x)) \equiv A_{\mu\nu}(y(x)) = J^\alpha_\mu(y(x)) J^\beta_\nu(y(x)) A_{\alpha\beta}(x) \ . \tag{4.24}$$

We already know one example of such a tensor, namely the metric tensor $g_{\alpha\beta}$ (which happens to be a symmetric tensor).

In order to avoid unnecessary clutter and to make tensorial equations more readable, I will from now on typically use a shorthand notation in which I drop not only the bars, overlines or other decorations on the transformed objects but in which I also omit the argument x or $y(x)$. In this notation, the above equation would then simply become

$$A_{\mu\nu} = J^\alpha_\mu J^\beta_\nu A_{\alpha\beta} \ , \tag{4.25}$$

which is more user-friendly.

5. Contravariant 2-Tensors

Likewise we define a contravariant 2-tensor (or a (2,0)-tensor) to be an object $B^{\alpha\beta}$ that transforms like the product of two vectors,

$$B^{\mu\nu} = J^{\mu}_{\alpha} J^{\nu}_{\beta} B^{\alpha\beta} \quad . \tag{4.26}$$

An example is the inverse metric tensor $g^{\alpha\beta}$.

6. (p, q)-Tensors

   It should now be clear how to define a general (p, q)-tensor - namely as an object with p contravariant (upper) and q covariant (lower) indices which under a coordinate transformation transforms like a product of p vectors and q covectors,

$$T^{\mu_1 \cdots \mu_p}_{\nu_1 \ldots \nu_q} = J^{\mu_1}_{\alpha_1} \ldots J^{\mu_p}_{\alpha_p} \; J^{\beta_1}_{\nu_1} \ldots J^{\beta_q}_{\nu_q} \; T^{\alpha_1 \cdots \alpha_p}_{\beta_1 \cdots \beta_q} \quad . \tag{4.27}$$

Remarks:

1. Note that, in particular, a tensor is zero (at a point) in one coordinate system if and only if the tensor is zero (at the same point) in another coordinate system.

   Thus, any law of nature (field equation, equation of motion) expressed in terms of tensors, say in the form $T^{\alpha_1 \cdots \alpha_p}_{\beta_1 \ldots \beta_q} = 0$, preserves its form under coordinate trasformations and is therefore automatically generally covariant,

$$T^{\alpha_1 \cdots \alpha_p}_{\beta_1 \cdots \beta_q}(x) = 0 \quad \forall x \quad \Leftrightarrow \quad T^{\mu_1 \cdots \mu_p}_{\nu_1 \ldots \nu_q}(y) = 0 \quad \forall y \quad . \tag{4.28}$$

2. An important special example of a tensor is the Kronecker tensor δ^{α}_{β}. That this is indeed a tensor follows from the fact that it can e.g. be written as a contraction of a product of tensors, namely the metric tensor and its inverse,

$$\delta^{\alpha}_{\beta} = g^{\alpha\gamma} g_{\gamma\beta} \quad , \tag{4.29}$$

   and the calculational rules of tensor algebra to be established in section 4.3 below. Calculating explicitly the components with respect to the new coordinates $y^{\mu} = \bar{x}^{\mu}$, one finds

$$\bar{\delta}^{\mu}_{\nu} = J^{\mu}_{\alpha} J^{\beta}_{\nu} \; \delta^{\alpha}_{\beta} = J^{\mu}_{\alpha} J^{\alpha}_{\nu} = \delta^{\mu}_{\nu} \quad . \tag{4.30}$$

   Thus the Kronecker tensor has the same components in all coordinate systems. This is reassuring but should not be too surprising. Together with scalars and products of scalars and Kronecker tensors it is the only tensor whose components are the same in all coordinate systems.

3. A covariant 2-tensor $T_{\alpha\beta}$, say, is said to be symmetric if $T_{\alpha\beta} = T_{\beta\alpha}$ and anti-symmetric if $T_{\alpha\beta} = -T_{\beta\alpha}$. This is well-defined because it is a generally covariant notion: a tensor is symmetric in all coordinate system iff it is symmetric in one coordinate system, etc.

This definition can be extended to any or all pairs of covariant indices or pairs of contravariant indices. Thus e.g. a tensor $T^{\alpha_1\ldots\alpha_p}$ is called totally symmetric (or totally anti-symmetric) if it is symmetric (anti-symmetric) under the exchange of any pair of indices.

On the other hand, it is not meaningful to talk of the symmetry of a (1,1)-tensor, say, as an equation like $T^{\alpha}_{\beta} = T^{\beta}_{\alpha}$ does not make any sense.

Symmetrisation and anti-symmetrisation of tensors will be discussed in section 4.3 below.

4. The number of independent components of a general (p, q)-tensor is 4^{p+q}. The number of independent components is reduced if the tensor has some symmetry properties. Thus a symmetric (0,2)- or (2,0)-tensor has $4 \times 5/2 = 10$ independent components, an anti-symmetric (0,2)- or (2,0)-tensor has $4 \times 3/2 = 6$ independent components, and a totally anti-symmetric $(0, 4)$-tensor $T_{\beta_1\ldots\beta_4}$ has only got one independent component, namely T_{0123} (all the others being determined by anti-symmetry).

5. Important examples of non-tensors are the Christoffel symbols. Another important example is the the ordinary partial derivative of a (p, q)-tensor, $\partial_\gamma T^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ which is *not* a $(p, q+1)$-tensor unless $p = q = 0$. This failure of the partial derivative to map tensors to tensors will motivate us below to introduce a *covariant derivative* which generalises the usual notion of a partial derivative and has the added virtue of mapping tensors to tensors.

4.3 Tensor Algebra

Tensors can be added, multiplied and contracted in certain obvious ways. The basic algebraic operations are the following:

1. Linear Combinations

   Given two (p, q)-tensors $A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ and $B^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$, their sum

   $$C^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} = A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} + B^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} \tag{4.31}$$

   is also a (p, q)-tensor.

2. Direct Products

   Given a (p, q)-tensor $A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ and a (p', q')-tensor $B^{\gamma_1\ldots\gamma_{p'}}_{\delta_1\ldots\delta_{q'}}$, their direct product

   $$A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} B^{\gamma_1\ldots\gamma_{p'}}_{\delta_1\ldots\delta_{q'}} \tag{4.32}$$

   is a $(p + p', q + q')$-tensor,

3. Contractions

   Given a (p,q)-tensor with p and q non-zero, one can associate to it a $(p-1,q-1)$-tensor via contraction of (i.e. summation over) one covariant and one contravariant index, for example

$$A^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q} \quad \to \quad B^{\alpha_1\ldots\alpha_{p-1}}_{\beta_1\ldots\beta_{q-1}} = A^{\alpha_1\ldots\alpha_{p-1}\gamma}_{\beta_1\ldots\beta_{q-1}\gamma} \ . \tag{4.33}$$

   This operation of contraction generalises the trace. In general, there are in total $(p \times q)$ distinct possibilities to do this, by contracting the 1st oder 2nd or ... or p'th upper index with the 1st or 2nd or ... or q'th lower index.

   That this indeed leads to a $(p-1,q-1)$-tensor follows from the fact that the Jacobi matrices corresponding to the two contracted indices are inverses to each other, and are multiplied (by the contraction), resulting in the identity matrix. This then leaves only $(p-1)$ Jacobi matrices and $(q-1)$ inverse Jacobi matrices, corresponding to the transformation behaviour of a $(p-1,q-1)$-tensor.

   For example, for a $(1,1)$-tensor T^α_β one has the unique contraction

$$T^\alpha_\beta \to T^\alpha_\alpha \ . \tag{4.34}$$

   This is just the standard trace of the linear tranformation represented by the $(1,1)$-tensor T^α_β. Under the coordinate transformation $x^\alpha \to y^\mu$ this trace transforms as

$$T^\mu_\mu = J^\mu_\alpha J^\beta_\mu T^\alpha_\beta = \delta^\beta_\alpha T^\alpha_\beta = T^\alpha_\alpha \ . \tag{4.35}$$

   Thus the trace is invariant (i.e. a scalar or $(0,0)$-tensor). Clearly this cancellation between the two Jacobi matrices corresponding to two contracted indices also takes place when the tensor T carries other non-contracted indices, and therefore the contraction of a (p,q)-tensor is a $(p-1,q-1)$-tensor, as claimed.

4. Raising and Lowering of Indices

   These operations can of course be combined in various ways. A particular important operation is, given a metric tensor, the lowering of indices with the metric, and the raising of indices with the inverse metric.

   In particular, by direct product with the metric and subsequent contraction, we can construct a covector from a vector by

$$V^\alpha \to g_{\alpha\beta} V^\beta \ . \tag{4.36}$$

   When it is clear from the context which metric is being used, it is common and convenient to simply call this covector V_α,

$$V_\alpha \equiv g_{\alpha\beta} V^\beta \ . \tag{4.37}$$

Likewise, to a covector one can associate a vector by using the inverse metric,

$$A_\alpha \to g^{\alpha\beta} A_\beta \equiv A^\alpha \quad . \tag{4.38}$$

REMARKS:

(a) Interpreted in terms of covectors as linear functions on vectors, this perhaps somewhat obscure convention and notation has a perfectly natural interpretation. Namely, recall that even though a vector space $\mathbb{V}$ and its dual $\mathbb{V}^*$ are isomorphic (in finite dimensions), there is no natural isomorphism between them, i.e. no natural identification of vectors in $\mathbb{V}$ with covectors in $\mathbb{V}^*$. However, if one has a scalar product (metric) $< v, w >$ on $\mathbb{V}$, then this provides an identification of $\mathbb{V}$ and $\mathbb{V}^*$ through

$$v \in \mathbb{V} \mapsto \alpha_v \in \mathbb{V}^* : \quad \alpha_v(w) =< v, w > \quad . \tag{4.39}$$

Thus a metric allows one to associate a covector to a vector, and in the notation of tensor algebra favoured by physicists this is conveniently just written as $V^\alpha \mapsto V_\alpha$.

(b) This convention of using the metric and its inverse to lower and raise indices on a tensor can of course be extended to higher rank tensors, but the result will in general depend on which index is lowered or raised. One then needs to make sure that the notation is sufficiently unambiguous to keep track of this. For example, the two (in general distinct) (1,1)-tensors one obtains from the lowering of one index of a (2,0)-tensor might be denoted amd distinguished by

$$g_{\gamma\alpha} T^{\alpha\beta} = T_\gamma{}^\beta \quad , \quad g_{\gamma\beta} T^{\alpha\beta} = T^\alpha{}_\gamma \quad . \tag{4.40}$$

(c) Finally note that this notation of raising and lowering indices with the metric is consistent with denoting the inverse metric by raised indices, i.e. it is indeed true that

$$g^{\alpha\beta} = g^{\alpha\gamma} g^{\beta\delta} g_{\gamma\delta} \tag{4.41}$$

(both indices of $g_{\gamma\delta}$ raised with two inverse metrics). That this is indeed correct follows from

$$g^{\alpha\gamma} g^{\beta\delta} g_{\gamma\delta} = g^{\alpha\gamma} \delta^\beta_\gamma = g^{\alpha\beta} \quad . \tag{4.42}$$

5. Symmetrisation and anti-Symmetrisation

Given any $(0,2)$-tensor $T_{\alpha\beta}$, say, one can decompose it into its symmetric and anti-symmetric parts as

$$T_{\alpha\beta} = \tfrac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}) + \tfrac{1}{2}(T_{\alpha\beta} - T_{\beta\alpha}) \equiv T_{(\alpha\beta)} + T_{[\alpha\beta]} \quad . \tag{4.43}$$

The decomposition into symmetric and anti-symmetric parts is invariant under coordinate transformations. In particular, when $T_{\alpha\beta}$ is a tensor, also $T_{(\alpha\beta)}$ and

$T_{[\alpha\beta]}$ are tensors, and thus (anti-)symmetrisation is yet another linear operation that one can perform on tensors.

The factor $\frac{1}{2}$ is chosen such that the symmetrisation of a symmetric tensor is the same as the original tensor,

$$T_{\alpha\beta} = T_{\beta\alpha} \quad \Rightarrow \quad T_{(\alpha\beta)} = T_{\alpha\beta} \quad , \quad T_{[\alpha\beta]} = 0 \tag{4.44}$$

(and likewise for the anti-symmetrisation of anti-symmetric tensors).

This can be generalised to the (anti-)symmetrisation of any pair of (contravariant or covariant) indices; e.g.

$$T_{(\alpha\beta)\gamma} = \tfrac{1}{2}(T_{\alpha\beta\gamma} + T_{\beta\alpha\gamma}) \tag{4.45}$$

is the symmetrisation of $T_{\alpha\beta\gamma}$ in its first and second index. It can also be generalised to the total (anti-)symmetrisation of a higher-rank tensor; e.g.

$$T_{(\alpha\beta\gamma)} \equiv \tfrac{1}{3!}(T_{\alpha\beta\gamma} + T_{\beta\alpha\gamma} + T_{\gamma\beta\alpha} + T_{\beta\gamma\alpha} + T_{\alpha\gamma\beta} + T_{\gamma\alpha\beta}) \tag{4.46}$$

is totally symmetric, i.e. symmetric under the exchange of any pair of indices, and

$$T_{[\alpha\beta\gamma]} \equiv \tfrac{1}{3!}(T_{\alpha\beta\gamma} - T_{\beta\alpha\gamma} - T_{\gamma\beta\alpha} + T_{\beta\gamma\alpha} - T_{\alpha\gamma\beta} + T_{\gamma\alpha\beta}) \tag{4.47}$$

is totally anti-symmetric. The prefactor $\frac{1}{6}$ is again there to ensure that the total symmetrisation of a totally symmetric tensor is the original tensor (and likewise for the total anti-symmetrisation of totally anti-symmetric tensors). This generalises in an evident way to higher rank p tensors, with the combinatorial prefactor $1/p!$.

An observation we will frequently make use of to recognise when some object is a tensor is the following (occasionally known as the *quotient theorem* or quotient lemma):

For example, say that in an equation of the form

$$A_\alpha = B_{\alpha\beta} C^\beta \tag{4.48}$$

you know that A_α transforms as a covector for any vector C^β. Then it follows that $B_{\alpha\beta}$ has to be a tensor. Likewise for higher rank tensors and more contractions, as in

$$A_{\alpha\beta} = B_{\alpha\beta\gamma\delta} C^{\gamma\delta} \quad . \tag{4.49}$$

Also in that case one has the statement that if A transforms as a tensor for every tensor C, then B itself has to be a tensor.

An elementary and ham-handed proof of these kinds of statements can be obtained by contradiction: assume that B does not transform as a tensor and write its transformation, as in (4.3), as

$$B' = (\ldots)B + \text{junk} \quad . \tag{4.50}$$

If "junk" $\neq 0$, then there will be some C such that "junk" contributes to the contraction $B'C'$. That means that "junk" contributes to A', the transformed A, contradicting the premise that A is a tensor.

4.4 Generally Covariant Integration and Volume Elements

While tensors are the objects which, in a sense, transform in the nicest and simplest possible way under coordinate transformations, they are not the only relevant objects. An important class of non-tensors (but "almost" tensors) are so-called *tensor densities*. They will play a crucial role for us in order to have a generally-covariant notion of integration at our disposal, and thus ultimately also a way of writing down generally covariant action principles for fields etc.

In this section we will address the issue of generally covariant integration in a space-time equipped with a metric. This will be accomplished with the help of a particular tensor density constructed from the metric. Having thus established that tensor densities are objects of legitimate interest in their own right, we will then discuss their properties in more generality in section 4.5 below.

To set the stage, consider once again first the situation in special relativity. In that case, the integral of a Lorentz scalar $f(\xi)$ with respect to the volume element $d^4\xi$ (or $d^{27}\xi$ ...) is itself a Lorentz scalar, i.e. independent of the inertial reference frame in which the integral is evaluated,

$$\int d^4\xi \; f(\xi) = \int d^4\bar{\xi} \; \bar{f}(\bar{\xi}) \qquad (\bar{\xi}^a = L^a_b \xi^b) \tag{4.51}$$

The reasons for this are that

1. f is a scalar by assumption,
$$\bar{f}(\bar{\xi}) = f(\xi) \;\; , \tag{4.52}$$

2. by the fundamental theorem of integral calculus, under an arbitrary coordinate transformation $\xi \to \bar{\xi} = \bar{\xi}(\xi)$ the volume element transforms with the *Jacobian*, the (absolute value of the) determinant of the Jacobi matrix,
$$d^4\bar{\xi} = \left| \det \frac{\partial \bar{\xi}}{\partial \xi} \right| d^4\xi \;\; , \tag{4.53}$$

3. for a Lorentz transformation one has
$$\left| \det \frac{\partial \bar{\xi}}{\partial \xi} \right| = |\det L| = 1 \;\; . \tag{4.54}$$
and thus the volume element is invariant under Lorentz transformations,
$$d^4\bar{\xi} = d^4\xi \;\; . \tag{4.55}$$

Turning now to general relativity and general covariance, it is immediately apparent that the integral of a scalar $\int d^4x \; f(x)$ will *not* be generally covariant, i.e. under a general coordinate transformation $x \to y = y(x)$ generically one has

$$\int d^4x \; f(x) \neq \int d^4y \; \bar{f}(y)(y) \tag{4.56}$$

because of the non-trivial Jacobian,

$$d^4y = \left|\det\left(\frac{\partial y}{\partial x}\right)\right| d^4x \ . \tag{4.57}$$

One way out would be to abandon the idea that one should integrate scalars and to require that the integrand $\rho(x)$ should transform in such a way that it cancels the Jacobian arising from the measure, namely as

$$\bar{\rho}(y) = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-1} \rho(x) \ . \tag{4.58}$$

This is indeed an option, and we will return to this below (see remark 1 in section 4.5), but at this stage this is rather unintuitive and not particularly useful, in particular because it is not clear how one should go about finding or constructing such objects in the first place.

A first simplification arises from the fact that (4.58) implies that

- if $\rho(x)$ satisfies (4.58), then so will $f(x)\rho(x)$ for any scalar $f(x)$;
- and conversely, if $\rho_1(x)$ and $\rho_2(x)$ satisfy (4.58), then $\rho_2(x)/\rho_1(x) \equiv f(x)$ is a scalar.

Thus, in order to find all objects that satisfy (4.58), we just need to find a single one, and with respect to that one as our measure we can then integrate any scalar function $f(x)$ in a generally covariant way,

$$\bar{\rho}(y) = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-1} \rho(x) \quad \Rightarrow \quad \int d^4y\ \bar{\rho}(y)\bar{f}(y) = \int d^4x\ \rho(x)f(x) \tag{4.59}$$

In order to find one candidate $\rho(x)$, let us approach this question in a different way. Integrals are used to calculate or measure volumes (or areas, or lenghts, or ...). Such integrals should have a coordinate-independent meaning, but they should depend on the prescription one uses for *measuring* volumes, areas, lenghts, ... These prescriptions are concisely encoded in the *metric*. Thus it is plausible that in order to define a generally covariant notion of integration one may need to specify the metric, but that this is all that one should need to know (while the Jacobian between two coordinate systems should fundamentally be irrelevant and be considered to be a red herring).

With this in mind, let us recall the standard tensorial transformation behaviour of the metric under coordinate transformations,

$$\bar{g}_{\alpha\beta}(y) = \frac{\partial x^\mu}{\partial y^\alpha}\frac{\partial x^\nu}{\partial y^\beta} g_{\mu\nu}(x) \ . \tag{4.60}$$

It follows from this that the *absolute value of the determinant of the metric*

$$g := |\det(g_{\mu\nu}(x))| \tag{4.61}$$

does not transform like a scalar or some other tensor at all, but instead transforms as

$$\bar{g} = \left|\det\left(\frac{\partial x}{\partial y}\right)^2\right| g = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-2} g \ . \tag{4.62}$$

In particular, its square-root $\sqrt{g}$ transforms as

$$\sqrt{\bar{g}} = \left|\det\left(\frac{\partial y}{\partial x}\right)\right|^{-1} \sqrt{g} \ . \tag{4.63}$$

Thus it satisfies (4.58) and provides us with our required reference object

$$\rho(x) = \sqrt{g(x)} \ . \tag{4.64}$$

In particular, the combined expression $\sqrt{g}d^4x$ is *invariant* under general coordinate transformations,

$$\sqrt{\bar{g}}d^4y = \sqrt{g}d^4x \ , \tag{4.65}$$

and can therefore be used to define integrals of scalars in a generally covariant (but metric-dependent) way,

$$\int \sqrt{\bar{g}}d^4y \ \bar{f}(y) = \int \sqrt{g}d^4x \ f(x) \ . \tag{4.66}$$

This will of course be important in order to formulate action principles etc. in a space-time equipped with a metric in a generally covariant way.

REMARKS:

1. This is also frequently the quickest way to determine the volume element in non-Cartesian coordinates in Euclidean space. In Cartesian coordinates one has $\sqrt{g} = 1$, so the correct integration measure is just the familiar

$$\sqrt{g}d^3x = d^3x \ . \tag{4.67}$$

To now determine what is the volume element in spherical coordinates $\{y^k\} = (r, \theta, \phi)$, say, instead of laboriously determining the Jacobi matrix for the coordinate transformation, and then (equally laboriously) calculating its determinant (which would be the standard uninspiring and uninspired procedure), all one needs to know is the metric in these coordinates (which one usually needs to determine anyway) to deduce

$$ds^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad \Rightarrow \quad \bar{g} = r^4 \sin^2\theta \tag{4.68}$$

and therefore

$$d^3x = \sqrt{\bar{g}} \ d^3y = r^2 \sin\theta \ dr \ d\theta \ d\phi \ . \tag{4.69}$$

This is of course the standard result.

2. As a variation of this theme, it is now also easy to construct a spherical (i.e. radial + angular) coordinate system z^k in which $\sqrt{g} = 1$, i.e. which is such that simply $d^3z = d^3x$ (without any Jacobian factors). To that end it suffices to let (4.69) suggest to introduce a new radial variable $\rho = \rho(r)$ by

$$d\rho = r^2 dr \quad \Rightarrow \quad \rho = r^3/3 \in (0, \infty) \tag{4.70}$$

and a new angular variable $\psi = \psi(\theta)$ by

$$d\psi = \sin\theta d\theta \quad \Rightarrow \quad \psi = -\cos\theta \in [-1, 1) \ . \tag{4.71}$$

In these coordinates, the Euclidean line element takes the form

$$\begin{aligned} ds^2 &= r(\rho)^{-4} d\rho^2 + r(\rho)^2 (d\psi^2/\sin^2\theta(\psi) + \sin^2\theta(\psi) d\phi^2) \\ &= (3\rho)^{-4/3} d\rho^2 + (3\rho)^{2/3} \left(\frac{d\psi^2}{1-\psi^2} + (1-\psi^2) d\phi^2 \right) \end{aligned} \tag{4.72}$$

It is now manifest that in these coordinates $\{z^k\} = (\rho, \psi, \phi)$ one has

$$\sqrt{g} = 1 \quad \Rightarrow \quad d^3x = d^3z = d\rho\, d\psi\, d\phi \ . \tag{4.73}$$

The Euclidean metric in these coordinates will make a brief appearance in the discussion of the derivation of the Schwarzschild metric in section 24.3 (cf. the discussion leading to (24.33)).

3. More generally, given any metric $g_{\alpha\beta}$, one can find a coordinate system in which $g = |\det(g_{\alpha\beta})| = 1$. Such a ("*unimodular*") coordinate system is highly non-unique. In fact, given any such coordinate system x, any other coordinate system y with unit Jacobi matrix $|\det(\partial y/\partial x)| = 1$ will also satisfy this condition. Such coordinate transformations form an infinite-dimensional subgroup of the group of all coordinate transformations, known as the group of *volume-preserving coordinate transformations*.

   In particular, the above transformation from Cartesian coordinates x^k to the coordinates z^k is a non-trivial example of a volume-preserving coordinate transformation.

4.5 Tensor Densities and Volume Elements

In the previous section we have encountered certain not strictly tensorial objects which nevertheless turned out to be useful. Having thus established the basic credentials of such objects, we will now formalise this somewhat.

Thus the prime example of what we will call a tensor density is the (absolute value of the) determinant $g := |\det g_{\mu\nu}|$ of the metric tensor, which, as we have seen, transforms as

$$\bar{g} = \left| \det\left(\frac{\partial y}{\partial x} \right) \right|^{-2} g \ . \tag{4.74}$$

An object which transforms in such a way under coordinate transformations is called a *scalar tensor density of weight* $w = +2$, and the square root of the determinant $\sqrt{g}$ transforms as, and hence is, a tensor density of weight $w = +1$.

In general, a tensor density of weight w is an object that transforms as a tensor, with an additional factor of $|\det(\partial y/\partial x)|^{-w}$. It thus transforms like a tensor multiplied by $g^{w/2}$, and one has

$$\tilde{T} \quad \text{a tensor density of weight } w \quad \Leftrightarrow \quad \tilde{T} = g^{w/2} T \quad \text{with } T \text{ a tensor} \ . \tag{4.75}$$

The algebraic rules for tensor densities are strictly analogous to those for tensors. Thus, for example, the sum of two (p, q) tensor densities of weight w (let us call this a $(p, q; w)$ tensor) is again a $(p, q; w)$ tensor, and the direct product of a $(p_1, q_1; w_1)$ and a $(p_2, q_2; w_2)$ tensor is a $(p_1 + p_2, q_1 + q_2; w_1 + w_2)$ tensor. Contractions and the raising and lowering of indices of tensor densities can also be defined just as for ordinary tensors.

Remarks:

1. Generalising the argument in section 4.4, we now learn that if ρ is any scalar density of weight $w = +1$, then its integral is well-defined and coordinate independent,

$$\int d^4x \ \rho(x) = \int d^4y \ \bar{\rho}(y) \ . \tag{4.76}$$

   See remark 4 at the end of this section for one way of constructing such objects without taking recourse to a metric.

2. There is one important tensor density which - like the Kronecker tensor - has the same components in all coordinate systems. This is the totally anti-symmetric Levi-Civita symbol $\in_{\alpha\beta\gamma\delta}$ (taking the values $0, \pm 1$) which is a tensor density of weight $w = -1$. Then $\sqrt{g} \in_{\alpha\beta\gamma\delta}$ is a tensor (strictly speaking it is a pseudo-tensor because of its behaviour under reversal of orientation - see below).

   To see this, recall first of all the definition of the Levi-Civita symbol: it is totally anti-symmetric,

$$\in_{\alpha\beta\gamma\delta} = \in_{[\alpha\beta\gamma\delta]} \ , \tag{4.77}$$

   and has therefore only got one independent component which we will normalise to be

$$\in_{0123} = +1 \ . \tag{4.78}$$

   Thus $\in_{\alpha\beta\gamma\delta} = +1$ if the indices $(\alpha\beta\gamma\delta)$ are an even permutation of (0123), $\in_{\alpha\beta\gamma\delta} = -1$ if the indices $(\alpha\beta\gamma\delta)$ are an odd permutation of (0123), and $\in_{\alpha\beta\gamma\delta} = 0$ iff any two indices are equal. This definition makes no reference to any coordinate system whatsoever, and thus tautologically the purely combinatorial object $\in_{\alpha\beta\gamma\delta}$ has the same components in all coordinate systems.

This evidently extends to other dimensions, and we will define the $D = (d+1)$-dimensional Levi-Civita symbol in the same way,

$$\in_{\alpha_1 \ldots \alpha_D} = \in_{[\alpha_1 \ldots \alpha_D]} \quad , \quad \in_{01\ldots d} = +1 \ . \tag{4.79}$$

Next, recall one possible definition of the determinant $\det M$ of a $(D \times D)$-matrix $M^\mu_{\ \nu}$, namely as the coefficient (proportionality factor) on the right-hand side of

$$\in_{\alpha_1 \ldots \alpha_D} \ M^{\alpha_1}_{\ \beta_1} \ldots M^{\alpha_D}_{\ \beta_D} = (\det M) \ \in_{\beta_1 \ldots \beta_D} \ . \tag{4.80}$$

Now choose M to be the Jacobi matrix $J^\mu_\alpha = (\partial y^\mu / \partial x^\alpha)$. Then the above equation can be rearranged to

$$\in_{\mu_1 \ldots \mu_D} = \det \left(\frac{\partial y}{\partial x} \right) J^{\alpha_1}_{\mu_1} \ldots J^{\alpha_D}_{\mu_D} \in_{\alpha_1 \ldots \alpha_D} \ . \tag{4.81}$$

This shows that $\in_{\alpha_1 \ldots \alpha_D}$ transforms as a tensor density of weight $w = -1$, provided that $\det(\partial y / \partial x) > 0$. The latter condition means that the coordinate transformation preserves the orientation. Thus, $\in_{\alpha_1 \ldots \alpha_D}$ transforms as a tensor density under orientation-preserving coordinate transformations but picks up a sign when the orientation is reversed. Strictly speaking $\in_{\alpha_1 \ldots \alpha_D}$ is then not a tensor density but something that is called a *pseudo-tensor density*.

Going back to 4 dimensions, it follows that

$$\epsilon_{\alpha\beta\gamma\delta} \equiv \sqrt{g} \in_{\alpha\beta\gamma\delta} \tag{4.82}$$

is a totally anti-symmetric $(0,4)$ (pseudo-)tensor. Likewise, the totally anti-symmetric symbol $\in^{\alpha\beta\gamma\delta}$ is a tensor density of weight $w = +1$ and

$$\epsilon^{\alpha\beta\gamma\delta} \equiv \frac{1}{\sqrt{g}} \in^{\alpha\beta\gamma\delta} \tag{4.83}$$

is a totally anti-symmetric $(4,0)$ (pseudo-)tensor. Here, as usual, we have raised the indices of the tensor on the left-hand side with the metric, and $\in^{\alpha\beta\gamma\delta}$ is totally anti-symmetric, with $\in^{0123} = -1$. The minus-sign arises because the contraction with the metrics on the left-hand side produces a factor of

$$\sqrt{g} \det(g^{\alpha\beta}) = \sqrt{g} (\det(g_{\alpha\beta}))^{-1} = -\frac{1}{\sqrt{g}} \ . \tag{4.84}$$

We could have chosen to not absorb the minus sign into the definition of $\in^{\alpha\beta\gamma\delta}$, at the expense of an explicit minus sign on the right-hand side of (4.83). The convention we have adopted is more convenient, however, in particular since it is compatible with the standard practice in special relativity to (tacitly) identify $\epsilon_{abcd} = \in_{abcd}$, the minus sign arising from raising the indices on $\in_{abcd}$ with the Minkowski metric η^{ab} with $\eta^{00} = -1$, so that $\in^{0123} = - \in_{0123}$.

3. There is an intimate relation between the preceding observations regarding the Levi-Civita symbol (remark 2) and those in section 4.4 above regarding invariant volume elements. Namely, the usual coordinate volume element d^4x can be written as

$$d^4x = \frac{1}{4!} \in_{\alpha\beta\gamma\delta} dx^\alpha dx^\beta dx^\gamma dx^\delta \quad . \tag{4.85}$$

This is not a tensor but transforms like a scalar density. On the other hand, if one works instead with the tensor $\epsilon_{\lambda\mu\nu\rho}$ one obtains a scalar, and this scalar is precisely the invariant volume element (4.65),

$$\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta} dx^\alpha dx^\beta dx^\gamma dx^\delta = \sqrt{g} d^4x \quad . \tag{4.86}$$

4. More generally (and without invoking a metric to provide the weight $w = +1$ density required for an invariant integration) this can be phrased in the following way: Let $A_{\alpha\beta\gamma\delta}$ be a totally anti-symmetric $(0,4)$-tensor. Thus it will be proportional to the metric dependent Levi-Civita tensor $\epsilon_{\alpha\beta\gamma\delta}$, but we will now not make use of this fact (which would return us to the setting of the previous remark). Rather, we consider its contraction with the contravariant Levi-Civita symbol $\in^{\alpha\beta\gamma\delta}$ (which exists independently of any additional structure like a metric),

$$A_{\alpha\beta\gamma\delta} \to \tilde{A} = \in^{\alpha\beta\gamma\delta} A_{\alpha\beta\gamma\delta} \quad . \tag{4.87}$$

This is now clearly a scalar density of weight $w = +1$. As a consequence, its integral (4.76) is well-defined and coordinate independent, without reference to any metric. Thus totally anti-symmetric $(0,4)$-forms provide natural 4-dimensional volume elements (and likewise for totally anti-symmetric $(0,p)$ tensors and p-dimensional volume elements).

4.6 Towards a Coordinate-Independent Interpretation of Tensors

There is a more invariant and coordinate-independent way of looking at tensors than we have developed so far. The purpose of this section (and the subsequent section 4.7) is to briefly explain this point of view, even though it is not indispensable for an understanding of the remainder of the course.

Consider first of all the differential

$$df = \partial_\alpha f(x) dx^\alpha \tag{4.88}$$

of a function (scalar field) $f = f(x)$. This is clearly a coordinate-independent object, because partial deriviatives and the differentials dx^α transform inversely to each other,

$$dy^\mu = J^\mu_\alpha dx^\alpha \quad , \quad \partial_\mu = J^\alpha_\mu \partial_\alpha \quad \Rightarrow \quad \partial_\alpha f(x) dx^\alpha = \partial_\mu \bar{f}(y) dy^\mu \quad . \tag{4.89}$$

This suggests that it is useful to regard the quantities $\partial_\alpha f$ as the coefficients of the coordinate independent object df in a particular coordinate system, namely when df is expanded in the basis $\{dx^\alpha\}$.

We can do the same thing for any covector A_α. If A_α is a covector (i.e. transforms like one under coordinate transformations), then $A := A_\alpha(x)dx^\alpha$ is coordinate-independent, and it is useful to think of the A_α as the coefficients of the covector A when expanded in a coordinate basis, $A = A_\alpha dx^\alpha$. Linear combinations of dx^α built in this way from covectors are known as *1-forms*.

From this point of view, we interpret the $\{A_\alpha\}$ simply as the (coordinate dependent) components of the (coordinate independent) 1-form A when expressed with respect to the (coordinate dependent) differentials $\{dx^\alpha\}$, considered as a basis of the space of covectors.

Something similar can be done for vector fields. Just as covectors transform inversely to coordinate differentials, vectors V^α transform inversely to partial derivatives ∂_α. Thus

$$V := V^\alpha(x)\frac{\partial}{\partial x^\alpha} \tag{4.90}$$

is coordinate-independent - a coordinate-independent linear first-order differential operator. One can thus always think of a vector field as a 1st order differential operator and this is a very fruitful point of view.

Acting on a function (scalar) f, V produces the derivative of f along V,

$$Vf = V^\alpha \partial_\alpha f \quad . \tag{4.91}$$

This is also a coordinate independent object, a scalar, arising from the contraction of a vector and a covector. And this is as it should be because, after all, both a function and a vector field can be specified on a space-time without having to introduce coordinates (e.g. by simply drawing the vector field and the profile of the function). Therefore also the change of the function along a vector field should be coordinate independent and, as we have seen, it is.

So far we have only discussed vectors and covectors. All this can, in principle, be extended to higher rank tensors, but at this point it would be very useful to introduce the notion (or at least the notation) of tensor products. I will briefly describe this in section 4.7 below.

For those who do not want to delve into this (and it is not required for the following): fact of the matter is that any (p, q)-tensor $T^{\alpha_1\ldots\alpha_p}_{\beta_1\ldots\beta_q}$ can be thought of as the collection of components of a coordinate independent object T when expanded in a particular coordinate basis in terms of the dx^α and $(\partial/\partial x^\alpha)$.

Any choice of coordinate system $\{x^\alpha\}$ gives rise to such a basis $\{dx^\alpha\}$, and such bases are known as *coordinate bases* or *natural bases*. This is not the only possible choice of basis, however, and we will return to this issue in section 4.8.

4.7 Multilinear Algebra and Tensors

In (multi-)linear algebra, the tensor product is used to describe multilinear maps. Let V be a vector space, and V^* its dual, consisting of the linear maps $V \to \mathbb{R}$, and denote the action of $a \in V^*$ on $v \in V$ by

$$a \in V^*, v \in V \quad \to \quad a(v) \in \mathbb{R} \ . \tag{4.92}$$

In components, with respect to a basis E_k in V and its dual basis e^k in V^*,

$$e^i(E_k) = \delta^i_k \ , \tag{4.93}$$

this would be written as

$$(a_i e^i)(v^k E_k) = a_i v^k e^i(E_k) = a_i v^k \delta^i_k = a_k v^k \ . \tag{4.94}$$

Then a bilinear map on the Cartesian product $V \times V$ can be considered as an element of $V^* \otimes V^*$, the tensor product being defined by

$$(a \otimes b)(v, w) = a(v)\ b(w) \ . \tag{4.95}$$

Note that this is not symmetric, i.e.

$$a \otimes b \neq b \otimes a \ . \tag{4.96}$$

Again in components this would read

$$(a \otimes b)(v, w) = a_i b_k v^i w^k \ , \tag{4.97}$$

and a general element of $V^* \otimes V^*$ can be written as an object with components a_{ik} (a covariant tensor),

$$a = a_{ik} e^i \otimes e^k \ , \tag{4.98}$$

acting as

$$a(v, w) = a_{ik} v^i w^k \ . \tag{4.99}$$

From these definitions it follows that the tensor product is evidently linear,

$$a \otimes (b + c) = a \otimes b + a \otimes c \tag{4.100}$$

(and likewise for the first factor), and $\mathbb{R}$-linear, i.e. for $r \in \mathbb{R}$ one has

$$r(a \otimes b) = (ra) \otimes b = a \otimes (rb) \ . \tag{4.101}$$

The $\mathbb{R}$-linearity is in a sense the characteristic feature of the tensor product $V \otimes W$ of two vector spaces (here $V^* \otimes V^*$) that sets it apart from the direct (Cartesian) product $V \times W$ of vector spaces (here $V^* \times V^*$), which consists of the pairs (v, w), and for which there is obviously no identification between (rv, w) and (v, rw) since these are just distinct points of the Cartesian product.

This can be straightforwardly extended to a description of general multilinear maps on vector spaces:

- Using the canonical isomorphism $(V^*)^* \cong V$ for finite-dimensional vector spaces,

$$v \in V \to \hat{v} \in (V^*)^* : \quad \hat{v}(a) = a(v) \ , \tag{4.102}$$

one can also in the same way define the tensor product $V \otimes V$ as the space of bilinear functions on $V^* \times V^*$,

$$(v \otimes w)(a, b) = a(v)\ b(w) \ , \tag{4.103}$$

and a general elelement of $V \otimes V$ can be represented in terms of its components T^{ik},

$$T = T^{ik} E_i \otimes E_k \ , \tag{4.104}$$

acting as

$$T(a, b) = T^{ik} a_i b_k \ . \tag{4.105}$$

- By the same token, the tensor product $V \otimes W$ is the space of bilinear maps on $V^* \otimes W^*$.

- Multilinear maps from $V \times \ldots \times V$ to $\mathbb{R}$ are elements of

$$\otimes^p V^* = \underbrace{V^* \otimes \ldots \otimes V^*}_{p \text{ times}} \tag{4.106}$$

(and can be represented as covariant tensors of rank p), etc.

- These multilinear maps can be added and multipled and thus form an algebra, the *tensor algebra* of V^*, denoted by $T(V^*)$. As a vector space, it consists of the sums of all the p-linear maps,

$$T(V^*) = \oplus_{p=0} \otimes^p V^* \ . \tag{4.107}$$

The tensor product can also be used to describe multilinear maps between vector spaces:

- An element $a \otimes v$ of $V^* \otimes V$ can be regarded as a linear map from V to itself via

$$(a \otimes v_1)(v_2) = a(v_2) v_1 \tag{4.108}$$

and a general element of $V^* \otimes V$ (a "matrix" $M^i_{\ k}$) can be written as a linear combination of such maps,

$$M = M^i_{\ k} E_i \otimes e^k : \quad M(v) = M^i_{\ k} e^k(v) E_i = (M^i_{\ k} v^k) E_i \ . \tag{4.109}$$

- Likewise a linear map from V to some other vector space W can be regarded as an element of $V^* \otimes W$.

- Multilinear maps from V to W ("W-valued multilinear maps") are elements of $V^* \otimes \ldots \otimes V^* \otimes W$, etc.

Clearly, in general, given a basis of V and a dual basis of V^*, the tensor product can be used to construct a basis

$$(E_{i_1} \otimes \ldots \otimes E_{i_p}) \otimes (e^{k_1} \otimes \ldots e^{k_q}) \tag{4.110}$$

in the space

$$T^{p,q} = \underbrace{(V \otimes \ldots \otimes V)}_{p \text{ times}} \otimes \underbrace{(V^* \otimes \ldots \otimes V^*)}_{q \text{ times}} \tag{4.111}$$

of (p,q)-tensors,

$$T \in T^{p,q} : T = T^{i_1 \ldots i_p}_{k_1 \ldots k_q} \ (E_{i_1} \otimes \ldots \otimes E_{i_p}) \otimes (e^{k_1} \otimes \ldots \otimes e^{k_q}) \ . \tag{4.112}$$

This is the way we will use the tensor product notation below, as a multilinear operation providing us with a basis for higher rank tensor fields.

Now, as we have seen above, in the standard component/index formulation of general relativity, say, a (p,q)-tensor is defined as an object with components $T^{\mu_1 \ldots \mu_p}_{\nu_1 \ldots \nu_q}$ which transforms multi-linearly with the Jacobi matrix under coordinate transformations, i.e. under $x^\mu \to y^\alpha$ one has

$$T^{\alpha_1 \ldots \alpha_p}_{\beta_1 \ldots \beta_q}(y) = J^{\alpha_1}_{\mu_1} \ldots J^{\alpha_p}_{\mu_p} \ J^{\nu_1}_{\beta_1} \ldots J^{\nu_q}_{\beta_q} \ T^{\mu_1 \ldots \mu_p}_{\nu_1 \ldots \nu_q}(x(y)) \tag{4.113}$$

where $J^\alpha_\mu = \partial y^\alpha / \partial x^\mu$ is the Jacobi matrix and $J^\mu_\alpha = \partial x^\mu / \partial y^\alpha$ is its inverse.

The reason for introducing and working with tensors, defined in this way, is that tensorial equations have the virtue that they are *generally covariant*, i.e. that they are satisfied in all coordinate system if and only if they are satisfied in one coordinate system. The emphasis in this formulation is thus not on tensors as multilinear maps but on how they transform under coordinate transformations. This seems to be somewhat at odds with the definition of tensors in multilinear algebra, but as we will see below this is simply due to the choice of a particular class of bases (coordinate bases), with respect to which multilinear maps indeed transform in this way under changes of the coordinate basis, i.e. under changes of coordinates.

We had already noted above, that there is a more coordinate independent way of looking at covector fields and vector fields, by associating to them the objects

$$A_\mu(x) \to A(x) = A_\mu(x) dx^\mu \quad , \quad V^\alpha(x) \to V(x) = V^\alpha(x) \partial_\alpha \ . \tag{4.114}$$

which are completely invariant under coordinate transformations, with the dx^μ and the ∂_α providing a basis for the space of covector and vector fields respectively.

This perspective can now be extended to higher-rank and mixed tensors. In particular, associated with the metric $g_{\mu\nu}(x)$ we have the coordinate independent line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \ . \tag{4.115}$$

which we can now also think of as the tensor

$$g = g_{\mu\nu} dx^\mu \otimes dx^\nu \quad . \tag{4.116}$$

where $\otimes$ is now again the *tensor product*.

Since we are now dealing with tensor fields rather than just with tensors (multilinear maps at a given point), the tensor product in this context is required to be multilinear not just over $\mathbb{R}$, but over functions (scalars) so that e.g.

$$dx^\mu \otimes (f(x) dx^\nu) = (f(x) dx^\mu) \otimes dx^\nu \quad . \tag{4.117}$$

Now let us return to (4.116). If one wants to emphasise that the metric is a symmetric (0,2)-tensor, one can also expand it with respect to the symmetrised basis as

$$g = g_{\mu\nu} \, (dx^\mu \otimes dx^\nu + dx^\nu \otimes dx^\mu)/2 \quad , \tag{4.118}$$

but for the metric the tensor-product is often omitted and one simply writes it as the line element (4.115).

If one has a non-symmetric $(0,2)$-tensor $T_{\mu\nu}$, say, then one can also group these coefficients into the components of a coordinate-invariant object, but now the tensor product notation

$$T_{\mu\nu} \to T = T_{\mu\nu} dx^\mu \otimes dx^\nu \tag{4.119}$$

is more useful than just writing $T_{\mu\nu} dx^\mu dx^\nu$, simply to emphasise the fact that all components of $T_{\mu\nu}$, not just the symmetric part of $T_{\mu\nu}$, contribute to T because $dx^\mu \otimes dx^\nu$ is not symmetric,

$$dx^\mu \otimes dx^\nu \neq dx^\nu \otimes dx^\mu \quad , \tag{4.120}$$

(whereas just writing $dx^\mu dx^\nu$ might lead one to believe that dx^μ and dx^ν commute).

More generally, to a $(0,p)$-tensor we can associate the object

$$T = T_{\mu_1 \dots \mu_p} dx^{\mu_1} \otimes \dots \otimes dx^{\mu_p} \quad . \tag{4.121}$$

If $T_{\mu_1 \cdots \mu_p}$ is totally anti-symmetric, the resulting object is also referred to as a *p-form*. As we have already seen above, such p-forms provide natural and invariant p-dimensional volume elements. In particular, applying this to the Levi-Civita tensor discussed in section 4.5, we reproduce the statement (4.65) that (4.86)

$$\frac{1}{4!} \epsilon_{\alpha\beta\gamma\delta} dx^\alpha dx^\beta dx^\gamma dx^\delta = \sqrt{g} d^4x \tag{4.122}$$

is a space-time volume element that is invariant under coordinate transformations.

The tensor product notation is also useful for higher-rank contravariant or mixed tensors. Given a $(2,0)$-tensor with components $T^{\mu\nu}$, say, one really does not want to write the corresponding coordinate-invariant object as $T^{\mu\nu} \partial_\mu \partial_\nu$, say, because this may be

interpreted as a second order differential operator whereas what one really means is a bilinear first order differential operator, which one writes as

$$T = T^{\mu\nu}\, \partial_\mu \otimes \partial_\nu \ , \tag{4.123}$$

and whose components with respect to the basis $\partial_\mu \otimes \partial_\nu$ are the $T^{\mu\nu}$.

In general, we can thus think of a (p, q)-tensor field, as given in (4.113), as the components of a coordinate-independent object

$$T = T^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q}(x)\ (\partial_{\mu_1} \otimes \dots \partial_{\mu_p}) \otimes (dx^{\nu_1} \otimes \dots \otimes dx^{\nu_q}) \ , \tag{4.124}$$

when expanded with respect to the coordinate basis in the space of tensor fields generated by dx^μ and $\partial_\mu = \partial_{x^\mu}$.

4.8 Vielbeins and Orthonormal Frames

As we saw in section 4.6, a choice of coordinates provides one with a choice of basis for vectors, covectors and other tensors, and a quantity like V^μ is then interpreted as the collection of components of an object $V = V^\mu \partial_\mu$ with respect to the coordinate basis ∂_μ. In classical tensor calculus one always works in such a basis, and with the components of tensors with respect to such a basis. This is very convenient and natural, but this is now clearly not the only choice.

Indeed, the above point of view suggests a reformulation and generalisation that is extremely natural and useful (but that I will nevertheless hardly ever make use of in these notes).

Namley, let $\{e^m_\mu(x)\}$ be such that it is an invertible matrix for every point x. Then another possible choice of basis for the space of covectors are the linear combinations

$$e^m := e^m_\mu dx^\mu \ . \tag{4.125}$$

A general such basis is called a *vielbein*, which is German for *multileg*, quite appropriate actually, as one should visualise this as a bunch of linearly independent (co-)vectors at every point of space-time.

In two, three, and four dimensions these are also known more specifically as *zweibeins, dreibeins* and *vierbeins* respectively. In four dimensions, the Greek word *tetrads* is also commonly used. The e^m are sometimes also referred to as *frame fields*, mostly in the context of *orthonormal frames* (see below).

In general, this new basis is *not* a coordinate basis, i.e. there does not exist a coordinate system $\{y^m\}$ such that $e^m = dy^m$. If such a coordinate system does exist, then one has

$$\begin{aligned} e^m = dy^m \quad &\Rightarrow \quad e^m_\mu = \frac{\partial y^m}{\partial x^\mu} \\ &\Rightarrow \quad \partial_\nu e^m_\mu = \partial_\mu e^m_\nu \ , \end{aligned} \tag{4.126}$$

and locally also the converse is true. In particular, if

$$\partial_\nu e^m_\mu - \partial_\mu e^m_\nu \neq 0 \quad \Rightarrow \quad e^m \quad \text{is not a coordinate basis} \ . \tag{4.127}$$

For many purposes, bases other than coordinate bases can also be extremely useful and natural, in particular the orthonormal bases we will introduce below.

The inverse relation to (4.125) is

$$dx^\mu = e^\mu_{\ m} e^m \ , \tag{4.128}$$

where $e^\mu_{\ m}(x)$ is (pointwise) the inverse matrix of $e^m_\mu(x)$,

$$e^m_\mu e^\mu_n = \delta^m_n \qquad e^\mu_m e^m_\nu = \delta^\mu_\nu \ . \tag{4.129}$$

With respect to this basis, one can expand a covector A as

$$A = A_\mu dx^\mu = A_\mu e^\mu_{\ m} e^m \equiv A_m e^m \ , \tag{4.130}$$

so that the components of A with respect to the new basis $\{e^m\}$ are

$$A_m = A_\mu e^\mu_{\ m} \ . \tag{4.131}$$

Likewise, the vielbeins allow us to pass from a natural (or coordinate) basis for vector fields, the $\{\partial_\mu\}$, to another basis

$$E_m = e^\mu_{\ m} \partial_\mu \ , \tag{4.132}$$

allowing us to write the coordinate independent vector field

$$V = V^\mu(x)\partial_\mu = V^m E_m \tag{4.133}$$

with

$$V^m = e^m_\mu V^\mu \ . \tag{4.134}$$

Note that, unlike the ∂_μ, the E_m do not commute in general, i.e.

$$[E_m, E_n] \neq 0 \ . \tag{4.135}$$

In fact, a 'dual' characterization of a coordinate basis is that the corresponding E_m do commute. This is clearly a necessary condition and, as above, locally it is also sufficient to ensure that there is a coordinate system y^m such that $E_m = \partial/\partial y^m$.

We can apply the same reasoning to any other tensor field, e.g. to the metric tensor itself. We can write the invariant line element as

$$ds^2 = g_{\mu\nu} dx^\mu dx^\mu = g_{\mu\nu} e^\mu_{\ m} e^\nu_{\ n} e^m e^n \equiv g_{mn} e^m e^n \ , \tag{4.136}$$

so that the components of the metric with respect to the new basis are

$$g_{mn} = g_{\mu\nu} e^\mu_{\ m} e^\nu_{\ n} \ . \tag{4.137}$$

Given a metric, there is a preferred class of bases $\{e^a\}$ which are such that the corresponding matrices $e^a_{\ \mu}(x)$ diagonalise (and normalise) the metric at every point x, i.e. which are such that $g_{ab} = \eta_{ab}$ or

$$g_{ab} = \eta_{ab} \quad \Leftrightarrow \quad g_{\mu\nu}(x) e^\mu_{\ a}(x) e^\nu_{\ b}(x) = \eta_{ab} \ . \tag{4.138}$$

Such a basis e^a, with respect to which the components of the metric are the Minkowski metric η_{ab}, is known as an *orthonormal basis* or *orthonormal frame*.

In the more mathematical literature, the e^a are also referred to as *soldering forms* because they identify (solder, glue) an abstract space of (co-)vectors at each point x, labelled by $a, b, \ldots$ with the concrete space of (co-)vectors tangent to the space-time at the point x, labelled e.g. by the indices $\mu, \nu, \ldots$.

For a general metric, a basis which achieves this cannot be a coordinate basis (because this would mean that the metric is equivalent to the Minkowski metric by a coordinate transformation). However, clearly there is no obstacle to finding a more general basis which will do this: for every point x we can find a matrix $e^a_{\ \mu}(x)$ which achieves (4.138) As the metric varies smoothly with x, we can also choose the matrices $e^a_{\ \mu}(x)$ to vary smoothly with x, and hence we can put them together to define the smooth matrix-valued function $e^a_{\ \mu}(x)$ for all x. [I am ignoring some global (topological) issues here. We will not need to worry about them here.]

The reason why I referred to a "class of bases" above is that, clearly, such an orthonormal basis is not unique. At every point x it is determined up to a Lorentz transformation

$$\begin{aligned} & e^a(x) \to \Lambda^a_{\ b}(x) e^b(x) \\ & \Lambda^a_{\ b}(x) \Lambda^c_{\ d}(x) \eta_{ac} = \eta_{bd} \ . \end{aligned} \tag{4.139}$$

Thus a given metric does not determine a unique orthonormal basis, but only an orthonormal basis up to Lorentz transformations

$$e^a(x) \to \Lambda^a_{\ b}(x) e^b(x) \ . \tag{4.140}$$

Conversely, however, an orthonormal basis uniquely determines a metric via

$$ds^2 = \eta_{ab} e^a(x) e^b(x) \ . \tag{4.141}$$

If one wants the components of the metric in a given coordinate system $\{x^\mu\}$, one expands the orthonormal basis e^a in terms of the natural basis dx^μ as above as

$$e^a(x) = e^a_{\ \mu}(x) dx^\mu \ , \tag{4.142}$$

to find, as above,

$$g_{\mu\nu}(x) = e^a_{\ \mu}(x) e^b_{\ \nu}(x) \eta_{ab} \ . \tag{4.143}$$

Thus instead of the metric one can choose orthonormal vielbeins as the basic variables of General Relativity. In that case one has to demand not only general covariance but

also invariance under local Lorentz transformations (acting on the orthonormal indices $a, b, \ldots$). [One could also allow for general vielbeins, in which case one would have to replace Lorentz transformations by the larger group of general linear transformations.]

EXAMPLES:

Here are a few examples to illustrate that orthonormal frames are not something mysterious but can usually be read off very easily from the metric in a coordinate basis.

1. The 2-Sphere Metric (2.16)

   The standard metric on a sphere of radius R is

   $$ds^2 = R^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{4.144}$$

   Now define

   $$e^1 = Rd\theta \quad , \quad e^2 = R\sin\theta d\phi \ , \tag{4.145}$$

   i.e.

   $$e^a = e^a{}_\alpha dx^\alpha \tag{4.146}$$

   with

   $$e^1_\theta = R \ , \ e^1_\phi = 0 \ , \ e^2_\theta = 0 \ , \ e^2_\phi = R\sin\theta \ . \tag{4.147}$$

   Then the metric can be written as

   $$ds^2 = e^1e^1 + e^2e^2 = \delta_{ab}e^ae^b \ , \tag{4.148}$$

   so the e^a are an orthonormal basis. They are obviously not a coordinate basis because (4.127)

   $$\partial_\theta e^2_\phi = R\cos\theta \neq \partial_\phi e^2_\theta = 0 \ . \tag{4.149}$$

   Likewise, we can introduce an orthonormal basis

   $$E_a = E^\alpha_a \partial_\alpha \tag{4.150}$$

   for vectors. A simple choice is

   $$E_1 = R^{-1}\partial_\theta \quad , \quad E_2 = (R\sin\theta)^{-1}\partial_\phi \ , \tag{4.151}$$

   which satisfies

   $$g_{\alpha\beta}E^\alpha_a E^\beta_b = \delta_{ab} \ . \tag{4.152}$$

   That this is not a coordinate basis is reflected in the fact that the commutator $[E_1, E_2] \neq 0$,

   $$[E_1, E_2] = R^{-2}(\partial_\theta(\sin\theta)^{-1})\partial_\phi = -R^{-1}\cot\theta \ E_2 \ . \tag{4.153}$$

2. The Schwarzschild Metric (2.38)

   The metric is

$$ds^2 = -(1-2m/r)dt^2 + (1-2m/r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{4.154}$$

   With

$$(e^0, e^1, e^2, e^3) = \left((1-\frac{2m}{r})^{1/2}dt, (1-\frac{2m}{r})^{-1/2}dr, rd\theta, r\sin\theta d\phi\right) \tag{4.155}$$

   the metric can be written as

$$ds^2 = \eta_{ab}e^a e^b \ , \tag{4.156}$$

   so the e^a are an orthonormal basis for the Schwarzschild metric.

3. The Kaluza-Klein Metric (section 44)

   Here is an example of a non-diagonal metric. The five-dimensional Kaluza-Klein metric is

$$d\widehat{s}^2_{KK} = g_{\mu\nu}dx^\mu dx^\nu + (dx^5 + A_\mu dx^\mu)^2 \ . \tag{4.157}$$

   Let $e^a_{\ \mu}$, $a = 0, 1, 2, 3$, be vielbeins (tetrads) for the four-dimensional space-time metric $g_{\mu\nu}$. Then an orthonormal frame $\widehat{e}^A$ for the Kaluza-Klein metric is

$$\widehat{e}^a = e^a_{\ \mu}dx^\mu \quad , \quad \widehat{e}^5 = dx^5 + A_\mu dx^\mu \ . \tag{4.158}$$

Remarks:

1. Consider the timelike trajectory $x^\mu = x^\nu(\tau)$ of an observer, parametrised by proper time τ. Then his 4-velocity $u^\mu = dx^\mu/d\tau$ satisfies (2.55)

$$g_{\mu\nu}u^\mu u^\nu = -1 \ . \tag{4.159}$$

   Recalling the defining relation for an orthonormal frame,

$$g_{\mu\nu}e^\mu_{\ a}e^\nu_{\ b} = \eta_{ab} \quad \Rightarrow \quad g_{\mu\nu}e^\mu_{\ a=0}e^\nu_{\ b=0} = \eta_{00} = -1 \ , \tag{4.160}$$

   we see that the 4-velocity u^μ can be interpreted as the timelike component $e^\mu_{\ a=0}$ of an orthonormal frame along the worldline,

$$u^\mu = e^\mu_{\ a=0} \ , \tag{4.161}$$

   with the spacelike components providing an orthonormal laboratory reference system. At this point there is still considerable freeedom in the choice of the spatial components of the orthonormal frame. This freedom can be significantly reduced (to rigid τ-independent rotations) by adopting a particular "parallel transport"

condition of these vectors along the worldline, such as the Fermi-Walker parallel transport to be discussed in section 5.10.

In any case, however the laboratory system is defined, the frame components

$$V^a = e^a_{\ \mu} V^\mu \tag{4.162}$$

of a vector now acquire the physical interpretation as the components of V as measured with respect to the observer's proper time and his laboratory frame. Note that this generalises the fact, already used in our second derivation of the gravitational redshift in section 3.5, that the frequency of a wave with wave vector k^μ as measured by an observer with 4-velocity u^μ is (3.117)

$$\omega = -u^\mu k_\mu = -e^\mu_{\ a=0} k_\mu = e^{a=0}_{\ \mu} k^\mu \equiv k^{a=0} \quad . \tag{4.163}$$

2. The $e^a_{\ \mu}$ can in some sense be regarded as the square-root of the metric. In particular denoting the determinant of the matrix $e^a_{\ \mu}$ by

$$e(x) := \det(e^a_{\ \mu}(x)) \quad , \tag{4.164}$$

(4.138) implies

$$g(x) := |\det(g_{\alpha\beta}(x))| = e(x)^2 \quad \Leftrightarrow \quad |e(x)| = \sqrt{g(x)} \quad . \tag{4.165}$$

3. Coordinate indices can, as usual, be raised and lowered with the space-time metric $g_{\mu\nu}$ and its inverse, and Minkowski (tangent space) indices with the Minkowski metric η_{ab} and its inverse.

Note that this is consistent with the notation for $e^a_{\ \mu}$ and its inverse $e^\mu_{\ a}$ because

$$e^\mu_{\ a} = g^{\mu\nu} \eta_{ab} e^b_{\ \nu} \quad . \tag{4.166}$$

One also has other fairly evident relations like

$$g^{\mu\nu} = \eta^{ab} e^\mu_{\ a} e^\nu_{\ b} \quad , \tag{4.167}$$

etc. The reason why I have called the basis of vector fields in a general frame E_m rather than e_m is that e^m and E_m are of course not related just by lowering or raising the indices of the metric, $E_m \neq g_{mn} e^n$. The former are linear combinations of the dx^μ, the latter linear combinations of the ∂_μ, so they are very different objects.

One could now go ahead and develop the entire machinery of tensor calculus (covariant derivatives, curvature, ...) that we are about to develop in the following sections in terms of vielbeins as the basic variables instead of the metric. This is rather straightforward. For example, given the expression for the Christoffel symbols in terms of the

metric, and for the metric in terms of the vielbeins, one can express the Christoffel symbols (and hence covariant derivatives and curvatures) in terms of vielbeins, but the resulting expressions are rather unenlightning and not of much use in practice.

The real power of the vielbein formalism emerges when one combines it with the formalism of *differential forms*. And in practice the most useful and efficient alternative to working in components in a coordinate basis is working with differential forms in an orthonormal basis.

I do most of my (curvature) calculations in the latter framework (and e.g. only then translate them into coordinate components for the purposes of inserting them into these notes), but this is (for the time being) not something I will develop further here.[10]

4.9 Epilogue: Indices? Indices!

Having reached this point, you may have the impression that the notation we have introduced for tensors, $T^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q}$ say, and which, as you might have noticed by looking ahead, we will continue to use in these notes, with its morass of indices, is somewhat cumbersome and unelegant. And perhaps you might prefer to at the very least see everything written in terms of the index-free coordinate-invariant objects like $V = V^\mu\partial_\mu$ or $A = A_\mu dx^\mu$ introduced in section 4.6.

I cannot disagree with the sentiment that using all these indices does not appear to be particularly elegant. Mathematicians abhor it. Physicists, however, are pragmatists by nature - they will use whatever turns out to be useful or efficient for what they want to achieve, regardless of whether or not it is considered or perceived to be beautiful or elegant according to some external criteria.

In particular, in the case at hand, the index-laden notation would not be that commonly used and widespread if it did not have some distinct advantages over other options. Indeed, this notation is an extremely useful and informative bookkeeping device that conveys a lot of information in a very compact way. In particular, as we have seen, the index notation allows one to reliably read off what kind of tensor one is dealing with, along the lines of "if it has p upper and q lower indices, it transform like, hence is, a (p, q)-tensor". Moreover, as we will see below, it provides one with a much more concise and informative way of describing and performing algebraic manipulations of tensors than some index-free notation is capable of.

Let me first make clear what the issue is and what it is not when one writes something like V^μ or $V^\mu(x)$, as this can be interpreted in (at least) 2 distinct ways:

[10] See e.g. W. Thirring, *Classical Mathematical Physics* for a presentation of general relativity entirely in the coordinate-independent formalism of differential forms, and N. Straumann, *General Relativity*, where differential forms are used whenever it is convenient or useful (and also occasionally when it is not ...).

1. On the one hand, V^μ may refer to the numerical values of the components of a specific vector V in a specific coordinate system.

2. On the other hand, the notation V^μ may be used to indicate that the object V transforms like a vector.

The first use of V^μ is completely uncontentious: if one wants to write down the components of some object with respect to some basis, one has to write down the components of that object with respect to that basis, there is no way around that.

It is mainly the second use and interpretation of the notation that is at stake, and it is also mainly in this sense that the index notation is used for tensor algebra and tensor calculus in general and in these notes in particular.

To a somewhat lesser extent the fact that the notation itself does not indicate whether one has in mind the first or the second interpretation is also an issue (even though this is usually clear from the context). It is actually not so much an issue (if desired this is something that can easily be remedied - I will come back to this at the end of this section) as possibly the source of a major misunderstanding between mathematicians and physicists - namely that a dislike of the index notation arises from the (false!) belief that it means that one is always writing down objects with respect to a particular basis. If this were the case, this would indeed be clumsy and silly, and quite contrary to the spirit of general covariance. However, as interpretation 2 indicates, this is absolutely not what is meant.

Returning to the use of indices as a way to indicate tensorial type and tensorial operations (like contractions), let us consider the alternatives. If one wants to indicate in symbols that some object V is a vector field, then as a mathematician one might write something like $V \in \Gamma(TM)$, stating that V is a section of the tangent bundle of the space or space-time (manifold) M. This is fine, but if the space M is clear from the context, why not declare once and for all that writing V^μ means the same thing? And perhaps use different kinds of indices to refer to tensors on different spaces?

If this were all then this would hardly be an issue and even physicists could be convinced to write "$V \in \Gamma(TM)$", at least when talking to mathematicians. Where the index notation really pays off, however, is when it comes to algebraic manipulations such as those discussed in section 4.3 (and even more so when it comes to tensor analysis, which is the subject of section 5, but tensor algebra will be enough to illustrate this).

As examples consider the contractions of a $(1,2)$ tensor T, say, with itself and with a vector V. With indices one would write $T^\mu{}_{\nu\lambda}$ and V^μ and the possible contractions would be written as

$$\begin{aligned} T^\mu{}_{\nu\lambda} \quad &\to \quad T^\mu{}_{\mu\lambda} \quad , \quad T^\mu{}_{\nu\mu} \\ (T^\mu{}_{\nu\lambda}, V^\mu) \quad &\to \quad T^\mu{}_{\nu\lambda} V^\nu \quad , \quad T^\mu{}_{\nu\lambda} V^\lambda \quad , \end{aligned} \tag{4.168}$$

the first line indicating the two distinct covectors one obtains as contractions of T itself, and the second the two distinct possibilities of contracting T and V to obtain a $(1,1)$-tensor. In an index-free notation one would have to invent some operation like C^m_n to indicate a contraction over the m'th upper and n'th lower index.[11] In this notation, the four objects above would then be written as

$$\begin{aligned} T \quad &\to \quad C^1_1(T) \quad , \quad C^1_2(T) \\ (T,V) \quad &\to \quad C^2_1(T\otimes V) \quad , \quad C^2_2(T\otimes V) \ . \end{aligned} \tag{4.169}$$

Is this superior? It does not even allow one to read off the tensor type of the resulting objects unless one remembers what the tensor types of T and V were to begin with, whereas this is completely manifest in (4.168).

Moreover, imagine how untransparent this would become were one to perform even the simplest sequence of such elementary operations: compare

$$A_{\alpha\beta}V^\alpha W^\beta \quad \longleftrightarrow \quad C^1_1C^2_2(A\otimes V\otimes W) \ . \tag{4.170}$$

If you prefer the right-hand side, or some variant of it, feel free to use it. However, you should be aware of the fact that the left-hand side contains an equivalent amount of information, simply packaged in a more digestible way that is both more informative ("it's a scalar!") and easier to manipulate. For most intents and purposes the index notation is really extremely convenient and it is for this reason that we will continue to make use of it in these notes.

One other reason for concern may be that by exclusively working with local coordinates and coordinate bases one may be missing some global aspects of a space or space-time. This is certainly true to a certain extent but is not primarily a notational issue. Rather, it means that in addition one needs to make use of more advanced notions from topology, global analysis etc. This is not something I will attempt here (cf. the book by Hawking and Ellis in the previous footnote for a description of the groundbreaking early applications of global analysis to general relativity). One related, but more elementary, issue is the introduction and use of the term *manifold* when referring to spaces or space-times of the kind we are dealing with in these notes. This is something I will very briefly come back to in section 5.11 below.

Let me, to conclude this ~~rant~~ section, come back to the issue of the notational ambiguity when one writes something like V^μ, which can occasionally be a source of confusion. Even though, as mentioned above, usually it is clear from the context what one means, one might imagine wanting to write down a couple of equations with indices which are only valid in spherical coordinates, say, and are therefore not to be understood as tensorial equations. Then it might be helpful to have a notation which reveals that information as well.

[11] I am not making this up - see e.g. section 2.2 of *The large scale structure of space-time* by S. Hawking and G. Ellis, in all other respects a wonderful book.

This can for instance be accomplished by inventing a new notation like $\stackrel{*}{=}$ (or whatever) to indicate an equality only in a special or specified coordinate system, but while this may add clarity it does not address the fundamental issue that just writing V^α does not unambiguously specify what one has in mind.

Alternatively, and more elegantly and attractively, this can e.g. be accomplished with very little effort with the help of what is known as the *Penrose abstract index notation*. The idea is to still indicate the tensor type of an object by a certain kind of indices, but with these indices only serving that purpose and not simultaneously referring to any particular kind of basis. Thus for example, one would indicate a vector by an object V^a, where the fact that one has a single upper index a just means that this is a $(1,0)$-tensor, and nothing else (exactly as in interpretation 2 above). For the components of this vector with respect to some basis (coordinates x^μ) one could then continue to use the traditional V^μ.

The advantage of this "abstract index" notation is that for tensorial operations one never needs to specify a basis anyway, so they can all be performed at the level of the abstract indices and tensorial equations look identical when written with these abstract indices or when written with concrete component indices. Thus $V^a W_a$ is used to indicate the scalar one obtains by contraction of a vector V^a with a covector W_a. Likewise, instead of $T^\mu_{\ \mu\lambda}$ (which may look basis dependent) one would write $T^a_{\ ab}$, and this is completely equivalent to writing something like $C^1_1(T)$,

$$T^a_{\ ab} \quad \longleftrightarrow \quad C^1_1(T) \tag{4.171}$$

but much more informative and user-friendly, and all the usual rules of tensor algebra apply to these abstract indices.

Whenever one wants or needs to specify a basis or coordinate system, this can be accomplished by using other kinds of indices. Thus g_{ab} could e.g. be used to refer to the metric tensor in general, while $g_{\mu\nu}$ could then be used to refer to its components in the basis x^μ. From this we see that

> "[...] the distinction between the index notation and the component notation is much more one of spirit (i.e., how one thinks of the quantities appearing) than of substance (i.e., the physical form the equations take)."[12]

While I will not make use of the abstract index notation in these notes (with the hope that this will not cause any confusion), the use of abstract indices appears to be an ideal ("eat the cake and have it too") compromise combining the best of both worlds

[12] R. Wald, *General Relativity*. See section 2.4 of this book for a more detailed explanation of the abstract index notation, which is systematically used throughout the book. For a detailed treatment of the abstract index notation and a discussion of some minor subtleties with this notation see R. Penrose, W. Rindler, *Spinors and Space-Time*, Vol. 1: *Two-Spinor Calculus and Relativistic Fields*.

and should actually keep both camps happy. It does not yet appear to have found widespread acceptance among mathematicians, however.

An alternative compromise solution is the already mentioned use of differential forms (in an orthonormal basis, say), which is manifestly covariant and minimises clutter, displaying only the (essential and informative) Lorentz Lie algebra indices while suppressing the component indices of forms (anti-symmetric tensors).

5 Tensor Analysis (Generally Covariant Differentiation)

Tensors transform in a nice and simple way under general coordinate transformations. Thus these appear to be the right objects to construct equations from that satisfy the Principle of General Covariance.

However, the laws of physics are differential equations, so we need to know how to differentiate tensors. This is not an issue of particular concern in Special Relativity, because (cf. (1.49)) the partial derivative of a Lorentz tensor

$$T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \quad \to \quad \partial_a T^{a_1\ldots a_p}_{c_1\ldots c_q}(\xi) \tag{5.1}$$

is again a Lorentz tensor. This relies on 2 facts, namely first that the partial derivative transforms as a covector under Lorentz transformations and secondly that the associated Jacobi matrix of Lorentz transformations is constant.

The former generalises to arbitrary coordinate transformations and implies, in particular, that the partial derivative of a scalar field is a covector field (4.18). However, because in general the Jacobi matrix is not constant, the ordinary partial derivative does not map tensors to tensors.

This is easy to see: take for example a vector V^μ. Under a coordinate transformation $x^\mu \to y^\alpha$, its partial derivative transforms as

$$\begin{aligned} \partial_\beta V^\alpha &= \frac{\partial x^\nu}{\partial y^\beta}\frac{\partial}{\partial x^\nu}\frac{\partial y^\alpha}{\partial x^\mu}V^\mu \\ &= \frac{\partial x^\nu}{\partial y^\beta}\frac{\partial y^\alpha}{\partial x^\mu}\partial_\nu V^\mu + \frac{\partial x^\nu}{\partial y^\beta}\frac{\partial^2 y^\alpha}{\partial x^\mu \partial x^\nu}V^\mu \quad . \end{aligned} \tag{5.2}$$

The appearance of the second term shows that the partial derivative of a vector is not a tensor.

As the second term is zero for linear transformations, you see that partial derivatives transform in a tensorial way e.g. under Lorentz transformations, so that partial derivatives are all one usually needs in special relativity.

We also see that the lack of covariance of the partial derivative is very similar to the lack of covariance of the equation $\ddot{x}^\mu = 0$, and this suggests that the problem can be cured in the same way - by introducing Christoffel symbols. This is indeed the case.

5.1 Covariant Derivative for Vector Fields

Let us define the *covariant derivative* $\nabla_\nu V^\mu$ of a vector field V^μ by

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\lambda} V^\lambda \quad . \tag{5.3}$$

It follows from the non-tensorial behaviour (2.78), (2.79) of the Christoffel symbols under coordinate transformations $x^\mu \to y^\alpha$ that $\nabla_\nu V^\mu$, as defined above, is indeed a $(1,1)$ tensor.

In order to establish this, we transform

$$\nabla_\beta V^\alpha = \partial_\beta V^\alpha + \Gamma^\alpha_{\beta\gamma} V^\gamma \tag{5.4}$$

to x-coordinates, using the tensorial transformation behaviour of ∂_α and V^α and the non-tensorial transformation behaviour (2.79) of the Christoffel symbols, to arrive at

$$\nabla_\beta V^\alpha = J^\mu_\beta \partial_\mu (J^\alpha_\rho V^\rho) + (J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \Gamma^\mu_{\ \nu\lambda} + J^\alpha_\mu \partial_\beta J^\mu_\gamma) J^\gamma_\rho V^\rho \ . \tag{5.5}$$

The obstructions to tensoriality are the 2 terms involving the derivatives of the Jacobi matrix, but these cooperatively combine to give

$$\begin{aligned} J^\mu_\beta \partial_\mu J^\alpha_\rho + J^\alpha_\mu (\partial_\beta J^\mu_\gamma) J^\gamma_\rho &= J^\mu_\beta \partial_\rho J^\alpha_\mu + J^\alpha_\mu (\partial_\gamma J^\mu_\beta) J^\gamma_\rho \\ &= J^\mu_\beta \partial_\rho J^\alpha_\mu - (\partial_\gamma J^\alpha_\mu) J^\mu_\beta J^\gamma_\rho \\ &= J^\mu_\beta \partial_\rho J^\alpha_\mu - (\partial_\rho J^\alpha_\mu) J^\mu_\beta = 0 \ . \end{aligned} \tag{5.6}$$

Here we have used the symmetry

$$\partial_\mu J^\alpha_\rho = \frac{\partial^2 y^\alpha}{\partial x^\mu \partial x^\rho} = \partial_\rho J^\alpha_\mu \tag{5.7}$$

and

$$J^\alpha_\mu J^\mu_\beta = \delta^\alpha_\beta \quad \Rightarrow \quad J^\alpha_\mu (\partial_\gamma J^\mu_\beta) = -(\partial_\gamma J^\alpha_\mu) J^\mu_\beta \ . \tag{5.8}$$

The remaining terms in (5.5) then just give rise to the tensorial transformation of a (1,1)-tensor. Thus we have shown that (5.3) indeed defines a tensor. Moreover, in a locally inertial coordinate system this reduces to the ordinary partial derivative, and we have thus, as desired, arrived at an appropriate tensorial generalisation of the partial derivative operator.

REMARKS:

1. Analysing the above argument for the tensoriality of the covariant derivative, we see that it relies exclusively on the specific non-tensorial form of the transformation behaviour of the Christoffel symbols, not on the explicit form of the Christoffel symbols themselves.

   Thus any other object $\tilde{\Gamma}^\mu_{\ \nu\lambda}$ could also be used to define a covariant derivative (generalising the partial derivative and mapping tensors to tensors) provided that it transforms in the same way as the Christoffel symbols, i.e. provided that one has

   $$\tilde{\Gamma}^\alpha_{\ \beta\gamma} = J^\alpha_\mu J^\nu_\beta J^\lambda_\gamma \tilde{\Gamma}^\mu_{\ \nu\lambda} + J^\alpha_\mu \partial_\beta J^\mu_\gamma \ . \tag{5.9}$$

This implies (and is equivalent to the fact) that the difference

$$C^{\mu}_{\ \nu\lambda} = \tilde{\Gamma}^{\mu}_{\ \nu\lambda} - \Gamma^{\mu}_{\ \nu\lambda} \tag{5.10}$$

transforms as a tensor. Thus, any such $\tilde{\Gamma}$ is of the form

$$\tilde{\Gamma}^{\mu}_{\ \nu\lambda} = \Gamma^{\mu}_{\ \nu\lambda} + C^{\mu}_{\ \nu\lambda} \tag{5.11}$$

where $C^{\mu}_{\ \nu\lambda}$ is a (1,2)-tensor, and could be used to define a corresponding covariant derivative $\tilde{\nabla}_{\mu}$.

Therefore the question arises if the covariant derivative defined in terms of the Christoffel symbols is somehow singled out or preferred. Indeed it is, and we will return to this question on various occasions below, in particular in section 5.4.

2. We could have arrived at the above definition of the covariant derivative (using the Christoffel symbols) in a somewhat more systematic way by appealing to the equivalence principle and/or general covariance. Namely, let $\{\xi^a\}$ be an inertial coordinate system. In an inertial coordinate system we can just use the ordinary partial derivative $\partial_b V^a$. We now define the new (improved, covariant) derivative $\nabla_\nu V^\mu$ in any other coordinate system $\{x^\mu\}$ by demanding that it transforms as a (1,1)-tensor, i.e. we define

$$\nabla_\nu V^\mu := \frac{\partial x^\mu}{\partial \xi^a}\frac{\partial \xi^b}{\partial x^\nu}\partial_b V^a \ . \tag{5.12}$$

By a straightforward calculation one finds that

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \gamma^{\mu}_{\ \nu\lambda} V^\lambda \ , \tag{5.13}$$

where $\gamma^{\mu}_{\ \nu\lambda}$ is our old friend (1.98)

$$\gamma^{\mu}_{\ \nu\lambda} = \frac{\partial x^\mu}{\partial \xi^a}\frac{\partial^2 \xi^a}{\partial x^\nu \partial x^\lambda} \ . \tag{5.14}$$

One would then also be led to adopt (5.13) with $\gamma \to \Gamma$ as a definition of the *covariant derivative* in a general metric space or space-time (with the Christoffel symbols calculated from the metric in the usual way).

That $\nabla_\mu V^\nu$, defined in this way, is indeed a $(1,1)$ tensor, now follows directly from the way we arrived at the definition of the covariant derivative. Indeed, imagine transforming from inertial coordinates to another coordinate system $\{y^\alpha\}$. Then (5.12) is replaced by

$$\nabla_\beta V^\alpha := \frac{\partial y^\alpha}{\partial \xi^a}\frac{\partial \xi^b}{\partial y^\beta}\partial_b V^a \ . \tag{5.15}$$

Comparing this with (5.12), we see that the two are related by

$$\nabla_\beta V^\alpha := \frac{\partial y^\alpha}{\partial x^\mu}\frac{\partial x^\nu}{\partial y^\beta}\nabla_\nu V^\mu \ , \tag{5.16}$$

as required.

3. Notation: frequently, the covariant derivative $\nabla_\nu V^\mu$ is also denoted by a semi-colon,

$$\nabla_\nu V^\mu = V^\mu{}_{;\nu} \quad . \tag{5.17}$$

Since covariant derivatives do not necessarily commute (as we will discuss in detail in section 8), when using this notation one has to pay attention to the order (and reversal of the order) of indices,

$$\nabla_\lambda \nabla_\nu V^\mu = V^\mu{}_{;\nu\,;\lambda} \quad . \tag{5.18}$$

One can also define the covariant directional derivative of a vector field V along another vector field X^μ by

$$\nabla_X V^\mu \equiv X^\nu \nabla_\nu V^\mu \quad . \tag{5.19}$$

4. The appearance of the Christoffel-term in the definition of the covariant derivative may at first sight appear a bit unusual (even though it also appears when one just transforms Cartesian partial derivatives to polar coordinates etc.). There is a more invariant way of explaining the appearance of this term, related to the more coordinate-independent way of looking at tensors explained in section 4.6. Namely, since the $V^\mu(x)$ are really just the coefficients of the vector field $V(x) = V^\mu(x)\partial_\mu$ when expanded in the basis ∂_μ, a meanigful definition of the derivative of a vector field must take into account not only the change in the coefficients but must also include a prescription how bases at (infinitesimally) neighbouring points are related (or *connected*). Such a prescription is provided by the *Levi-Civita connection* $\Gamma^\mu_{\nu\lambda}$ (or a general *connection* $\tilde{\Gamma}^\mu_{\nu\lambda}$).

Indeed, writing

$$\begin{aligned} \nabla_\nu V &= \nabla_\nu (V^\mu \partial_\mu) \\ &= (\partial_\nu V^\mu)\partial_\mu + V^\lambda (\nabla_\nu \partial_\lambda) \quad , \end{aligned} \tag{5.20}$$

we see that the covariant derivative of the coordinate basis vector ∂_λ (i.e. $V^\lambda = 1$, $V^\mu = 0$ otherwise), is the linear transformation (a prescription for a change of basis)

$$\nabla_\nu \partial_\lambda = \Gamma^\mu_{\nu\lambda} \partial_\mu \quad . \tag{5.21}$$

5.2 Extension of the Covariant Derivative to Other Tensor Fields

So far we have defined the covariant derivative for vector fields, and we now want to extend the definition of the covariant derivative to other tensor fields. In order to achieve this, we now adopt a more systematic and axiomatic approach.

Our basic postulates for the covariant derivative are the following:

1. Linearity and Tensoriality

   ∇_μ is a linear operator that maps (p,q)-tensors to $(p,q+1)$-tensors

2. Generalisation of the Partial Derivative

   On scalars ϕ, the covariant derivative ∇_μ reduces to the ordinary partial derivative (since $\partial_\mu\phi$ is already a covector),

$$\nabla_\mu\phi = \partial_\mu\phi \ . \tag{5.22}$$

3. Leibniz Rule (or Product Rule)

   Acting on the direct product of tensors, ∇_μ satisfies a generalised Leibniz rule,

$$\nabla_\mu(A^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q}B^{\rho_1\ldots\rho_r}_{\lambda_1\ldots\lambda_s}) = \nabla_\mu(A^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q})B^{\rho_1\ldots\rho_r}_{\lambda_1\ldots\lambda_s} + A^{\mu_1\ldots\mu_p}_{\nu_1\ldots\nu_q}\nabla_\mu B^{\rho_1\ldots\rho_r}_{\lambda_1\ldots\lambda_s} \tag{5.23}$$

We will now see that, demanding the above properties, in particular the Leibniz rule, there is a unique extension of the covariant derivative on vector fields to a differential operator on general tensor fields, mapping (p,q)- to $(p,q+1)$-tensors.

To define e.g. the covariant derivative for covectors U_μ, we note that $U_\mu V^\mu$ is a scalar for any vector V^μ so that

$$\nabla_\mu(U_\nu V^\nu) = \partial_\mu(U_\nu V^\nu) = (\partial_\mu U_\nu)V^\nu + U_\nu(\partial_\mu V^\nu) \tag{5.24}$$

(since the partial derivative satisfies the Leibniz rule), and we demand

$$\nabla_\mu(U_\nu V^\nu) = (\nabla_\mu U_\nu)V^\nu + U_\nu\nabla_\mu V^\nu \ . \tag{5.25}$$

As we know $\nabla_\mu V^\nu$, these two equations determine $\nabla_\mu U_\nu$ uniquely to be

$$\nabla_\mu U_\nu = \partial_\mu U_\nu - \Gamma^\lambda_{\mu\nu}U_\lambda \ . \tag{5.26}$$

That this is indeed a $(0,2)$-tensor can either be checked directly or, alternatively, is a consequence of the quotient theorem.

The extension to other (p,q)-tensors is now immediate. Here are two ways to proceed:

1. If the (p,q)-tensor is the direct product of p vectors and q covectors, then we already know its covariant derivative (using the Leibniz rule again). We simply adopt the same resulting formula for an arbitrary (p,q)-tensor.

2. Alternatively, contract the (p,q)-tensor with p covectors and q vectors to turn it into a scalar, and proceed as above for a covector.

Either way, the result is that the covariant derivative of a general (p,q)-tensor is the sum of the partial derivative, a Christoffel symbol with a positive sign for each of the p upper indices, and a Christoffel with a negative sign for each of the q lower indices. In equations

$$\begin{aligned}\nabla_\mu T^{\nu_1\cdots\nu_p}{}_{\rho_1\cdots\rho_q} &= \partial_\mu T^{\nu_1\cdots\nu_p}{}_{\rho_1\cdots\rho_q} \\ &+ \underbrace{\Gamma^{\nu_1}_{\mu\lambda} T^{\lambda\nu_2\cdots\nu_p}{}_{\rho_1\cdots\rho_q} + \ldots + \Gamma^{\nu_p}_{\mu\lambda} T^{\nu_1\cdots\nu_{p-1}\lambda}{}_{\rho_1\cdots\rho_q}}_{p \text{ terms}} \\ &- \underbrace{\Gamma^{\lambda}_{\mu\rho_1} T^{\nu_1\cdots\nu_p}{}_{\lambda\rho_2\cdots\rho_q} - \ldots - \Gamma^{\lambda}_{\mu\rho_q} T^{\nu_1\cdots\nu_p}{}_{\rho_1\cdots\rho_{q-1}\lambda}}_{q \text{ terms}}\end{aligned} \tag{5.27}$$

Having defined the covariant derivative for arbitrary tensors, we are also ready to define it for tensor densities. For this we recall that if T is a $(p,q;w)$ tensor density, then $g^{-w/2}T$ is a (p,q)-tensor. Thus $\nabla_\mu(g^{-w/2}T)$ is a $(p,q+1)$-tensor. To map this back to a tensor density of weight w, we multiply this by $g^{w/2}$, arriving at the definition

$$\nabla_\mu T := g^{+w/2}\nabla_\mu(g^{-w/2}T) \ . \tag{5.28}$$

Working this out explictly, one finds

$$\nabla_\mu T = -\frac{w}{2g}(\partial_\mu g)T + \nabla^{\text{tensor}}_\mu T \ , \tag{5.29}$$

where $\nabla^{\text{tensor}}_\mu$ just means the usual covariant derivative for (p,q)-tensors defined above. For example, for a scalar density ϕ one has

$$\nabla_\mu \phi = \partial_\mu \phi - \frac{w}{2g}(\partial_\mu g)\phi \ . \tag{5.30}$$

In particular, since the determinant g is a scalar density of weight $+2$, it follows that

$$\nabla_\mu g = 0 \ , \tag{5.31}$$

which obviously simplifies integrations by parts in integrals defined with the measure $\sqrt{g}d^4x$. However, it should be kept in mind that the crucial property that makes an integral like $\int \sqrt{g}\nabla_\alpha(\ldots)^\alpha$ a total derivative is *not* this fact but the fact that in this expression the $\sqrt{g}$ cancels and the integrand becomes an ordinary total derivative (cf. the discussion of the Gauss Theorem (5.61) below).

5.3 Main Properties of the Covariant Derivative

The main properties of the covariant derivative, in addition to those that were part of our postulates (like linearity and the Leibniz rule) are the following:

1. ∇_μ Commutes with Contraction

   This means that if A is a (p,q)-tensor and B is the $(p-1,q-1)$-tensor obtained by contraction over two particular indices, then the covariant derivative of B is the same as the covariant derivative of A followed by contraction over these two indices. This comes about because of a cancellation between the corresponding two Christoffel symbols with opposite signs. Consider e.g. a (1,1)-tensor $A^\nu{}_\rho$ and its contraction $A^\nu{}_\nu$. The latter is a scalar and hence its covariant derivative is just the partial derivative. This can also be obtained by taking first the covariant derivative of A,

$$\nabla_\mu A^\nu{}_\rho = \partial_\mu A^\nu{}_\rho + \Gamma^\nu{}_{\mu\lambda} A^\lambda{}_\rho - \Gamma^\lambda{}_{\mu\rho} A^\nu{}_\lambda \ , \tag{5.32}$$

   and then contracting:

$$\nabla_\mu A^\nu{}_\nu = \partial_\mu A^\nu{}_\nu + \Gamma^\nu{}_{\mu\lambda} A^\lambda{}_\nu - \Gamma^\lambda{}_{\mu\nu} A^\nu{}_\lambda = \partial_\mu A^\nu{}_\nu \ . \tag{5.33}$$

   The most transparent way of stating this property is that the Kronecker delta is covariantly constant, i.e. that

$$\nabla_\mu \delta^\nu{}_\lambda = 0 \ . \tag{5.34}$$

   To see this, we use the Leibniz rule to calculate

$$\begin{aligned} \nabla_\mu A^{\nu\ldots}{}_{\nu\ldots} &= \nabla_\mu (A^{\nu\ldots}{}_{\rho\ldots} \delta^\rho{}_\nu) \\ &= (\nabla_\mu A^{\nu\ldots}{}_{\rho\ldots}) \delta^\rho{}_\nu + A^{\nu\ldots}{}_{\rho\ldots} \nabla_\mu \delta^\rho{}_\nu \\ &= (\nabla_\mu A^{\nu\ldots}{}_{\rho\ldots}) \delta^\rho{}_\nu \end{aligned} \tag{5.35}$$

   which is precisely the statement that covariant differentiation and contraction commute. To establish that the Kronecker delta is covariantly constant, we follow the rules to find

$$\begin{aligned} \nabla_\mu \delta^\nu{}_\lambda &= \partial_\mu \delta^\nu{}_\lambda + \Gamma^\nu{}_{\mu\rho} \delta^\rho{}_\lambda - \Gamma^\rho{}_{\mu\lambda} \delta^\nu{}_\rho \\ &= \Gamma^\nu{}_{\mu\lambda} - \Gamma^\nu{}_{\mu\lambda} = 0 \ . \end{aligned} \tag{5.36}$$

   This property does not rely on the specific form of the $\Gamma^\mu{}_{\nu\lambda}$, and is thus true for any covariant derivative defined by some choice of connection $\tilde{\Gamma}^\mu{}_{\nu\lambda}$,

2. The Metric is Covariantly Constant: $\nabla_\mu g_{\nu\lambda} = 0$

   This is one of the key properties of the covariant derivative ∇_μ we have defined. I will give two arguments to establish this:

   (a) Since $\nabla_\mu g_{\nu\lambda}$ is a tensor, we can choose any coordinate system we like to establish if this tensor is zero or not at a given point x. Choose an inertial coordinate system at x. Then the partial derivatives of the metric and the Christoffel symbols are zero there. Therefore the covariant derivative of the metric is zero. Since $\nabla_\mu g_{\nu\lambda}$ is a tensor, this is then true in every coordinate system.

(b) The other argument is by direct calculation. Recalling the identity

$$\partial_\mu g_{\nu\lambda} = \Gamma_{\nu\lambda\mu} + \Gamma_{\lambda\nu\mu} \ , \tag{5.37}$$

we calculate

$$\begin{aligned} \nabla_\mu g_{\nu\lambda} &= \partial_\mu g_{\nu\lambda} - \Gamma^\rho{}_{\mu\nu} g_{\rho\lambda} - \Gamma^\rho{}_{\mu\lambda} g_{\nu\rho} \\ &= \Gamma_{\nu\lambda\mu} + \Gamma_{\lambda\nu\mu} - \Gamma_{\lambda\mu\nu} - \Gamma_{\nu\mu\lambda} \\ &= 0 \ . \end{aligned} \tag{5.38}$$

3. ∇_μ Commutes with Raising and Lowering of Indices

   This is really a direct consequence of the covariant constancy of the metric. For example, if V_μ is the covector obtained by lowering an index of the vector V^μ, $V_\mu = g_{\mu\nu} V^\nu$, then

   $$\nabla_\lambda V_\mu = \nabla_\lambda (g_{\mu\nu} V^\nu) = g_{\mu\nu} \nabla_\lambda V^\nu \ . \tag{5.39}$$

4. Covariant Derivatives Commute on Scalars

   This is of course a familiar property of the ordinary partial derivative, but it is also true for the second covariant derivatives of a scalar and is a consequence of the symmetry of the Christoffel symbols in the second and third indices and is also knowns as the *no torsion* property of the covariant derivative. Namely, we have

   $$\begin{aligned} \nabla_\mu \nabla_\nu \phi - \nabla_\nu \nabla_\mu \phi &= \nabla_\mu \partial_\nu \phi - \nabla_\nu \partial_\mu \phi \\ &= \partial_\mu \partial_\nu \phi - \Gamma^\lambda{}_{\mu\nu} \partial_\lambda \phi - \partial_\nu \partial_\mu \phi + \Gamma^\lambda{}_{\nu\mu} \partial_\lambda \phi = 0 \ . \end{aligned} \tag{5.40}$$

   Note that *the second covariant derivatives on higher rank tensors do not commute* - we will come back to this in our discussion of the curvature tensor later on.

5.4 Uniqueness of the Levi-Civita Connection (Christoffel symbols)

We noted before that the postulates for a covariant derivative (a linear tensorial operator reducing to the partial derivative on scalars and satisfying the Leibniz rule) do not determine it uniquely but only up to the addition of a tensor to the connection,

$$\Gamma^\mu{}_{\nu\lambda} \to \tilde{\Gamma}^\mu{}_{\nu\lambda} = \Gamma^\mu{}_{\nu\lambda} + C^\mu{}_{\nu\lambda} \ , \tag{5.41}$$

where $C^\mu{}_{\nu\lambda}$ is a (1,2)-tensor. Is there anything special or preferred about the Levi-Civita connection using the Christoffel symbols?

In some sense, the answer is an immediate yes because it is this particular covariant derivative (or connection) that enters in determining the paths of freely falling particles (the geodesics which extremise proper time).

Not unrelated to this is the fact that it is the unique connection that can be built from only the metric and its 1st derivatives (and which thus vanishes in an inertial coordinate system in Minkowski space or at the origin of an inertial coordinate system in an arbitrary gravitational field).

Moreover, as we have seen, this covariant derivative has two important properties, namely that

1. the metric is covariantly constant, $\nabla_\mu g_{\nu\lambda} = 0$, and
2. the torsion is zero, i.e. the second covariant derivatives of a scalar commute.

In fact, it turns out that these two conditions uniquely determine the $\tilde{\Gamma}$ to be the Christoffel symbols. The second condition implies that the $\tilde{\Gamma}^{\mu}_{\ \nu\lambda}$ are symmetric in the two lower indices,

$$[\tilde{\nabla}_\mu, \tilde{\nabla}_\nu]\phi = 0 \quad \Leftrightarrow \quad \tilde{\Gamma}^{\lambda}_{\ \mu\nu} = \tilde{\Gamma}^{\lambda}_{\ \nu\mu} \ . \tag{5.42}$$

The first condition now allows one to express the $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$ in terms of the derivatives of the metric, leading uniquely to the familiar expression for the Christoffel symbols $\Gamma^{\mu}_{\ \nu\lambda}$: First of all, by definition / construction one has (e.g. from demanding the Leibniz rule for $\tilde{\nabla}_\mu$)

$$\tilde{\nabla}_\mu g_{\nu\lambda} = \partial_\mu g_{\nu\lambda} - \tilde{\Gamma}^{\rho}_{\ \nu\mu} g_{\rho\lambda} - \tilde{\Gamma}^{\rho}_{\ \lambda\mu} g_{\nu\rho} \equiv \partial_\mu g_{\nu\lambda} - \tilde{\Gamma}_{\lambda\nu\mu} - \tilde{\Gamma}_{\nu\lambda\mu} \ . \tag{5.43}$$

Requiring that this be zero implies in particular that

$$\begin{aligned} 0 &= \tilde{\nabla}_\mu g_{\nu\lambda} + \tilde{\nabla}_\nu g_{\mu\lambda} - \tilde{\nabla}_\lambda g_{\mu\nu} \\ &= \partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu} - \tilde{\Gamma}_{\lambda\nu\mu} - \tilde{\Gamma}_{\nu\lambda\mu} - \tilde{\Gamma}_{\lambda\mu\nu} - \tilde{\Gamma}_{\mu\lambda\nu} + \tilde{\Gamma}_{\nu\mu\lambda} + \tilde{\Gamma}_{\mu\nu\lambda} \\ &= 2(\Gamma_{\lambda\mu\nu} - \tilde{\Gamma}_{\lambda\mu\nu}) \end{aligned} \tag{5.44}$$

(where the cancellations are entirely due to the assumed symmetry of the coefficients in the last two indices). Thus $\tilde{\Gamma} = \Gamma$. This unique metric-compatible and torsion-free connection is also known as the *Levi-Civita connection*. It is the connection canonically associated to a space-time (manifold) equipped with a metric tensor, and it is the connection used in general relativity.

It is possible to relax either of the conditions (1) or (2), or both of them and this will be discussed in section 11.5, and subsequently also in section 20.7.

5.5 Tensor Analysis: Some Special Cases

In this section we will look at some common and useful special cases of the Levi-Civita covariant derivative (simply "*the* covariant derivative" in the following), such as the covariant curl and divergence etc.

1. The Covariant Curl of a Covector

   One has

   $$\nabla_\mu U_\nu - \nabla_\nu U_\mu = \partial_\mu U_\nu - \partial_\nu U_\mu \ , \tag{5.45}$$

   because the symmetric Christoffel symbols drop out in this anti-symmetric linear combination. Thus in particular the Maxwell field strength

   $$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{5.46}$$

   is a tensor under general coordinate transformations, no metric or covariant derivative is needed to make it a tensor in a general space-time. The fact that the (ordinary) curl of a covector is, i.e. transforms like, a tensor under general coordinate transformations can of course also be checked directly (and then "explains" the above identity). We will come back to this below.

2. The Covariant Curl of an Antisymmetric Tensor

   Let $A_{\nu\lambda\ldots}$ be completely anti-symmetric. Then, as for the curl of covectors, the metric and Christoffel symbols drop out of the expression for the curl, i.e. one has

   $$\nabla_{[\mu} A_{\nu\lambda\cdots]} = \partial_{[\mu} A_{\nu\lambda\cdots]} \ . \tag{5.47}$$

   Here the square brackets on the indices denote complete anti-symmetrisation. In particular, the Bianchi identity for the Maxwell field strength tensor is independent of the metric also in a general metric space-time.

3. The Covariant Divergence of a Vector

   By the covariant divergence of a vector field one means the scalar

   $$\nabla_\mu V^\mu = \partial_\mu V^\mu + \Gamma^\mu_{\ \mu\lambda} V^\lambda \ . \tag{5.48}$$

   Now a useful identity for the contracted Christoffel symbol is

   $$\Gamma^\mu_{\ \mu\lambda} = \frac{1}{\sqrt{g}} \partial_\lambda \sqrt{g} \ . \tag{5.49}$$

   I will give a proof of this identity in an appendix to this section (subsection 5.6).

   Thus the covariant divergence can be written compactly as

   $$\nabla_\mu V^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} V^\mu) \ , \tag{5.50}$$

   and one only needs to calculate g and its derivative, not the Christoffel symbols themselves, to calculate the covariant divergence of a vector field.

   This formula is also useful (and provides the quickest way of arriving at the result) if one just wants to write the ordinary flat space divergence of vector calculus on $\mathbb{R}^3$ in, say, polar or cylindrical coordinates.

In Cartesian coordinates (x^1, x^2, x^3), the divergence of a 3-vector $\vec{V}$ is of course given by the familiar expression

$$\mathsf{div}\vec{V} = \partial_1 V^1 + \partial_2 V^2 + \partial_3 V^3 \quad . \tag{5.51}$$

However, as you also know, e.g. in spherical coordinates (r, θ, ϕ) the divergence is not simply of this form,

$$\mathsf{div}\vec{V} \neq \partial_r V^r + \partial_\theta V^\theta + \partial_\phi V^\phi \quad . \tag{5.52}$$

Rather, going through the coordinate transformation and Jacobians etc., one finds that calculating the divergence in spherical coordinates one picks up additional terms, the result taking the somewhat unintuitive form

$$\mathsf{div}\vec{V} = \partial_r V^r + \partial_\theta V^\theta + \partial_\phi V^\phi + \frac{2}{r} V^r + \cot\theta V^\theta \quad . \tag{5.53}$$

The easy and quick way to obtain this, which provides a rationale for and explanation of the origin of these additional terms, is from the result (5.50). Using $\sqrt{g} = r^2 \sin\theta$, one has

$$\begin{aligned} \mathsf{div}\vec{V} &= \frac{1}{r^2 \sin\theta} \left[\partial_r (r^2 \sin\theta V^r) + \partial_\theta (r^2 \sin\theta V^\theta) + \partial_\phi (r^2 \sin\theta V^\phi) \right] \\ &= \partial_r V^r + \partial_\theta V^\theta + \partial_\phi V^\phi + \frac{2}{r} V^r + \cot\theta V^\theta \quad . \end{aligned} \tag{5.54}$$

This thus produces the correct result on the nose and with very little effort.

4. The Covariant Laplacian of a Scalar

   How should the Laplacian be defined? Well, the obvious guess (something that is covariant and reduces to the ordinary Laplacian for the Minkowski metric) is $\Box = g^{\mu\nu}\nabla_\mu \nabla_\nu$, which can alternatively be written as

$$\Box = g^{\mu\nu}\nabla_\mu \nabla_\nu = \nabla^\mu \nabla_\mu = \nabla_\mu \nabla^\mu = \nabla_\mu g^{\mu\nu} \nabla_\nu \tag{5.55}$$

   etc. Note that, even though the covariant derivative on scalars reduces to the ordinary partial derivative, so that one can write

$$\Box\phi = \nabla_\mu g^{\mu\nu} \partial_\nu \phi \quad , \tag{5.56}$$

   it makes no sense to write this as $\nabla_\mu \partial^\mu \phi$: since ∂_μ does not commute with the metric in general, the notation ∂^μ is at best ambiguous as it is not clear whether this should represent $g^{\mu\nu}\partial_\nu$ or $\partial_\nu g^{\mu\nu}$ or something altogether different. This ambiguity does not arise for the Minkowski metric, but of course it is present in general.

A compact yet explicit expression for the Laplacian follows from the expression for the covariant divergence of a vector:

$$\begin{aligned} \Box\phi &:= g^{\mu\nu}\nabla_\mu\nabla_\nu\phi \\ &= \nabla_\mu(g^{\mu\nu}\partial_\nu\phi) \\ &= g^{-1/2}\partial_\mu(g^{1/2}g^{\mu\nu}\partial_\nu\phi) \ . \end{aligned} \tag{5.57}$$

Again, this formula is also useful (and provides the quickest way of arriving at the result) if one just wants to write the ordinary flat space Laplacian on $\mathbb{R}^3$ in, say, polar or cylindrical coordinates.

To illustrate this, let us calculate the Laplacian for the standard metric on $\mathbb{R}^{n+1}$ in polar coordinates. The standard procedure would be to first determine the coordinate transformation $x^i = x^i(r, \text{angles})$, then calculate $\partial/\partial x^i$, and finally assemble all the bits and pieces to calculate $\Delta = \sum_i (\partial/\partial x^i)^2$. This is a pain.

To calculate the Laplacian, we do not need to know the coordinate transformation, all we need is the metric. In polar coordinates, this metric takes the form

$$ds^2(\mathbb{R}^{n+1}) = dr^2 + r^2 d\Omega_n^2 \ , \tag{5.58}$$

where $d\Omega_n^2$ is the standard line-element on the unit n-sphere S^n. The determinant of this metric is $g \sim r^{2n}$ (times a function of the coordinates (angles) on the sphere). Thus, for $n = 1$ one has $ds^2 = dr^2 + r^2 d\phi^2$ and therefore

$$\Delta = r^{-1}\partial_\mu(r g^{\mu\nu}\partial_\nu) = r^{-1}\partial_r(r\partial_r) + r^{-2}\partial_\phi^2 = \partial_r^2 + r^{-1}\partial_r + r^{-2}\partial_\phi^2 \ . \tag{5.59}$$

In general, denoting the angular part of the Laplacian, i.e. the Laplacian of S^n, by Δ_{S^n}, one finds analogously

$$\Delta = \partial_r^2 + n r^{-1}\partial_r + r^{-2}\Delta_{S^n} \ . \tag{5.60}$$

I hope you agree that this method is superior to the standard procedure.

5. The Covariant Form of the Gauss Theorem

Let V^μ be a vector field, $\nabla_\mu V^\mu$ its divergence and recall that integrals in curved space are defined with respect to the integration measure $\sqrt{g}d^4x$. Thus one has

$$\int \sqrt{g}d^4x \nabla_\mu V^\mu = \int d^4x \partial_\mu(\sqrt{g}V^\mu) \ . \tag{5.61}$$

Now the integrand on the right-hand side is an ordinary total derivative and thus, by the usual fundamental theorem of calculus, the integral of this over some region R can be written as an integral over the boundary ∂R of that region,

$$\int_R \sqrt{g}d^4x \nabla_\mu V^\mu = \int_{\partial R} d^3x(\ldots) \ . \tag{5.62}$$

A somewhat more precise statement of this theorem, including the precise boundary contribution to the integral (involving the component of V^μ normal to the boundary), will be given in section 16.3.

In particular, if

- either the integral is over all of space-time and V^μ vanishes sufficiently rapidly at infinity,
- or the integral is over some region R with (finite) boundary ∂R, and the integrand vanishes on ∂R,

one has

$$\int \sqrt{g} d^4x \nabla_\mu V^\mu = 0 \ . \tag{5.63}$$

In these circumstances, one also has the integration by parts formula

$$\int \sqrt{g} d^4x \nabla_\mu (f V^\mu) = 0 \quad \Rightarrow \quad \int \sqrt{g} d^4x f \nabla_\mu V^\mu = -\int \sqrt{g} d^4x (\nabla_\mu f) V^\mu \tag{5.64}$$

for a scalar f and a vector field V^μ.

6. The Covariant Divergence of an Antisymmetric Tensor

For a $(p,0)$-tensor $T^{\mu\nu\cdots}$ one has

$$\begin{aligned} \nabla_\mu T^{\mu\nu\cdots} &= \partial_\mu T^{\mu\nu\cdots} + \Gamma^\mu_{\mu\lambda} T^{\lambda\nu\cdots} + \Gamma^\nu_{\mu\lambda} T^{\mu\lambda\cdots} + \dots \\ &= g^{-1/2} \partial_\mu (g^{1/2} T^{\mu\nu\cdots}) + \Gamma^\nu_{\mu\lambda} T^{\mu\lambda\cdots} + \dots \ . \end{aligned} \tag{5.65}$$

In particular, if $A^{\mu\nu\cdots}$ is completely anti-symmetric, the Christoffel terms disappear and one is left with

$$\nabla_\mu A^{\mu\nu\cdots} = g^{-1/2} \partial_\mu (g^{1/2} A^{\mu\nu\cdots}) \ . \tag{5.66}$$

7. The Lie derivative of the Metric

In section 3.2 we had encountered the expression (3.34) for the variation of the metric under an infinitesimal coordinate transformation $\delta x^\alpha = \epsilon V^\alpha$,

$$\delta_V g_{\alpha\beta} = V^\gamma \partial_\gamma g_{\alpha\beta} + (\partial_\alpha V^\gamma) g_{\gamma\beta} + (\partial_\beta V^\gamma) g_{\alpha\gamma} \ . \tag{5.67}$$

While we saw that this expression could be understood and deduced from the requirement that the variation of the metric is itself a tensorial object that transforms like the metric, the tensorial nature of the above expression is far from manifest. However, it has a very nice and simple expression in terms of covariant derivatives of V, namely

$$\delta_V g_{\alpha\beta} = \nabla_\alpha V_\beta + \nabla_\beta V_\alpha \tag{5.68}$$

(as is easily verified).

Thus a vector field K^α generates a symmetry of the metric (such vectors are called *Killing vectors*) if it satisfies the *Killing equation*

$$K \quad \text{Killing Vector} \quad \Leftrightarrow \quad \nabla_\alpha K_\beta + \nabla_\beta K_\alpha = 0 \ . \tag{5.69}$$

We can also obtain this condition as the covariantisation of the statement that in a particular coordinate system the coefficients of the metric do not depend on one of these coordinates, say y,

$$\partial_y g_{\alpha\beta} = 0 \ , \tag{5.70}$$

so that the metric is then manifestly invariant under translations in y. In such a coordinate system adapted to the symmetry at hand, these translations are generated by $K = \partial_y$, and for a vector of this form (in particular, thus, with constant coefficients) one has

$$\begin{aligned} K = \partial_y \quad &\Rightarrow \quad \nabla_\alpha K^\beta = \Gamma^\beta_{\alpha y} \\ &\Rightarrow \quad \nabla_\alpha K_\beta = \Gamma_{\beta\alpha y} \\ &\Rightarrow \quad \nabla_\alpha K_\beta + \nabla_\beta K_\alpha = \partial_y g_{\alpha\beta} \end{aligned} \tag{5.71}$$

(where in the last step the basic relation (2.68) was used). Thus we find that the fact that the metric is y-translation invariant can be characterised covariantly as the statement that $K = \partial_y$ satisfies

$$\partial_y g_{\alpha\beta} = 0 \quad \Leftrightarrow \quad \nabla_\alpha K_\beta + \nabla_\beta K_\alpha = 0 \ . \tag{5.72}$$

This is again the Killing equation (5.69). As this equation is now tensorial it is valid in any coordinate system, in particular independently of whether or not the coordinate system is adapted to K in the way described above.

The expressions (5.68) and (5.72) will be rederived (and placed into the general context of Lie derivatives and Killing vectors) in section 9 - see in particular section 9.5.

You will have noticed that many equations simplify considerably for completely anti-symmetric tensors. In particular, their curl can be defined in a tensorial way without reference to any metric. This observation is at the heart of the coordinate independent calculus of *differential forms*. In this context, the curl is known as the *exterior derivative*.

Indeed, it is also straightforward to show directly, i.e. without going through the illogical loop of introducing the covariant derivative in order to obtain something manifestly tensorial only to find it disappear again from the final expression, that $\partial_{[\mu} A_{\mu_1 \ldots \mu_p]}$ is a tensor, i.e. transforms as a tensor under coordinate transformations: what happens is that the possible obstructions to the tensorial behaviour, namely derivatives of Jacobians, drop out after anti-symmetrisations because they are are really 2nd partial

derivatives of the coordinates, which are symmetric and thus do not survive the anti-symmetrisation.

To see this completely explicitly, consider a covector $A_\mu(x)$ and a coordinate transformation $x^\mu = x^\mu(y^\alpha)$, with Jacobi matrix

$$J^\mu_\alpha = \frac{\partial x^\mu}{\partial y^\alpha} \quad . \tag{5.73}$$

As a covector, A_μ transforms as $A_\alpha = J^\mu_\alpha A_\mu$, and therefore its derivative transforms as (using $\partial_\beta = J^\nu_\beta \partial_\nu$)

$$A_\alpha = J^\mu_\alpha A_\mu \quad \Rightarrow \quad \partial_\beta A_\alpha = J^\mu_\alpha J^\nu_\beta \partial_\nu A_\mu + (\partial_\beta J^\mu_\alpha) A_\mu \quad . \tag{5.74}$$

Because of

$$\partial_\beta J^\mu_\alpha = \frac{\partial^2 x^\mu}{\partial y^\alpha \partial y^\beta} = \partial_\alpha J^\mu_\beta \quad , \tag{5.75}$$

for the anti-symmetrised derivative one finds the tensorial transformation behaviour

$$\partial_\beta A_\alpha - \partial_\alpha A_\beta = J^\mu_\alpha J^\nu_\beta (\partial_\nu A_\mu - \partial_\mu A_\nu) \quad . \tag{5.76}$$

Likewise, Lie derivatives of tensors in general (section 9) are, as the special case of the Lie derivative of the metric mentioned above - see (5.68), automatically tensorial objects (and one can, but need not, make their tensorial nature manifest by writing these derivatives in terms of covariant derivatives).

5.6 Appendix: A Formula for the Variation of the Determinant

Here is an elementary proof of the identity (5.49), and a useful more general formula for the variation of the determinant of the metric, namely

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu} \quad \text{or} \quad g^{-1} \delta g = g^{\mu\nu} \delta g_{\mu\nu} \quad . \tag{5.77}$$

This proof is based on the standard cofactor or minor expansion of the determinant of a matrix (an alternative standard proof can, as also outlined below, be based on the identity $\det G = \exp \operatorname{tr} \log G$ and its derivative or variation). The cofactor expansion formula for the determinant is

$$g = \sum_\nu (-1)^{\mu+\nu} g_{\mu\nu} |m_{\mu\nu}| \quad , \tag{5.78}$$

for a fixed (but arbitrary) value of the index μ. Here $|m_{\mu\nu}|$ is the determinant of the minor of $g_{\mu\nu}$, i.e. of the matrix one obtains by removing the μ'th row and ν'th column from $g_{\mu\nu}$.

As a consequence of (5.78) one also has

$$\sum_\nu (-1)^{\mu+\nu} g_{\lambda\nu} |m_{\mu\nu}| = 0 \qquad \lambda \neq \mu \quad , \tag{5.79}$$

since this is, in particular, the determinant of a matrix with $g_{\mu\nu} = g_{\lambda\nu}$, i.e. of a matrix with two equal rows. Together, these two results can be written as

$$\sum_\nu (-1)^{\mu+\nu} g_{\lambda\nu} |m_{\mu\nu}| = \delta_{\lambda\mu} g \quad . \tag{5.80}$$

This shows that the coefficients of the inverse metric $g^{\mu\nu}$ are given by

$$g^{\nu\mu} = (-1)^{\mu+\nu} \frac{|m_{\mu\nu}|}{g} \quad , \tag{5.81}$$

a formula that should also be familiar from linear algebra.

In order to now determine the variation (or derivative) of the determinant with respect to the matrix elements $g_{\mu\nu}$, for each value of μ we can use (5.78), and we then have to also perform the summation over μ. Noting that, by construction, $m_{\mu\nu}$ does not depend on $g_{\mu\nu}$, one then finds

$$\delta g = \sum_\mu \sum_\nu (-1)^{\mu+\nu} \delta g_{\mu\nu} |m_{\mu\nu}| = g g^{\nu\mu} \delta g_{\mu\nu} \quad . \tag{5.82}$$

For a symmetric matrix, in particular for the metric, this reduces to the formula (5.77) we set out to establish. Here are some variations and applications of this formula:

1. When the determinant g is viewed as a (smooth) function of the coefficients $g_{\mu\nu}$, this shows that
$$\frac{\partial g}{\partial g_{\mu\nu}} = g g^{\nu\mu} \quad . \tag{5.83}$$

2. It also implies
$$\delta\sqrt{g} = \tfrac{1}{2}\sqrt{g} g^{\nu\mu} \delta g_{\mu\nu} \quad , \tag{5.84}$$
a particularly useful result that we will repeatedly make use of.

3. An equally useful variant of this equation is an expression for the variation of $\sqrt{g}$ expressed in terms of the variations $\delta g^{\mu\nu}$ of the components of the inverse metric. As a consequence of
$$g^{\mu\nu} g_{\nu\lambda} = \delta^\mu_\lambda \quad \Rightarrow \quad \delta g^{\mu\nu} = -g^{\mu\lambda} \delta g_{\lambda\rho} g^{\rho\nu} \tag{5.85}$$
or
$$g^{\mu\nu} g_{\mu\nu} = 4 \quad \Rightarrow \quad (\delta g^{\mu\nu}) g_{\mu\nu} = -g^{\mu\nu} \delta g_{\mu\nu} \tag{5.86}$$
one can equivalently write (5.84) as
$$\delta\sqrt{g} = \tfrac{1}{2}\sqrt{g} g^{\nu\mu} \delta g_{\mu\nu} = -\tfrac{1}{2}\sqrt{g} g_{\nu\mu} \delta g^{\mu\nu} \quad . \tag{5.87}$$

4. It follows from (5.83) that if the variation is the partial derivative ("how does the determinant $g = g(x)$ of the metric vary with x?") one has

$$\partial_\lambda g = \frac{\partial g}{\partial g_{\mu\nu}} \partial_\lambda g_{\mu\nu} = g g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ . \tag{5.88}$$

or

$$g^{-1} \partial_\lambda g = g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ , \tag{5.89}$$

and therefore also

$$\partial_\lambda \sqrt{g} = \tfrac{1}{2} \sqrt{g} g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ . \tag{5.90}$$

5. On the other hand, the contracted Christoffel symbol is

$$\Gamma^\mu_{\ \mu\lambda} = \frac{1}{2} g^{\mu\nu} \partial_\lambda g_{\mu\nu} \ . \tag{5.91}$$

Therefore

$$\Gamma^\mu_{\ \mu\lambda} = \frac{1}{\sqrt{g}} \partial_\lambda \sqrt{g} \tag{5.92}$$

which establishes the identity (5.49).

The result (5.82) can also be written in matrix form, with G denoting the matrix with components $(G)_{\mu\nu} = g_{\mu\nu}$, as

$$\delta \log \det G = \operatorname{tr} G^{-1} \delta G \ . \tag{5.93}$$

In this form, the result can also be derived from variation of the remarkably useful identity

$$\det G = \mathrm{e}^{\operatorname{tr} \log G} \tag{5.94}$$

This identity, in turn, can be derived in an elementary way for diagonalisable G by noting that it holds trivially for diagonal matrices, and therefore, by the conjugation invariance of det and tr, also for diagonalisable matrices (like the metric). [And if desired, this can in turn be extended to all matrices by topological arguments involving extensions of continuous functionals from the dense set of diagonalisable matrices to the space of all matrices ...]

5.7 Covariant Differentiation Along a Curve

So far, we have defined covariant differentiation for tensors defined everywhere in space-time. Frequently, however, one encounters tensors that are only defined on curves - like the momentum of a particle which is only defined along its world line. In this section we will see how to define covariant differentiation along a curve. Thus consider a curve $x^\mu(\tau)$ (where τ could be, but need not be, proper time) and the tangent vector field $X^\mu(x(\tau)) = \dot{x}^\mu(\tau)$. Now define the covariant derivative D_τ along the curve, covariantising $d/d\tau$, by

$$\frac{d}{d\tau} = \dot{x}^\mu \partial_\mu \quad \to \quad D_\tau = X^\mu \nabla_\mu = \dot{x}^\mu \nabla_\mu \ . \tag{5.95}$$

Frequently one also uses the (suggestive, but ugly) notation

$$D_\tau = D/D\tau \quad \text{or} \quad D/d\tau \ . \tag{5.96}$$

For example, for a vector one has

$$\begin{aligned} D_\tau V^\mu &= \dot{x}^\nu \partial_\nu V^\mu + \dot{x}^\nu \Gamma^\mu_{\ \nu\lambda} V^\lambda \\ &= \frac{d}{d\tau} V^\mu(x(\tau)) + \Gamma^\mu_{\ \nu\lambda}(x(\tau)) \dot{x}^\nu(\tau) V^\lambda(x(\tau)) \ . \end{aligned} \tag{5.97}$$

For this to make sense, V^μ needs to be defined only along the curve and not necessarily everywhere in space-time.

This notion of covariant derivative along a curve permits us, in particular, to define the (covariant) acceleration a^μ of a curve $x^\mu(\tau)$ as the covariant derivative of the velocity $u^\mu = \dot{x}^\mu$ along the curve,

$$a^\mu = D_\tau \dot{x}^\mu = \ddot{x}^\mu + \Gamma^\mu_{\ \nu\lambda} \dot{x}^\nu \dot{x}^\lambda = u^\nu \nabla_\nu u^\mu \ . \tag{5.98}$$

Thus we can characterise (affinely parametrised) geodesics as those curves whose covariant acceleration is zero,

$$\text{Geodesics:} \quad a^\mu = u^\nu \nabla_\nu u^\mu = 0 \ , \tag{5.99}$$

a reasonable and natural statement regarding the movement of freely falling particles. If they are not affinely parametrised, as in (2.136), then instead of $u^\nu \nabla_\nu u^\mu = 0$ one has

$$u^\nu \nabla_\nu u^\mu = \kappa u^\mu \ . \tag{5.100}$$

5.8 Parallel Transport and Geodesics

We now come to the important notion of *parallel transport* of a tensor along a curve. Note that, in a general (curved) metric space-time, it does not make sense to ask if two vectors defined at points x and y are parallel to each other or not. However, given a metric and a curve connecting these two points, one can compare the two by dragging one along the curve to the other using the covariant derivative.

We say that a tensor $T^{\cdots}_{\ \ \cdots}$ is *parallel transported along the curve* $x^\mu(\tau)$ if

$$D_\tau T^{\cdots}_{\ \ \cdots} = 0 \ . \tag{5.101}$$

Here are some immediate consequences of this definition:

1. In a locally inertial coordinate system along the curve, this condition reduces to $dT/d\tau = 0$, i.e. to the statement that the tensor does not change along the curve. Thus the above is indeed an appropriate tensorial generalisation of the intuitive notion of parallel transport to a general metric space-time.

2. The parallel transport condition is a first order differential equation along the curve and thus defines $T^{\cdots}_{\cdots}(\tau)$ given an initial value $T^{\cdots}_{\cdots}(\tau_0)$.

3. Taking T to be the tangent vector $u^\mu = \dot{x}^\mu$ to the curve itself, the condition for parallel transport becomes

$$D_\tau u^\mu = 0 \quad \Leftrightarrow \quad \ddot{x}^\mu + \Gamma^\mu_{\nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{5.102}$$

   i.e. precisely the geodesic equation. We have already seen that geodesics are precisely the curves with zero acceleration. We can now equivalently characterise them by the property that their tangent vectors are parallel transported (do not change) along the curve. For this reason geodesics are also known as *autoparallels*.

4. Since the metric is covariantly constant, it is parallel along any curve. Thus, in particular, if V^μ is parallel transported, also its length remains constant along the curve,

$$D_\tau V^\mu = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\mu\nu}V^\mu V^\nu) = D_\tau(g_{\mu\nu}V^\mu V^\nu) = 0 \ . \tag{5.103}$$

   In particular, we rediscover the fact claimed in (2.100) that the quantity $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ is constant along a geodesic,

$$D_\tau \dot{x}^\mu = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu) = 0 \ . \tag{5.104}$$

5. Now let $x^\mu(\tau)$ be a geodesic and V^μ parallel along this geodesic. Then, as one might intuitively expect, also the angle between V^μ and the tangent vector to the curve u^μ remains constant. This is a consequence of the fact that both the norm of V and the norm of u are constant along the curve and that

$$\frac{d}{d\tau}(g_{\mu\nu}u^\mu V^\nu) = D_\tau(g_{\mu\nu}u^\mu V^\nu) = g_{\mu\nu}(D_\tau u^\mu)V^\nu + g_{\mu\nu}u^\mu D_\tau V^\nu = 0 \tag{5.105}$$

5.9 Example: Parallel Transport on the 2-Sphere

As usual, the simplest non-trivial example is provided by the 2-sphere with its standard line element

$$ds^2 = d\Omega^2 = d\theta^2 + \sin^2\theta \, d\phi^2 \ , \tag{5.106}$$

with the non-zero Christoffel symbols (determined e.g. from the geodesic equation, as in (3.16) - (3.21))

$$\Gamma^\theta_{\phi\phi} = -\sin\theta \, \cos\theta \quad , \quad \Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta \ . \tag{5.107}$$

Let us consider parallel transport of some vector $V = V^\alpha\partial_\alpha$ along a circle with $\theta = \theta_0$ constant, choosing the angle ϕ to parametrise the curve, i.e. we consider the family of paths

$$x^\alpha(\tau) = (\theta(\tau), \phi(\tau)) = (\theta_0, \tau) \tag{5.108}$$

with tangent vector

$$\dot{x}^\alpha = (0,1) \ . \tag{5.109}$$

Note that this is not normalised in the standard way,

$$g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = \sin^2\theta_0 \ , \tag{5.110}$$

so proper distance would be measured not by ϕ but, as is also pictorially evident, by $s = \phi\sin\theta_0$ (which agrees with ϕ on the equator $\theta_0 = \pi/2$).

A vector $V^\alpha\partial_\alpha$ with coordinate components (V^θ, V^ϕ) parallel transported along such a curve thus satisfies the equations

$$\begin{aligned} 0 &= \partial_\phi V^\alpha + \Gamma^\alpha_{\beta\gamma}\dot{x}^\beta V^\gamma = \partial_\phi V^\alpha + \Gamma^\alpha_{\phi\gamma}V^\gamma \\ &= \partial_\phi V^\alpha + \Gamma^\alpha_{\phi\phi}V^\phi + \Gamma^\alpha_{\phi\theta}V^\theta \ . \end{aligned} \tag{5.111}$$

Using the explicit form of the Christoffel symbols, the parallel transport equations are thus

$$\begin{aligned} 0 &= \partial_\phi V^\theta - \sin\theta\ \cos\theta\ V^\phi \\ 0 &= \partial_\phi V^\phi + \cot\theta\ V^\theta \ . \end{aligned} \tag{5.112}$$

Differentiating once more, these equations can be decoupled and take the form of harmonic oscillator equations with frequency $\cos\theta_0$,

$$(\partial_\phi^2 + \cos^2\theta_0)V^\alpha = 0 \ . \tag{5.113}$$

The general solution of this 2nd order differential equation is of course

$$V^\alpha = A^\alpha\sin(\phi\cos\theta_0) + B^\alpha\cos(\phi\cos\theta_0) \ . \tag{5.114}$$

Plugging this into the 1st order equations to reduce the spurious 4 to 2 integration constants, and relating them to the intial values at $\phi = 0$, say,

$$V^\alpha(\theta_0, \phi = 0) = v^\alpha \ , \tag{5.115}$$

one finally finds the result

$$\begin{aligned} V^\theta(\theta_0,\phi) &= v^\phi\sin\theta_0\ \sin(\phi\cos\theta_0) + v^\theta\ \cos(\phi\cos\theta_0) \\ V^\phi(\theta_0,\phi) &= -(v^\theta/\sin\theta_0)\ \sin(\phi\cos\theta_0) + v^\phi\ \cos(\phi\cos\theta_0) \ . \end{aligned} \tag{5.116}$$

Remarks:

1. In the special case of parallel transport along the equator $\theta_0 = \pi/2$, one has $\cos\theta_0 = 0$, and therefore

$$\theta_0 = \pi/2 \quad \Rightarrow \quad V^\alpha(\pi/2,\phi) = v^\alpha \ . \tag{5.117}$$

In other words, the components are constant under parallel transport along the equator. This is inuitively obvious on the basis of spherical symmetry. Since among the family of constant $\theta = \theta_0$ curves only the equator is a geodesic (great circle), this is also in agreement with the general results obtained above, which imply that upon parallel transport along the equator the angle between the vector and the equator remains constant. In 2 dimensions, this condition, together with the fact that the lenght of a vector remains invariant under parallel transport in general, is sufficient to imply that the parallel transported components are constant along the path.

2. While the above is not unexpected, perhaps the most interesting consequence of the above result (5.116) is that, in general, not only are the components not constant but that actually, after having completed the 2π-circuit along the path to return to the starting point, the parallel transported vector will not agree with the initial vector. Indeed, the components at $\phi = 2\pi$ are related to the components v^α at $\phi = 0$ by

$$\begin{aligned} V^\theta(\theta_0, \phi = 2\pi) &= v^\phi \sin\theta_0 \ \sin(2\pi \ \cos\theta_0) + v^\theta \ \cos(2\pi \ \cos\theta_0) \\ V^\phi(\theta_0, \phi = 2\pi) &= -(v^\theta / \sin\theta_0) \ \sin(2\pi \ \cos\theta_0) + v^\phi \ \cos(2\pi \ \cos\theta_0) \ . \end{aligned} \tag{5.118}$$

3. As we will see in section 11.1, this fact that parallel transport along closed paths is non-trivial (equivalently that parallel transport from one point to another depends on the path) can be directly attributed to (and is the smoking gun of) the presence of *curvature*.

4. If desired, the result can be written in terms of proper distance s along the circle, rather than the angle ϕ, by the substitution

$$\phi \ \cos\theta_0 = s \ \cot\theta_0 \ . \tag{5.119}$$

5. The result (5.116) takes on a more transparent form when written in terms of the components of V and v with respect to an orthonormal basis (section 4.8) E_α rather than the coordinate basis ∂_α. Such an orthonormal basis is provided by

$$E_\theta = \partial_\theta \quad , \quad E_\phi = (\sin\theta)^{-1}\partial_\phi \ , \tag{5.120}$$

since one evidently has

$$g_{\alpha\beta} E^\alpha_\theta E^\beta_\theta = g_{\alpha\beta} E^\alpha_\phi E^\beta_\phi = 1 \quad , \quad g_{\alpha\beta} E^\alpha_\theta E^\beta_\phi = 0 \ . \tag{5.121}$$

The components with respect to this orthonormal basis are related to the coordinate components by

$$V = V^\alpha \partial_\alpha = \hat{V}^\alpha E_\alpha \quad \Rightarrow \quad \hat{V}^\theta = V^\theta \quad , \quad \hat{V}^\phi = \sin\theta \, V^\phi \tag{5.122}$$

(and likewise for $v = v^\alpha \partial_\alpha = \hat{v}^\alpha E_\alpha$). Then (5.116) can be written in matrix form as a rotation (orthogonal transformation)

$$\begin{pmatrix} \hat{V}^\theta(\theta_0, \phi) \\ \hat{V}^\phi(\theta_0, \phi) \end{pmatrix} = \begin{pmatrix} \cos(\phi \cos\theta_0) & \sin(\phi \cos\theta_0) \\ -\sin(\phi \cos\theta_0) & \cos(\phi \cos\theta_0) \end{pmatrix} \begin{pmatrix} \hat{v}^\theta \\ \hat{v}^\phi \end{pmatrix} \tag{5.123}$$

by the angle

$$\alpha(\phi) = \phi \cos\theta_0 \quad . \tag{5.124}$$

Thus parallel transport amounts to a continuous rotation of the orthonormal components along the path.

6. In particular, the angle that one picks up after a 2π-rotation,

$$\alpha(2\pi) = 2\pi \cos\theta_0 \quad , \tag{5.125}$$

is known as the *deficit angle* or *holonomy* of the parallel transport along the given loop. With this terminology we can say that the holonomy along the equator is trivial.

7. At the other extreme, we see that there is a non-trivial holonomy as $\theta_0 \to 0$, i.e. for parallel transport along an infinitesimal loop around the north pole, along which the parallel transported vector performs a complete 2π-rotation, $\alpha(2\pi) = 2\pi$. As shown in section 11.1, parallel transport along infinitesimal loops at or around a point provides a precise measure of the curvature at that point.

8. Curiously, as shown by Rothman, Ellis and Murugan, the holonomy along circular equatorial orbits in the Schwarzschild geometry (such orbits are geodesics at the critical points of the effective potential for geodesic motion, to be discussed in section 25.6), is non-trivial, even though again intuitive reasoning based on spherical symmetry might have led one to expect a trivial result (and would thus have led one astray).[13]

5.10 Fermi-Walker Parallel Transport

The properties of parallel transport established in section 5.8 show that this is a natural prescription for transporting tensorial objects along a geodesic. However, it is important to keep in mind that this is just one possible description, obtained by imposing the differential equation (5.101), e.g. for a vector

$$D_\tau V^\alpha = 0 \quad . \tag{5.126}$$

[13] T. Rothman, G. Ellis, J. Murugan, *Holonomy in the Schwarzschild-Droste Geometry*, `arXiv:gr-qc/0008070`.

If the curve is not a geodesic,

$$a^\alpha = D_\tau u^\alpha = \dot{x}^\beta \nabla_\beta \dot{x}^\alpha \neq 0 \ , \tag{5.127}$$

however, this prescription has some shortcomings. For example, parallel transport of a tangent vector to the curve at a point to another point at the curve will not give rise to the tangent vector at the second point, simply because $D_\tau V^\alpha = 0$ with initial condition $V^\alpha(\tau_0) = u^\alpha(\tau_0)$, say (parallel transport) is evidently not the same as $D_\tau u^\alpha = a^\alpha$ (the equation satisfied by the tangent vector). Likewise, the scalar product between the tangent vector to the (non-geodesic) curve and some parallel-transported vector along it will not remain constant in general,

$$D_\tau u^\alpha = a^\alpha \quad , \quad D_\tau V^\alpha = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\alpha\beta} u^\alpha V^\beta) = a_\alpha V^\alpha \ . \tag{5.128}$$

A vivid illustration of this is provided by the example of the previous section:

- As we have seen, parallel transporting a pair of orthonormal vectors along a circle $\theta = \theta_0 \neq \pi/2$ results in a continuous rotation of these two basis vectors.
- On the other hand, it is clearly possible to transport an orthonormal basis along the circle in such a way that, for example, one basis vector always points forwards along the latitude (a tangent vector to the curve), and the other always points northwards along the longitude.

The latter procedure appears to be much more natural in this case than rotating one's basis as one goes around the sphere. Analogously, for an observer along a timelike curve it would be desirable to be able to set up once and for all a local reference system on the worldline, consisting of the (unit) tangent vector $E_0 = u^\mu \partial_\mu$ in the time-direction, and three orthogonal and mutually orthogonal vectors E_k in the spatial directions (the laboratory system of the observer), regardless of whether the observer is in free fall or not (indeed, for perfectly good reasons most laboratories are not ...).

This procedure can be formalised by replacing the parallel transport condition (5.126) along a timelike curve by the *Fermi-Walker Transport* prescription

$$F_\tau V^\alpha \equiv D_\tau V^\alpha + \mathcal{F}^\alpha_{\ \beta} V^\beta = 0 \ , \tag{5.129}$$

with

$$\mathcal{F}^\alpha_{\ \beta} = a^\alpha u_\beta - u^\alpha a_\beta \ . \tag{5.130}$$

Indeed, parallel transport according to this prescription has the following desirable features:

1. Fermi-Walker transport evidently reduces to parallel transport if the curve is a geodesic, i.e. for $a^\alpha = 0$. It obligingly does this even when the geodesic is not affinely parametrised, i.e. if one has $a^\alpha \sim u^\alpha$,

$$a^\alpha = u^\beta \nabla_\beta u^\alpha \sim u^\alpha \quad \Rightarrow \quad \mathcal{F}^\alpha_{\ \beta} = 0 \ . \tag{5.131}$$

2. The tangent vector to a curve is Fermi-Walker transported for any curve,

$$F_\tau u^\alpha = 0 \ . \tag{5.132}$$

Proof:

$$\begin{aligned} F_\tau u^\alpha &= D_\tau u^\alpha + \mathcal{F}^\alpha_{\ \beta} u^\beta \\ &= a^\alpha + (a^\alpha u_\beta - u^\alpha a_\beta) u^\beta = a^\alpha - a^\alpha = 0 \end{aligned} \tag{5.133}$$

because $u_\beta u^\beta = -1$ and $a_\beta u^\beta = 0$. Thus the solution to the Fermi-Walker transport prescription for $V^\alpha(\tau_0) = u^\alpha(\tau_0)$ is just the tangent vector u^α,

$$F_\tau V^\alpha = 0 \quad , \quad V^\alpha(\tau_0) = u^\alpha(\tau_0) \quad \Rightarrow \quad V^\alpha(\tau) = u^\alpha(\tau) \ . \tag{5.134}$$

3. If V^α is Fermi-Walker transported along the curve, then instead of (5.128) one obtains

$$D_\tau u^\alpha = a^\alpha \quad , \quad D_\tau V^\alpha = -\mathcal{F}^\alpha_{\ \beta} V^\beta \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\alpha\beta} u^\alpha V^\beta) = 0 \ . \tag{5.135}$$

Proof:

$$\begin{aligned} \frac{d}{d\tau}(g_{\alpha\beta} u^\alpha V^\beta) = D_\tau(g_{\alpha\beta} u^\alpha V^\beta) &= g_{\alpha\beta}(a^\alpha V^\beta - u^\alpha \mathcal{F}^\beta_{\ \gamma} V^\gamma) \\ &= a_\alpha V^\alpha - a_\alpha u^\alpha u_\gamma V^\gamma + u_\alpha u^\alpha a_\gamma V^\gamma = a_\alpha V^\alpha - a_\gamma V^\gamma = 0 \ , \end{aligned} \tag{5.136}$$

beause $a_\alpha u^\alpha = 0$ and $u_\alpha u^\alpha = -1$.

4. Similarly, if V and W are Fermi-Wallker transported, one has

$$F_\tau V^\alpha = F_\tau W^\alpha = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(g_{\alpha\beta} V^\alpha W^\beta) = 0 \ . \tag{5.137}$$

Proof:

$$\begin{aligned} \frac{d}{d\tau}(g_{\alpha\beta} V^\alpha W^\beta) &= g_{\alpha\beta}(-\mathcal{F}^\alpha_{\ \gamma} V^\gamma W^\beta - \mathcal{F}^\beta\ _\gamma V^\alpha W^\gamma) \\ &= -(\mathcal{F}_{\alpha\gamma} + \mathcal{F}_{\gamma\alpha}) V^\alpha W^\gamma = 0 \end{aligned} \tag{5.138}$$

because $\mathcal{F}_{\alpha\gamma}$ is anti-symmetric.

Remarks:

1. The signs chosen here are appropriate for timelike curves with $u^\alpha u_\alpha = -1$. As the proofs of the above statements show, in the spacelike case one needs to replace $\mathcal{F}^\alpha_{\ \beta} \to -\mathcal{F}^\alpha_{\ \beta}$.

2. The above manipulations can be formalised (and then subsequently trivialised) by

   - extending the action of F_τ to arbitrary rank tensors in the same way as the covariant derivative, i.e. by requiring that on scalars it reduces to the ordinary derivative,

$$F_\tau f = \frac{d}{d\tau} f \ , \tag{5.139}$$

- extending it to arbitrary tensors by requiring the Leibniz rule, so that e.g. on covectors one has

$$F_\tau A_\beta = D_\tau A_\beta - \mathcal{F}^\alpha_{\ \beta} A_\alpha \quad , \tag{5.140}$$

- and then showing that as a consequence

$$\begin{aligned} F_\tau g_{\alpha\beta} &= D_\tau g_{\alpha\beta} - \mathcal{F}^\gamma_{\ \alpha} g_{\gamma\beta} - \mathcal{F}^\gamma_{\ \beta} g_{\alpha\gamma} \\ &= -(\mathcal{F}_{\alpha\beta} + \mathcal{F}_{\beta\alpha}) = 0 \quad . \end{aligned} \tag{5.141}$$

Then assertions like (5.135),

$$F_\tau u^\alpha = F_\tau V^\alpha = 0 \quad \Rightarrow \quad \frac{d}{d\tau}(u_\alpha V^\alpha) = 0 \quad . \tag{5.142}$$

or (5.137) become a triviality.

3. Note that the properties 2-4 in the above list rely on the 3 properties

$$\mathcal{F}^\alpha_{\ \beta} u^\beta = -a^\alpha \quad , \quad u_\alpha \mathcal{F}^\alpha_{\ \beta} = a_\beta \quad , \quad \mathcal{F}_{\alpha\beta} + \mathcal{F}_{\beta\alpha} = 0 \tag{5.143}$$

of $\mathcal{F}^\alpha_{\ \beta}$ respectively. These conditions determine $\mathcal{F}^\alpha_{\ \beta}$ up to rotations in the plane orthogonal to u^α, i.e. up to the ambiguity

$$\mathcal{F}^\alpha_{\ \beta} \to \mathcal{F}^\alpha_{\ \beta} + \omega^\alpha_{\ \beta} \tag{5.144}$$

with

$$u^\alpha \omega_{\alpha\beta} = \omega_{\alpha\beta} u^\beta = 0 \quad , \quad \omega_{\alpha\beta} + \omega_{\beta\alpha} = 0 \quad . \tag{5.145}$$

Since there is no such rotation term in the prescription for Fermi-Walker transport, and no natural candidate for it either with only u^α and a^α at one's disposal, it is natural to think of Fermi-Walker transport as a prescription for transporting objects in a non-rotating way.

4. In particular, if one uses the Fermi-Walker prescription to construct an orthonormal basis (E_0, E_k) along the worldline, the spatial vectors can be interpreted as providing a *non-rotating* choice of axes.[14]

5.11 Epilogue: Manifolds? Think Globally, Act Locally!

In section 4.9 I had already briefly discussed some issues regarding the use of indices (and thus in some sense of local coordinates), and had advocated them as a useful bookkeeping device that also provides a transparent way of performing algebraic operations (tensor algebra). In the meantime we have seen that this extends to tensor analysis, and I can only reiterate that for most purposes and in most cases it is much more convenient to

[14] And according to S. Hawking, G. Ellis, *The large-scale structure of space-time*, section 4.1, these "could be realised physically by small gyroscopes pointing in the direction of each vector".

perform calculations in this notation than in some supposedly more elegant index-free notation.

There is one issue, however, that is worth commenting upon, and that in the end actually provides further justification for being allowed to adopt this procedure. Namely, in using local (Cartesian, say) coordinates x^μ to describe a space or space-time (I will use "space" in the following) one is implicitly assuming the following 3 things:

1. first of all, that one can always locally introduce Cartesian coordinates on that space (so as to then be able to perform tensor algebra, tensor analysis etc.);
2. secondly, that different choices of local coordinates will give compatible descriptions of that space;
3. and finally, that in principle one can obtain complete information about the space by covering it with such local coordinate systems.

When these assumptions are satisfied, then one is justified in using local coordinates to describe such a space. The point of this brief section is just to point out that (modulo some topological fine-points) these conditions amount precisely to the definition of a (differentiable or smooth) *manifold* in mathematics.

Thus while I could have started off these notes with an introduction to and definition of smooth manifolds (and numerous textbooks do), for all local intents and purposes this is then really equivalent to (consistently) working in local coordinates, as we have done and will continue to do. It is true that the notion of manifolds, of vector bundles on them etc. becomes indispensable for certain more advanced questions dealing with the global structure of a space-time, or theorems about the existence and uniqueness of solutions to differential equations on some manifold, say, but these are not topics that will be addressed in these notes.

The idea of a manifold is that an n-dimensional manifold is a sufficiently nice topological space that locally looks like (i.e. can be modelled on) the simple and nice topological space $\mathbb{R}^n$, and that this allows one to do calculus on this space by importing the relevant concepts from $\mathbb{R}^n$.

The usual textbook definition of a manifold consists essentially of the following steps:[15]

1. Topological Spaces

   A topological space is a set S together with a collection of subsets U of S (called open sets) which includes S and the empty set, and which is closed under union

[15] This presentation is adapted from the concise and clear description in S. Mukhi, N. Mukunda, *Lectures on Advanced Mathematical Methods for Physicists*, but equivalent descriptions can be found in many other places.

and finite intersection. This set of open sets defines the topology of the space and a corresponding notion of continuous maps (the inverse image of any open set is open) and homeomorphisms (bijective maps ϕ such that both ϕ and ϕ^{-1} are continuous) between topological spaces. In particular there is a notion of continuity for (real-valued, say) functions

$$f: \quad S \to \mathbb{R} \tag{5.146}$$

(with $\mathbb{R}$ equipped with its standard topology).

2. Charts

   However, in this context there is no notion of differentiability or differentiation. In order to have such things at one's disposal one needs topological spaces that locally "look like" $\mathbb{R}^n$. The essential building blocks of such a topological space are "charts":

   A chart C on a topological space S is the pair $C = (U, \phi)$ where $U \subset S$ is an open set of S and ϕ is a homeomorphism

   $$U \subset S \to \phi(U) \subset \mathbb{R}^n \ . \tag{5.147}$$

   The homeomorphism condition implies in particular that $\phi(U)$ is open in $\mathbb{R}^n$. The integer n is then known as the dimension of U (it does not depend on ϕ).

3. Topologial Manifolds

   A topological manifold is a topological space M that is locally homeomorphic to $\mathbb{R}^n$ in the sense that for each point p there is a chart $C = (U, \phi)$ with $p \in U$ (and that satisfies some further topological regularity conditions we are not interested in, such as Hausdorff and usually either second countable or paracompact). Equivalently, a topological space has the structure of a topological manifold when it possesses a covering by open sets U_a with charts $C_a = (U_a, \phi_a)$.

4. Local Coordinates and Local Coordinate Transformations

   The notion of a chart allows (and is equivalent to and the formalisation of) the introduction of local coordinates on the open set $U \subset M$. The coordinates of a point $p \in U$ in this chart are by definition simply the Cartesian coordinates $\vec{x}_p$ of the point $\phi(p) \in \mathbb{R}^n$.

   If one has two charts on M, $C_1 = (U_1, \phi_1)$ and $C_2 = (U_2, \phi_2)$, and $U_1 \cap U_2 \neq \emptyset$, then the "transition functions"

   $$\begin{aligned} \phi_1 \circ \phi_2^{-1}: &\quad \phi_2(U_1 \cap U_2) \to \phi_1(U_1 \cap U_2) \\ \phi_2 \circ \phi_1^{-1}: &\quad \phi_1(U_1 \cap U_2) \to \phi_2(U_1 \cap U_2) \end{aligned} \tag{5.148}$$

   are automatically continous maps between open subsets of $\mathbb{R}^n$. These can be interpreted as local coordinate transformations,

   $$(\phi_2 \circ \phi_1^{-1})(\vec{x}_p^1) = \phi_2(p) = \vec{x}_p^2 \ . \tag{5.149}$$

5. Local Functions and Differentiation

   In particular, with the help of charts we can express functions on M in terms of "local oordinates" on $\mathbb{R}^n$. More precisely, given a (continuous) function

$$f: \quad M \to \mathbb{R} \tag{5.150}$$

   and a chart $C = (U, \phi)$, we can associate to the restriction of f to U the function

$$f_U = f \circ \phi^{-1}: \quad \phi(U) \subset \mathbb{R}^n \to \mathbb{R} \ . \tag{5.151}$$

   i.e.

$$p \in U \quad \Rightarrow \quad f(p) = f_U(\vec{x}_p) \ . \tag{5.152}$$

   For such functions on $\mathbb{R}^n$ we now not only have a notion of continuity at our disposal, but also the notions of differentiability, smoothness, differentiation etc.

   On the intersection of 2 charts we can represent the function f in 2 different ways in terms of local coordinates, namely by the functions $f_{U_a} \equiv f_a$ for $a = 1, 2$,

$$\begin{aligned} \text{on } \ U_1 \cap U_2: \quad f = f_1 \circ \phi_1 = f_2 \circ \phi_2 \quad \Rightarrow \quad & f_2 = f_1 \circ (\phi_1 \circ \phi_2^{-1}) \\ & f_1 = f_2 \circ (\phi_2 \circ \phi_1^{-1}) \end{aligned} \tag{5.153}$$

   This is just the change of variables formula for a function (scalar), namely

$$f_2(\vec{x}_p^2) = f_1(\vec{x}_p^1) \ . \tag{5.154}$$

6. Compatibility of Charts

   In order to be able to extend the notion of smoothness (C^∞-differentiability), say, of a function from a local chart consistently to all of M, we need to impose compatibility conditions on intersecting charts.

   It is evident from (5.153) that the notion of smoothness of a function around a point p will only be independent of the chart if the transition functions $\phi_1 \circ \phi_2^{-1}$ and $\phi_2 \circ \phi_1^{-1}$ (i.e. the coordinate transformations) are also smooth. Thus we define 2 charts to be smoothly compatible if either $U_1 \cap U_2$ is empty or, otherwise, if these maps are smooth.

   Note that for topological manifolds and the condition of continuity any 2 charts are automatically compatible since the transition functions are continuous.

7. Smooth Atlas and Compatibility and Equivalence of Atlases

   A smooth atlas $\mathcal{A}(M)$ of M is now naturally a family of charts $C_a = (U_a, \phi_a)$ which cover M and such that all charts are mutually smoothly compatible.

   2 smooth atlases $\mathcal{A}_1(M)$ and $\mathcal{A}_2(M)$ for the same topological manifold M are said to be compatible with each other if all the charts of $\mathcal{A}_1$ are compatible with all the charts of $\mathcal{A}_2$. This defines an equivalence relation on atlases.

8. Smooth Structure and Smooth Manifold

   A smooth structure on a topological manifold M is an equivalence class

   $$\mathcal{S}(M) = [\mathcal{A}(M)] \tag{5.155}$$

   of smooth atlases on M. A smooth manifold is a topological manifold M equipped with a smooth structure $\mathcal{S}$.

9. Smooth Functions and Smooth Maps

   A function

   $$f: \quad M \to \mathbb{R} \tag{5.156}$$

   on a smooth manifold M is then said to be smooth if all its local coordinate representatives

   $$f_a = f \circ \phi_a^{-1}: \quad \phi(U_a) \to \mathbb{R} \tag{5.157}$$

   are smooth, and a map

   $$\mu: \quad M \to N \tag{5.158}$$

   from a smooth manifold M (with charts (U_a, ϕ_a)) of dimension m to a smooth manifold N (with charts $(V_{b'}, \psi_{b'})$) of dimension n is said to be smooth if all of its local coordinate representatives

   $$\mu_{ab'} = \psi_{b'} \circ \mu \circ \phi_a^{-1}: \quad \phi_a(U_a) \subset \mathbb{R}^m \to \psi_{b'}(V_{b'}) \subset \mathbb{R}^n \tag{5.159}$$

   are smooth. Such smooth functions can be differentiated by differentiating their local coordinate representatives and mapping the result back to M using the charts (and likewise for maps).

Topological fine-points aside we see that a smooth manifold is by definition a space on which one can consistently do calculus in local coordinates. Hence in these notes we were, are and will be dealing with (smooth) manifolds, regardless of whether or not we state this explicitly.

Analogously one can define C^k-differentiable manifolds (transition functions are required to be of degree C^k), real analytic manifolds (transition functions are required to be real analytic), complex manifolds (modelled on open subsets of $\mathbb{C}^n$, with holomorphic transition functions), etc., as well as submanifolds (modelled on subspaces of $\mathbb{R}^n$), manifolds with boundary (modelled on the half-space $\mathbb{R}^n_+$) etc.

6 Physics in a Gravitational Field and Minimal Coupling

6.1 Principle (or Algorithm) of Minimal Coupling

Recall that the Principle of General Covariance (section 4.1) says that, by virtue of the Einstein Equivalence Principle, a generally covariant equation will be valid in an arbitrary gravitational field provided that it is valid in Minkowski space in inertial coordinates (i.e. in the absence of gravity and/or acceleration).

We now have all the tools at our disposal to construct such equations. In particular, the fact that the covariant derivative ∇ maps tensors to tensors and reduces to the ordinary partial derivative in a locally inertial coordinate system suggests the following procedure or algorithm for obtaining equations that satisfy the Principle of General Covariance:

1. Write down the Lorentz invariant equations or expressions of Special Relativity you are interested in (e.g. those of relativistic mechanics, Maxwell theory, relativistic hydrodynamics, ...) in terms of inertial coordinates ξ^a, the Minkowski metric η_{ab} and other Lorentz tensors $T^{a\cdots}_{b\cdots}$.

2. Replace the coordinates ξ^a by arbitrary coordinates x^μ,
$$\xi^a \mapsto x^\mu \ . \tag{6.1}$$

3. Wherever the Minkowski metric η_{ab} appears, replace it by the metric $g_{\mu\nu}$ describing the gravitational field,
$$\eta_{ab} \mapsto g_{\mu\nu}(x) \ . \tag{6.2}$$

4. Promote the Lorentz tensors $T^{a\cdots}_{b\cdots}$ to tensors $T^{\mu\cdots}_{\nu\cdots}$ under general coordinate transformations,
$$T^{a\cdots}_{b\cdots}(\xi) \mapsto T^{\mu\cdots}_{\nu\cdots}(x) \ . \tag{6.3}$$

5. Wherever a partial derivative $\partial_a = \partial_{\xi^a}$ appears, replace it by the covariant derivative ∇_μ,
$$\partial_a \mapsto \nabla_\mu \ . \tag{6.4}$$

6. In particular, for the proper-time derivative along a curve this entails replacing $d/d\tau$ by D_τ,
$$\frac{d}{d\tau} \mapsto D_\tau = \dot{x}^\mu \nabla_\mu \ . \tag{6.5}$$

7. Wherever an integral $\int d^4\xi$ appears, replace it by $\int \sqrt{g} d^4x$,
$$\int d^4\xi \mapsto \int \sqrt{g} d^4x \ . \tag{6.6}$$

By construction, the resulting equations or expressions are tensorial (generally covariant) and true in the absence of gravity and hence satisfy the conditions for the Principle of General Covariance to apply. As a consequence they will be true in the presence of gravitational fields, at least on scales small compared to those of the gravitational fields. This procedure can thus be regarded as providing us with a prescription how to couple matter (particles, fields) to the gravitational field.

Remarks:

1. This procedure is analogous to the perhaps more familiar "minimal coupling" algorithm for the coupling of matter to gauge fields ("replace partial by gauge covariant derivatives"), and hence also in the current context this procedure is referred to as minimal coupling.

2. The reasons for the "at least on small scales" *caveat* in the paragraph above is that if one considers higher derivatives of the metric tensor then there are other equations that one can write down, involving e.g. the curvature tensor, that are tensorial but reduce to the same equations in the absence of gravity.

3. Thus "minimal coupling", as formulated here, is not a unique and unambiguous description, but it is nevertheless a pragmatic and effective procedure. We will see an example of the ambiguity in the minimal coupling prescription in the discussion of Maxwell theory in a gravitational field in section 6.6, and we will briefly return to the issue in section 8.10.

6.2 Particle Mechanics in a Gravitational Field Revisited

We can see the power of the formalism we have developed so far by rederiving the laws of particle mechanics in a general gravitational field. In Special Relativity (SR), the motion of a free particle with mass m is governed by the equation

$$\text{SR:} \quad a^a = \frac{du^a}{d\tau} = 0 \ , \tag{6.7}$$

where $u^a = d\xi^a/d\tau$ is the 4-velocity and a^a the 4-acceleration. Thus, using the principle of minimal coupling, the equation of motion of a free particle in a general gravitational field is

$$\text{GR:} \quad a^\mu = D_\tau u^\mu = 0 \quad \Leftrightarrow \quad \ddot{x}^\mu + \Gamma^\mu_{\nu\lambda}\dot{x}^\nu\dot{x}^\lambda = 0 \ , \tag{6.8}$$

where $u^\mu = dx^\mu/d\tau$. Thus we rediscover the familiar geodesic equation, but we see that it follows much faster from demanding general covariance (as made precise by the principle of minimal coupling) than from our previous somewhat more convoluted and roundabout considerations based e.g. on the equivalence principle.

We could also have arrived at this equation for a free particle in a gravitational field by applying the minimal coupling description not at the level of the equations of motion but rather (and perhaps conceptually more satisfactorily) at the level of the action, i.e. by replacing

$$S = -m\int d\tau = -m\int \sqrt{-\eta_{ab}d\xi^a d\xi^b} \quad \to \quad -m\int d\tau = -m\int \sqrt{-g_{\mu\nu}dx^\mu dx^\nu} \ , \tag{6.9}$$

and this is exactly what we already did back in section 2.3 where we showed that this also leads to the geodesic equation (6.8).

6.3 Klein-Gordon Scalar Field in a Gravitational Field

Here is where the formalism we have developed really pays off. We will see once again that, using the minimal coupling rule, we can immediately rewrite the equations for a scalar field (here) and the Maxwell equations (in section 6.6 below) in a form in which they are valid in an arbitrary gravitational field.

1. The action for a (real) free massive scalar field ϕ in Special Relativity is

$$\text{SR:} \qquad S[\phi] = \int d^4\xi \left[-\tfrac{1}{2}\eta^{ab}\partial_a\phi\partial_b\phi - \tfrac{1}{2}m^2\phi^2\right] \ . \tag{6.10}$$

   To covariantise this, we replace $d^4\xi \to \sqrt{g}d^4x$, $\eta^{ab} \to g^{\alpha\beta}$, and we can replace ∂_a by ∇_α or ∂_α (since this makes no difference on scalars). Therefore, the covariant action in a general gravitational field is

$$\text{GR:} \qquad S[\phi, g_{\alpha\beta}] = \int \sqrt{g}d^4x \left[-\tfrac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}m^2\phi^2\right] \ . \tag{6.11}$$

   Here I have also indicated the dependence of the action on the metric $g_{\alpha\beta}$. This is not (yet) a dynamical field, though, just the gravitational background field.

2. The equations of motion for ϕ one derives from this are

$$\frac{\delta}{\delta\phi}S[\phi, g_{\alpha\beta}] = 0 \quad \Rightarrow \quad \left(\Box_g - m^2\right)\phi = 0 \tag{6.12}$$

   where $\Box_g = g^{\alpha\beta}\nabla_\alpha\nabla_\beta$, the Laplacian associated to the metric $g_{\alpha\beta}$. This is precisely what one would have obtained by applying the minimal coupling description to the Minkowski Klein-Gordon equation $(\Box_\eta - m^2)\phi = 0$.

   Remarks:

   (a) A comment on how to derive this: if one thinks of the ∂_α in the action as covariant derivatives, $\partial_\alpha \to \nabla_\alpha$, then one can use the covariant intgration by

parts formula (5.64) to conclude (dropping the boundary term, as usual in variational calculus)

$$-\int \sqrt{g}d^4x\ (\nabla_\alpha\delta\phi)g^{\alpha\beta}\nabla_\beta\phi = \int \sqrt{g}d^4x\ \delta\phi\ g^{\alpha\beta}\nabla_\alpha\nabla_\beta\phi = \int \sqrt{g}d^4x\ \delta\phi\ \Box_g\phi\ . \tag{6.13}$$

Thus essentially the calculation is identical to that in Minkowski space. If one sticks with the ordinary partial derivatives, then upon the usual integration by parts one picks up a term $\sim \partial_\alpha(\sqrt{g}g^{\alpha\beta}\partial_\beta\phi)$ which then evidently leads to the Laplacian in the form (5.57).

(b) If the relative sign of $\Box_\eta$ (or $\Box_g$) and m^2 in the Klein-Gordon equation looks unfamiliar to you, then this is probably due to the fact that in a course where you first encountered the Klein-Gordon equation the opposite (particle physicists') sign convention for the Minkowski metric was used, with its negative definite spatial metric.

(c) All of this generalises in a straightforward way to (self-)interacting scalar fields, described by a potential $V(\phi)$. In particular, the action is

$$S[\phi, g_{\alpha\beta}] = \int \sqrt{g}d^4x\ \left[-\tfrac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - V(\phi)\right]\ . \tag{6.14}$$

Logically the next thing to discuss would be the energy-momentum tensor, e.g. the minimally coupled counterpart of the special relativistic (Noether) energy-momentum tensor

$$\text{SR:}\quad T_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L \tag{6.15}$$

and its properties. However, it turns out that there is more to say about this than meets the eye, and we will therefore return to this issue in more detail in section 7.

6.4 Interlude: General Covariance in Minkowski Space?

Before turning to our next example, I want to briefly comment on the issue of general covariance in Minkowski space, as this tends to generate quite a bit of confusion and unnecessary debates. I will discuss this issue in the context of the above example of a scalar field, but the discussion is valid more generally.

On the one hand, the action (6.10) is generally considered to be invariant (only) under Lorentz or Poincaré transformations, while by construction the action (6.11) is invariant under arbitrary coordinate transformations. Does this really mean that the theory of a scalar field in a non-trivial gravitational background has more invariances than that in a Minkowski background?

On the other hand, certainly nothing prevents one from using e.g. spherical (and thus in particular non-inertial) coordinates in Minkowski space to write down the Klein-Gordon

equation or action. But does this mean that the action (6.10) is actually (secretly) invariant also under such non-Lorentz transformations?

Well, that depends ... While this sounds like (and generally is correctly considered to be) a somewhat unsatisfactory answer, I can be more specific:

- it depends on what one means by "invariance" (or "covariance")
- and it depends on how one treats or regards the Minkowski metric.

From the current point of view, the natural answer is that the action (6.11) is generally covariant in *any* gravitational field, in particular therefore also in the absence of a true gravitational field, i.e. in a purely fictitious gravitational field or, equivalently, in Minkowski space. If we specialise the action (6.11) to such a gravitational field, i.e. to the Minkowski metric written in some perhaps non-inertial coordinates, we get

$$S[\phi, \eta_{\alpha\beta}] = \int \sqrt{\eta} d^4x \left[-\tfrac{1}{2}\eta^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}m^2\phi^2 \right] \quad . \tag{6.16}$$

Here it is important to keep in mind that $\eta_{\alpha\beta}$ refers to the components of the Minkowski metric in the not-necessarily inertial coordinates x^α, as in

$$\eta_{\alpha\beta} = \frac{\partial\xi^a}{\partial x^\alpha}\frac{\partial\xi^b}{\partial x^\beta}\eta_{ab} \quad . \tag{6.17}$$

As a consequence, also $\sqrt{\eta}$ is not necessarily equal to 1. This action is invariant under arbitrary coordinate transformations, provided that one transforms the fields *and the metric* appropriately.

If one now chooses to write this action in inertial coordinates, $x^\alpha \to \xi^a$, with $\eta_{\alpha\beta} \to \eta_{ab}$ and thus $\sqrt{\eta} \to 1$, then the action (6.16) appears to reduce to the special relativistic action (6.10). So is this action, which is simply a generally covariant action written in some particular coordinates, invariant under Lorentz (or Poincaré) transformations only or under all coordinate transformations?

1. If one looks for the transformations of the coordinates ξ^a and the fields ϕ that leave the action invariant (with fixed metric components η_{ab}) then none too surprisingly one finds that the action is invariant under Poincaré transformations of the coordinates provided that the scalar fields transform as scalars, but not under more general transformations.

2. If one looks for the transformations of the coordinates ξ^a and the fields ϕ *and the metric* η_{ab} that leave the action invariant, then one finds that the action is invariant under arbitrary coordinate transformations

   - provided that one also transforms $\eta_{ab} \to \eta_{\alpha\beta}$ like a (0,2)-tensor

- and provided that one either thinks of $d^4\xi$ as the invariant volume element $\sqrt{\eta}d^4\xi$, or equivalently one treats the Lagrangian L as a scalar density $\sqrt{\eta}L$.

Sometimes option (1) is taken to define the *invariance group* (Poincaré transformations) while option (2) refers to the *covariance group*. In this sense, special relativity is *invariant* under Poincaré transformations but is at the same time *generally covariant*. In philosophy of science or epistemological terms whether one has option (1) or option (2) is related to the question whether or not the Minkowski metric is regarded as an *absolute element* of the theory. With η_{ab} promoted to an absolute element, general covariance is reduced to Poincaré invariance (those transformations that, from the generally covariant "$\eta_{\alpha\beta}$ transforms" point of view, leave η_{ab} invariant).[16] Unfruitful discussions ensue when tacitly conflicting assumptions are made about what are considered to be the absolute elements of a theory.

6.5 Lorentz-Covariant Formulation of Maxwell Theory (Review)

In order to discuss the formulation of Maxwell theory in a gravitational field, we will need to quickly recall the Lorentz-covariant formulation of Maxwell theory in Minkowski space. This will also fix our conventions for Maxwell theory.

In the traditional non-covariant formulation one has

1. the homogeneous equations

$$\vec{\nabla}\cdot\vec{B}=0 \quad , \quad \vec{\nabla}\times\vec{E}+\partial_t\vec{B}=0 \tag{6.18}$$

2. the inhomogeneous equations

$$\vec{\nabla}\cdot\vec{E}=\rho/\epsilon_0 \quad , \quad \vec{\nabla}\times\vec{B}-\frac{1}{c^2}\partial_t\vec{E}=\mu_0\vec{J} \tag{6.19}$$

3. the ensuing continuity equation

$$\partial_t\rho+\vec{\nabla}.\vec{J}=0 \tag{6.20}$$

4. the vector and scalar potentials $\vec{A}$ and ϕ,

$$\vec{B}=\vec{\nabla}\times\vec{A} \quad , \quad \vec{E}=-\vec{\nabla}\phi-\partial_t\vec{A} \tag{6.21}$$

5. and the corresponding gauge transformations leaving $\vec{E}$ and $\vec{B}$ invariant,

$$\vec{A}\to\vec{A}+\vec{\nabla}\Psi \quad , \quad \phi\to\phi-\partial_t\Psi \quad \Rightarrow \quad \vec{E}\to\vec{E} \quad , \quad \vec{B}\to\vec{B} \ . \tag{6.22}$$

[16] For more erudite discussions of these and related issues, see e.g. section 2.5 ("On Covariance and Invariance") of N. Straumann, *General Relativity* or D. Giulini, *Some remarks on the notions of general covariance and background independence*, `arXiv:gr-qc/0603087`.

The charge density and current can be packaged into a Lorentz vector

$$J^a = (c\rho, \vec{J}) \tag{6.23}$$

(note that in signature (-+++) one has to choose whether to identify J^0 or $J_0 = -J^0$ with the charge density, here we choose the former), and the continuity equation can be written in the manifestly Lorentz-invariant form

$$\partial_a J^a = 0 \ . \tag{6.24}$$

Likewise, the scalar and vector potential can be packaged into a Lorentz covector

$$A_a = (-\phi/c, \vec{A}) \ , \tag{6.25}$$

and the gauge transformations can be compactly written as

$$A_a \to A_a + \partial_a \Psi \ . \tag{6.26}$$

The gauge invariant Maxwell field strength tensor F_{ab} is defined by

$$F_{ab} = \partial_a A_b - \partial_b A_a \tag{6.27}$$

and has the components

$$F_{0k} = -F_{k0} = -E_k/c \quad , \quad F_{ik} = \epsilon_{ik\ell} \, B_\ell \tag{6.28}$$

or, in matrix form,

$$(F_{ab}) = \begin{pmatrix} 0 & -E_1/c & -E_2/c & -E_3/c \\ +E_1/c & 0 & +B_3 & -B_2 \\ +E_2/c & -B_3 & 0 & +B_1 \\ +E_3/c & +B_2 & -B_1 & 0 \end{pmatrix} \tag{6.29}$$

and

$$(F^{ab}) = \begin{pmatrix} 0 & +E_1/c & +E_2/c & +E_3/c \\ -E_1/c & 0 & +B_3 & -B_2 \\ -E_2/c & -B_3 & 0 & +B_1 \\ -E_3/c & +B_2 & -B_1 & 0 \end{pmatrix} \tag{6.30}$$

In terms of these Lorentz tensors, the homogeneous Maxwell equations can be written as

$$\partial_{[a} F_{bc]} = 0 \quad \Leftrightarrow \quad \partial_a F_{bc} + \partial_c F_{ab} + \partial_b F_{ca} = 0 \ , \tag{6.31}$$

and these equations are identically satisfied if F_{ab} derives from a potential,

$$F_{ab} = \partial_a A_b - \partial_b A_a \quad \Rightarrow \quad \partial_{[a} F_{bc]} = 0 \ . \tag{6.32}$$

The inhomogeneous Maxwell equations can (suppressing μ_0, i.e. setting $\mu_0 = 1$) be written as

$$\partial_a F^{ab} = -J^b \quad \Leftrightarrow \quad \Box A_a - \partial_a (\partial_b A^b) = -J_a \ . \tag{6.33}$$

These equations can be derived from the Lorentz-invariant action

$$S[A,J] = S[A] + S_I[A,J] = -\tfrac{1}{4}\int d^4\xi\ F_{ab}F^{ab} + \int d^4\xi\ A_a J^a \ , \tag{6.34}$$

with the Maxwell Lagrangian

$$-\tfrac{1}{4}F_{ab}F^{ab} = -\tfrac{1}{2}F_{0k}F^{0k} - \tfrac{1}{4}F_{ik}F^{ik} = \tfrac{1}{2}(\vec{E}^2/c^2 - \vec{B}^2) \ . \tag{6.35}$$

This is essentially all we will need (some facts regarding the Noether versus covariant energy-momentum tensor of Maxwell theory will be recalled below).

6.6 Maxwell Theory in a Gravitational Field

Mutatis mutandis we can now proceed in the same way as for a scalar field.

1. The basic dynamical field is the vector potential A_a. Given the vector potential A_μ, the Maxwell field strength tensor in Special Relativity is

$$\text{SR:} \quad F_{ab} = \partial_a A_b - \partial_b A_a \ . \tag{6.36}$$

   Therefore in a general metric space-time (gravitational field) one is led to (or tempted to) define the field strength tensor as

$$\text{GR:} \quad F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu \ . \tag{6.37}$$

   - A cautionary remark: Actually, this is a bit misleading. The field strength tensor (two-form) in any, Abelian or non-Abelian, gauge theory is always given in terms of the gauge-covariant exterior derivative of the vector potential (i.e. it is the curvature of the connection), and as such has nothing whatsoever to do with a metric on space-time. So you should not really regard the first equality in the above equation as the definition of $F_{\mu\nu}$, but you should regard the second equality as a proof that $F_{\mu\nu}$, *always* defined by $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, is a tensor.

     The mistake of adopting $\nabla_\mu A_\nu - \nabla_\nu A_\mu$ as the definition of $F_{\mu\nu}$ in a curved space-time has led some poor souls to believe, and even claim in published papers, that in a space-time with torsion, for which the second equality does not hold, the Maxwell field strength tensor is invariably modified by the torsion. This is nonsense.

     If there is torsion, one is of course free to consider non-minimal couplings of the torsion tensor to other tensor fields (like the Maxwell field strength tensor), but this is not required by either gauge invariance or general covariance.

2. In Special Relativity, the Maxwell equations read

$$\begin{aligned} \text{SR:} \qquad & \partial_a F^{ab} = -J^b \\ & \partial_{[a} F_{bc]} = 0 \ . \end{aligned} \tag{6.38}$$

Thus in a general gravitational field (curved space-time) these equations become

$$\begin{aligned} \text{GR:} \quad & \nabla_\mu F^{\mu\nu} = -J^\nu \\ & \nabla_{[\mu} F_{\nu\lambda]} = 0 \ , \end{aligned} \tag{6.39}$$

where now of course all indices are raised and lowered with the metric $g_{\mu\nu}$,

$$F^{\mu\nu} = g^{\mu\lambda} g^{\nu\rho} F_{\lambda\rho} \ . \tag{6.40}$$

Remarks:

(a) Regarding the use of the covariant derivative in the second equation, the same *caveat* as above applies.

(b) In particular, using the results derived in section 5.5, we can rewrite these two equations as

$$\begin{aligned} \text{GR:} \qquad & \partial_\mu (\sqrt{g} F^{\mu\nu}) = -\sqrt{g} J^\nu \\ & \partial_{[\mu} F_{\nu\lambda]} = 0 \ . \end{aligned} \tag{6.41}$$

(c) It is clear from the first of these equations that the Maxwell equations imply that the current is covariantly conserved: since

$$\partial_\nu \partial_\mu (\sqrt{g} F^{\mu\nu}) = 0 \tag{6.42}$$

by anti-symmetry of $F^{\mu\nu}$, it follows that

$$\partial_\nu (\sqrt{g} J^\nu) = 0 \quad \Leftrightarrow \quad \nabla_\nu J^\nu = 0 \ . \tag{6.43}$$

From the covariant version $\nabla_\mu F^{\mu\nu} = -J^\nu$ this follows in the seemingly more roundabout way from the identity (8.47) for the commutator of covariant derivatives that we will establish later, in the context of our discussion of the Riemann curvature tensor in section 8.

(d) In Special Relativity, the inhomogeneous Maxwell equations can be decoupled by imposing the Loren(t)z (see footnote 65 in section 23.5) gauge condition $\partial_a A^a = 0$,

$$\partial_a A^a = 0 \quad \Rightarrow \quad \partial_a F^{ab} = -J^b \ \rightarrow \ \Box A_b = -J_b \ . \tag{6.44}$$

This gauge condition has the virtue of preserving Lorentz invariance. Similarly, its covariantised version

$$\nabla_\mu A^\mu = 0 \tag{6.45}$$

has the virtue of preserving general covariance, because $\nabla_\mu A^\mu$ is a scalar. However, the inhomogeneous Maxwell equations in this gauge do not take the form $\Box A_\nu = -J_\nu$ one might perhaps have anticipated on the basis of minimal coupling. Rather, using the covariant Lorenz gauge condition (6.45), the covariant divergence of the Maxwell field strength tensor can be written as

$$\nabla_\mu A^\mu = 0 \quad \Rightarrow \quad \nabla_\mu F^{\mu\nu} = \nabla_\mu(\nabla^\mu A^\nu - \nabla^\nu A^\mu) = \Box A^\nu - [\nabla_\mu, \nabla_\nu] A^\mu \ , \tag{6.46}$$

where $\Box A_\mu = \nabla^\nu \nabla_\nu A_\mu$ is the "naive" Laplacian on scalars. The second term would of course be zero in Minkowski space, but here it is not. Indeed, as we will see in section 8, the quintessence of a non-trivial geometry is that covariant derivatives do not commute on tensors other than scalars. In particular, here one finds that as a consequence of (8.40) the Maxwell equations in the covariant Lorenz gauge can be written as

$$\Box A^\nu - R^\nu_{\ \mu} A^\mu = -J^\nu \ , \tag{6.47}$$

where $R_{\mu\nu}$ is the *Ricci tensor*, a particular contraction of the Riemann curvature tensor, constructed from the second derivatives of the metric.

(e) Thus, these equations appear to display a *non-minimal coupling to the gravitational field*, even though we started off with the minimally coupled equations which we can also derive, see below, from the minimally coupled action. We will return to a discussion of this issue in section 8.10.

3. The electromagnetic force acting on a particle of charge e is given in Special Relativity by the Lorentz force

$$\text{SR:} \quad f^a = e F^a_{\ b} \dot{\xi}^b \ . \tag{6.48}$$

Thus in General Relativity it becomes

$$\text{GR:} \quad f^\mu = e g^{\mu\lambda} F_{\lambda\nu} \dot{x}^\nu \ . \tag{6.49}$$

4. The Lorentz-invariant action of (vacuum) Maxwell theory is

$$\text{SR:} \quad S[A_a] = -\tfrac{1}{4} \int d^4\xi \ F_{ab} F^{ab} \tag{6.50}$$

in Special Relativity, and thus becomes

$$\text{GR:} \quad S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4} \int \sqrt{g} d^4x \ F_{\mu\nu} F^{\mu\nu} \equiv -\tfrac{1}{4} \int \sqrt{g} d^4x \ g^{\mu\lambda} g^{\nu\rho} F_{\mu\nu} F_{\lambda\rho} \tag{6.51}$$

in General Relativity.

As for the scalar field, depending on whether one writes the field strength tensor as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ or as $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$, by varying this action with respect to the A_μ one obtains the vacuum Maxwell equations $\nabla_\mu F^{\mu\nu} = 0$ in either of the 2 forms

$$\frac{\delta}{\delta A_\nu} S[A_\alpha, g_{\alpha\beta}] = 0 \quad \Rightarrow \quad \left\{ \begin{array}{c} \partial_\mu(\sqrt{g} F^{\mu\nu}) = 0 \\ \nabla_\mu F^{\mu\nu} = 0 \end{array} \right. \tag{6.52}$$

REMARKS:

(a) Writing out explicitly the Lagrangian in terms of its components (with respect to some coordinate system $x^\alpha = (t, x^k)$) one finds

$$\begin{aligned} -\tfrac{1}{4} F_{\alpha\beta} F^{\alpha\beta} = & -\tfrac{1}{2} F_{0i} F_{0k} (g^{00} g^{ik} - g^{0i} g^{0k}) \\ & -\tfrac{1}{4} F_{ij} F_{kl} g^{ik} g^{jl} \\ & -\tfrac{1}{2} F_{0i} F_{kl} (g^{0k} g^{il} - g^{0l} g^{ik}) \end{aligned} \tag{6.53}$$

While the 1st and 2nd lines look just like "gravitationally dressed" standard terms $\sim \vec{E}^2$ and $\sim \vec{B}^2$, the last line appears to suggest a gravitationally induced coupling between the electric and magnetic fields. This, however, is misleading and simply not a meaningful way of expressing things. After all, even in Minkowski space the decomposition of the electro-magnetic field into electric and magnetic fields depends on the choice of inertial reference system.

(b) In order to add sources, one can add $\int \sqrt{g} d^4x \; A_\mu J^\mu$ to the Maxwell action, thus coupling the matter current to the Maxwell gauge field. Instead of just adding such a (phenomenological) source-term by hand, a more coherent microscopic approach (which also provides the sources with their own dynamics) is to consider a matter action (minimally) coupled to the Maxwell field,

$$S_M[\phi] \to S_M[\phi, A_\alpha] \quad . \tag{6.54}$$

The combined Maxwell + matter action will then give rise to the Maxwell equations with a source provided that one defines the current J^α as the variation of the matter action with respect to the gauge field,

$$J^\alpha \sim \frac{\delta S_M[\phi, A_\alpha]}{\delta A_\alpha} \quad . \tag{6.55}$$

As in the case of scalar fields, we will postpone a discussion of the energy-momentum tensor and how to properly define it (something that is already an issue in Minkowski space because for Maxwell theory the Noether energy-momentum tensor turns out to be neither symmetric nor gauge-invariant!) to section 7.

In anticipation of this I just want to point out that, by the same rationale as that leading to (6.55), perhaps we should define the source term for the gravitational field by the variation of the gravitationally minimally coupled matter action with respect to the metric. If we now call this source term the energy-momentum tensor, then we have a candidate definition of the energy-momentum tensor which is natural and appropriate from the gravitational point of view. We will pursue this point of view in section 7.6.

6.7 Minimal Coupling and (quasi-)Topological Couplings

In all the cases considered so far, the minimal coupling prescription resulted in a minimally coupled matter action that depends explicitly on the metric - this is as it should be and is not a surprise. What would be more of a surprise would be to find minimally coupled and hence generally covariant contributions to an action that do not depend on the metric, but such examples do indeed exist (and play an important role in many branches of physics and even mathematics, ranging from the strong-CP problem in QCD to high-T_c superconductors to topology). Such terms in the action are usually referred to as "topological terms" in the physics literature but as they need not be (and usually are not) purely topological in the mathematics sense, for lack of a better name I refer to them as "quasi-topological".

Here are 2 prototypical examples illustrating this phenomenon:

1. Axionic Coupling in (3+1) Dimensions

   The first toy-model we will consider consists of Maxwell-theory coupled to a neutral scalar field through what is known as an axionic coupling only (with analogous considerations for the more interesting case of a non-Abelian Yang-Mills field),

   $$S[\phi, A] = S_s[\phi] + S_m[A] + S_a[\phi, A] \quad , \tag{6.56}$$

   with

   $$S_s[\phi] = \int d^4x \; L_s(\phi, \partial_\alpha \phi) \tag{6.57}$$

   some arbitrary standard scalar field action (of the type already discussed), $S_m[A]$ the usual Maxwell action,

   $$S_m[A] = \int d^4x \; L_m(\partial_\alpha A_\beta) = -\tfrac{1}{4} \int d^4x \; F^{\alpha\beta} F_{\alpha\beta} \tag{6.58}$$

   and the axionic coupling term is

   $$S_a[\phi, A] = -\tfrac{1}{4} \int d^4x f(\phi) \tilde{F}^{\alpha\beta} F_{\alpha\beta} \tag{6.59}$$

   where

   $$\tilde{F}^{\alpha\beta} = \tfrac{1}{2} \in^{\alpha\beta\gamma\delta} F_{\gamma\delta} \tag{6.60}$$

with $\in^{\alpha\beta\gamma\delta} = 0, \pm 1$ the Levi-Civita symbol, a tensor density (cf. remark 2 in section 4.5) and $f(\phi)$ some function of the scalar field ϕ. Note that for $f(\phi) = 1$ (or in the absence of a scalar field) the axionic term would be (locally) a total derivative and would hence not contribute to the equations of motion. For non-trivial $f(\phi)$, on the other hand, the axionic term is itself non-trivial.

Minimal coupling for the first two (standard) terms proceeds as already discussed above. For the third, axionic term, we make the usual replacement $d^4x \to \sqrt{g}d^4x$ and recall from (4.83) that

$$\epsilon^{\alpha\beta\gamma\delta} \equiv \frac{1}{\sqrt{g}} \in^{\alpha\beta\gamma\delta} \tag{6.61}$$

is a (4,0) tensor, so that the generally covariant generalisation of the axionic action is

$$\begin{aligned} S_a[\phi, A, g_{\alpha\beta}] &= -\tfrac{1}{8} \int \sqrt{g} d^4x f(\phi) \epsilon^{\alpha\beta\gamma\delta} F_{\gamma\delta} F_{\alpha\beta} \\ &= -\tfrac{1}{8} \int d^4x f(\phi) \in^{\alpha\beta\gamma\delta} F_{\gamma\delta} F_{\alpha\beta} = S_a[\phi, A] \end{aligned} \tag{6.62}$$

We see that, as announced, the metric dependence drops out of the minimally coupled generally covariant action. The reason for this is that the axionic Lagrangian is already all by itself a scalar density of weight $w = +1$, and that therefore its integral (4.76) is well-defined and generally covariant without having to take recourse to a metric to construct an auxiliary weight-one object like $\sqrt{g}$.

2. Maxwell - Chern-Simons Theory in (2+1) Dimensions

The second prominent example involves the addition of what is known as an Abelian Chern-Simons term to the Maxwell action in (2+1) dimensions (with analogous considerations for the more interesting case of a non-Abelian Yang-Mills field). The Minkowski space Lagrangian of this model is

$$L = L_m + k\, L_{cs} = -\tfrac{1}{4} F^{\alpha\beta} F_{\alpha\beta} + \tfrac{1}{2} k \;\in^{\alpha\beta\gamma} A_\alpha F_{\beta\gamma} \;\; . \tag{6.63}$$

Minimal coupling for the first term is standard and for the 2nd term one finds, as above, that the generally covariant minimally coupled Chern-Simons action is actually metric independent (since the Chern-Simons Lagrangian is a density of weight $w = 1$),

$$\begin{aligned} S_{cs}[A, g_{\alpha\beta}] &= \tfrac{1}{2} k \int \sqrt{g} d^3x \, \epsilon^{\alpha\beta\gamma} A_\alpha F_{\beta\gamma} \\ &= \tfrac{1}{2} k \int d^3x \;\in^{\alpha\beta\gamma} A_\alpha F_{\beta\gamma} = S_{cs}[A] \end{aligned} \tag{6.64}$$

As an aside note that the above theory is also known as topologically massive Maxwell theory, since the CS term provides a gauge-invariant mass term for the photon. One quick way to see this is to note that the equations of motion are

$$\partial_\alpha F^{\alpha\beta} + k \;\in^{\beta\gamma\delta} F_{\gamma\delta} = 0 \;\; . \tag{6.65}$$

and that in terms of the dual field strength

$$G^\beta = \tfrac{1}{2} \in^{\beta\gamma\delta} F_{\gamma\delta} \tag{6.66}$$

the equations of motion and the Bianchi identity take the form

$$\partial_\alpha G_\beta - \partial_\beta G_\alpha = 2k \ \in_{\alpha\beta\gamma} G^\gamma \quad , \quad \partial_\beta G^\beta = 0 \tag{6.67}$$

respectively. Acting with ∂^α on the equation of motion and using the Bianchi identity and again the equation of motion one finds

$$\begin{aligned} \Box G_\beta &= 2k \ \in_{\alpha\beta\gamma} \partial^\alpha G^\gamma = k \ \in_{\alpha\beta\gamma} (\partial^\alpha G^\gamma - \partial^\gamma G^\alpha) \\ &= 2k^2 \ \in_{\alpha\beta\gamma} \in^{\alpha\gamma\delta} \ G_\delta = 4k^2 G_\beta \end{aligned} \tag{6.68}$$

so that the theory describes excitations of mass $m^2 = 4k^2$.

These quasi-topological terms modify the equations of motion. Moreover, since they depend on the derivatives of the fields, they will contribute to the canonical Noether energy-momentum tensor. On the other hand, since they do not depend on the metric, they do not contribute to the covariant energy-momentum tensor, defined in section 7 in terms of the variation of the matter action with respect to the metric (and as such playing the role of the source term for the Einstein gravitational field equations).

How it nevertheless conspires that this tensor is conserved on-shell (meaning: for a solution to the matter equations of motion) even though the equations of motion have been modified and how the improved canonical energy-momentum tensor nevertheless ends up agreeing with the covariant energy-momentum tensor on-shell will be explored and explained in section 22.5.

6.8 Conserved Charges from Covariantly Conserved Currents

In Special Relativity a conserved current J^a is characterised by the vanishing of its divergence, i.e. by $\partial_a J^a = 0$. It leads to a conserved charge Q by integrating J^a over a spacelike hypersurface, say the one described by $t = t_0$,

$$Q = \int_{t=t_0} d^3x \ J^0 \ . \tag{6.69}$$

That Q is conserved, i.e. independent of t_0, is a consequence of the fact that by virtue of the Gauss theorem

$$Q(t_1) - Q(t_0) = \int_V d^4\xi \ \partial_a J^a = 0 \ , \tag{6.70}$$

where V is the four-volume $\mathbb{R}^3 \times [t_0, t_1]$. This holds provided that J vanishes at spatial infinity.

Now in General Relativity, the conservation law will be replaced by the covariant conservation law $\nabla_\mu J^\mu = 0$, and one may wonder if this also leads to some conserved charges in the ordinary sense. The answer is yes because, recalling the formula for the covariant divergence of a vector,

$$\nabla_\mu J^\mu = g^{-1/2}\partial_\mu(g^{1/2}J^\mu) \ , \tag{6.71}$$

we see that

$$\nabla_\mu J^\mu = 0 \Leftrightarrow \partial_\mu(g^{1/2}J^\mu) = 0 \ , \tag{6.72}$$

so that $g^{1/2}J^\mu$ is a conserved current in the ordinary sense. We then obtain conserved quantities in the ordinary sense by integrating J^μ over a spacelike hypersurface Σ. We will develop a more precise formula for this, an appropriate version of the Gauss theorem for hypersurfaces in curved space-times, in section 16.3.

The factor $g^{1/2}$ apearing in the current conservation law can be understood physically. To see what it means, split J^μ into its space-time direction u^μ, with $u^\mu u_\mu = -1$, and its magnitude ρ as

$$J^\mu = \rho u^\mu \ . \tag{6.73}$$

This defines the average four-velocity of the conserved quantity represented by J^μ and its density ρ measured by an observer moving at that average velocity (rest mass density, charge density, number density, ...). Since u^μ is a vector, in order for J^μ to be a vector, ρ has to be a scalar. Therefore this density is defined as per unit proper volume. The factor of $g^{1/2}$ transforms this into density per coordinate volume and this quantity is conserved (in a comoving coordinate system where $J^0 = \rho, J^i = 0$).

We will come back to this in the context of cosmology later on in this course, but for now just think of the following picture (Figure 44 in section 34): take a balloon, draw lots of dots on it at random, representing particles or galaxies. Next choose some coordinate system on the balloon and draw the coordinate grid on it. Now inflate or deflate the balloon. This represents a time dependent metric, roughly of the form $ds^2 = r^2(t)(d\theta^2 + \sin^2\theta d\phi^2)$. You see that the number of dots per coordinate volume element (area element in this case) does not change, whereas the number of dots per unit proper volume (area) will.

7 Energy-Momentum Tensor I: Basics

7.1 Introduction

Newton's gravitational field equation for the gravitational potential ϕ is the Poisson equation $\Delta\phi = 4\pi G_N \mu$, with μ the mass density. Thus in Newton's theory, mass is the source of gravity. We can also more usefully, and thinking relativistically, write this in terms of the energy density $\rho = \mu c^2$ as

$$\Delta\phi = \frac{4\pi G_N}{c^2}\rho \overset{(c=1)}{=} 4\pi G_N \ \rho \ . \tag{7.1}$$

Now we already noted in section 1.1 that in Special Relativity ρ is not a scalar but rather just one component of a tensor, the energy-momentum tensor

$$T_{ab}: \ \ T_{00} = \rho \ , \tag{7.2}$$

with the components T_{ab} transforming into each other under Lorentz transformations according to the transformation rules for Lorentz tensors.

It is therefore entirely plausible that in a relativistic theory of gravity the source of gravity should be the entire energy-momentum tensor. In particular, also the other components of T_{ab}, T_{0k} ($\sim$ energy flux), T_{k0} ($\sim$ momentum density) and T_{ik} ($\sim$ stresses or pressure) are a source of gravity. Clearly, therefore, the notion of energy-momentum tensor will play a crucial role in the following. This then immediately raises the question how to find or define an energy-momentum tensor.

Within the framework of special relativity and relativistic field theories there are (at least) 2 common approaches to constructing or defining an energy-momentum tensor, namely

1. a Macroscopic Phenomenological Description
2. a Microscopic Lagrangian Prescription

and we will now briefly discuss these in turn.

7.2 Perfect Fluid Energy-Momentum Tensor in Special Relativity

A macroscopic phenomenological description is useful when one does not know (or does not care about) the microscopic description of the matter one is dealing with but rather tries to characterise its properties in terms of the specification of some macroscopic (thermodynamic, hydrodynamic) parameters such as energy (density), pressure, viscosity etc. For many purposes this is the appropriate language for describing e.g. gases or fluids.

In this case, one constructs the energy-momentum tensor in such a way that it encodes the physics one is trying to describe (primarily conservation laws and dynamics). As a simple example of this (not by coincidence the one which is of most relevance for gravitational physics and thus also later on in these notes), we consider a *perfect fluid*.

By definition, a perfect fluid is one in which a comoving observer (i.e. an observer in a local rest-frame of the fluid) sees the fluid around him as isotropic (rotation-invariant). This means that in this reference system the components of the energy-momentum tensor have the form (any non-zero T_{0k} would break rotation invariance, and δ_{ik} is the unique rotation-invariant symmetric $(0,2)$-tensor)

$$T_{00} = \rho \quad , \quad T_{0k} = 0 \quad , \quad T_{ik} = p\delta_{ik} \ . \tag{7.3}$$

Here ρ and p are any functions of the coordinates, interpreted as the energy density and the pressure of the fluid .

To specify the kind of fluid one is working with, one should supplement this by an *equation of state* which provides a relation between ρ and p. Typically this amounts to specifying p as a function of ρ,

$$\text{Equation of State:} \quad p = p(\rho) \tag{7.4}$$

(and possibly other parameters).

In terms of the 4-velocity u^a of the fluid, which in the local rest frame has the components

$$u^a = (1,0,0,0) \ , \tag{7.5}$$

one can combine the components of the energy-momentum tensor into the expression,

$$T_{ab} = (\rho + p)u_a u_b + p\eta_{ab} \ , \tag{7.6}$$

(note that energy density and pressure = force per unit area have the same dimensions). As this is now a tensorial equation it is now valid in any inertial system. It defines the energy-momentum tensor of a perfect fluid. The conditions

$$\partial^a T_{ab} = 0 \tag{7.7}$$

imply a continuity equation and (as we will see below) a relativistic generalisation of the Euler equations for a perfect fluid. These are usually supplemented by a further continuity equation for the fluid density current

$$j^a = nu^a \tag{7.8}$$

with n e.g. the number density or particle density, say, namely

$$\partial_a j^a = 0 \ . \tag{7.9}$$

Now let us look at the consequences of these equations. Since

$$u^a u_a = -1 \quad \Rightarrow \quad (\partial_a u_b) u^b = \partial_a (u_b u^b)/2 = 0 \ , \tag{7.10}$$

the u-component of (7.7) can be written as

$$(\partial^a T_{ab}) u^b = 0 \quad \Leftrightarrow \quad u^a \partial_a \rho + (\rho + p) \partial_a u^a = 0 \ . \tag{7.11}$$

With the help of the current conservation equation, this equation can be recast into the form

$$\begin{aligned} 0 &= u^a \partial_a \rho + (\rho + p) \partial_a (j^a / n) \\ &= u^a \partial_a \rho + (\rho + p) j^a \partial_a (1/n) \\ &= u^a [\partial_a \rho + (\rho + p) n \partial_a (1/n)] \\ &= n u^a [p \partial_a (1/n) + \partial_a (\rho/n)] \ . \end{aligned} \tag{7.12}$$

The point of rewriting the equation in this way is that (assuming a situation of thermodynamic equilibrium) the 2nd law of thermodynamics says that pressure p, energy density ρ and the volume per particle $(1/n)$ are related by

$$T ds = p d(1/n) + d(\rho/n) \ . \tag{7.13}$$

where T is the temperature and s the specific entropy, i.e. the entropy per particle.[17] Thus the above equation says that the specific entropy s is constant along the flow,

$$u^a \partial_a s = 0 \ . \tag{7.14}$$

The significance of the spatial (transverse to u) components of (7.7) is easier to decipher if one writes the equations non-covariantly by setting

$$u^0 = \gamma(v) \quad , \quad u^i = \gamma(v) v^i = v^i u^0 \ , \tag{7.15}$$

so that

$$u^a \partial_a = \gamma(v) (\partial_t + \vec{v}.\vec{\nabla}) \tag{7.16}$$

is ($\gamma(v)$ times) the usual *convective derivative* or comoving time-derivative, and the above equation for the conservation of the specific entropy can be written as

$$(\partial_t + \vec{v}.\vec{\nabla}) s = 0 \ . \tag{7.17}$$

Moreover, the continuity equation for the current j^a with components

$$j^0 = \gamma(v) n \quad , \quad j^i = \gamma(v) n v^i \tag{7.18}$$

becomes

$$\partial_t (\gamma(v) n) + \vec{\nabla}.(\gamma(v) n \vec{v}) = 0 \ , \tag{7.19}$$

[17] See e.g. J. van Holten, *Relativistic Fluid Dynamics*, `http://www.nikhef.nl/~t32/relhyd.pdf` for a derivation of this and further discussion.

and the time-component of (7.7) can be written as

$$\partial_t(p-\gamma(v)^2(\rho+p))-\vec{\nabla}.[\gamma(v)^2(\rho+p)\vec{v}]=0 \ . \tag{7.20}$$

Using this equation the spacelike components of (7.7) can then be written as

$$\gamma(v)^2(\rho+p)(\partial_t\vec{v}+\vec{v}.\vec{\nabla}\vec{v})+\vec{v}\partial_t p+\vec{\nabla}p=0 \ . \tag{7.21}$$

In a suitable non-relativistic limit ($v \ll 1, p \ll \rho$), this latter equation reduces to

$$\rho(\partial_t\vec{v}+\vec{v}.\vec{\nabla}\vec{v})+\vec{\nabla}p=0 \ , \tag{7.22}$$

which is the non-relativistic Euler equation for a perfect fluid.

As we will see, it is straightforward to promote such a perfect fluid energy-momentum tensor by minimal coupling to the energy-momentum tensor describing a perfect fluid in a gravitational field, and we will come back to this below.

For a covariant rendition and elementary covariant derivation of the ensuing equations of motion in a general gravitational field from the conservation of the energy-momentum tensor, see e.g. the derivation of (35.81) and (35.82) in sections 35.4 and 35.5.

7.3 Noether Energy-Momentum Tensor in Special Relativity (Review)

A microscopic Lagrangian description is the method of choice when one has a Poincaré-invariant Lagrangian field theory description of the matter one is trying to describe. In particular, this applies to the scalar and Maxwell field theories we have already discussed and, more generally, to the modern microscopic and action-based description of the fundamental interactions of particle physics.

In this case, there is a canonical procedure for constructing an energy-momentum tensor, namely from Noether's theorem applied to translations, resulting in what is then appropriately known as the *Noether energy-momentum tensor* or the *canonical energy-momentum tensor* Θ_{ab}.

For a Lagrangian $L = L(\phi, \partial_a\phi)$ depending on some fields ϕ and their 1st derivatives (these could be scalar, vector, ... fields), this tensor is defined by

$$\Theta^a_{\ b}=-\frac{\partial L}{\partial(\partial_a\phi)}\partial_b\phi+\delta^a_{\ b}L \tag{7.23}$$

(sign conventions are such that Θ_{00} rather than $\Theta^0_{\ 0}$ is the energy density). It is built from the 4 Noether currents

$$\Theta^a_{\ b}\equiv J^a_{(b)} \tag{7.24}$$

associated to translation invariance in the x^b-direction, $\delta_{(b)}\phi=\partial_b\phi$. By calculating its divergence, one finds

$$\partial_a\Theta^a_{\ b}=\frac{\delta L}{\delta\phi}\partial_b\phi \ , \tag{7.25}$$

where $\delta L/\delta\phi$ is the Euler-Lagrange variational derivative,

$$\frac{\delta L}{\delta\phi} = \frac{\partial L}{\partial\phi} - \partial_a \frac{\partial L}{\partial(\partial_a\phi)} \ . \tag{7.26}$$

Thus Θ_{ab} is on-shell (meaning: for a solution to the matter equations of motion) conserved,

$$\partial_a \Theta^a_{\ b} = 0 \quad \text{on-shell} \ , \tag{7.27}$$

and leads to the conserved energy-momentum 4-vector

$$P_b = \int d^3x \ J^0_{(b)} = \int d^3x \ \Theta^0_{\ b} \ . \tag{7.28}$$

This procedure and prescription is perfectly adequate and sufficient for scalar (spin 0) fields, but it turns out to be far from satisfactory and far from the end of the story for other fields (e.g. for Maxwell theory, for which Θ_{ab} turns out to be neither symmetric nor gauge invariant). In this more general situation one is then required to "improve" this prescription in order to obtain an energy-momentum tensor T_{ab} with the desired properties.

As a first example where everything works out nicely, consider the energy-momentum tensor of a Klein-Gordon scalar field in Minkowski space. In this case,

$$\Theta_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L = \partial_a\phi\partial_b\phi - \tfrac{1}{2}\eta_{ab}\left(\eta^{cd}\partial_c\phi\partial_d\phi + m^2\phi^2\right) \tag{7.29}$$

with

$$\Theta_{00} = \tfrac{1}{2}(\dot\phi^2 + (\vec\nabla\phi)^2 + m^2\phi^2) \ . \tag{7.30}$$

This energy-momentum tensor is conserved for ϕ a solution to the equations of motion,

$$\left(\Box_\eta - m^2\right)\phi = 0 \quad \Rightarrow \quad \partial^a\Theta_{ab} = 0 \ . \tag{7.31}$$

This energy-momentum tensor is also manifestly symmetric (off-shell, i.e. without using the equations of motion),

$$\Theta_{ba} = \Theta_{ab} \ . \tag{7.32}$$

In particular, this implies that the angular momentum current associated to an infinitesimal Lorentz transformation (1.28) with parameters $\omega_{bc} = -\omega_{cb}$, namely

$$L^a = \tfrac{1}{2}\omega_{bc}L^{abc} \tag{7.33}$$

with

$$L^{abc} = x^b\Theta^{ac} - x^c\Theta^{ab} \ , \tag{7.34}$$

is on-shell conserved,

$$\partial_a L^{abc} = \Theta^{bc} - \Theta^{cb} = 0 \ . \tag{7.35}$$

Since Θ_{ab} is symmetric (and gauge invariance is not an issue), in this example there is no need to "improve" the Noether energy-momentum tensor, and we thus denote it by T_{ab},

$$T_{ab} = \Theta_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L \tag{7.36}$$

As we will see below, it is also straightforward to promote this tensor by minimal coupling to a (covariantly conserved) energy-momentum tensor of a scalar field in a gravitational field,

Now let us take a look at Maxwell theory in Minkowski space. In this case the canonical Noether energy-momentum tensor is

$$\Theta_{ab} = -\frac{\partial L}{\partial(\partial^a A_c)}\partial_b A_c + \eta_{ab}L = F_a{}^c\partial_b A_c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \ . \tag{7.37}$$

It is of course on-shell conserved by construction,

$$\partial^a\Theta_{ab} = 0 \quad \text{on-shell} \tag{7.38}$$

(note that both sets of Maxwell equations are required to derive this), but it is neither symmetric nor gauge-invariant. In particular, therefore, the angular momentum current (7.34) is not conserved (even though Maxwell theory is Lorentz invariant), and the expression for the energy-density is not gauge-invariant and does not agree with the standard expression

$$\Theta_{00} \neq \tfrac{1}{2}(\vec{E}^2 + \vec{B}^2) \ . \tag{7.39}$$

This can be rectified by manipulating Θ_{ab} as

$$\begin{aligned}\Theta_{ab} &= F_a{}^c(\partial_b A_c - \partial_c A_b) - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} + F_{ac}\partial^c A_b \\ &= F_a{}^c F_{bc} - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} + F_{ac}\partial^c A_b \ ,\end{aligned} \tag{7.40}$$

and noting that the last term can be written as a sum of two terms,

$$F_{ac}\partial^c A_b = \partial^c(F_{ac}A_b) - (\partial^c F_{ac})A_b \tag{7.41}$$

the first of which is identically conserved because of $F_{ac} = -F_{ca}$,

$$\partial^a\partial^c(F_{ac}A_b) = 0 \ , \tag{7.42}$$

and the second of which vanishes on-shell,

$$(\partial^c F_{ac})A_b = 0 \quad \text{on-shell.} \tag{7.43}$$

Therefore one can redefine the energy-momentum tensor in a first step to

$$\hat{\Theta}_{ab} = \Theta_{ab} - \partial^c(F_{ac}A_b) \tag{7.44}$$

and notes that this energy-momentum tensor is still conserved on-shell,

$$\partial^a\hat{\Theta}_{ab} = 0 \quad \text{on-shell} \ , \tag{7.45}$$

as well as on-shell gauge invariant,

$$\begin{aligned}\hat{\Theta}_{ab} &= F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} - (\partial^c F_{ac})A_b \\ &= F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \quad \text{on-shell} \ .\end{aligned} \tag{7.46}$$

Therefore one can define the "improved" energy-momentum tensor

$$T_{ab} = F_{ac}F_b{}^c - \tfrac{1}{4}\eta_{ab}F_{cd}F^{cd} \ , \tag{7.47}$$

in such a way that

- T_{ab} is still on-shell conserved,

$$\partial^a T_{ab} = 0 \quad \text{on-shell} \tag{7.48}$$

  (again both sets of Maxwell equations are required to establish this; with an external source,

$$\partial_{[a}F_{bc]} = 0 \quad , \quad \partial_a F^{ab} = -J^b \tag{7.49}$$

  one has the non-conservation law

$$\partial^a T_{ab} = J^a F_{ab} \tag{7.50}$$

  instead, which becomes a conservation law when one adds to T_{ab} the energy-momentum tensor of the source fields + interaction terms);

- T_{ab} is off-shell symmetric,

$$T_{ab} = T_{ba} \tag{7.51}$$

- T_{ab} is gauge-invariant and correctly gives the gauge-invariant and positive-definite energy-density as

$$T_{00} = \tfrac{1}{2}(\vec{E}^2 + \vec{B}^2) \ . \tag{7.52}$$

Moreover, the components of T_{0k} are the components of the Poynting vector and the spatial components T_{ik} are the components of the Maxwell stress tensor. Thus T_{ab} is *the* correct energy-momentum tensor of Maxwell theory.

This procedure to obtain T_{ab} from Θ_{ab} can be understood in a more general and systematic way, via the so-called Belinfante improvement (or symmetrisation) procedure. A brief synopsis of this construction will be provided in section 7.4 below.

One of the many useful properties of a symmetric, conserved energy-momentum tensor, and one that is occasionally used in general relativity, e.g. in the discussion of the energy and energy flux of gravitational waves, is the *Laue Theorem* (or *tensor virial theorem*). It states that for such an energy-momentum tensor and a localised source (so that one can integrate by parts with impunity) one has the relation

$$\left.\begin{array}{c}\partial^a T_{ab} = 0 \quad , \quad T_{ab} = T_{ba} \\ \text{localised source}\end{array}\right\} \quad \Rightarrow \quad \int d^3x \, T^{ik} = \tfrac{1}{2}(\partial_0)^2 \int d^3x \, T_{00} x^i x^k \tag{7.53}$$

between the integrated spatial components T_{ik} and the "quadrupole moments"

$$Q^{ik}(t) = \int d^3x\; T_{00} x^i x^k \tag{7.54}$$

of the energy density T_{00},

$$\int d^3x\; T^{ik} = \tfrac{1}{2}\ddot{Q}^{ik} \quad . \tag{7.55}$$

The proof of this identity is a straightforward repeated application of the conservation law and integration by parts. Indeed, using the symmetry, the time and space components of the conservation law

$$\partial^a T_{a0} = 0 \quad \Leftrightarrow \quad \partial_0 T_{00} = \partial^i T_{i0} \quad , \quad \partial^a T_{ak} = 0 \quad \Leftrightarrow \quad \partial_0 T_{0i} = \partial^k T_{ki} \tag{7.56}$$

and discarding boundary terms, one calculates

$$\begin{aligned}
\tfrac{1}{2}(\partial_0)^2 \int d^3x\; T_{00} x^i x^k &= +\tfrac{1}{2}\partial_0 \int d^3x\; \partial_0 T_{00} x^i x^k \\
&= +\tfrac{1}{2}\partial_0 \int d^3x\; (\partial_j T^j_{\;0}) x^i x^k \\
&= -\tfrac{1}{2}\partial_0 \int d^3x\; (T^i_{\;0} x^k + T^k_{\;0} x^i) \\
&= -\tfrac{1}{2} \int d^3x\; (\partial_0 T^i_{\;0} x^k + \partial_0 T^k_{\;0} x^i) \\
&= -\tfrac{1}{2} \int d^3x\; ((\partial_j T^{ij}) x^k + (\partial_j T^{kj}) x^i) \\
&= + \int d^3x\; T^{ik} \quad .
\end{aligned} \tag{7.57}$$

7.4 Synopsis of the Belinfante Improvement Procedure (Review)

The procedure to obtain a symmetric and conserved T_{ab} from the canonical Noether energy-momentum tensor Θ_{ab} of a Poincaré-invariant field theory, illustrated above in the case of Maxwell theory, can be understood in a more general and systematic, but also somewhat round-about way by appealing to the Lorentz-invariance of the action and taking into account the non-trivial transformation behaviour of the fields with spin $\neq 0$ under Lorentz transformations. This recipe is known as the *Belinfante improvement procedure*.[18] Here is, just for reference purposes, a brief description of the general features of this construction:

[18] This is explained in many places, with varying degree of comprehensibility or comprehension. For a detailed explanation, geared also towards applications to general relativity, see section 2 of T. Ortin, *Gravity and Strings*; for a succinct description, and an extension of the usual procedure to Lagrangians depending also on second derivatives of the fields, see section II of D. Bak, D. Cangemi, R. Jackiw, *Energy-Momentum Conservation in General Relativity*, `arXiv:hep-th/9310025`.

- In general (with the exception of spin zero scalar fields), $\Theta_{ab} = \eta_{ac}\Theta^c{}_b$ is not symmetric,
$$\Theta_{ab} \neq \Theta_{ba} \quad . \tag{7.58}$$

- As a consequence, the would-be angular momentum current (7.34) is now not on-shell conserved,
$$\partial_a L^{abc} = \Theta^{bc} - \Theta^{cb} \neq 0 \quad . \tag{7.59}$$

- By Lorentz invariance of the action and Noether's theorem, the total (orbital + spin) angular momentum should be conserved, and the above (purely orbital) angular momentum current fails to be conserved because it does not take into account the spin, i.e. the fact that the ϕ are possibly non-trivial Lorentz tensors (an irrelevant fact as far as the translational symmetries and hence the Noether energy-momentum tensor are concerned).

- This can be rectified by constructing the conserved total angular momentum current J^{abc} directly from Noether's theorem applied to Lorentz transformations $\delta\phi = \delta_L\phi$ of the fields and coordinates. This gives rise to an additional (spin) contribution to the current, schematically of the form
$$J^a = J^a_{\text{orbit}} + \frac{\partial L}{\partial(\partial_a\phi)}\delta_L\phi \quad , \quad J^{a[bc]}_{\text{orbit}} = L^{abc} \quad . \tag{7.60}$$
From the conservation of this current one can then via some gymnastics deduce and extract a candidate energy-momentum tensor $\hat{\Theta}_{ab}$ which is such that the total angular momentum current J^a takes the form
$$J^{abc} = x^b\hat{\Theta}^{ac} - x^c\hat{\Theta}^{ab} \quad . \tag{7.61}$$
Note that the spin-contribution to the total angular momentum has in this way been transformed into an orbital contribution with respect to the new energy-momentum tensor $\hat{\Theta}_{ab}$.

- This tensor $\hat{\Theta}_{ab}$ turns out to differ from the canonical energy-momentum tensor Θ_{ab} by an identically conserved term,
$$\hat{\Theta}_{ab} = \Theta_{ab} + \partial_c\Psi^{cab} \quad , \tag{7.62}$$
with
$$\Psi^{cab} = -\Psi^{acb} \quad \Rightarrow \quad \partial_a\partial_c\Psi^{cab} \equiv 0 \tag{7.63}$$
so that
$$\partial_a\Theta^a{}_b = 0 \quad \text{on-shell} \quad \Rightarrow \quad \partial_a\hat{\Theta}^a{}_b = 0 \quad \text{on-shell} \quad . \tag{7.64}$$

- Addition of such a term to the energy-momentum tensor is always possible as it does not violate the conservation law. While this changes the definition of the

local energy and momentum densities, with suitable fall-off conditions on the Ψ^{abc} this has no effect on the total energy-momentum P_b (7.28),

$$\begin{aligned} P^b \to P^b + \int d^3x\, \partial_c \Psi^{c0b} &= P^b + \int d^3x\, \partial_k \Psi^{k0b} \quad . \\ &= P^b + \oint dS_k \Psi^{k0b} \quad . \end{aligned} \tag{7.65}$$

- Angular momentum conservation together with (7.64) now implies

$$\partial_a J^{abc} = 0 \quad \text{on-shell} \quad \Rightarrow \quad \hat{\Theta}_{ab} = \hat{\Theta}_{ba} \quad \text{on-shell} \quad . \tag{7.66}$$

- Thus on-shell $\hat{\Theta}_{ab}$ agrees with a tensor T_{ab}, which can be chosen to be symmetric (off-shell) and on-shell conserved,

$$\begin{aligned} \hat{\Theta}_{ab} \to T_{ab} : \quad & T_{ab} = T_{ba} \quad \text{off-shell} \\ & \partial_a T^a_{\ b} = 0 \quad \text{on-shell} \quad . \end{aligned} \tag{7.67}$$

- This tensor T_{ab} (or occasionally just $\hat{\Theta}_{ab}$) is known as the *Belinfante improvement* of the energy-momentum tensor, and $\partial_c \Psi^{cab}$ as the (identically conserved) *improvement term*,

$$T_{ab} = \{\text{Improvement of } \Theta\}_{ab} \quad . \tag{7.68}$$

T_{ab} is then generally considered to be the "correct" choice of energy-momentum tensor for the Lagrangian field thory at hand, but it should be clear from the above discussion that this somewhat round-about procedure for finding and obtaining it leaves something to be desired (to put it mildly), already in the framework of Special Relativity.

7.5 Energy-Momentum Tensor from Minimal Coupling?

Given the success of the minimal coupling prescription, it is natural to try to define the matter energy-momentum tensor in a gravitational field in the same way. While this is certainly possible to a certain extent (as the examples will show), this procedure also leaves something to be desired (as the examples will also show).

Let us start by considering the "phenomenological" perfect fluid energy-momentum tensor (7.6),

$$T_{ab} = (\rho + p) u_a u_b + p \eta_{ab} \quad . \tag{7.69}$$

Following the minimal coupling rules, we promote this to the energy-momentum tensor

$$T_{\alpha\beta} = (\rho + p) u_\alpha u_\beta + p g_{\alpha\beta} \quad , \tag{7.70}$$

where u^α denotes the proper-time normalised velocity field of the fluid, $g_{\alpha\beta} u^\alpha u^\beta = -1$.

The covariantisation of the conservation law (7.7) evidently reads

$$\partial^a T_{ab} = 0 \quad \rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \ . \tag{7.71}$$

This generalises the continuity equation and the relativistic Euler equations to a fluid moving in a gravitational field and reduces to the special relativistic laws at the origin of a freely falling coordinate system, as it should.

There are neither conceptual nor technical complications in this example, and we will adopt this perfect fluid energy-momentum tensor, supplemented by an appropriate equation of state, to model the interior of a star (section 24.7) and the matter content of the universe (in our discussion of cosmology). In both of these examples, such a phenomenological description is quite appropriate and sufficient (although for more detailed investigations one may need to go beyond the perfect fluid approximation). For a detailed analysis of the conservation equations in the context of cosmology, see sections 35.4 and 35.6.

Let us now turn to energy-momentum tensors for Lagrangian field theories, starting with the example of the Klein-Gordon scalar field. As we saw above, in Minkowski space its (Noether = improved) energy-momentum tensor is given by

$$T_{ab} = \partial_a\phi\partial_b\phi + \eta_{ab}L = \partial_a\phi\partial_b\phi - \tfrac{1}{2}\eta_{ab}\left(\eta^{cd}\partial_c\phi\partial_d\phi + m^2\phi^2\right) \ . \tag{7.72}$$

The corresponding minimally coupled energy-momentum tensor in a gravitational field is then evidently

$$T_{\alpha\beta} = \partial_\alpha\phi\partial_\beta\phi + g_{\alpha\beta}L = \partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}g_{\alpha\beta}\left(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + m^2\phi^2\right) \ , \tag{7.73}$$

and it is easy to check that it is covariantly conserved for ϕ a solution to the equations of motion in a gravitational background,

$$\left(\Box_g - m^2\right)\phi = 0 \quad \Rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \ . \tag{7.74}$$

For the action (6.14) with a potential $V(\phi)$, the energy-momentum tensor of course also has the form (7.73) with $m^2\phi^2/2$ unsurprisingly replaced by $V(\phi)$,

$$T_{\alpha\beta} = \partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2}g_{\alpha\beta}\, g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - g_{\alpha\beta}V(\phi) \ , \tag{7.75}$$

with

$$\Box_g\phi = V'(\phi) \quad \Rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \ . \tag{7.76}$$

So far so good. However, the significance of this energy-momentum tensor outside the realm of special relativity is not clear. In special relativity, it encodes the conserved quantities associated to translation invariance, but in a general gravitational field there is no translation invariance (or other symmetry). In particular, in a general gravitational field

- one cannot even derive the energy-momentum tensor (7.73) from Noether's theorem applied to translations
- and, related to this is the fact that one does not obtain an ordinary conservation law but the covariant conservation law $\nabla_\alpha T^{\alpha\beta} = 0$.

Regarding the first point, it is fair to wonder if $T_{\alpha\beta}$ possesses an intrinsic gravitational significance beyond being merely the non-conserved minimally coupled counterpart of something that happens to have a significance in the absence of gravity. We will see below that, yes indeed, it is precisely the *source of gravity* arising from scalar fields.

Regarding the second point, we will see in sections 7.9 and 10.1 below that to any continous symmetry of a gravitational field (metric) and the covariantly conserved energy-momentum tensor one *can* associate a covariantly conserved current and thus also (as discussed in section 6.8) a conserved charge.

Now let us turn to Maxwell theory. Here the situation is a priori a bit murkier, because in principle we have both the canonical Noether energy-momentum tensor Θ_{ab} (7.37),

$$\Theta_{ab} = F_a{}^c \partial_b A_c - \tfrac{1}{4}\eta_{ab} F_{cd} F^{cd} \quad , \tag{7.77}$$

and its Belinfante-improved symmetric gauge-invariant sibling T_{ab} (7.47),

$$T_{ab} = F_{ac} F_b{}^c - \tfrac{1}{4}\eta_{ab} F_{cd} F^{cd} \quad , \tag{7.78}$$

at our disposal. Let us start with the latter, not only because it is the nicer object but also because it turns out to give the "correct" result. Applying the rules of minimal coupling, one finds the tensor

$$T_{\alpha\beta} = F_{\alpha\gamma} F_\beta{}^\gamma - \tfrac{1}{4} g_{\alpha\beta} F_{\gamma\delta} F^{\gamma\delta} \quad , \tag{7.79}$$

where indices of the (metric independent) field strength tensor $F_{\alpha\beta}$ are of course raised with the aid of the inverse metric $g^{\alpha\beta}$. This object turns out to have all the right properties to qualify as a candidate energy-momentum tensor of Maxwell theory in a gravitational field. In particular, it is off-shell symmetric and moreover on-shell covariantly conserved,

$$\nabla_\alpha T^{\alpha\beta} = 0 \quad \text{on-shell} \quad , \tag{7.80}$$

where "on-shell" of course refers to the equations of motion (6.39) in a gravitational field. Again both sets of vacuum Maxwell equations are required to verify this. In the presence of an external current, this is modified to

$$\nabla_\alpha T^{\alpha\beta} = J_\alpha F^{\alpha\beta} \quad \text{on-shell} \quad . \tag{7.81}$$

While one may have anticipated these last two equations on the basis of the minimal coupling recipe, it is important (and a useful exercise) to verify by direct calculation that

they indeed hold. The point of this verification is to make sure that no commutators of covariant derivatives, i.e. "curvature terms", arise in and mess up this equation, as they will in the calculation below involving the Noether energy-momentum tensor.

So let us take a brief look at the covariantised or minimally coupled Noether energy-momentum tensor, namely

$$\Theta_{\alpha\beta} = F_{\alpha}^{\ \gamma}\nabla_{\beta}A_{\gamma} - \tfrac{1}{4}g_{\alpha\beta}F_{\gamma\delta}F^{\gamma\delta} \quad . \tag{7.82}$$

While the canonical energy-momentum tensor in Minkowski space had some undesirable properties, its one redeeming feature was that it was on-shell conserved. In contrast to this, $\Theta_{\alpha\beta}$ is neither on-shell conserved nor on-shell covariantly conserved. In order to establish $\partial^a\Theta_{ab} = 0$ in Minkowski space, one uses the fact that partial derivatives commute. Thus, analogously, in calculating $\nabla^{\alpha}\Theta_{\alpha\beta}$ one encounters the *commutator of covariant derivatives*. Explicitly on-shell one finds

$$\nabla_{\alpha}\Theta^{\alpha}_{\ \beta} = \tfrac{1}{2}F^{\alpha\gamma}[\nabla_{\alpha},\nabla_{\gamma}]A_{\beta} \quad , \tag{7.83}$$

However, as we will discuss at length in section 8, the characteristic and defining feature of a non-trivial curved space-time is that these covariant derivatives do not commute when acting on tensors other than scalars (their commutator defining the curvature tensor of the space-time).

Likewise the covariant version of the improvement term in

$$\hat{\Theta}_{ab} = \Theta_{ab} - \partial^{c}(F_{ac}A_{b}) \quad , \tag{7.84}$$

namely $\nabla^{\gamma}(F_{\alpha\gamma}A_{\beta})$ is not identically conserved anymore, rather one has

$$\nabla^{\alpha}\nabla^{\gamma}(F_{\alpha\gamma}A_{\beta}) = \tfrac{1}{2}F^{\alpha\gamma}[\nabla_{\alpha},\nabla_{\gamma}]A_{\beta} \quad , \tag{7.85}$$

so that it would not qualify as an "improvement term" in the standard sense. Nevertheless, subtracting this term from the (non-conserved) Noether energy-momentum tensor, one finds that this indeed cancels the commutator term arising form (7.83), thus giving rise to an on-shell covariantly conserved $\hat{\Theta}_{\alpha\beta}$ or $T_{\alpha\beta}$. From the present perspective, however, this must be considered to be somewhat of a miracle or fluke. For some more comments on this, see section 22.2.

Thus we adopt (7.79),

$$T_{\alpha\beta} = F_{\alpha\gamma}F_{\beta}^{\ \gamma} - \tfrac{1}{4}g_{\alpha\beta}F_{\gamma\delta}F^{\gamma\delta} \quad , \tag{7.86}$$

as our (preliminary) definition of the energy-momentum tensor of Maxwell theory in a gravitational field, but we now face the same issue as in the case of scalar fields, namely the question what, if any, is the intrinsic gravitational significance of this energy-momentum tensor.

7.6 Covariant Energy-Momentum Tensor: the Source of Gravity

As we have seen, there are some irritating conceptual and technical issues associated with the "Noether + minimal coupling" procedure in general. These irritants turn out to be a good thing, though, because they motivate us to rethink this issue from scratch, and this will now lead us to a much more compelling and both conceptually and technically perfectly satisfactory general definition of the energy-momentum tensor of any Lagrangian field theory in a gravitational field.

Thus let us think about this issue from a Lagrangian, action-based, perspective. So far we have discussed what is the appropriate form of the action for matter fields in a gravitational field, namely a generally covariant action

$$S_{matter} = S_M[\phi; g_{\alpha\beta}] = \int \sqrt{g} d^4x \; L_M(\phi, \partial_\alpha \phi, \ldots, g_{\alpha\beta}, \ldots) \tag{7.87}$$

for the matter fields ϕ in a gravitational background $g_{\alpha\beta}$, obtained e.g. by the minimal coupling description and thus describing the dynamics of the fields in a gravitational background and encoding the coupling of the matter fields to gravity. Ultimately, this action should then be one part of the total gravitational + matter action describing the dynamics of the matter fields *and* of the gravitational field,

$$S_{total} = S_{gravity} + S_{matter} \quad . \tag{7.88}$$

Since the gravitational field is described by the (now dynamical) variables $g_{\alpha\beta}(x)$, we can write this marginally more explicitly as

$$S[g_{\alpha\beta}, \phi] = S_g[g_{\alpha\beta}] + S_M[\phi; g_{\alpha\beta}] \quad . \tag{7.89}$$

The precise form of the gravitational action S_g will not be relevant here - this is something that we will discuss at length in section 20. All we need to keep in mind is that this action is to provide us with the gravitational part of the gravitational field equations, i.e. with the appropriate tensorial generalisation of the left-hand side $\Delta\phi$ of the Newtonian field equation $\Delta\phi = 4\pi G_N \rho$.

Variation of this total action with respect to the matter fields ϕ is equivalent to the variation of the matter action S_M alone with respect to the matter fields,

$$\frac{\delta S[g_{\alpha\beta}, \phi]}{\delta \phi} = 0 \quad \Leftrightarrow \quad \frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta \phi} = 0 \quad , \tag{7.90}$$

and will thus simply give rise to the equations of motion of the matter fields in a gravitational field, as required.

Now let us consider the variation of the total action with respect to the gravitational dynamical variables $g_{\alpha\beta}$,

$$\frac{\delta S[g_{\alpha\beta}, \phi]}{\delta g_{\alpha\beta}} = \frac{\delta S_g[g_{\alpha\beta}]}{\delta g_{\alpha\beta}} + \frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta g_{\alpha\beta}} \stackrel{!}{=} 0 \tag{7.91}$$

Variation of the gravitational action with respect to the gravitational field $g_{\alpha\beta}$ will give us the gravitational part of the field equations. Thus variation of the matter action with respect to the gravitational field will give us the source term for the gravitational field equations provided by the matter fields,

$$\frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta g_{\alpha\beta}} = \text{Source of Gravity} \quad . \tag{7.92}$$

On the other hand, as recalled in the introduction to this section (section 7.1), we expect the energy-momentum tensor to act as the source of gravity. Therefore we should simply define the energy-momentum tensor by this relation,

$$\begin{aligned} & T^{\alpha\beta} := \text{Source of Gravity} \\ \Rightarrow \quad & T^{\alpha\beta} \sim \frac{\delta S_M[\phi; g_{\alpha\beta}]}{\delta g_{\alpha\beta}} \quad . \end{aligned} \tag{7.93}$$

We will fix the proportionality factor momentarily.

Note that this is precisely analogous to the way a source term for the Maxwell equations, a current J^α, arises from the variation of the coupled matter-Maxwell action with respect to the gauge field A_α (6.55),

$$J^\alpha \sim \frac{\delta S_M[\phi, A_\alpha]}{\delta A_\alpha} \quad . \tag{7.94}$$

In order to test this suggestion, let us take a look at our two standard examples, a scalar field and Maxwell theory. For a scalar field, the minimally coupled action is (6.14)

$$S[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \left(-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right) \quad . \tag{7.95}$$

Since the action depends explicitly on the inverse metric, it is more convenient to determine the variation of the action under variations

$$g^{\alpha\beta} \to g^{\alpha\beta} + \delta g^{\alpha\beta} \tag{7.96}$$

of the inverse metric. Under such a variation, the volume factor $\sqrt{g}$ varies as (5.87)

$$\delta\sqrt{g} = -\tfrac{1}{2}\sqrt{g} g_{\alpha\beta} \delta g^{\alpha\beta} \quad . \tag{7.97}$$

Thus the metric-variation of the scalar field action is

$$\delta S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int \sqrt{g} d^4x \left(\partial_\alpha \phi \partial_\beta \phi + g_{\alpha\beta} (-\tfrac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi)) \right) \delta g^{\alpha\beta} \quad , \tag{7.98}$$

Comparison with the minimally coupled energy-momentum tensor (7.75) of a scalar field,

$$T_{\alpha\beta} = \partial_\alpha \phi \partial_\beta \phi - \tfrac{1}{2} g_{\alpha\beta}\, g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - g_{\alpha\beta} V(\phi) \quad , \tag{7.99}$$

shows that this is precisely what we have obtained from the metric-variation of the matter action,

$$\delta S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int \sqrt{g} d^4x\, T_{\alpha\beta} \delta g^{\alpha\beta} \quad . \tag{7.100}$$

Now let us look at Maxwell theory, our litmus test. In this case, the action is (6.51)

$$S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4}\int \sqrt{g} d^4x \; g^{\mu\lambda} g^{\nu\rho} F_{\mu\nu} F_{\lambda\rho} \quad . \tag{7.101}$$

The variation of $\sqrt{g}$ is as before, and as regards the variation of the inverse metric, there is now an additional relative factor of two compared with the calculation for the scalar fields because the action depends quadratically on the inverse metric. Thus one has

$$\delta S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^4x \; \left(g^{\nu\rho} F_{\alpha\nu} F_{\beta\rho} - \tfrac{1}{4} g_{\alpha\beta} F_{\mu\nu} F^{\mu\nu} \right) \delta g^{\alpha\beta} \quad . \tag{7.102}$$

Comparing with (7.79),

$$T_{\alpha\beta} = F_{\alpha\gamma} F_\beta{}^\gamma - \tfrac{1}{4} g_{\alpha\beta} F_{\gamma\delta} F^{\gamma\delta} \quad , \tag{7.103}$$

we see that we once again have

$$\delta S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^4x \; T_{\alpha\beta} \delta g^{\alpha\beta} \quad . \tag{7.104}$$

Thus the metric variation of the matter action has given us on the nose the symmetric, gauge invariant, on-shell conserved energy-momentum tensor of Maxwell theory, without any need to appeal to any improvement procedures!

Thus, when it comes to defining the energy-momentum tensor for Maxwell theory, the above approach based on the variation of the matter action with respect to the metric wins hands down over the painful canonical definition based on Noether's theorem for translations and the Belinfante improvement procedure combined with minimal coupling.

Encouraged by this, we now define the energy-momentum tensor $T_{\alpha\beta}$ in general by

$$\delta_{\text{metric}} S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^4x \; T_{\alpha\beta} \delta g^{\alpha\beta} \quad , \tag{7.105}$$

or, equivalently,

$$T_{\alpha\beta} := -\frac{2}{\sqrt{g}} \frac{\delta}{\delta g^{\alpha\beta}} S_M[\phi, g_{\alpha\beta}] \quad . \tag{7.106}$$

Even though, as we have seen, there are other definitions of the energy-momentum tensor, this is the modern, and by far the most useful, definition of the energy-momentum tensor, namely as the response of the matter action to a variation of the metric (equivalently, as the source of gravity).

Moreover, crucially for the present context, whatever the virtues of other definitions may be, from the variational principle for general relativity it is this energy-momentum tensor that plays the role of the source term for the Einstein equations.

REMARKS:

1. The energy-momentum tensor as defined by (7.105) or (7.106) is frequently called the *metric* energy-momentum (or stress-energy) tensor, or also the *Hilbert* or

Rosenfeld energy-momentum tensor. It is sometimes also referred to as the gravitational energy-momentum tensor, but that is confusing as it does not describe the energy-momentum of the gravitational field itself, a more mysterious and elusive quantity we will briefly look at and for in section 22.6.

I prefer the attribute *covariant*, to distinguish it from what is usually called the *canonical* Noether energy-momentum tensor. Thus, even though this terminology is not standard, I will henceforth refer to $T_{\alpha\beta}$ as defined by (7.105) or (7.106), as the *Covariant Energy-Momentum Tensor*.

2. One of the many advantages of this definition is that it automatically and in general gives a symmetric and gauge invariant tensor (no improvement terms or similar gymnastics required). This is obvious from the definition.

3. This energy-momentum tensor turns out to also automatically be covariantly conserved (on-shell, i.e. for matter fields satisfying their Euler-Lagrange equations of motion). We will establish this latter fact in section 20.6 below where we will see that this is simply a consequence of the general covariance of the matter action S_M,

$$\text{general covariance of } S_M \quad \Rightarrow \quad \nabla^\alpha T_{\alpha\beta} = 0 \quad \text{on-shell} \ . \tag{7.107}$$

4. When the minimally coupled matter Lagrangian depends only on the metric and not on the first derivatives of the metric (i.e. not on the Christoffel symbols),

$$L_M(x) = L_M(\phi(x), \partial_\mu\phi(x), g_{\mu\nu}(x)) \ , \tag{7.108}$$

as in the case of scalar or Maxwell gauge fields, then more explicitly the covariant energy-momentum tensor can be written as (and calculated from)

$$T_{\mu\nu}(x) = -\frac{2}{\sqrt{g}}\frac{\partial(\sqrt{g}L_M(x))}{\partial g^{\mu\nu}(x)} = -2\frac{\partial L_M(x)}{\partial g^{\mu\nu}(x)} + g_{\mu\nu}(x)L_M(x) \tag{7.109}$$

or

$$T^{\mu\nu}(x) = 2\frac{\partial L_M(x)}{\partial g_{\mu\nu}(x)} + g^{\mu\nu}(x)L_M(x) \ . \tag{7.110}$$

Here the sign change is due to the fact that $\delta g^{\mu\nu}$ denotes the variation of the inverse metric, not the contravariant components of $\delta g_{\mu\nu}$. Thus it is not the same as $g^{\mu\lambda}g^{\nu\rho}\delta g_{\lambda\rho}$, but rather minus this expression,

$$\delta g^{\mu\nu} = -g^{\mu\lambda}g^{\nu\rho}\delta g_{\lambda\rho} \ , \tag{7.111}$$

as can be seen by varying $g^{\mu\nu}g_{\nu\lambda} = \delta^\mu_\lambda$,

$$0 = \delta(g^{\mu\nu}g_{\nu\lambda}) = (\delta g^{\mu\nu})g_{\nu\lambda} + g^{\mu\rho}\delta g_{\rho\lambda} \quad \Leftrightarrow \quad \delta g^{\mu\nu} = -g^{\mu\lambda}g^{\nu\rho}\delta g_{\lambda\rho} \ . \tag{7.112}$$

5. The definition (7.105) or the explicit expression (7.109) also provides an efficient strategy to determine the energy-momentum tensor even if one is just interested in Poincaré-invariant field theories in Minkowski space:

   In order to determine a symmetric, gauge invariant, and on-shell conserved energy-momentum tensor T_{ab} for such a theory, one

   - temporarily minimally couples the theory to a metric $g_{\alpha\beta}(x)$,
   - uses (7.105) or (7.106) or (7.109) to determine $T_{\alpha\beta}$,
   - and then replaces $g_{\alpha\beta} \to \eta_{ab}$ etc. again at the end.

   In equations, one defines T_{ab} by

   $$T_{ab} := (T_{\alpha\beta})|_{x^\alpha \to \xi^a, g_{\alpha\beta} \to \eta_{ab}} \quad . \tag{7.113}$$

   It can be shown that for fields of any spin this energy-momentum tensor agrees on-shell with what one could have also obtained by invoking the Belinfante improvement procedure of the Noether energy-momentum tensor,

   $$\hat{\Theta}_{ab} = T_{ab} \quad \text{on-shell} \tag{7.114}$$

   (see the discussion and references in section 7.4).

6. When the minimally coupled matter action depends also on the first derivatives of the metric, through the covariant derivative $\nabla_\mu \psi$ of some (non-scalar) field ψ, say, by the usual rules of variational calculus there will be additional contributions to the energy-momentum tensor, arising from an integration by parts of

   $$\int \sqrt{g} d^4x \, \left(-2 \frac{\partial L_M(x)}{\partial \nabla_\mu \psi(x)} (\delta \nabla_\mu \psi)(x) \right)$$

   where $\delta(\nabla_\mu \psi) = (\delta \nabla_\mu)\psi$ denotes the variation of the covariant derivative induced by the metric-variation (e.g. via the corresponding variation (20.14) of the Christoffel symbols). The precise form of the resulting contribution to the energy-momentum tensor depends on the tensorial type of ψ, is rarely needed, and it is unedifying to attempt to write down a general formula for this.

7.7 On the Energy-Momentum Tensor for Weyl-invariant Actions

Another general feature of the energy-momentum tensor that is readily understood by adopting the definition

$$T_{\alpha\beta} = -\frac{2}{\sqrt{g}} \frac{\delta S_{\text{matter}}}{\delta g^{\alpha\beta}} \quad . \tag{7.115}$$

is the relation between Weyl invariance and the trace of the energy-momentum tensor.

We consider the situation where the minimally coupled matter action happens to be invariant under *Weyl rescalings*, i.e. under rescalings of the metric

$$g_{\alpha\beta}(x) \to \mathrm{e}^{2\omega(x)} g_{\alpha\beta}(x) \tag{7.116}$$

by a positive definite function, or infinitesimally

$$\delta_\omega g_{\alpha\beta}(x) = 2\omega(x) g_{\alpha\beta}(x) \ . \tag{7.117}$$

In particular, thus, we consider the (admittedly very special) situation where one has such a symmetry without any accompanying transformation of the matter fields. The discussion can be extended to the case where also a transformation of the matter fields is required, but for present purposes this special case is good enough (see the end of this section for a comment on the general case).

Examples of such actions are e.g. the action of a massless scalar field (6.11) in $D = 2$ (space-time) dimensions

$$S[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int d^2x \ \sqrt{g} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi \tag{7.118}$$

and that of Maxwell theory (6.51) in $D = 4$ dimensions,

$$S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4} \int d^4x \ \sqrt{g} g^{\alpha\beta} g^{\gamma\delta} F_{\alpha\gamma} F_{\beta\delta} \ . \tag{7.119}$$

Indeed, in that case the metric dependence of the action is precisely such that the combination of of the determinant $\sqrt{g}$ and the inverse metric that appears is invariant under Weyl rescalings,

$$g_{\alpha\beta} \to \mathrm{e}^{2\omega} g_{\alpha\beta} \quad \Rightarrow \quad \begin{cases} D = 2 & \sqrt{g} g^{\alpha\beta} \to \sqrt{g} g^{\alpha\beta} \\ D = 4 & \sqrt{g} g^{\alpha\beta} g^{\gamma\delta} \to \sqrt{g} g^{\alpha\beta} g^{\gamma\delta} \end{cases} \tag{7.120}$$

This is reflected in the fact that the corresponding energy-momentum tensor is traceless precisely in these dimensions: from (7.73) and (7.79) one finds

$$\begin{aligned} T_{\alpha\beta} = \partial_\alpha \phi \partial_\beta \phi - \tfrac{1}{2} g_{\alpha\beta} \left(g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right) \quad &\Rightarrow \quad T^\alpha_{\ \alpha} = -\tfrac{1}{2}(D-2) g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \\ T_{\alpha\beta} = F_{\alpha\lambda} F_\beta^{\ \lambda} - \tfrac{1}{4} g_{\alpha\beta} F_{\lambda\sigma} F^{\lambda\sigma} \quad &\Rightarrow \quad T^\alpha_{\ \alpha} = -\tfrac{1}{4}(D-4) F_{\mu\nu} F^{\mu\nu} \ . \end{aligned} \tag{7.121}$$

The relation between these two observations / assertions is provided by noting that if the matter action is invariant under Weyl rescalings one has

$$\begin{aligned} 0 = \delta_\omega S_{\text{matter}} &= -\tfrac{1}{2} \int \sqrt{g} d^D x \ T_{\alpha\beta}(x) \delta_\omega g^{\alpha\beta}(x) \\ &= \int \sqrt{g} d^D x \ T_{\alpha\beta}(x) g^{\alpha\beta}(x) \omega(x) = \int \sqrt{g} d^D x \ T^\alpha_{\ \alpha}(x) \omega(x) \ . \end{aligned} \tag{7.122}$$

Since this is to be zero for all functions $\omega(x)$, this proves

$$\text{invariance under Weyl rescalings of the metric} \quad \Rightarrow \quad T^\alpha_{\ \alpha} = 0 \ . \tag{7.123}$$

In the special case that we have considered here (invariance under scalings of the metric alone, without transforming the matter fields), this is true off-shell, i.e. without using the equations of motion for the matter fields. In the more general case of an invariance under joint Weyl rescalings of the metric and accompanying scalings of the matter fields, in the above chain of arguments one would need to also vary the matter action with respect to the matter fields to establish the invariance of the action. The term arising from the variation of the matter fields is evidently proportional to the Euler-Lagrange equations of the matter fields, and therefore in that case one could only conclude that $T^\alpha_\alpha = 0$ on-shell,

$$\left.\begin{array}{c}\text{invariance under joint Weyl rescalings}\\ \text{of the metric and the matter fields}\end{array}\right\} \quad \Rightarrow \quad T^\alpha_\alpha = 0 \ \text{ on-shell.} \tag{7.124}$$

An example of this is provided by the so-called *conformally coupled scalar field*. This conformal coupling involves a space-time dependent mass term that represents a non-minimal coupling of the scalar field to the scalar curvature (a contraction of the Riemann curvature tensor to be introduced in section 8), and understanding the Weyl invariance of this model requires a formula for the variation of the scalar curvature with respect to the metric which we will derive in section 20.2. Therefore we will need to postpone a discussion of this model to section 22.3.

7.8 Klein-Gordon Scalar Field in (1+1) Minkowski and Rindler Space

As an aside, but as a concrete, and the simplest non-trivial, example, and an illustration of the above remarks regarding Weyl invariance, let us consider a massless scalar field in (1+1)-dimensions, in either the usual Minkowski coordinates, or in the Rindler coordinates discussed in sections 1.3 and 3.4 (we will in particular make use of the results in section 3.4).

In inertial coordinates in (1+1)-dimensional Minkowski space-time, $ds^2 = -dt^2 + dx^2$, the action and equation of motion of a massless scalar field are

$$S_M[\phi] = -\tfrac{1}{2}\int dtdx\ \eta^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi \quad \Rightarrow \quad \Box\phi = (-\partial_t^2 + \partial_x^2)\phi = 0\ . \tag{7.125}$$

Thus a natural basis of solutions to this equation is provided by the plane waves $f_k \sim \exp(-i\omega t + ikx)$, with $k^2 = \omega^2$, i.e. $k = \pm\omega, \omega > 0$, and their complex conjugates. For a given ω there are thus two linearly-independent positive frequency solutions,

$$\begin{aligned} f_\omega(t,x) &= \frac{1}{(4\pi\omega)^{1/2}}\mathrm{e}^{-i\omega(t-x)} \\ g_\omega(t,x) &= \frac{1}{(4\pi\omega)^{1/2}}\mathrm{e}^{-i\omega(t+x)} \end{aligned} \tag{7.126}$$

(the normalisation factors are inserted for QFT-pedantry reasons only and are irrelevant for the following). Thus the basis of solutions splits into *right-movers* or *right-moving*

modes f_ω and *left-movers* g_ω. It is thus convenient to introduce the corresponding null coordinates $u_M = t - x, v_M = t + x$ as in (3.94), in terms of which the solutions can be written as

$$\begin{aligned} f_\omega = f_\omega(u_M) &= \frac{1}{(4\pi\omega)^{1/2}} \mathrm{e}^{-i\omega u_M} \\ g_\omega = g_\omega(v_M) &= \frac{1}{(4\pi\omega)^{1/2}} \mathrm{e}^{-i\omega v_M} \quad . \end{aligned} \tag{7.127}$$

That the solutions split in this way could have also been deduced from the form of the wave operator in these lightcone (null) coordinates, namely $\Box = -4\partial_{u_M}\partial_{v_M}$, and the ensuing solutions to the equation of motion,

$$\Box\phi = 0 \quad \Rightarrow \quad \phi = f(u_M) + g(v_M) \quad . \tag{7.128}$$

Here f and g can now be arbitrary wave packets constructed from the solutions f_ω and g_ω respectively.

The energy-density $\rho_M = T_{tt}$ of the scalar field with respect to Minkowski time is

$$\rho_M = \tfrac{1}{2}((\partial_t\phi)^2 + (\partial_x\phi)^2) \tag{7.129}$$

and in terms of lightcone coordinates this splits into a sum of left-moving and right-moving contributions,

$$\rho_M = (\partial_{u_M}\phi)^2 + (\partial_{v_M}\phi)^2 \quad , \tag{7.130}$$

with $f(u_M)$ evidently only contributing to the former and $g(v_M)$ to the latter.

Now let us consider the same issue in Rindler coordinates. In terms of the coordinates (η, ξ) (3.89), the metric takes the form (3.87)

$$ds^2 = \mathrm{e}^{2a\xi}(-d\eta^2 + d\xi^2) \quad . \tag{7.131}$$

Note that, as mentioned in section 3.4, the metric in these coordinates is conformally flat. Thus, by the reasoning above, in section 7.7, in particular the discussion around equation (7.120), we know that the action and equation of motion for a scalar field in Rindler coordinates will look just like those in Minkowski coordinates, with the replacement $(t, x) \to (\eta, \xi)$,

$$S_R[\phi] = -\tfrac{1}{2}\int \sqrt{g}d\eta d\xi \; g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi = -\tfrac{1}{2}\int d\eta d\xi \; \eta^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi \quad , \tag{7.132}$$

and

$$\Box_g\phi = 0 \quad \Leftrightarrow \quad (-\partial_\eta^2 + \partial_\xi^2)\phi = 0 \quad . \tag{7.133}$$

Thus by the same reasoning as above, the solutions can be split into left- and right-movers and are conveniently written in terms of the Rindler lightcone coordinates (3.95)

$$(u_R, v_R) = \eta \mp \xi \quad , \tag{7.134}$$

i.e. one has

$$\Box_g \phi = 0 \quad \Leftrightarrow \quad \partial_{u_R}\partial_{v_R}\phi = 0 \quad \Leftrightarrow \quad \phi = f(u_R) + g(v_R) \ . \tag{7.135}$$

The energy-density $\rho_R = T_{\eta\eta}$ of the scalar field with respect to Rindler time is

$$\rho_R = \tfrac{1}{2}((\partial_\eta \phi)^2 + (\partial_\xi \phi)^2) \tag{7.136}$$

and in terms of lightcone coordinates this splits into a sum of left-moving and right-moving contributions,

$$\rho_R = (\partial_{u_R}\phi)^2 + (\partial_{v_R}\phi)^2 \ , \tag{7.137}$$

with $f(u_R)$ evidently only contributing to the former and $g(v_R)$ to the latter.

The interest in these (fairly trivial) considerations lies in the fact that the exponential relation (3.97) between the Minkowski and Rindler null coordinates

$$u_M = -a^{-1}\mathrm{e}^{-au_R} \quad , \quad v_M = +a^{-1}\mathrm{e}^{+av_R} \ , \tag{7.138}$$

reflecting the exponential redshift of a Rindler relative to an inertial observer (and vice-versa) has a number of non-trivial and remarkable implications. I will just mention 2 of them here:

1. The exponential redshift expressed by (7.138) implies that the right-moving energy densities in Minkowski and Rindler coordinates are related by

$$\frac{\partial u_M}{\partial u_R} = -au_M \quad \Rightarrow \quad (\partial_{u_M}\phi)^2 = \frac{1}{a^2 u_M^2}(\partial_{u_R}\phi)^2 \tag{7.139}$$

   (and likewise for the left-movers). Thus essentially any classical solution that is regarded as regular by the Rindler observer (finite and non-zero ρ_R) corresponds to a divergent Minkowski energy-density as $u_M \to 0$, i.e. on the future boundary (horizon) $t = x$ of the Rindler wedge.

2. The exponential redshift expressed by (7.138) also implies that the notions of *positive frequency* with respect to Minkowski and Rindler time are inequivalent, e.g. in the sense that $f_\omega(u_M)$, restricted to the right Rindler-wedge $u_M < 0$, say, cannot be written as a superposition of Rindler right-moving positive frequency waves alone,

$$f_\omega(u_M) \neq \int_0^\infty d\omega' \alpha(\omega,\omega') f_{\omega'}(u_R) \ . \tag{7.140}$$

   Of course, the $f_\omega(u_R)$ and their complex conjugates $f^*_\omega(u_R)$ provide a basis of solutions for the right-moving modes (in the right Rindler-wedge), so that one can certainly expand the Minkowski plane waves as

$$f_\omega(u_M) = \int_0^\infty d\omega' \ \left(\alpha(\omega,\omega') f_{\omega'}(u_R) + \beta(\omega,\omega') f^*_{\omega'}(u_R)\right) \ , \tag{7.141}$$

but necessarily with some of the $\beta(\omega, \omega') \neq 0$.

If you know a little bit of quantum field theory, you will be able to anticipate that this means that the notions of creation and annihilation operators are inequivalent, and that therefore what is the vacuum, say, for an inertial observer, will not be seen as the vacuum by the accelerating observer (and vice-versa).

Combining the two facts, one also arrives at the conclusion that the "Rindler vacuum" is singular both at the future horizon (from right-movers) and at the past horizon (from left-movers).

In the spirit of the equivalence principle ("before studying gravity, let us study accelerations in flat space"), this *Unruh Effect* is a fascinating and rewarding first step towards understanding (or appreciating the difficulties encountered by) quantum field theory in curved space-times, i.e. in non-trivial gravitational fields. For more on this see the references given in section 27.7.

As further examples of scalar fields in particular gravitational backgrounds, in section 26.8 we will consider scalar fields in the Schwarzschild space-time, and in section 34.10 we will look at the equations of motion of scalar fields in a cosmological gravitational background.

7.9 Conserved Currents from the Energy-Momentum Tensor?

In section 6.8 we had discussed how to obtain conserved charges from covariantly conserved currents. Now in special relativity one can construct conserved currents (corresponding to the generators of Poincaré transformations) from the conserved energy-momentum tensor, and hence from there the corresponding conserved charges like energy, momentum and angular momentum. In this section we will take a first look at the question if or to which extent we can also obtain such conserved currents from the covariantly conserved energy-momentum tensor in a gravitational field.

To set the stage, recall that in Special Relativity, if T^{ab} is the energy-momentum tensor of a physical system, it generally satisfies an equation of the form

$$\partial_a T^{ab} = G^b \quad , \tag{7.142}$$

where G^b represents the density of the external forces acting on the system. In particular, if there are no external forces, the divergence of the energy-momentum tensor is zero. For example, in the case of Maxwell theory and a current corresponding to a charged particle we have

$$G^b = J_a F^{ab} = -F^a_{\ b} J^b \sim -F^a_{\ b} \dot{\xi}^b \quad , \tag{7.143}$$

which is indeed the relevant external (Lorentz) force density (in writing this I have suppressed the δ-function that localises the current to the worldline $\xi^a = \xi^a(\tau)$ of the particle).

When there are no external forces, i.e. when one has taken into account the complete matter action, the total energy-momentum tensor is conserved. In that case, $T^{ab} = J^{(b)a}$ defines four conserved currents, more or less (modulo Belinfante improvement terms, see e.g. the discussion in sections 7.4 and 22.2 and the references given there) the currents associated to translation invariance of the action via Noether's theorem. One is thus in the setting of conserved currents of section 6.8, and one can define conserved quantities like total energy and momentum, P^a, and angular momentum J^{ab}, by integrals of T^{0a} or $\xi^a T^{0b} - \xi^b T^{0a}$ (the latter being conserved if T_{ab} is symmetric) over spacelike hypersurfaces.

The situation in general relativity is somewhat different (exactly how different it is perceived to be is partly a matter of personal preconceptions or desires). In particular, in general relativity, and assuming that $T_{\mu\nu}$ is the complete matter energy-momentum tensor (otherwise we certainly cannot expect to derive any conservation law), we will have a "conservation law" of the form

$$\nabla_\mu T^{\mu\nu} = g^{-1/2}\partial_\mu(g^{1/2}T^{\mu\nu}) + \Gamma^\nu{}_{\mu\lambda}T^{\mu\lambda} = 0 \ . \tag{7.144}$$

We see that, due to the second term, this does not define four conserved currents in the ordinary or covariant sense (and we will return to the interpretation of this equation, and the related issue of energy and energy density of the gravitational field, in section 22.6).

Nevertheless, in analogy with special relativity, one might like to attempt to define conserved quantities like total energy and momentum, P^μ, and angular momentum $J^{\mu\nu}$, by integrals of $T^{0\mu}$ or $x^\mu T^{0\nu} - x^\nu T^{0\mu}$ over spacelike hypersurfaces. However, these quantities are rather obviously not covariant, and nor are they conserved.

This should perhaps not be too surprising because, after all, for a Poincaré-invariant field theory in Minkowski space these quantities are preserved as a consequence of Poincaré invariance, i.e. because of the symmetries (isometries) of the Minkowski metric (as well as of the action).

A generic metric has no isometries whatsoever (the explicit examples of metrics in these notes not withstanding, all of which exhibit at least some symmetries). As it has no symmetries, we have no reason to expect to find associated conserved quantities in general.

However, if there are symmetries then one should indeed be able to define conserved quantities (think of Noether's theorem again), one for each symmetry generator. In order to implement this we need to understand how to define and detect isometries of

the metric. For this we need the concepts of Lie derivatives and Killing vectors. These already made occasional brief appearances in previous sections and will be discussed more systematically in section 9, the corresponding conserved charges then being the subject of section 10.

Alternatively, one might try to just go ahead optimistically and attempt to construct a covariant current-like object (with a corresponding conservation law and the ensuing possibility to define conserved charges) by contracting the energy-momentum tensor not with the coordinates but with a vector field V^λ, along the lines of

$$J^\mu_V = T^\mu_\lambda V^\lambda \ . \tag{7.145}$$

At least this now has the merit of clearly being a vector field, but is it conserved? Calculating its covariant divergence, and using the fact that $T^{\mu\nu}$ is symmetric and conserved, one finds

$$\nabla_\mu J^\mu_V = \tfrac{1}{2} T^{\mu\nu}(\nabla_\mu V_\nu + \nabla_\nu V_\mu) \ . \tag{7.146}$$

Thus we would have a conserved current (and associated conserved charge by the previous section) for any conserved energy-momentum tensor if the vector field V^λ were such that it satisfies

$$\nabla_\mu V_\nu + \nabla_\nu V_\mu = 0 \quad \Rightarrow \quad \nabla_\mu (T^\mu_\lambda V^\lambda) = 0 \ . \tag{7.147}$$

The link between this observation and the one in the preceding paragraph regarding symmetries is that this is precisely the condition characterising (infinitesimal) symmetries of metric:

- First of all, this is the condition we already found and encountered in (3.35), as reformulated in (5.68), for the infinitesimal coordinate transformation $\delta x^\mu = \epsilon V^\mu$ to generate a symmetry of the metric, thus leading to a conserved charge for geodesics.
- More generally, as we will discuss in detail in section 9 below, vector fields satisfying the equation $\nabla_\mu V_\nu + \nabla_\nu V_\mu = 0$ are indeed in one-to-one correspondence with infinitesimal generators of continuous symmetries of a metric (isometries).

Thus this gives a satisfactory and coherent overall picture of symmetries and conservation laws in a gravitational field.

8 Curvature I: The Riemann Curvature Tensor

8.1 Curvature: Preliminary Remarks

We now come to one of the most important concepts of General Relativity and Riemannian Geometry, that of curvature and how to describe it in tensorial terms. Among other things, this will finally allow us to decide unambiguously if a given metric is just the (flat) Minkowski metric in disguise or the metric of a genuinely curved space (but a proof of this statement is postponed to section 11). More importantly (for present purposes) it will allow us to construct tensors that depend on the 2nd derivatives of the metric and will thus allow us to construct tensorial (generally covariant) differential equations for the metric. In particular, this will then lead us fairly directly to the Einstein equations (section 19), i.e. to the field equations for the gravitational field.

Recall that the equations that describe the behaviour of particles and fields in a gravitational field involve the metric and the Christoffel symbols determined by the metric. Thus the equations for the gravitational field should be generally covariant (tensorial) differential equations for the metric.

At first, here we seem to face a dilemma. How can we write down covariant differential equations for the metric when the covariant derivative of the metric is identically zero? Having come to this point, Einstein himself reached an impasse and required the help of his mathematician friend Marcel Grossmann ("Grossmann, you have to help me, or else I'll go crazy!") whom he had asked to investigate if there were any tensors that could be built from the second derivatives of the metric.

Grossmann soon found that this problem had indeed been addressed and solved in the mathematics literature, in particular by Riemann (generalising work of Gauss on curved surfaces), Ricci-Curbastro and Levi-Civita. It was shown by them that there are indeed non-trivial tensors that can be constructed from (ordinary) derivatives of the metric. These can then be used to write down covariant differential equations for the metric.[19]

The most important among these are the Riemann curvature tensor and its various contractions. In fact, it is known that these are the only tensors that can be constructed from the metric and its first and second derivatives, and they will therefore play a central role in all that follows.

Technically the most straightforward way of introducing the Riemann curvature tensor is via the commutator of covariant derivatives. In this section we will adopt this pragmatic (and relatively streamlined) approach, as it is sufficient to

[19] Of course, the story is not as simple and straightforward as that. For an account of Marcel Grossmann's (often overlooked) contributions to tensor calculus and the development of general relativity, see T. Sauer, *Marcel Grossmann and his contribution to the general theory of relativity*, `arXiv:1312.4068 [physics.hist-ph]`.

- determine the most important algebraic and differential properties of the curvature tensor (symmetries and Bianchi identities)
- assess its physical significance (gravitational tidal forces) via the influence of the curvature tensor on the motion of (families of) freely falling particles
- and to thus provide us with all the information and ingredients we need to then discuss the Einstein equations (section 19) and their formulation in terms of an action principle (section 20).

However, this is not geometrically the most intuitive way to introduce the concept of curvature, and it downplays the extent to which the curvature tensor reflects and encodes the geometric properties of space-time and, more generally, does not do justice to the fundamental differential geometric notion and significance of curvature. Some of these aspects are discussed in Part B of these notes, in particular in sections 11, 12, 13 and 14.

8.2 Riemann Tensor from the Commutator of Covariant Derivatives

As mentioned before, second covariant derivatives do not commute on (p,q)-tensors unless $p = q = 0$. However, the fact that they *do* commute on scalars has the pleasant consequence that e.g. the commutator of covariant derivatives acting on a vector field V^μ does not involve any derivatives of V^μ. In fact, I will first show, without actually calculating the commutator, that

$$[\nabla_\mu, \nabla_\nu](\phi V^\lambda) = \phi[\nabla_\mu, \nabla_\nu]V^\lambda \tag{8.1}$$

for any scalar field ϕ. This implies that $[\nabla_\mu, \nabla_\nu]V^\lambda$ cannot depend on derivatives of V because if it did it would also have to depend on derivatives of ϕ.

Hence, the commutator can be expressed purely algebraically in terms of V. As the dependence on V is clearly linear, the commutator of covariant derivatives must then act like a linear transformation. There must therefore be an object $R^\lambda{}_{\sigma\mu\nu}$ such that

$$[\nabla_\mu, \nabla_\nu]V^\lambda = R^\lambda{}_{\sigma\mu\nu}V^\sigma \quad . \tag{8.2}$$

This can of course also be verified by a direct calculation, and we will come back to this below. For now let us just note that, since the left hand side of this equation is clearly a tensor for any V, the quotient theorem implies that the quantities $R^\lambda_{\sigma\mu\nu}$ are the components of a tensor.

Let us first verify (8.1). We have

$$\nabla_\mu\nabla_\nu\phi V^\lambda = (\nabla_\mu\nabla_\nu\phi)V^\lambda + (\nabla_\nu\phi)(\nabla_\mu V^\lambda) + (\nabla_\mu\phi)(\nabla_\nu V^\lambda) + \phi\nabla_\mu\nabla_\nu V^\lambda \quad . \tag{8.3}$$

Thus, upon taking the commutator the 2nd and 3rd terms drop out (because the 3rd is the symmetrisation of the 2nd), and we are left with

$$\begin{aligned}[\nabla_\mu, \nabla_\nu]\phi V^\lambda &= ([\nabla_\mu, \nabla_\nu]\phi)V^\lambda + \phi[\nabla_\mu, \nabla_\nu]V^\lambda \\ &= \phi[\nabla_\mu, \nabla_\nu]V^\lambda \ , \end{aligned} \tag{8.4}$$

where the last line follows from the fact that 2nd covariant derivatives do commute on scalars. Thus we have established (8.1).

By explicitly calculating the commutator, one can confirm the structure displayed in (8.2). This explicit calculation shows that the *Riemann-Christoffel Curvature Tensor* (or Riemann tensor for short) is given by

$$\boxed{R^\lambda{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\lambda{}_{\sigma\nu} - \partial_\nu \Gamma^\lambda{}_{\sigma\mu} + \Gamma^\lambda{}_{\mu\rho}\Gamma^\rho{}_{\nu\sigma} - \Gamma^\lambda{}_{\nu\rho}\Gamma^\rho{}_{\mu\sigma}} \tag{8.5}$$

REMARKS:

1. Note how useful the quotient theorem is in this case. It would be quite unpleasant to have to verify the tensorial nature of this expression by explicitly checking its behaviour under coordinate transformations.

2. Note also that this tensor is clearly zero for the Minkowski metric written in Cartesian coordinates. Hence it is also zero for the Minkowski metric written in any other coordinate system. We will prove the converse, that vanishing of the Riemann curvature tensor implies that the metric is (locally) equivalent to the Minkowski metric, in section 11.2.

3. In the above we have defined the Riemann tensor by the relation (8.2) and then deduced the explicit expression (8.5). While this is, pragmatically speaking, a useful way of proceeding, it may be more logical to initially define the Riemann tensor in a different way, e.g. directly by (8.5) (for instance because by painful calculations one has discovered that this particular combination of non-tensorial objects miraculously happens to transform as a tensor). In that case, (8.2) is a result rather than a definition, known as the *Ricci identity*.

It is straightforward to extend the above to an action of the commutator $[\nabla_\mu, \nabla_\nu]$ on arbitrary tensors. For covectors we have, since we can raise and lower the indices with the metric with impunity,

$$\begin{aligned}[\nabla_\mu, \nabla_\nu]V_\rho &= g_{\rho\lambda}[\nabla_\mu, \nabla_\nu]V^\lambda \\ &= g_{\rho\lambda}R^\lambda{}_{\sigma\mu\nu}V^\sigma \\ &= R_{\rho\sigma\mu\nu}V^\sigma \\ &= R_\rho{}^\sigma{}_{\mu\nu}V_\sigma \ . \end{aligned} \tag{8.6}$$

We will see later that the Riemann tensor is anti-symmetric in its first two indices. Hence we can also write

$$[\nabla_\mu, \nabla_\nu] V_\rho = -R^\sigma{}_{\rho\mu\nu} V_\sigma \quad . \tag{8.7}$$

The extension to arbitrary (p, q)-tensors now follows the usual pattern, with one Riemann curvature tensor, contracted as for vectors, appearing for each of the p upper indices, and one Riemann curvature tensor, contracted as for covectors, for each of the q lower indices. Thus, e.g. for a $(2, 0)$-tensor $T^{\alpha\beta}$ one has

$$[\nabla_\mu, \nabla_\nu] T^{\alpha\beta} = R^\alpha{}_{\gamma\mu\nu} T^{\gamma\beta} + R^\beta{}_{\gamma\mu\nu} T^{\alpha\gamma} \tag{8.8}$$

and for a (1,1)-tensor A^λ_ρ one has

$$[\nabla_\mu, \nabla_\nu] A^\lambda_\rho = R^\lambda{}_{\sigma\mu\nu} A^\sigma_\rho - R^\sigma{}_{\rho\mu\nu} A^\lambda_\sigma \quad . \tag{8.9}$$

I will give two other versions of the fundamental formula (8.2) which are occasionally useful and used.

1. Instead of looking at the commutator $[\nabla_\mu, \nabla_\nu]$ of two derivatives in the coordinate directions x^μ and x^ν, we can look at the commutator $[\nabla_X, \nabla_Y]$ of two directional covariant derivatives. Evidently, in calculating this commutator one will pick up new terms involving $\nabla_X Y^\mu - \nabla_Y X^\mu$. Denoting this vector field by $[X, Y]^\mu$ (the rationale for this notation will be explained in section 9.3), we can also write the formula for the curvature tensor as

$$([\nabla_X, \nabla_Y] - \nabla_{[X,Y]}) V^\lambda = R^\lambda{}_{\sigma\mu\nu} X^\mu Y^\nu V^\sigma \quad . \tag{8.10}$$

2. Secondly, one can consider a net of curves $x^\mu(s_1, s_2)$ parametrising, say, a two-dimensional surface, and look at the commutators of the covariant derivatives along the s_1- and s_2-curves. The formula one obtains in this case (it can be obtained from (8.10) by noting that X and Y commute in this case) is

$$(D_{s_1} D_{s_2} - D_{s_2} D_{s_1})\, V^\lambda = R^\lambda{}_{\sigma\mu\nu} \frac{dx^\mu}{ds_1} \frac{dx^\nu}{ds_2} V^\sigma \quad , \tag{8.11}$$

   where D_{s_k} denotes the covariant derivative along the curve parametrised by s_k, i.e. (section 5.7)

$$D_{s_k} = \frac{\partial x^\mu(s_1, s_2)}{\partial s_k} \nabla_\mu \quad . \tag{8.12}$$

8.3 Symmetries and Algebraic Properties of the Riemann Tensor

A priori, the Riemann tensor has $256 = 4^4$ components in 4 dimensions. However, because of a large number of symmetries, the actual number of independent components is much smaller.

In general, to read off all the symmetries from the formula (8.5) is difficult. One way to simplify things is to look at the Riemann curvature tensor at the origin x_0 of a Riemann normal coordinate system (or some other inertial coordinate system). In that case, all the first derivatives of the metric disappear and only the first two terms of (8.5) contribute. One finds

$$\begin{aligned} R_{\alpha\beta\gamma\delta}(x_0) &= g_{\alpha\lambda}(\partial_\gamma \Gamma^\lambda_{\beta\delta} - \partial_\delta \Gamma^\lambda_{\beta\gamma})(x_0) \\ &= (\partial_\gamma \Gamma_{\alpha\beta\delta} - \partial_\delta \Gamma_{\alpha\beta\gamma})(x_0) \\ &= \tfrac{1}{2}(g_{\alpha\delta,\beta\gamma} + g_{\beta\gamma,\alpha\delta} - g_{\alpha\gamma,\beta\delta} - g_{\beta\delta,\alpha\gamma})(x_0) \ . \end{aligned} \tag{8.13}$$

In principle, this expression is sufficiently simple to allow one to read off all the symmetries of the Riemann tensor. However, it is more insightful to derive these symmetries in a different way, one which will also make clear *why* the Riemann tensor has these symmetries.

1. Anti-symmetry in the second pair of indices:

$$R_{\alpha\beta\gamma\delta} = -R_{\alpha\beta\delta\gamma} \tag{8.14}$$

   This is obviously true from the definition or by construction.

2. Anti-symmetry in the first pair of indices:

$$R_{\alpha\beta\gamma\delta} = -R_{\beta\alpha\gamma\delta} \tag{8.15}$$

   This is a consequence of the fact that the metric is covariantly constant. In fact, we can calculate

$$\begin{aligned} 0 &= [\nabla_\gamma, \nabla_\delta] g_{\alpha\beta} \\ &= R_{\alpha\ \gamma\delta}^{\ \lambda} g_{\lambda\beta} + R_{\beta\ \gamma\delta}^{\ \lambda} g_{\alpha\lambda} \\ &= (R_{\alpha\beta\gamma\delta} + R_{\beta\alpha\gamma\delta}) \ . \end{aligned} \tag{8.16}$$

   As mentioned before, this implies that we can write the commutator of covariant derivatives on a covector as

$$[\nabla_\mu, \nabla_\nu] V_\rho = R_{\rho\ \mu\nu}^{\ \sigma} V_\sigma = -R^\sigma_{\ \rho\mu\nu} V_\sigma \ . \tag{8.17}$$

3. Cyclic permutation symmetry (or first Bianchi identity)

$$R_{\alpha[\beta\gamma\delta]} = 0 \quad \Leftrightarrow \quad R_{\alpha\beta\gamma\delta} + R_{\alpha\delta\beta\gamma} + R_{\alpha\gamma\delta\beta} = 0 \tag{8.18}$$

   This Bianchi identity is a consequence of the fact that there is no torsion. In fact, applying $[\nabla_\gamma, \nabla_\delta]$ to the covector $\nabla_\beta \phi$, ϕ a scalar, one has

$$\nabla_{[\gamma} \nabla_\delta \nabla_{\beta]} \phi = 0 \quad \Rightarrow \quad R^\lambda_{\ [\beta\gamma\delta]} \nabla_\lambda \phi = 0 \ . \tag{8.19}$$

As this has to be true for all scalars ϕ, this implies $R_{\alpha[\beta\gamma\delta]} = 0$ (to see this you could e.g. choose the (locally defined) coordinate functions $\phi^{(\mu)}(x) = x^\mu$ with $\nabla_\lambda \phi^{(\mu)} = \delta^\mu_\lambda$).

In turn this identity now implies that for any covector (not necessarily a gradient) one has the identity (also called Bianchi identity)

$$\nabla_{[\gamma}\nabla_\delta V_{\beta]} = -R^\alpha_{\ [\beta\gamma\delta]} V_\alpha = 0 \ . \tag{8.20}$$

4. Symmetry under exchange of the two pairs of indices

$$R_{\alpha\beta\gamma\delta} = R_{\gamma\delta\alpha\beta} \tag{8.21}$$

This identity, stating that the Riemann tensor is symmetric in its two pairs of indices, is not an independent symmetry but can be deduced from the three other symmetries by some not particularly interesting algebraic manipulations. One (quite possibly not optimal or minimal) possibility is

$$\begin{aligned} R_{\gamma\delta\alpha\beta} &\overset{(3)}{=} -R_{\gamma\alpha\beta\delta} - R_{\gamma\beta\delta\alpha} \\ &\overset{(2)}{=} R_{\alpha\gamma\beta\delta} + R_{\beta\gamma\delta\alpha} \\ &\overset{(3)}{=} -R_{\alpha\delta\gamma\beta} - R_{\alpha\beta\delta\gamma} - R_{\beta\alpha\gamma\delta} - R_{\beta\delta\alpha\gamma} \\ &\overset{(1,2)}{=} 2R_{\alpha\beta\gamma\delta} + R_{\delta\alpha\gamma\beta} + R_{\delta\beta\alpha\gamma} \\ &\overset{(3)}{=} 2R_{\alpha\beta\gamma\delta} - R_{\delta\gamma\beta\alpha} \\ &\overset{(1,2)}{=} 2R_{\alpha\beta\gamma\delta} - R_{\gamma\delta\alpha\beta} \ , \end{aligned} \tag{8.22}$$

from which the claim follows.

Slightly more elegant (but equally obtuse) is the following argument.[20] Consider the matrix

$$\mathcal{R} = \begin{pmatrix} R_{\alpha\beta\gamma\delta} & R_{\alpha\delta\beta\gamma} & R_{\alpha\gamma\delta\beta} \\ R_{\delta\alpha\beta\gamma} & R_{\delta\gamma\alpha\beta} & R_{\delta\beta\gamma\alpha} \\ \vdots & \dots & \dots \end{pmatrix} \tag{8.23}$$

where the first column consists of the 4 cyclic permutations of all 4 indices of the Riemann tensor, while each row consists of the 3 cyclic permutations of the last 3 indices. Thus the sum σ_k of the entries of the k'th row is zero for all k (by symmetry (3)),

$$\sigma_k = \sum_{l=1}^{3} \mathcal{R}_{kl} \overset{(3)}{=} 0 \ , \tag{8.24}$$

while

$$\sigma_1 + \sigma_2 - \sigma_3 - \sigma_4 \overset{(1,2)}{=} 2R_{\alpha\beta\gamma\delta} - 2R_{\gamma\delta\alpha\beta} \ . \tag{8.25}$$

[20] See e.g. D. Bleecker, *Gauge Theory and Variational Principles*.

We can now count how many independent components the Riemann tensor really has. (1) implies that the second pair of indices can only take $N = (4\times 3)/2 = 6$ independent values. (2) implies the same for the first pair of indices. (4) thus says that the Riemann curvature tensor behaves like a symmetric (6×6) matrix and therefore has $(6\times 7)/2 = 21$ components. We now come to the remaining condition (3): if two of the indices in (3) are equal, (3) is equivalent to (4) and (4) we have already taken into account. With all indices unequal, (3) then provides one and only one more additional constraint. We conclude that the total number of independent components is 20.

REMARKS:

1. Note that this agrees precisely with our previous counting in section 3.6 of how many of the second derivatives of the metric cannot be set to zero by a coordinate transformation: the second derivative of the metric has 100 independent components, to be compared with the $4 \times (4 \times 5 \times 6)/(2 \times 3) = 80$ components of the third derivatives of the coordinates. This also leaves 20 components. We thus see very explicitly that the Riemann curvature tensor contains all the coordinate independent information about the geometry up to second derivatives of the metric. In fact, it can be shown that in a Riemann normal coordinate system one has

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + 0 + \tfrac{1}{3} R_{\mu\lambda\sigma\nu}(x_0)(x-x_0)^\lambda (x-x_0)^\sigma + \mathcal{O}((x-x_0)^3) \quad . \tag{8.26}$$

2. Just for the record, I note here that in general dimension $D = d+1$ the Riemann tensor has $D^2(D^2-1)/12$ independent components. This number arises as

$$\begin{aligned} \frac{D^2(D^2-1)}{12} &= \frac{N(N+1)}{2} - \binom{D}{4} \\ N &= \frac{D(D-1)}{2} \end{aligned} \tag{8.27}$$

and describes (as above) the number of independent components of a symmetric $(N\times N)$-matrix, now subject to $\binom{D}{4}$ conditions which arise from all the possibilities of choosing 4 out of D possible distinct values for the indices in (3). Just as for $D = 4$, this number of components of the Riemann tensor coincides with the number of second derivatives of the metric minus the number of independent components of the third derivatives of the coordinates determined in (3.179),

$$\frac{D(D+1)}{2} \times \frac{D(D+1)}{2} - D \times \frac{D(D+1)(D+2)}{2\times 3} = \frac{D^2(D^2-1)}{12} \quad . \tag{8.28}$$

For $D = 2$ this formula predicts one independent component, and this is as it should be. Rather obviously the only independent non-vanishing component of the Riemann tensor in this case is R_{1212}. We will discuss curvature in 2 dimensions in more detail in sections 8.6 and 11.3 below.

Finally, a word of warning: there are a large number of sign conventions involved in the definition of the Riemann tensor (and its contractions we will discuss below), so whenever reading a book or article, in particular when you want to use results or equations presented there, make sure what conventions are being used and either adopt those or translate the results into some other convention. As a check: the conventions used here are such that $R_{\phi\theta\phi\theta}$ as well as the curvature scalar (to be introduced below) are *positive* for the standard metric on the two-sphere.

8.4 Tidal Forces: Influence of Curvature on Particle Trajectories

In a certain sense the main effect of curvature (or gravity) is that initially parallel trajectories of freely falling non-interacting particles (dust, pebbles,...) do not remain parallel, i.e. that gravity is an attractive force that has the tendency to focus matter. This statement find its mathematically precise formulation in equations describing the influence of space-time curvature on the behaviour of (families of) geodesics.

Let us, as we will need this later anyway, recall first the situation in the Newtonian theory. One particle moving under the influence of a gravitational field is governed by the equation

$$\tfrac{d^2}{dt^2}x^i = -\partial^i\phi(x) \ , \tag{8.29}$$

where ϕ is the potential. Now consider a family of particles, or just two nearby particles, one at $x^i(t)$ and the other at $x^i(t) + \delta x^i(t)$. The other particle will of course obey the equation

$$\tfrac{d^2}{dt^2}(x^i + \delta x^i) = -\partial^i\phi(x + \delta x) \ . \tag{8.30}$$

From these two equations one can deduce an equation for δx itself, namely

$$\tfrac{d^2}{dt^2}\delta x^i = -\partial^i\partial_j\phi(x)\delta x^j \ . \tag{8.31}$$

It describes the effect of *gravitational tidal forces* (the gradient of the gravitational force) on a family of particles moving in a gravitational field.

In particular, when there is no gravitational force, and the trajectories are straight lines, one has

$$\tfrac{d^2}{dt^2}\delta x^i = 0 \quad \Rightarrow \quad \delta x^i = (\delta x^i)_0 + (\delta v^i)t \ . \tag{8.32}$$

Thus one recovers Euclid's parallel axiom, that two straight lines intersect at most once (for suitable choices of $\delta v^i \neq 0$) and that they never intersect when they are initially parallel ($\delta v^i = 0$). Any departure from this equation or its Minkowskian counterpart

$$\tfrac{d^2}{d\tau^2}\delta\xi^a = 0 \ , \tag{8.33}$$

will therefore indicate a departure from Euclidean geometry!

It is the counterpart of (8.31) that we will be seeking in the context of General Relativity. One derivation of this can be modelled on the Newtonian derivation above. It is elementary but looks non-covariant (and therefore somewhat messy) at intermediate stages of the calculation (see section 12.1 for a manifestly covariant derivation).

The starting point is of course the geodesic equation for x^μ and for its nearby partner $x^\mu + \delta x^\mu$,

$$\frac{d^2}{d\tau^2}x^\mu + \Gamma^\mu_{\ \nu\lambda}(x)\frac{d}{d\tau}x^\nu \frac{d}{d\tau}x^\lambda = 0 \quad , \tag{8.34}$$

and

$$\frac{d^2}{d\tau^2}(x^\mu + \delta x^\mu) + \Gamma^\mu_{\ \nu\lambda}(x+\delta x)\frac{d}{d\tau}(x^\nu + \delta x^\nu)\frac{d}{d\tau}(x^\lambda + \delta x^\lambda) = 0 \quad . \tag{8.35}$$

As above, from these one can deduce an equation for δx, namely

$$\frac{d^2}{d\tau^2}\delta x^\mu + 2\Gamma^\mu_{\ \nu\lambda}(x)\frac{d}{d\tau}x^\nu \frac{d}{d\tau}\delta x^\lambda + \partial_\rho \Gamma^\mu_{\ \nu\lambda}(x)\delta x^\rho \frac{d}{d\tau}x^\nu \frac{d}{d\tau}x^\lambda = 0 \quad . \tag{8.36}$$

Now this does not look particularly covariant. Thus instead of in terms of $d/d\tau$ we would like to rewrite this in terms of the covariant operator D_τ, with

$$D_\tau \delta x^\mu = \frac{d}{d\tau}\delta x^\mu + \Gamma^\mu_{\ \nu\lambda}\frac{dx^\nu}{d\tau}\delta x^\lambda \quad . \tag{8.37}$$

Calculating $(D_\tau)^2\delta x^\mu$, replacing $\ddot{x}^\mu$ appearing in that expression by $-\Gamma^\mu_{\ \nu\lambda}\dot{x}^\nu\dot{x}^\lambda$ (because x^μ satisfies the geodesic equation) and using (8.36), one eventually finds the nice covariant *geodesic deviation equation*

$$(D_\tau)^2\delta x^\mu = R^\mu_{\ \nu\lambda\rho}\dot{x}^\nu\dot{x}^\lambda\delta x^\rho \tag{8.38}$$

REMARKS:

1. This shows very clearly that curvature, as captured by the Riemann curvature tensor, leads to non-Euclidean geometry in which e.g. the parallel axiom is not necessarily satisifed.

2. In general, solutions to the geodesic deviation equation are called *Jacobi fields*. They describe the difference between the given geodesic and a (hypothetical) infinitely close neighbouring geodesic.

3. Clearly the present derivation of this result leaves something to be desired. It is also possible to give a manifestly covariant, and thus perhaps slightly more satisfactory, derivation of the above geodesic deviation equation, and we will return to this in section 12.1.

8.5 Contractions of the Riemann Tensor: Ricci Tensor and Ricci Scalar

The Riemann tensor, as we have seen, is a four-index tensor. For many purposes this is not the most useful object, but we can create new tensors by contractions of the Riemann tensor. Due to the symmetries of the Riemann tensor, there is essentially only one possibility, namely the *Ricci tensor*

$$R_{\mu\nu} := R^{\lambda}{}_{\mu\lambda\nu} = g^{\lambda\sigma} R_{\sigma\mu\lambda\nu} \quad . \tag{8.39}$$

It arises naturally from the definition (8.2) of the Riemann tensor in terms of commutators of covariant derivatives, when one considers a contracted commutator,

$$[\nabla_\mu, \nabla_\nu] V^\lambda = R^{\lambda}{}_{\sigma\mu\nu} V^\sigma \quad \Rightarrow \quad [\nabla_\mu, \nabla_\nu] V^\mu = R^{\mu}{}_{\sigma\mu\nu} V^\sigma \equiv R_{\sigma\nu} V^\sigma \quad . \tag{8.40}$$

In particular, this identity explains why the Maxwell equations in the covariant Lorenz gauge (6.46) take the non-minimally coupled form (6.47).

It follows from the symmetries of the Riemann tensor that $R_{\mu\nu}$ is symmetric. Indeed

$$R_{\nu\mu} = g^{\lambda\sigma} R_{\sigma\nu\lambda\mu} = g^{\lambda\sigma} R_{\lambda\mu\sigma\nu} = R^{\sigma}{}_{\mu\sigma\nu} = R_{\mu\nu} \quad . \tag{8.41}$$

Thus, for $D = 4$, the Ricci tensor has 10 independent components, for $D = 3$ it has 6, while for $D = 2$ there is only 1 because there is only one independent component of the Riemann curvature tensor to start off with.

There is one more contraction of the Riemann tensor we can perform, namely on the Ricci tensor itself, to obtain what is called the *Ricci scalar* or *curvature scalar*

$$R := g^{\mu\nu} R_{\mu\nu} \quad . \tag{8.42}$$

Remarks:

1. One might have thought that at least in four dimensions there is another way of constructing a (pseudo-)scalar, by contracting the Riemann tensor with the Levi-Civita tensor, but
$$\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = 0 \tag{8.43}$$
because of the Bianchi identity (cyclic symmetry of the Riemann tensor).

2. Note that for $D = 2$ the Riemann curvature tensor has as many independent components as the Ricci scalar, namely one, and that for $D = 3$ the Ricci tensor has as many components as the Riemann tensor, namely 6. Thus in $D = 2$ one can express the entire Riemann tensor in terms of the Ricci scalar (and the metric) alone, and one has
$$D = 2 : \quad R_{\alpha\beta\gamma\delta} = \tfrac{1}{2}(g_{\alpha\gamma} g_{\beta\delta} - g_{\alpha\delta} g_{\beta\gamma}) R \tag{8.44}$$

(we will establish this relation in section 11.3, see (11.29)), while in $D = 3$ one has

$$D = 3: \quad R_{\alpha\beta\gamma\delta} = (g_{\alpha\gamma}R_{\beta\delta} + R_{\alpha\gamma}g_{\beta\delta} - g_{\alpha\delta}R_{\beta\gamma} - R_{\alpha\delta}g_{\beta\gamma}) + \tfrac{1}{2}(g_{\alpha\delta}g_{\beta\gamma} - g_{\alpha\gamma}g_{\beta\delta})R \tag{8.45}$$

(and we will prove this in section 11.4).

3. It is thus only in four (and more) dimensions that there are strictly less components of the Ricci tensor than of the Riemann tensor. This has profound implications for the dynamics of gravity in these dimensions. In fact, we will see that it is only in dimensions $D > 3$ that gravity becomes truly dynamical, where empty space can be curved, where gravitational waves can exist etc.

4. Contracting (8.8), one consequence of the symmetry of the Ricci tensor is the useful general result

$$[\nabla_\mu, \nabla_\nu]T^{\mu\nu} = R_{\mu\nu}(T^{\mu\nu} - T^{\nu\mu}) = 0 \tag{8.46}$$

for any tensor $T^{\mu\nu}$. If $T^{\mu\nu} = F^{\mu\nu}$ is anti-symmetric, $F^{\mu\nu} = -F^{\nu\mu}$, it is not necessary to take the commutator, so one also has

$$F^{\mu\nu} = -F^{\nu\mu} \quad \Rightarrow \quad \nabla_\mu(\nabla_\nu F^{\mu\nu}) = \tfrac{1}{2}[\nabla_\mu, \nabla_\nu]F^{\mu\nu} = 0 \ . \tag{8.47}$$

Note that this can also be deduced (without knowing anything about curvature in general or the Ricci tensor in particular) from the general expression (5.66) for the divergence of an anti-symmetric tensor,

$$\nabla_\mu F^{\mu\nu} = g^{-1/2}\partial_\mu(g^{1/2}F^{\mu\nu}) \quad \Rightarrow \quad \nabla_\nu\nabla_\mu F^{\mu\nu} = g^{-1/2}\partial_\nu\partial_\mu(g^{1/2}F^{\mu\nu}) = 0 \ . \tag{8.48}$$

This is how we had shown in section 6.6 that the Maxwell equations imply covariant current conservation,

$$\nabla_\mu F^{\mu\nu} = -J^\nu \quad \Rightarrow \quad \nabla_\nu J^\nu = 0 \ . \tag{8.49}$$

Now we see that we can alternatively directly use the identity (8.47) to arrive at this result.

5. There are other scalars that can be built from the curvature tensor, but these are necessarily of higher order in the curvature tensor, such as (trivially) R^2 or (somewhat less trivially) $R_{\mu\nu}R^{\mu\nu}$ or the square of the Riemann tensor, the so-called *Kretschmann scalar*

$$K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \ . \tag{8.50}$$

Analogously, scalars can be built from higher powers of the Riemann tensor and or from powers of covariant derivatives of the Riemann tensor ($\Box R$ being the simplest example).

6. Such scalars are useful in analysing a given metric because, since they are scalars they are invariant under coordinate transformations. Thus they directly provide coordinate-invariant information about a metric. For instance if K is singular at some point in some coordinate system then it will be singular at that point in all coordinate systems, and thus such a singularity is not an artefact of a bad choice of coordinate system but a property of the space(-time) itself described by that metric. A prominent example is the singularity at the origin $r = 0$ of the Schwarzschild metric, unambiguously unveiled by the singularity of its Kretschmann scalar (27.163).

7. Contracting (8.40) with V^ν, one finds

$$V^\nu \nabla_\mu \nabla_\nu V^\mu - V^\nu \nabla_\nu \nabla_\mu V^\mu = R_{\mu\nu} V^\mu V^\nu \ . \tag{8.51}$$

Rewriting the first term as

$$V^\nu \nabla_\mu \nabla_\nu V^\mu = \nabla_\mu (V^\nu \nabla_\nu V^\mu) - (\nabla_\mu V^\nu)(\nabla_\nu V^\mu) \tag{8.52}$$

this identity can be written as

$$V^\nu \nabla_\nu (\nabla_\mu V^\mu) + (\nabla_\mu V_\nu)(\nabla^\nu V^\mu) - \nabla_\mu (V^\nu \nabla_\nu V^\mu) + R_{\mu\nu} V^\mu V^\nu = 0 \ . \tag{8.53}$$

This is a very useful and versatile "master equation" which provides valuable information about the relation between vector fields and curvature when specialised e.g. to geodesic vector fields, $V^\nu \nabla_\nu V^\mu = 0$, or Killing vector fields, $\nabla_\mu V_\nu = -\nabla_\nu V_\mu$ and $\nabla_\mu V^\mu = 0$. Various specialisations of this equation will therefore appear later on in these notes, and even though we will then usually rederive them from scratch in the case at hand, it is good to keep in mind that e.g. (12.22) (our starting point for the discussion of the Raychaudhuri equation in section 12.2) and (13.12) (a useful identity relating Killing vectors and curvature) are special cases of (8.53).

8. As an a(far)side, and as an illustration of what one can do with (8.53), assume that V is such that its curl $\nabla_\mu V_\nu - \nabla_\nu V_\mu = 0$ and its divergence $\nabla_\mu V^\mu = 0$ are zero. Locally, the first condition has the solution $V_\mu = \partial_\mu f$, and then the second condition says that $\Box f = 0$, i.e. that f is *harmonic*. Therefore let us call a V^μ *harmonic* if it satisfies the above two conditions (to the mathematically more sophisticated: yes, I know that this is backwards, but we will specialise to the compact Riemannian case below ...).

For V harmonic in this sense, (8.53) reduces to

$$(\nabla_\mu V_\nu)(\nabla^\mu V^\nu) + R_{\mu\nu} V^\mu V^\nu = \nabla_\mu (V^\nu \nabla_\nu V^\mu) \ . \tag{8.54}$$

The simplest (albeit perhaps not of most direct relevance for physics) situation where one can deduce something of substance from this equation is when one

has a Riemannian (i.e. positive-definite) metric and the space one is dealing with is compact, without boundary. Then (a) the first term is non-negative, and (b) upon integration over the space the total derivative term on the right-hand side gives zero upon use of the Gauss theorem (5.63) (discussed in some more detail in section 16.3).

This implies that for a harmonic V to exist on such a space, the integral of $R_{\mu\nu}V^\mu V^\nu$ must be non-positive. In particular,

- if the Ricci tensor is positive (as a quadratic form), there are no harmonic vector fields at all,
- and if $R_{\mu\nu}V^\mu V^\nu$ =0, then a harmonic vector field is necessarily covariantly constant, $\nabla_\mu V_\nu = 0$.

In more mathematical terms this means that the first Betti number of a compact manifold admitting a metric with positive Ricci curvature is equal to zero. A variant of this kind of argument for Killing vectors will be given in section 13.3.[21]

8.6 Example: Curvature Tensor of the 2-Sphere

To see how calculations of the curvature tensor can be done in practice, let us work out the example of the two-sphere of unit radius, i.e. with line element

$$ds^2 = d\theta^2 + \sin^2\theta d\phi^2 \equiv g_{ab}dx^a dx^b \quad . \tag{8.55}$$

We already know that the non-zero Christoffel symbols necessarily have two ϕ-indices and one θ-index (from $g_{\phi\phi} = \sin^2\theta$), and are given by

$$\Gamma^\phi_{\phi\theta} = \cot\theta \quad , \qquad \Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta \quad . \tag{8.56}$$

We also know that the Riemann curvature tensor has only one independent component. Let us therefore work out $R^\theta_{\phi\theta\phi}$. From the definition we find

$$R^\theta_{\phi\theta\phi} = \partial_\theta\Gamma^\theta_{\phi\phi} - \partial_\phi\Gamma^\theta_{\theta\phi} + \Gamma^\theta_{\theta c}\Gamma^c_{\phi\phi} - \Gamma^\theta_{\phi c}\Gamma^c_{\theta\phi} \quad . \tag{8.57}$$

The second and third terms are manifestly zero, and we are left with

$$R^\theta_{\phi\theta\phi} = \partial_\theta(-\sin\theta\cos\theta) + \sin\theta\cos\theta\cot\theta = \sin^2\theta \quad . \tag{8.58}$$

Thus we have

$$\begin{aligned} R^\theta_{\phi\theta\phi} &= R_{\theta\phi\theta\phi} = \sin^2\theta \\ R^\phi_{\theta\phi\theta} &= 1 \quad . \end{aligned} \tag{8.59}$$

[21] Theorems of this kind, and proved with the help of these kinds of techniques, were pioneered by S. Bochner and K. Yano in the 1930s - 1950s. See e.g. S. Bochner, K. Yano, *Curvature and Betti Numbers* or W. Poor, *Differential Geometric Structures* for more details, rigour and applications.

Therefore the Ricci tensor R_{ab} has the components

$$\begin{aligned} R_{\theta\theta} &= 1 \\ R_{\theta\phi} &= 0 \\ R_{\phi\phi} &= \sin^2\theta \ . \end{aligned} \tag{8.60}$$

These equations can succinctly be written as

$$R_{ab} = g_{ab} \ , \tag{8.61}$$

showing that the standard metric on the two-sphere is what we will later call an *Einstein metric*. The Ricci scalar R is

$$R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = 1 + \frac{1}{\sin^2\theta}\sin^2\theta = 2 \ . \tag{8.62}$$

In particular, we have here our first concrete example of a space with non-trivial, in fact positive, curvature.

The result for the Riemann tensor can be written succinctly as

$$R^a{}_{bcd} = \delta^a{}_c g_{bd} - \delta^a{}_d g_{bc} \ , \tag{8.63}$$

which also immediately implies (8.61),

$$R_{bd} = R^a{}_{bad} = g_{bd} \ . \tag{8.64}$$

We will see later on, in section 14, that this form of the curvature tensor, or its equivalent,

$$R_{abcd} = g_{ac}g_{bd} - g_{ad}g_{bc} \ , \tag{8.65}$$

is characteristic of the curvature tensor of the sphere in any dimension.

8.7 More Examples: Curvature Tensor and Polar/Spherical Coordinates

We now turn to some variations of the above theme (and some other generalisations are discussed in section 11.3 below).

1. First of all, let us address the question what is the curvature (scalar) of a sphere of radius L, i.e. of the space with line element

$$ds^2 = L^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{8.66}$$

   There are at least 3 ways to answer this question:

   - The first is to simply and blindly redo the above calculations in this case and to see what one gets.

- Alternatively, and somewhat more insightfully, rather than redoing the calculation in that case one can argue as follows. Let us observe first of all that the Christoffel symbols are invariant under constant rescalings of the metric because they are schematically of the form $g^{-1}\partial g$. Therefore the Riemann curvature tensor, which only involves derivatives and products of Christoffel symbols, is also invariant. Hence the Ricci tensor, which is just a contraction of the Riemann tensor, is also invariant:

$$g_{ab} \to L^2 g_{ab} \quad \Rightarrow \quad \Gamma^a_{\ bc} \to \Gamma^a_{\ bc} \quad \Rightarrow \quad R^a_{\ bcd} \to R^a_{\ bcd} \quad \Rightarrow \quad R_{bd} \to R_{bd} \ . \tag{8.67}$$

  However, to construct the Ricci scalar, one needs the inverse metric. This introduces an explicit L-dependence and the result is that the curvature scalar of a sphere of radius L is $R = 2/L^2$,

$$R(L^2 g_{ab}) = L^{-2} R(g_{ab}) = 2/L^2 \ . \tag{8.68}$$

  In particular, the curvature scalar of a large sphere is smaller than that of a small sphere, something which makes intuitve sense, a very large sphere locally "looking flatter" than a small sphere. However, one should use this intuition with care since, as we have seen, e.g. the Ricci tensor is independent of the size of the sphere.

- Finally, this result could also have been obtained on purely dimensional grounds. The curvature scalar is constructed from second derivatives of the metric. Hence it has length-dimension (-2). Therefore for a sphere of radius L, R has to be proportional to $1/L^2$. Comparing with the known result for $L = 1$ determines $R = 2/L^2$, as before.

2. Now let us consider, instead of the unit 2-sphere, the unit hyperboloid H^2 with metric (2.31)

$$ds^2(H^2) = d\sigma^2 + \sinh^2\sigma \, d\phi^2 \ . \tag{8.69}$$

   It is clear that, apart from a few sign changes here and there, the calculation of the Riemann curvature tensor is identical to that for S^2. These sign changes ultimately lead to the conclusion that the curvature scalar of H^2 is (-2). While the sphere is the prototypical example of a space with positive curvature, the hyperboloid is the prototypical example of a space with negative curvature.

   Instead of just doing the calculation for this specific example, it is slightly more instructive to do it for the class of metrics

$$ds^2 = dx^2 + f(x)^2 d\phi^2 \ , \tag{8.70}$$

   for some (for the time being unspecified) function $f = f(x)$. Denoting the derivative with respect to x by a prime, $f'(x) = df/dx$, one finds (this is a simple but

constructive exercise) that the Ricci tensor and Ricci scalar are

$$R_{ab} = -(f''/f)g_{ab} \quad , \quad R(x) = -2f''(x)/f(x) \ . \tag{8.71}$$

In particular, for the Euclidean metric and the standard metrics on the sphere and the hyperboloid one finds

$$f(x) = \begin{cases} x & (R^2) \\ \sin x & (S^2) \\ \sinh x & (H^2) \end{cases} \quad \Rightarrow \quad R = \begin{cases} 0 \\ +2 \\ -2 \end{cases} \tag{8.72}$$

In 2 dimensions, R is related to the *Gauss Curvature* K of a surface by $K = R/2$ so that $K = 0, \pm 1$ in these examples. See section 11.3 for some more information.

3. Now let us promote the constant radius L of S^2 to a new radial coordinate r and ask the question what is the curvature tensor of the 3-dimensional space with coordinates $(r, x^a) = (r, \theta, \phi)$ and line element

$$ds^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{8.73}$$

On the one hand, because one seems to have just added a trivial r-direction to the 2-sphere, one might be tempted to suspect that also this 3-dimensional space has non-trivial curvature. On the other hand, we recognise the above metric as the Euclidean metric on $\mathbb{R}^3$, written in spherical coordinates, and as such we expect its curvature (in fact, all components of the Riemann tensor) to be zero.

The latter expectation is of course borne out, but it is instructive to see explicitly how this cancellation occurs. In fact, it will be even more instructive to consider an apparently harmless and innocuous modification of the above metric which consists in replacing dr^2 by some constant multiple of dr^2,

$$ds^2 = p\, dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ . \tag{8.74}$$

Equivalently, up to a truly harmless overall constant factor, we can think of this as the Euclidean metric, but with the metric on the unit-sphere replaced by that of a sphere of radius $1/\sqrt{p} \neq 1$),

$$ds^2 = p\ \left(dr^2 + (r^2/p)(d\theta^2 + \sin^2\theta d\phi^2)\right) \ . \tag{8.75}$$

Proceeding in a pedestrian way, we thus have a metric $g_{\alpha\beta}$ with components

$$g_{rr} = p \quad , \quad g_{ar} = 0 \quad , \quad g_{ab} = r^2\gamma_{ab} \ , \tag{8.76}$$

with γ_{ab} in this example denoting the components of the metric on the unit sphere (and with $\gamma^a_{\ bc}$ and $r^a_{\ bcd}$ its associated Christoffel symbols and components of the Riemann curvature tensor determined in the previous section). From these we can deduce that for $r > 0$ the non-trivial Christoffel symbols are

$$\Gamma^r_{\ ab} = -p^{-1}r\gamma_{ab} \quad , \quad \Gamma^a_{\ br} = r^{-1}\delta^a_{\ b} \quad , \quad \Gamma^a_{\ bc} = \gamma^a_{\ bc} \ . \tag{8.77}$$

From this, in turn, one finds that all the components of the Riemann tensor involving at least one r-index are zero, whereas for the purely angular components one finds

$$R^a{}_{bcd} = r^a{}_{bcd} + \Gamma^a{}_{cr}\Gamma^r{}_{bd} - \Gamma^a{}_{dr}\Gamma^r{}_{bc} \ . \tag{8.78}$$

Using (8.63) and (8.77), one sees that

$$R^a{}_{bcd} = (1-p^{-1})r^a{}_{bcd} = (1-p^{-1})(\delta^a{}_c\gamma_{bd} - \delta^a{}_d\gamma_{bc}) \ . \tag{8.79}$$

Therefore precisely for $p = 1$ the two contributions to the curvature tensor indeed cancel and the curvature tensor is identically zero, as expected.

Equally interesting is the fact that for $p \neq 1$ the curvature is non-zero even away from $r = 0$ (in addition, there is a conical deficit angle singularity at $r = 0$, as in the next example below, but this shall not be our concern here). In particular it follows from the above result that the only non-vanishing components of the Ricci tensor of this 3-dimensional space are

$$R^a{}_{bcd} = (1-p^{-1})r^a{}_{bcd} \quad \Rightarrow \quad R_{bd} = (1-p^{-1})r_{bd} = (1-p^{-1})\gamma_{bd} \ . \tag{8.80}$$

Therefore also its Ricci scalar is non-zero,

$$R = g^{\alpha\beta}R_{\alpha\beta} = g^{ab}R_{ab} = 2(1-p^{-1})r^{-2} \ . \tag{8.81}$$

We also see from this that this space actually has a curvature singularity as $r \to 0$. Since the Ricci scalar is a scalar (under coordinate transformations), this divergence cannot be an artefact of a bad choice of coordinates, and indicates that there is a genuine geometric singularity for $r \to 0$.

Extended to a four-dimensional space-time metric via

$$ds^2 = -dt^2 + p\, dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ , \tag{8.82}$$

this describes the gravitational field outside a "monopole".[22]

4. As a final variation of this theme, we consider the above example in one dimension less, i.e. we look at the metric one obtains if one replaces the Euclidean metric on $\mathbb{R}^2$ written in polar coordinates by

$$dr^2 + r^2 d\phi^2 \to p\, dr^2 + r^2 d\phi^2 \ , \tag{8.83}$$

where the angle ϕ has period 2π.

In this case there is an interesting twist (pun intended) and the situation is somewhat different. Pulling out the factor of p, one sees that (up to this irrelevant overall constant factor) the metric can be written as

$$ds^2 = dr^2 + r^2 d(\phi/\sqrt{p})^2 \equiv dr^2 + C^2 r^2 d\phi^2 \ . \tag{8.84}$$

[22] M. Barriola, A. Vilenkin, *Gravitational Field of a Global Monopole*, Phys. Rev. Lett. 63 (1989) 341-343.

This would be the standard Euclidean metric on $\mathbb{R}^2$ either for $p = 1$ or if the angle ϕ had periodicity $2\pi\sqrt{p}$, but since ϕ has period 2π, this results in a misidentification of the points in a plane, like when one rolls up a flat piece of paper into a cone. Away from $r = 0$, this space is intrinsically flat (all the components of the Riemann curvature tensor are zero, as one can easily calculate - see section 11.1 for an explanation of this use of the word "intrinsic"). There is, however, a conical singularity at the tip of the cone $r = 0$, which can be thought of as a δ-function contribution to the curvature localised at $r = 0$. Extended to a four-dimensional space-time metric,

$$ds^2 = -dt^2 + dz^2 + dr^2 + C^2 r^2 d\phi^2 \quad , \tag{8.85}$$

it can be interpreted as the space-time metric of an idealised cosmic string extended in the z-direction.[23]

8.8 Bianchi Identities and the Einstein Tensor

So far, we have discussed algebraic properties of the Riemann tensor. The Riemann tensor also satisfies some differential identities which, in particular in their contracted form, will be of fundamental importance in the following.

The first identity is easy to derive. As a (differential) operator the covariant derivative clearly satisfies the Jacobi identity

$$[\nabla_{[\mu}, [\nabla_\nu, \nabla_{\lambda]}]] = 0 \tag{8.86}$$

(total anti-symmetrisation over all 3 indices). Since the commutator $[\nabla_\nu, \nabla_\lambda]$ is already anti-symmetric in the indices ν, λ, this anti-symmetrisation is equivalent to cyclic permutation of the 3 indices,

$$[\nabla_{[\mu}, [\nabla_\nu, \nabla_{\lambda]}]] = 0 \quad \Leftrightarrow \quad [\nabla_\mu, [\nabla_\nu, \nabla_\lambda]] + \circlearrowleft (\mu, \nu, \lambda) = 0 \ . \tag{8.87}$$

If you do not believe this identity (valid for any 3 associative linear operators), you can just write out the twelve relevant terms explicitly to see that there is indeed a complete

[23] It is far from straightforward, however, to find a formalism which allows one to calculate and derive the distributional Riemann tensor of this space-time - see R. Geroch, J. Traschen, *Strings and other distributional sources in general relativity*, Phys. Rev. D36 (1987) 1017-1031 for a general analysis of the problem and issues arising in this and related contexts, C. Clarke, J. Vickers, J. Wilson, *Generalized functions and distributional curvature of cosmic strings*, Class. Quantum Grav. 13 (1996) 2485-2498 for one approach (based on the Colombeau algebra of distributions), and D. Garfinkle, *Metrics with distributional curvature*, `arXiv:gr-qc/9906053` for a different approach. We will (mostly) stay away from distributional curvatures in these notes.

cancellation:

$$\begin{aligned}[\nabla_{[\mu},[\nabla_\nu,\nabla_{\lambda]}]] &\sim \nabla_\mu\nabla_\nu\nabla_\lambda - \nabla_\mu\nabla_\lambda\nabla_\nu - \nabla_\nu\nabla_\lambda\nabla_\mu + \nabla_\lambda\nabla_\nu\nabla_\mu \\ &+ \nabla_\lambda\nabla_\mu\nabla_\nu - \nabla_\lambda\nabla_\nu\nabla_\mu + \nabla_\nu\nabla_\mu\nabla_\lambda - \nabla_\mu\nabla_\nu\nabla_\lambda \\ &+ \nabla_\nu\nabla_\lambda\nabla_\mu - \nabla_\nu\nabla_\mu\nabla_\lambda - \nabla_\lambda\nabla_\mu\nabla_\nu + \nabla_\mu\nabla_\lambda\nabla_\nu \\ &= 0 \ . \end{aligned} \tag{8.88}$$

To determine the implications of this identity for the Riemann tensor, we apply it to a vector field V, say. The first term in (8.87) is

$$\begin{aligned}[\nabla_\mu,[\nabla_\nu,\nabla_\lambda]]V^\rho &= \nabla_\mu(R^\rho{}_{\sigma\nu\lambda}V^\sigma) - [\nabla_\nu,\nabla_\lambda](\nabla_\mu V^\rho) \\ &= (\nabla_\mu R^\rho{}_{\sigma\nu\lambda})V^\sigma + R^\rho{}_{\sigma\nu\lambda}\nabla_\mu V^\sigma - R^\rho{}_{\sigma\nu\lambda}\nabla_\mu V^\sigma + R^\sigma{}_{\mu\nu\lambda}\nabla_\sigma V^\rho \\ &= (\nabla_\mu R^\rho{}_{\sigma\nu\lambda})V^\sigma + R^\sigma{}_{\mu\nu\lambda}\nabla_\sigma V^\rho \ . \end{aligned} \tag{8.89}$$

Upon taking the cyclic permutations, the sum of the 2nd terms vanishes by the cyclic symmetry of the Riemann tensor, and therefore one finds

$$(\nabla_\mu R^\rho{}_{\sigma\nu\lambda})V^\sigma + \circlearrowleft(\mu,\nu,\lambda) = 0 \ . \tag{8.90}$$

Since this holds for any V, one deduces the *Bianchi identity*

$$\nabla_\mu R^\rho{}_{\sigma\nu\lambda} + \circlearrowleft(\mu,\nu,\lambda) = 0 \quad \Leftrightarrow \quad \nabla_{[\mu}R^\rho{}_{|\sigma|\nu\lambda]} = 0 \tag{8.91}$$

(where $|\sigma|$ indicates that this index is to be excluded from the anti-symmetrisation). Using the symmetry (IV) of the Riemann tensor, this can equivalently be written as

$$\nabla_{[\mu}R_{\nu\lambda]\rho\sigma} = 0 \ . \tag{8.92}$$

We will mainly be interested in a (double) contraction of this identity. To that end we write out (8.91) explicitly as

$$\nabla_\lambda R_{\alpha\beta\mu\nu} + \nabla_\nu R_{\alpha\beta\lambda\mu} + \nabla_\mu R_{\alpha\beta\nu\lambda} = 0 \ . \tag{8.93}$$

By contracting this with $g^{\alpha\mu}$ we obtain

$$\nabla_\lambda R_{\beta\nu} - \nabla_\nu R_{\beta\lambda} + \nabla_\mu R^\mu{}_{\beta\nu\lambda} = 0 \ . \tag{8.94}$$

This is not yet particularly useful. To also turn the last term into a Ricci tensor we contract once more, with $g^{\beta\lambda}$ to obtain the contracted Bianchi identity

$$\nabla_\lambda R^\lambda{}_\nu - \nabla_\nu R + \nabla_\mu R^\mu{}_\nu = 0 \ , \tag{8.95}$$

or

$$\nabla^\mu(R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R) = 0 \ . \tag{8.96}$$

The tensor appearing in this equation is the so-called *Einstein tensor* $G_{\mu\nu}$,

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R \ . \tag{8.97}$$

It is the unique divergence-free tensor that can be built from the metric and its first and second derivatives (apart from $g_{\mu\nu}$ itself, of course),

$$\nabla^\mu G_{\mu\nu} = 0 \quad , \tag{8.98}$$

and this is why it will play *the* central role in the Einstein equations for the gravitational field.

A minor *caveat* regarding the above statement about the uniqueness of the Einstein tensor is that, as it stands, it is only true in $D = 4$ space-time dimensions. In $D > 4$, there are other tensors with this property, but they are non-linear in 2nd derivatives of the metric. The uniqueness statement continues to be true for $D > 4$ if one adds the requirement that the tensor is linear in 2nd derivatives of the metric. I will briefly come back to this in the discussion of the action principle for general relativity in section 20.1.

8.9 Riemann Normal Coordinates Revisited

In section 3.6 we had introduced Riemann normal coordinates as a special class of inertial coordinate systems, based on geodesics. The main idea was to introduce new coordinates $x^\alpha \to \xi^a$ in such a way that the coordinate lines of the new coordinates ξ^a are geodesics passing through the point p at which one wants to erect this coordinate system.

In particular,

1. we considered the Taylor expansion (3.170)

   $$x^\alpha(\tau) = x_0^\alpha + \tau u_0^\alpha - \tfrac{1}{2}\tau^2 \Gamma^\alpha_{\beta\gamma}(x_0) u_0^\beta u_0^\gamma + \dots \quad , \tag{8.99}$$

   of a solution to the geodesic equation;

2. this led us to consider the coordinate transformation (3.172)

   $$x^\alpha(\xi) = x_0^\alpha + (\xi - \xi_0)^a e_a^\alpha - \tfrac{1}{2}(\xi - \xi_0)^b (\xi - \xi_0)^c \Gamma^\alpha_{\beta\gamma}(x_0) e_b^\beta e_c^\gamma + \dots \tag{8.100}$$

   which has the property that the lines

   $$\xi^a(\tau) = \xi_0^a + \tau \lambda^a \tag{8.101}$$

   are geodesics for any constant λ^a;

3. for these geodesics the geodesic equation reduces to

   $$\ddot{\xi}^a + \Gamma^a_{bc} \dot{\xi}^b \dot{\xi}^c = 0 \quad \Rightarrow \quad \Gamma^a_{bc}(\xi_0^a + \tau\lambda^a)\lambda^b\lambda^c = 0 \quad , \tag{8.102}$$

   implying at $\tau = 0$

   $$\Gamma^a_{bc}(\xi_0)\lambda^b\lambda^c = 0 \quad \forall \lambda^a \quad \Rightarrow \Gamma^a_{bc}(\xi_0) = 0 \tag{8.103}$$

   (and we will look at the implications of the next term in the Taylor expansion of (8.102) below).

Therefore the Taylor expansion of the metric around $\xi = \xi_0$ has the form

$$g_{ab}(\xi) = \eta_{ab} + \tfrac{1}{2}(\xi - \xi_0)^c(\xi - \xi_0)^d g_{ab,\,cd}(\xi_0) + \dots \quad , \tag{8.104}$$

and we will now determine the quadratic term in this expansion (and be able to express it in terms of the components of the Riemann tensor $R_{abcd}(\xi_0)$ at the point p in these coordinates). To that end we look at the next term in the Taylor expansion of (8.102). Thus we differentiate (8.102) along the geodesic, i.e. with respect to τ, and evaluate the results at $\tau = 0$ to deduce

$$\partial_d \Gamma^a{}_{bc}(\xi_0)\lambda^d\lambda^b\lambda^c = 0 \quad \forall \ \ \lambda^b \tag{8.105}$$

or, equivalently

$$\partial_{(d}\Gamma^a{}_{bc)}(\xi_0) = 0 \quad \Leftrightarrow \quad \partial_d \Gamma^a{}_{bc}(\xi_0) + \partial_b \Gamma^a{}_{cd}(\xi_0) + \partial_c \Gamma^a{}_{db}(\xi_0) = 0 \ . \tag{8.106}$$

This condition, and the analogous conditions

$$\partial_{(d\dots e}\Gamma^a{}_{bc)}(\xi_0) = 0 \tag{8.107}$$

arising from the higher-order terms in the Taylor expansion of (8.102) impose constraints on the Christoffel symbols and their derivatives that are satisfied in Riemann normal coordinates (but not in general inertial coordinate systems).

A useful way of reexpressing the condition (8.106) is the following (a certain amount of hindsight or trial-and-error is required for this): because $\Gamma^a{}_{bc}(\xi_0) = 0$, from the definition of the Riemann tensor we have

$$\begin{aligned} R^a{}_{bcd}(\xi_0) + R^a{}_{cbd}(\xi_0) &= \partial_c \Gamma^a{}_{bd}(\xi_0) - \partial_d \Gamma^a{}_{bc}(\xi_0) + \partial_b \Gamma^a{}_{cd}(\xi_0) - \partial_d \Gamma^a{}_{cb}(\xi_0) \\ &= \partial_c \Gamma^a{}_{bd}(\xi_0) + \partial_b \Gamma^a{}_{cd}(\xi_0) - 2\partial_d \Gamma^a{}_{bc}(\xi_0) \ , \end{aligned} \tag{8.108}$$

and using (8.106) this can be written as

$$\partial_d \Gamma^a{}_{bc}(\xi_0) = -\tfrac{1}{3}(R^a{}_{bcd}(\xi_0) + R^a{}_{cbd}(\xi_0)) \ . \tag{8.109}$$

On the other hand, from

$$g_{ab,\,c}(\xi) = \Gamma_{abc}(\xi) + \Gamma_{bac}(\xi) \tag{8.110}$$

we have

$$g_{ab,\,cd}(\xi) = \partial_d \Gamma_{abc}(\xi) + \partial_d \Gamma_{bac}(\xi) \tag{8.111}$$

and at ξ_0 we can use (8.109) and the symmetries of the Riemann tensor to deduce

$$\begin{aligned} g_{ab,\,cd}(\xi_0) &= -\tfrac{1}{3}(R_{abcd} + R_{acbd} + R_{bacd} + R_{bcad})(\xi_0) \\ &= -\tfrac{1}{3}(R_{acbd} + R_{bcad})(\xi_0) = -\tfrac{1}{3}(R_{acbd} + R_{adbc})(\xi_0) \ . \end{aligned} \tag{8.112}$$

We have thus found that, to quadratic order in a Taylor expansion of the metric around the origin of a Riemann normal coordinate system, the metric can be written as

$$\begin{aligned} g_{ab}(\xi) &= \eta_{ab} - \tfrac{1}{6}(R_{acbd}(\xi_0) + R_{adbc}(\xi_0))(\xi - \xi_0)^c(\xi - \xi_0)^d + \mathcal{O}(\xi^3) \\ &= \eta_{ab} - \tfrac{1}{3}R_{acbd}(\xi_0)(\xi - \xi_0)^c(\xi - \xi_0)^d + \mathcal{O}(\xi^3) \ . \end{aligned} \tag{8.113}$$

If required, higher order terms can be determined analogously with the help of the higher order terms in the Taylor expansion of (8.102), and (with a steady hand) can be expressed in terms of the covariant derivatives of the Riemann tensor at ξ_0.

8.10 Principle of Minimal Coupling Revisited

In sections 4.1 and 6.1 on the principles of general covariance and minimal coupling respectively, I mentioned that these do not necessarily fix the equations uniquely. In other words, there could be more than one generally covariant equation which reduces to a given equation in Minkowski space. Having the curvature tensor at our disposal now, we can construct examples of this kind.

Given some tensorial equation, obtained by the minimal coupling prescription, say, one can always contemplate the possibility to add additional terms to it involving the curvature tensor. Since such terms take the form of higher derivative corrections to the original equation, multiplied by appropriate dimensionful constants, one can usually get away with ignoring such terms when dealing with weak fields and other low-energy phenomena, and under such conditions the minimal coupling rule can usually be trusted. However, such terms are not negligible under extreme conditions involving e.g. very strong or strongly fluctuating gravitational fields.

An example which shows very clearly that the minimal coupling prescription, at least the way we have formulated it, is itself ambiguous is, as already briefly pointed out in section 6.6, provided by Maxwell theory. In that case, we saw that in the covariant Lorenz gauge one has (6.46)

$$\nabla_\mu A^\mu = 0 \quad \Rightarrow \quad \nabla_\mu F^{\mu\nu} = \nabla_\mu(\nabla^\mu A^\nu - \nabla^\nu A^\mu) = \Box A^\nu - [\nabla_\mu, \nabla^\nu]A^\mu \ , \tag{8.114}$$

where $\Box A^\nu = \nabla^\mu \nabla_\mu A^\nu$. It thus follows from (8.40) that the Maxwell equations in the covariant Lorenz gauge can be written as (6.47)

$$\nabla_\mu A^\mu = 0 \quad \Rightarrow \quad \nabla_\mu F^{\mu\nu} = -J^\nu \quad \rightarrow \quad \Box A^\nu - R^\nu{}_\mu A^\mu = -J^\nu \ . \tag{8.115}$$

What this shows is that "minimal coupling" all by itself is not a unique prescription, as we would have obtained (8.115) without the curvature terms by applying the minimal coupling prescription to the special relativity Maxwell equation in the Lorenz gauge, namely just $\Box A_\nu = -J_\nu$.

In the present situation, (6.47) is superior to the equation without the curvature term because

- it follows from a variational principle (involving the minimally coupled counterpart of the Maxwell action)

- and (related to this) because (8.115) implies that the current is covariantly conserved (as we had verified in section 6.6 in an arbitrary gauge), while for the equation without the curvature term covariant current conservation would then be violated by a curvature term (as can easily be verified).

Thus occasionally some such additional criteria can be used to eliminate (or reduce) the ambiguity in the minimal coupling prescription, but this need not always be the case.

As another example, consider the wave equation for a (massless, say) scalar field Φ. In Minkowski space, this is the Klein-Gordon equation which has the obvious curved space analogue (5.56)

$$\Box\Phi = 0 \tag{8.116}$$

obtained by the minimal coupling description. However, one could equally well postulate the equation

$$(\Box + \xi R)\Phi = 0 \ , \tag{8.117}$$

where ξ is a (dimensionless) constant and R is the scalar curvature. This equation is generally covariant, and reduces to the ordinary Klein-Gordon equation in Minkowski space, so this is an acceptable curved-space extension of the wave equation for a scalar field. This equation of motion arises (in D sapce-time dimensions) from the action

$$S_\xi[\phi, g_{\alpha\beta}] = -\tfrac{1}{2}\int \sqrt{g} d^D x \ \left(g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + \xi R\phi^2\right) \ , \tag{8.118}$$

exhibiting the non-minimal (yet generally covariant) coupling of the scalar field to gravity via the term $\xi R\phi^2$ (which acts as an x-dependent mass term for the scalar field).

Moreover, ξ is dimensionless, so one cannot argue that on dimensional grounds this ambiguity is irrelevant for weak fields. Indeed, one frequently postulates a specific non-zero value for ξ which makes the wave equation *conformally invariant* (invariant under position-dependent Weyl rescalings of the metric) for massless fields, and this criterion can be imposed to select a particular non-zero value for ξ (e.g. for a 4-dimensional space-time this turns out to be the value $\xi = 1/6$). This will be discussed and explained in section 22.3.

Thus in general such ambiguities are present and are something one has to live with.